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6
The control of enzyme activity

6.1 Introduction

In this chapter we shall consider the third of the remarkable properties of enzymes mentioned in Chapter 1, Section 1.3, namely the control of enzyme activity. It should be self-evident why enzyme activity needs to be controlled in living organisms, since if no such control existed all metabolic processes would tend towards states in which they were at equilibrium with their surroundings. Thus, for instance, storage of glycogen as an energy reserve in muscle would be impossible, since there is an enzyme (phosphorylase) catalysing the breakdown of glycogen into glucose units, which would then be acted upon to yield ultimately carbon dioxide and water.* Clearly, the existence of glycogen as an energy store implies that the activity of phosphorylase can be controlled in such a way as to allow the store to be called on as the situation demands. We shall see later (Section 6.4.2) that there are a number of elaborate mechanisms to effect this control in muscle, thereby allowing the rate of glycogen breakdown to vary by several orders of magnitude in different circumstances.

Regulation of enzyme activities can be brought about in two ways. Firstly, the amount of an enzyme can be altered as a result of changes in the rate of enzyme synthesis or degradation (or both). This type of regulation is suitable for long-term regulation, e.g. in response to changes in nutrients, and is discussed in more detail in Chapter 9. Secondly, the activities of enzymes already present in a cell or organism can be altered, permitting a rapid response to changes in conditions. This chapter will be confined to this second means of regulation.

We shall first outline in Section 6.2 some of the mechanisms that have been shown to be important in controlling the activity of single enzymes, and then proceed in Sections 6.3 and 6.4 to consider the control of a sequence of enzyme-catalysed reactions that constitute a metabolic pathway. In the cell, there are usually a number of interconnected metabolic pathways and the control of these presents more complex features (see Chapter 8, Section 8.3).

6.2 Control of the activities of single enzymes

The principal mechanisms that exist for control of enzyme activity can be broadly classified into two categories: (i) those mechanisms that involve a change in the covalent structure of an enzyme, and (ii) those mechanisms that involve conformational changes in an enzyme caused by the reversible binding of

* This would be the case in an oxidizing atmosphere, i.e. under aerobic conditions. Under anaerobic conditions, glucose would be converted to pyruvate and thence lactate (see Fig. 6.20).

'regulator' molecules. These categories are discussed in Sections 6.2.1 and 6.2.2, respectively.

There are various other mechanisms by which enzyme activity could be controlled, including the following.

(i) Specific inhibitor macromolecules
The best studied cases of such macromolecules act as inhibitors of proteases. We have already mentioned how bovine pancreatic trypsin inhibitor and soybean trypsin inhibitor bind at the active site of trypsin by mimicking the structure of the tetrahedral intermediate that occurs in the enzyme-catalysed reaction (see Chapter 5, Section 5.5.1.3).* The function of this type of inhibitor molecule will be mentioned in Section 6.2.1.1.

In most other cases the precise details of the binding of inhibitor to enzyme are less clear. However, the interactions can be of great physiological significance, as is illustrated by the role of the phosphatase inhibitor in the regulation of glycogen metabolism (Section 6.4.2.4). A compilation of proteins that act as inhibitors of intracellular enzymes has been made. It has been suggested that these proteins be termed 'regulatory subunits' rather than 'antizymes'.[3]

(ii) Availability of substrate or cofactor
The intracellular concentrations of the substrates for many enzymes are significantly lower than the values of the Michaelis constant (K_m). For example the concentration of D-glyceraldehyde 3-phosphate in muscle (approximately 3 $\mu mol\,dm^{-3}$) is well below the K_m of this substrate for glyceraldehyde-3-phosphate dehydrogenase (approximately 70 $\mu mol\,dm^{-3}$).[4] Under these conditions the rate of the enzyme-catalysed reaction will depend on the concentration of substrate (see Chapter 4, Fig. 4.1). If the prevailing concentration of substrate is much greater than the K_m, changes in the concentration of substrate will have little effect on the rate of reaction. This would be the case for fructose-bisphosphate aldolase in mouse brain, where the concentration of fructose-bisphosphate (approximately 200 $\mu mol\,dm^{-3}$) is well above its K_m (12 $\mu mol\,dm^{-3}$).[5]

When data on the intracellular concentrations of substrates are being used to decide whether a certain enzyme is essentially saturated, it is important to bear in mind that cells usually contain a number of distinct organelles, such as nuclei, mitochondria, lysosomes, etc. These organelles are bounded by membranes which often display selective permeability and thus enable an uneven distribution of a metabolite to exist within the various compartments of a cell. Many methods of determining the concentrations of metabolites give only an average intracellular concentration, rather than the concentration in a particular compartment where the relevant enzyme may be located. The consequences of this compartmentation within cells are discussed more fully in Chapter 8, Section 8.3.

* Some other examples of protease inhibitors include α_1-antichymotrypsin, α_1-antiprotease, and α_2-macroglobulin, which are all found in blood plasma. [1]There are also several protease inhibitors of bacterial origin, e.g. leupeptin and antipain, which are of relatively low M_r.[2]

(iii) Product inhibition
The product of an enzyme-catalysed reaction may act as an inhibitor of that reaction, so that under circumstances in which product accumulates, the enzyme is inhibited and the rate of product formation is decreased. (See for example the inhibition of hexokinase caused by D-glucose 6-phosphate discussed in Section 6.4.1.2.)

(iv) Non-enzyme-catalysed reactions
A fourth possibility that has as yet been little explored is that in certain cases the rate of non-enzyme-catalysed reactions may be rate limiting. Attention has been focused on the fact that several enzymes involved in carbohydrate metabolism display anomeric specificity, i.e. they act on only one anomer of a sugar.[6] 6-Phosphofructokinase acts on the β-anomer of D-fructose 6-phosphate (F6P), whereas fructose-bisphosphatase acts on the α-anomer of D-fructose 1,6-bisphosphate (FBP) (see Fig. 6.1).

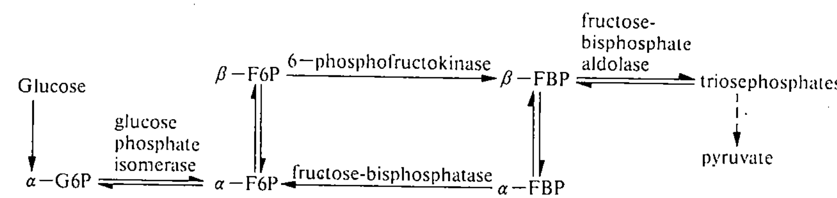


FIG. 6.1. Part of the pathways of glycolysis and gluconeogenesis, illustrating anomeric specificities of certain enzymes.[6] G6P, F6P, and FBP represent D-glucose 6-phosphate, D-fructose 6-phosphate, and D-fructose 1,6-bisphosphate, respectively.

The rates of anomerization (i.e. interconversion of α- and β-forms) of F6P and FBP are known. Calculations suggest that the non-enzyme-catalysed anomerization α-F6P $\rightarrow$ β-F6P, would not be rate limiting in glycolysis (glucose breakdown), but that the anomerization β-FBP $\rightarrow$ α-FBP might be so in gluconeogenesis (glucose synthesis).[6] These types of non-enzyme-catalysed reactions might well be rate limiting in other metabolic processes, especially when there are relatively high concentrations of the relevant enzymes, as is often the case in cells (see Chapter 8, Section 8.5).

Having briefly described these additional mechanisms of control, we shall now turn our attention to the principal means of regulating the activity of single enzymes.

6.2.1 Control of activity by changes in the covalent structures of enzymes

This category of control can be subdivided into those cases in which the change is essentially irreversible (Section 6.2.1.1) and those cases in which the change can be reversed provided that a source of energy, such as ATP, is available (Section 6.2.1.2).

6.2.1.1 *Control of activity by essentially irreversible changes in covalent structure*

It has been recognized for many years that a number of proteases are synthesized in the form of inactive precursors or zymogens, which can then be activated by the action of other proteases. The activation of chymotrypsinogen was discussed in detail in Chapter 5, Section 5.5.1.2. Some examples of enzymes that are activated in this way are given in Table 6.1. The first four enzymes mentioned in the table are synthesized as zymogens in the acinar cells of the pancreas and stored in zymogen granules. The release of the zymogens from these granules into the duodenum is under the control of hormones, primarily cholecystokinin-pancreozymin and secretin. Digestion of proteins requires the coordinated action of these various enzymes with their different specificities (see Chapter 5, Section 5.5.1.2); coordination is achieved by the common factor that the zymogens are all activated by the action of trypsin (Fig. 6.2.). In each case, the peptide bond cleaved during the activation of the zymogen lies on the C-terminal side of a lysine or arginine side-chain, in accordance with the known specificity of trypsin. For example, the activation of trypsinogen involves the removal of a hexapeptide Val-Asp-Asp-Asp-Asp-Lys from the N-terminus of the polypeptide chain. In order to account for the role of trypsin in initiating the activation of the zymogens, we must consider the involvement of another protease, enteropeptidase (previously referred to as enterokinase). Enteropeptidase, which is synthesized in the brush border of the epithelial cells of the small intestine, is an enzyme of high specificity catalysing the activation of trypsinogen as it enters the duodenum. The trypsin produced then catalyses the activation of the zymogens (Fig. 6.2). The scheme illustrates a number of important control features. It is essential that there is no premature activation of the zymogens, since this would

TABLE 6.1
Some enzymes activated by proteolytic action

Enzyme	Precursor	Function
Trypsin Chymotrypsin Elastase Carboxypeptidase	Trypsinogen Chymotrypsinogen Proelastase Procarboxypeptidase	Pancreatic secretions (see Section 6.2.1.1)
Phospholipase A_2[7]	Prophospholipase A_2	Pancreatic secretion
Pepsin	Pepsinogen	Secreted into gastric juice: most active in pH range 1–5
Thrombin	Prothrombin	Part of the blood coagulation system (see Section 6.2.1.1)
Clr̄	Clr	Part of the first component of the complement system (see Section 6.2.1.1)
Chitin synthase[8]	Zymogen	Involved in the formation of the septum during budding and cell division in yeast

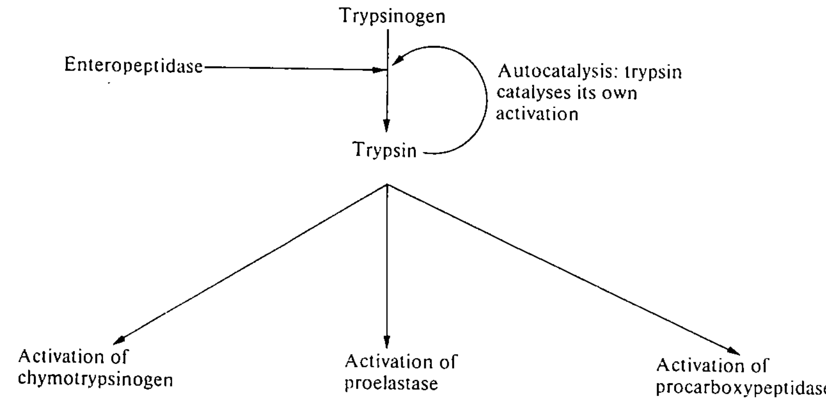


FIG. 6.2. Coordinated activation of the pancreatic proteases.

result in considerable damage to the pancreas.* In order to achieve this, the initial trigger provided by enteropeptidase is located at a site distinct from the site of production of the zymogens. The amount of active proteases produced is controlled by the amount of zymogens secreted into the duodenum, which is in turn under hormonal control. Eventually, the proteases digest themselves into amino acids and small peptides, which are reabsorbed from the intestinal tract and used for protein synthesis. The average daily production of the short-lived hydrolytic enzymes by the human pancreas is of the order of 10 g.[9]

An additional mechanism to prevent premature activation of trypsinogen is provided by the presence of a trypsin inhibitor protein in the pancreatic secretion. This protein, which is present to the extent of about 2 per cent of the protein content of the secretion (very much less in molar terms than the concentration of trypsinogen) binds very tightly to any active trypsin that might be present in the zymogen granules, and thus ensures that the activation of trypsinogen occurs only at the physiologically appropriate time and place, i.e. when it encounters enteropeptidase in the small intestine. The control of the production of enteropeptidase is less well understood and remains a problem for future work.

The scheme shown in Fig. 6.2 illustrates the principle of *signal amplification*, which will provide a recurrent theme in this Chapter. A small amount of enteropeptidase can produce rapidly a larger amount of trypsin, since one molecule of enzyme can act on many molecules of substrate in a short time. The trypsin, in turn, gives rise to an even larger quantity of active proteases.

This feature of amplification is also well illustrated by the systems involved in blood coagulation and complement activation. In the blood coagulation systems,[10] a series of enzymes and other protein factors act on each other in a

* In the condition known as acute pancreatitis, which can be lethal in severe cases, there is premature activation of the precursors of proteases and lipases synthesized in the pancreas.

sequential fashion, known as a *cascade*, eventually leading to the activation of the zymogen prothrombin to yield thrombin. Thrombin then catalyses the hydrolysis of a limited number of Arg-Gly bonds in fibrinogen to produce fibrin, which spontaneously forms an insoluble clot, strengthened by cross-linking between fibrin molecules. There are two pathways of activation of blood coagulation; extrinsic, which is triggered by a lipoprotein released from injured tissue, and intrinsic, which is triggered by contact of one of the factors with a surface such as the fibrous protein collagen that surrounds the damaged blood vessel. In both pathways, the sequence of enzyme-catalysed reactions provides a large amplification of the initial signal to ensure that sufficient fibrin is formed rapidly at the site of injury to restrict blood loss.

The complement system consists of a series of blood plasma proteins that cause damage to, and eventually lysis of, an invading organism, such as a bacterium.[11] The trigger for the system is usually provided by the aggregation of antibody molecules bound to the surface of the invading organism. This aggregation activates the components of the complement system by a sequence of activations of zymogens, leading to the rupture of the membrane of the invading organism.

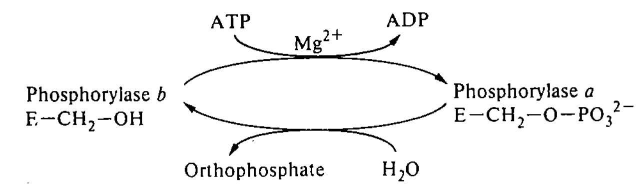
In the various cases discussed, it is clear that the sequences of enzyme-catalysed reactions provide a *rapid, amplified response* to a small initial signal. Once the zymogens have been activated and the resulting proteases have performed their particular functions, the active enzymes are degraded rather than being converted back to zymogens. It is obviously important that the systems are 'switched off' at an appropriate time, so that for instance fibrin is formed only at the site of an injury and not in the entire blood system. The 'switching off' aspect has so far received less attention than the 'switching on'.

6.2.1.2 *Control of activity by reversible changes in covalent structure*

It is now clear that a large number of enzymes exist in two forms (of different catalytic properties), which can be interconverted by the action of other enzymes. The classic example of this type of enzyme is phosphorylase, which catalyses the reaction:

$$(\text{Glycogen})_n + \text{orthophosphate} \rightleftharpoons (\text{glycogen})_{n-1} + \alpha\text{-D-glucose 1-phosphate}$$

The enzyme exists in two forms: phosphorylase *b*, which requires AMP (or a few other ligands) for activity, and phosphorylase *a*, which is active in the absence of AMP, although AMP does activate it to a small extent. The only difference in covalent structure between the *a* and *b* forms of phosphorylase is that the side-chain of Ser 14 is phosphorylated in the *a* form but not phosphorylated in the *b* form.[12] Comparison of the three-dimensional structures of the two forms shows that the conformations of the first 19 amino acids from the N-terminus are markedly different. This segment is flexible in the *b* form, but well ordered in the *a* form, with the phosphorylated side-chain of Ser 14 interacting with the positively charged side-chain of Arg 69.[12]



In the direction $b \rightarrow a$, the reaction requires ATP and Mg^{2+} and is catalysed by phosphorylase kinase; in the direction $a \rightarrow b$, the reaction is catalysed by phosphorylase phosphatase and liberates orthophosphate. If the reaction proceeds in a cyclical fashion, the net result would be the hydrolysis of ATP to ADP and orthophosphate.

A review published in 1986[13] lists over 140 enzymes and other proteins whose biological activity can be controlled by reversible covalent modification; some of these are listed in Table 6.2. (An indication of the growth of interest in this topic

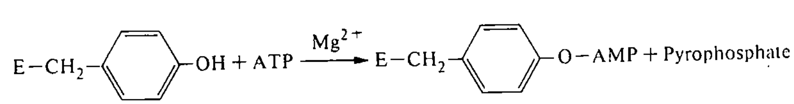
TABLE 6.2

Some enzymes and proteins whose activity can be controlled by reversible covalent modification

Enzyme or protein	Modification	Biological function
Phosphorylase	Phosphorylation	Glycogen metabolism (see Section 6.4.2)
Glycogen synthase	Phosphorylation	
Phosphorylase kinase	Phosphorylation	
Phosphatase inhibitor protein	Phosphorylation	
Fructose 2,6-bisphosphatase/ 6-phosphofructo-2-kinase	Phosphorylation	Regulation of glycolysis (see Section 6.4.1.1)
Pyruvate dehydrogenase complex (mammalian)	Phosphorylation	Entry of pyruvate into tricarboxylic-acid cycle (see Chapter 7, Section 7.7.4)
Branched chain 2-oxoacid dehydrogenase complex	Phosphorylation	Breakdown of leucine, isoleucine and valine
Acetyl-CoA carboxylase	Phosphorylation	Synthesis of fatty acids
Troponin-I	Phosphorylation	Muscular contraction
Myosin light chain	Phosphorylation	
Glutamine synthetase (mammalian)	ADP-ribosylation	glutamine acts as N donor in a wide range of biosynthetic reactions
Glutamine synthetase (*E. coli*)	Adenylylation	
RNA nucleotidyltransferase (*E. coli*)	ADP-ribosylation	On infection by T4 phage, an Arg side chain in the α-subunit of the enzyme becomes modified. This shuts off transcription of the host genes
G-protein	ADP-ribosylation	G-proteins act as transducing agents relaying the effect of hormone binding to the activation of adenylate cyclase (see Chapter 8, Section 8.4.5)

For further details, see reference 13.

can be judged from the fact that only 32 such examples were listed in a 1978 review[14].) The roles of a number of these enzymes and proteins will be discussed in connection with the regulation of glycogen metabolism (Section 6.4.2). The majority of the reversible covalent modifications involve a phosphorylation—dephosphorylation cycle, usually at a serine side-chain, but at a threonine side-chain in the case of the phosphatase-inhibitor protein (Section 6.4.2.4). In the case of glutamine synthetase, a tyrosine side-chain (Tyr 397) in each of the 12 subunits of the enzyme can be adenylylated.[15]



In addition to those listed in Table 6.2, other types of modification include methylation, acetylation, tyrosinolation and sulphation.[13] The significance of reversible covalent modification of enzymes has been discussed in a number of reviews.[14, 16, 17] Two of the more noteworthy aspects of these systems are the following.

1. Because the conversion of an enzyme from one form to another is enzyme catalysed, there can be a *rapid change* in the amount of active enzyme present, and a *large amplification* of an initial signal. The degrees of amplification that can be achieved in various circumstances are discussed in references 13 and 16.
2. Reversible modification permits much more controlled responses to different metabolic circumstances than is the case with irreversible covalent modification. In the former case, a system can be viewed as being 'poised' for response; in this state there is continual activation and inactivation of the enzyme.

Such 'poising' obviously consumes a certain amount of energy, usually in the form of ATP, but some calculations suggest that this represents only a small fraction of the total cellular turnover of ATP.[13, 18] However, this conclusion has been challenged by Goldbeter and Koshland,[19] who maintain that a significant fraction of the total energy expenditure is required for the large number of reactions that involve reversible covalent modification of proteins. Whatever the actual fraction, however, it is clear that this is the price that the organism pays in order to support its sophisticated control mechanism. Under the influence of an appropriate stimulus, a system can rapidly be activated to produce almost exclusively the active form of the required enzyme. When the stimulus is removed, the system can be converted back to its resting state (almost exclusively the inactive form of the enzyme) and it is also possible to generate a whole range of intermediate levels of response. By contrast, in the cases of irreversible covalent modification, such as blood coagulation and complement activation, there is a rapid amplified response to some emergency (e.g. injury or infection) but, after use, the whole cascade systems must be replenished, since the steps in the cascades operate in one direction only.

6.2.2 Control of activity by ligand-induced conformational changes in enzymes

We shall first describe (Section 6.2.2.1) some of the observations that led to the formulation of ideas about the role of conformational changes in controlling the activities of enzymes, and then proceed (Section 6.2.2.2) to outline some of the theoretical models that have been used to describe such control systems. A final section (6.2.2.3) will discuss the biological significance of this type of control.

6.2.2.1 *Experimental observations on regulatory enzymes**

One of the earliest studies that demonstrated the importance of ligands in controlling the activity of enzymes was that of Cori and her coworkers[20] who showed that AMP was required to bring about the breakdown of glycogen in muscle (as a result of the activation of phosphorylase *b*; see Section 6.4.2.1).

However, many of the ideas about the role of conformational changes in controlling the activities of enzymes were developed as a result of work on biosynthetic pathways in microorganisms. In the mid 1950s, for example, it was found that threonine dehydratase, the first enzyme in the pathway of isoleucine biosynthesis in *E. coli*, was strongly inhibited by the end product, isoleucine, which bears only a limited structural resemblance to the substrate or the product of the reaction[21] (Fig. 6.3). Only the first enzyme in the pathway shows this inhibition.

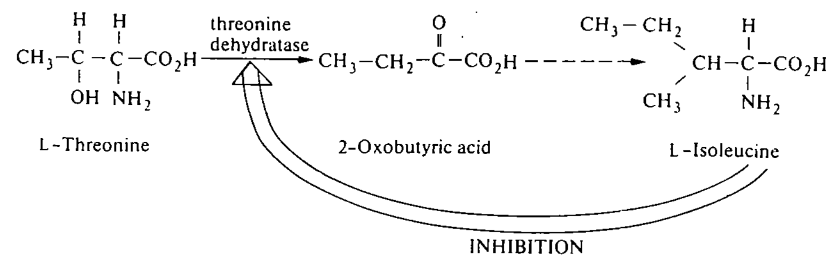
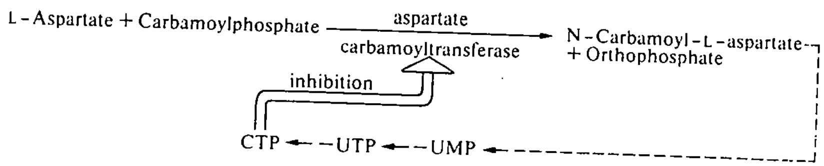


FIG. 6.3. Inhibition of threonine dehydratase, the first enzyme in the pathway of isoleucine biosynthesis in *E. coli*.

A more striking example was provided by aspartate carbamoyltransferase, the enzyme catalysing the first step in pyrimidine biosynthesis in *E. coli*, which is inhibited by the end products of the pathway, especially CTP.[22] It was also found that the purine nucleotide ATP could activate aspartate carbamoyltran-

* In one sense most, if not all, enzymes are regulatory, since their activities can be changed to some extent by, for example, changes in the concentration of substrates. etc. However, in this context, regulatory enzymes are those whose activity is controlled by conformational changes that accompany the binding of ligands. This definition includes those enzymes that exhibit sigmoid kinetics (see Fig. 6.4).



sferase, thus providing a mechanism for achieving a balance between the production of pyrimidine and purine nucleotides, which is required for the synthesis of nucleic acids.

By the early 1960s a number of examples had been described of this type of regulation, in which only the first enzyme in a pathway was subject to *feedback inhibition* by the end product of the pathway. In an incisive review[23] of these systems, Monod, Changeux, and Jacob noted several common features.

1. The ligands that were involved in regulation of enzyme activity (referred to as *regulator molecules* or *effectors*) were usually structurally distinct from the substrates or products of the relevant enzyme-catalysed reactions. It was therefore unlikely that an effector would bind at the active site of an enzyme. In most cases the inhibition caused by effectors was of a mixed type, i.e. did not conform to any of the three limiting cases of competitive, non-competitive, or uncompetitive inhibition (see Chapter 4, Section 4.3.2). This is also consistent with the proposal that the effector did not bind at the active site of the enzyme.

2. Many of the enzymes whose activity was controlled in this way did not show the normal type of kinetic behaviour, i.e. the plots of velocity against substrate concentration were of a sigmoidal (Fig. 6.4) rather than a hyperbolic (see Chapter 4, Fig. 4.1) nature. The sigmoidal shape implies that over a certain range of substrate concentration the velocity is more sensitive to substrate concentration than would be the case for an enzyme that showed normal kinetic behaviour.

3. It was often possible to distinguish between the binding of effector and the binding of substrate to a regulatory enzyme. Various physical or chemical treatments could lead to desensitization of the enzyme, i.e. loss of response to regulator molecules with no loss of catalytic activity. In the case of aspartate carbamoyltransferase, mild heat treatment yielded an active enzyme that was not inhibited by CTP. The desenstized enzyme showed normal, hyperbolic kinetic behaviour (Fig. 6.5). (It is now known that in the cases of aspartate carbamoyltransferase and cAMP-dependent protein kinase the binding sites for effectors and substrates are actually on different subunits, although this is not the case in most other regulatory enzymes.)

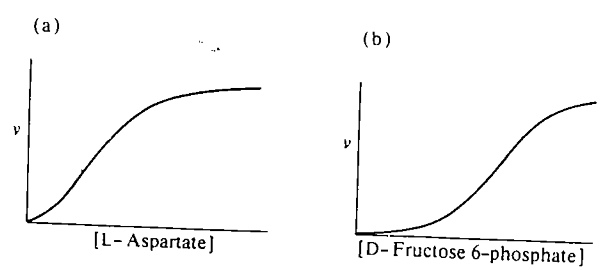


FIG. 6.4. Sigmoidal plots of velocity against substrate concentration for (a) aspartate carbamoyltransferase from *E. coli*, (b) 6-phosphofructokinase from rabbit skeletal muscle (assayed at high concentrations of ATP; see Fig. 6.21).

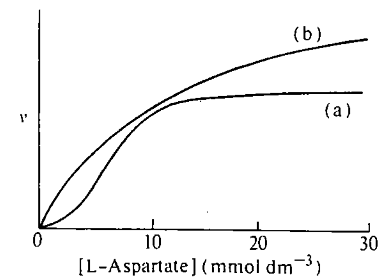


FIG. 6.5. Kinetic behaviour of aspartate carbamoyltransferase from *E. coli*. Curve (a) was obtained with enzyme assayed in the absence of CTP. Curve (b) was obtained with desensitized enzyme assayed in the absence or presence of CTP (0.2 mmol dm^{-3}).

4. In general, the regulatory enzymes were found to be oligomeric, i.e. they consisted of multiple subunits held together by non-covalent forces as described in Chapter 3, Section 3.6.

Consideration of these and other properties of regulatory enzymes led Monod, Changeux, and Jacob[23] to propose that in each case the regulator molecule (or effector) bound to a site distinct from the active site, so that the relationship between regulator molecule and substrate was an indirect or *allosteric* one (the term allosteric is derived from the Greek meaning 'different solid'). The binding of the regulator molecule to the regulator site induces a reversible conformational change in the enzyme that causes an alteration of the structure of the active site and a consequent change in the kinetic properties of the enzyme (Fig. 6.6).

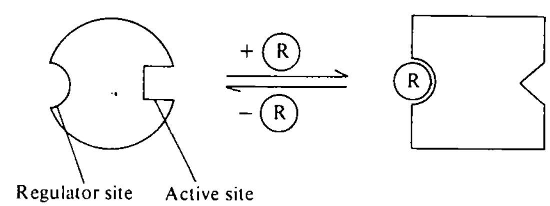


FIG. 6.6. Alteration of the structure of the active site of an enzyme by reversible binding of a regulator molecule R to a regulator site.

Although the point was not discussed at length in the 1963 review,[23] the sigmoidal nature of the kinetic plots (Fig. 6.4) was presumed to have its origin in conformational changes that were relayed between the subunits of multisubunit regulatory enzymes, i.e. the events at one active site could influence events at another. This subject will be discussed later when we consider the Monod–Wyman–Changeux model (Section 6.2.2.2).

6.2.2.2 *Theoretical models which account for the behaviour of regulatory enzymes*

A number of theoretical models have been proposed to try to account quantitatively for the behaviour of regulatory enzymes. In these models the major emphasis is placed on the interactions between the active sites on different subunits that can lead to sigmoidal kinetic plots of the type shown in Fig. 6.4. The models can then be extended to include the effects of regulator molecules on the kinetics. We shall describe four different models, those due to (i) Hill; (ii) Adair; (iii) Monod, Wyman, and Changeux; and (iv) Koshland, Némethy, and Filmer. Models (i) and (ii) are purely mathematical and make little reference to the behaviour of the enzyme or protein concerned, whereas models (iii) and (iv) make substantial reference to the behaviour of the enzyme or protein.

There are a number of other models that could account for sigmoidal kinetics, e.g. that of Rabin,[24] which involves slow conformational changes in an enzyme–substrate complex, and that of Ferdinand,[25] which applies to two-substrate reactions proceeding via random formation of a ternary complex (see Chapter 4, Section 4.3.5.1). The behaviour of glucokinase is discussed by Cornish-Bowden.[26]

Before considering these models it should be remembered that deviations from hyperbolic saturation curves had been detected in the early years of this century, when a study was made of the binding of oxygen to haemoglobin (Hb) and to myoglobin (Mb) (Fig. 6.7). It was later shown that myoglobin consisted of a single subunit with one oxygen-binding site, whereas haemoglobin consisted of four subunits each with an oxygen-binding site. The saturation curve for

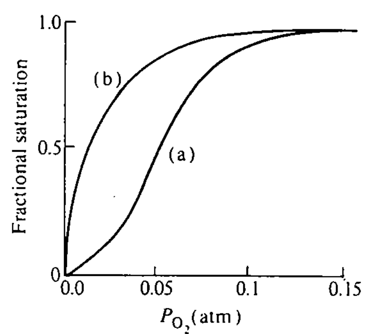


FIG. 6.7. Oxygen saturation curves for (a) haemoglobin and (b) myoglobin.

haemoglobin indicates that the binding of oxygen is *cooperative*, i.e. that binding of the first molecule of oxygen facilitates the binding of subsequent molecules of oxygen, thus giving rise to the sigmoid curve.

The hyperbolic saturation curve for myoglobin (note the analogy with the hyperbolic kinetic plot shown in Chapter 4, Fig. 4.1) can be accounted for as follows.

Consider the binding of a ligand, A, to a protein, P:

$$\mathrm{P+A} \rightleftharpoons \mathrm{PA}.$$

The dissociation constant, K, is given by

$$K=\frac{[\mathrm{P}][\mathrm{A}]}{[\mathrm{PA}]},$$

$$\therefore \quad [\mathrm{PA}]=\frac{[\mathrm{P}][\mathrm{A}]}{K}. \tag{6.1}$$

The fractional saturation, Y(i.e. the concentration of sites on the protein that are actually occupied, divided by the total concentration of such sites) is given by

$$Y=\frac{[\mathrm{PA}]}{[\mathrm{P}]+[\mathrm{PA}]}.$$

Substituting from (6.1),

$$Y=\frac{[\mathrm{A}]}{K+[\mathrm{A}]}. \tag{6.2}$$

The plot of Y against [A] is shown in Fig. 6.8.

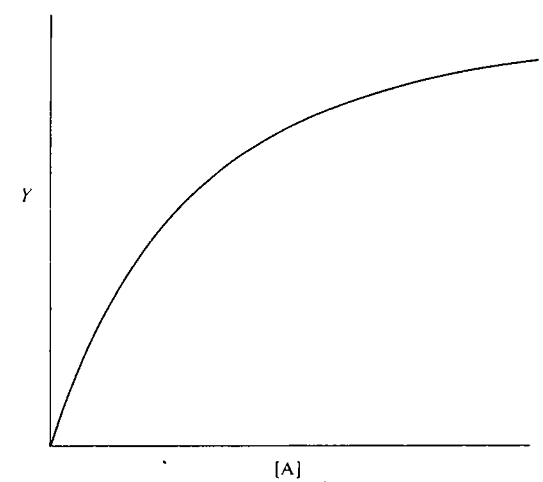


FIG. 6.8. Plot of fractional saturation, Y, against concentration of free ligand, [A], according to eqn (6.2).

(i) The Hill equation

In 1910, Hill[27] suggested the use of an equation that resembles (6.2) but has the concentration of free ligand, [A], raised to the power h:

$$Y=\frac{[A]}{K+[A]^h}. \tag{6.3}$$

An equation of this type can give rise to a sigmoidal plot of Y against [A] (see Fig. 6.9 for the case where $h=3$), similar to the observed behaviour of the haemoglobin–oxygen system (Fig. 6.7). This suggests that there may be some physical basis for the assumption underlying the derivation of the equation.

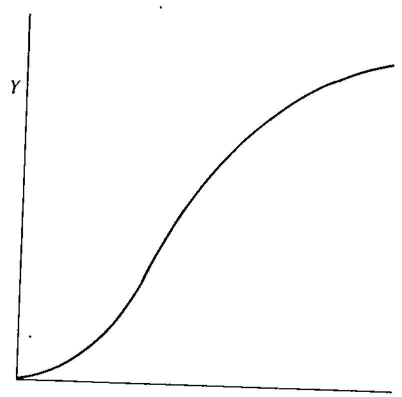


FIG. 6.9. Plot of fractional saturation, Y, against concentration of free ligand, [A], according to eqn (6.3) when $h=3$.

An equation of the type given in (6.3) can be derived by assuming that a protein, P, has n binding sites for a ligand, A, and that the binding of A is *extremely cooperative*. This means that, as soon as one site on P is occupied, the remaining sites are immediately occupied, so that essentially we can consider an equilibrium between only two species P and PA_n, with no significant contribution from intermediate complexes (e.g. PA_{n-1}):

$$P+nA \rightleftharpoons PA_n,$$

$$K=\frac{[P][A]^n}{[PA_n]}. \tag{6.4}$$

The fractional saturation, Y, is given by

$$Y=\frac{[PA_n]}{[P]+[PA_n]}.$$

Substituting from (6.4), this becomes

$$Y=\frac{[A]^n}{K+[A]^n} \quad \text{(which is of the same form as eqn (6.3))}$$

In eqn (6.3), h is known as the Hill coefficient. The value of h can be derived from experimental measurements of Y as a function of [A]. Rearranging eqn (6.3):

$$\frac{Y}{1-Y}=\frac{[A]^h}{K},$$

$$\therefore \quad \log\left[\frac{Y}{1-Y}\right]=h\log[A]-\log K. \tag{6.5}$$

Thus a plot of $\log[Y/(1-Y)]$ against $\log[A]$ is linear with a slope equal to h (Fig. 6.10).

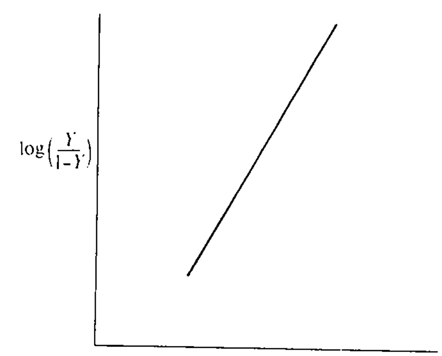


FIG. 6.10. Plot of $\log(Y/1-Y)$ against log [A] according to eqn (6.5). The slope of the line gives the value of h, the Hill coefficient.

For the binding of oxygen to haemoglobin, a straight line is obtained in this type of plot over most of the range of saturation of binding sites (at very low and very high degrees of saturation, deviations from linearity are observed because of a breakdown of the assumption underlying the derivation of eqn (6.3)). From the slope of the linear part of the plot, a value of $h=2.6$ can be derived. The value of h can be taken as a measure of the strength of the cooperativity in ligand binding. If h is found to be equal to the number of binding sites, n, (four in the case of haemoglobin), this would imply extreme cooperativity in ligand binding, i.e. that intermediate species of the type PA_{n-1} make no significant contribution. In most cases, h is found to be less than the number of binding sites and is not necessarily integral. A value of h greater than 1 but less than the number of binding sites implies that the binding of ligand is cooperative (but not extremely cooperative). When h is equal to 1, there is no cooperativity in ligand binding and the saturation curve is hyperbolic (see eqn (6.2) and Fig. 6.8). If h is found to be less than 1, the binding could be described as *negatively cooperative*, i.e. the binding of the first ligand molecule discourages the binding of subsequent molecules of ligand.

An equation analogous to the Hill equation can be used to analyse enzyme kinetic data. In this case a plot of $\log[v/(V_{max}-v)]$ against log[substrate] gives a straight line, the slope of which equals h.

In summary, it can be stated that although the assumption underlying the derivation of eqn (6.3), i.e. that there is extreme cooperativity in ligand binding, is highly questionable in most cases, because intermediate complexes do make a significant contribution, the Hill coefficient, h, is nevertheless widely used as an empirical measure of the strength of cooperativity of ligand binding.

(ii) The Adair equation

In 1952, Adair[28] suggested that a more realistic description of the binding of oxygen to haemoglobin would include the various intermediate complexes. There would therefore be four equilibria, each with a characteristic dissociation constant:

$$\begin{aligned} \mathrm{Hb}+\mathrm{O_2} &\rightleftharpoons \mathrm{HbO_2} && K_1 \\ \mathrm{HbO_2}+\mathrm{O_2} &\rightleftharpoons \mathrm{Hb(O_2)_2} && K_2 \\ \mathrm{Hb(O_2)_2}+\mathrm{O_2} &\rightleftharpoons \mathrm{Hb(O_2)_3} && K_3 \\ \mathrm{Hb(O_2)_3}+\mathrm{O_2} &\rightleftharpoons \mathrm{Hb(O_2)_4} && K_4 \end{aligned}$$

The fractional saturation of sites (Y) can be expressed in terms of the concentration of free ligand ([A]) and the four dissociation constants (eqn (6.6)). For a derivation of this equation, see Appendix 6.1.

$$Y=\frac{\dfrac{[\mathrm{A}]}{K_1}+\dfrac{2[\mathrm{A}]^2}{K_1K_2}+\dfrac{3[\mathrm{A}]^3}{K_1K_2K_3}+\dfrac{4[\mathrm{A}]^4}{K_1K_2K_3K_4}}{4\left[1+\dfrac{[\mathrm{A}]}{K_1}+\dfrac{[\mathrm{A}]^2}{K_1K_2}+\dfrac{[\mathrm{A}]^3}{K_1K_2K_3}+\dfrac{[\mathrm{A}]^4}{K_1K_2K_3K_4}\right]}. \qquad (6.6)$$

From the experimental values of Y as a function of [A], it is possible to deduce the 'best fit' values of the four dissociation constants by curve-fitting procedures. The relative values of these dissociation constants (strictly speaking, of the *intrinsic* dissociation constants*) give a clue to the cooperativity of ligand binding. For a further discussion of the Adair equation references 29 and 30 should be consulted.

* In considering the binding of ligand to more than one site on a protein, statistical factors have to be taken into account. The dissociation constants (K_1, etc.) are related to intrinsic dissociation constants (K'_1, etc.) for a protein containing four sites, as follows:

$$K_1=K'_1/4, \qquad K_2=2K'_2/3, \qquad K_3=3K'_3/2, \qquad K_4=4K'_4 \quad \text{(see Appendix 6.1).}$$

If the *intrinsic* dissociation constants followed a sequence $K'_1>K'_2>K'_3>K'_4$, the binding would be said to show (positive) cooperativity at each stage.

(iii) The model of Monod, Wyman, and Changeux

In 1965, Monod, Wyman, and Changeux[31] proposed a model to account for the behaviour of allosteric proteins and enzymes, i.e. those that showed allosteric, or indirect, interactions between ligand-binding sites.* Starting from the observation that most allosteric proteins are oligomeric, that is they consist of more than one subunit, the model makes four principal statements about the symmetry and the conformation of an allosteric protein.

(a) The subunits occupy equivalent positions, so that within the oligomer, there is at least one axis of symmetry. (It now appears that the dimeric enzyme hexokinase from yeast may be an exception to this statement.[32])
(b) The conformation of each subunit depends on its interaction with the other subunits.
(c) There are two conformational states available to the oligomer. These are designated R (relaxed) and T (tense) and they differ in affinity for a given ligand.
(d) When the conformation of the protein changes from R to T (or *vice versa*), the symmetry of the oligomer is conserved.

If the conformations of the subunits in the T and R states are denoted by circles and squares, respectively, then statements (c) and (d) would mean that for a dimeric protein there is an equilibrium:

$$\underset{\mathrm{T}}{\bigcirc\!\bigcirc} \;\rightleftharpoons\; \underset{\mathrm{R}}{\square\!\square}$$

There would be no 'hybrid' species ($\square\!\bigcirc$), since in such species the symmetry of the oligomer would be lost. In other words, all the subunits change conformation in a 'concerted' fashion—this gives rise to the alternative name 'concerted' for this model.

The various equilibria involved in binding of a ligand, A, to a dimeric protein are illustrated in Fig. 6.11.

Monod, Wyman, and Changeux[31] defined two parameters ($\bar{Y}$ and $\bar{R}$) to describe the state of the allosteric system. $\bar{Y}$ is the fractional saturation of ligand sites on the protein, and $\bar{R}$ is the fraction of protein that is in the R state. The values of both $\bar{Y}$ and $\bar{R}$ are in the range from 0 to 1.

Using the equilibrium expressions, $\bar{Y}$ and $\bar{R}$ were evaluated in terms of the number of ligand-binding sites, n, the concentration of free ligand, α,† and two

* Monod, Wyman, and Changeux[31] introduced terms to define the two classes of allosteric effects. *Homotropic* effects are interactions between identical ligands, e.g. between substrate molecules bound to different subunits in an enzyme. *Heterotropic* effects are interactions between different ligands, e.g. between substrate and regulator molecule.

† For convenience, the concentration of free ligand [A] is divided by the dissociation constant of the ligand from the R state, K_R, to give a dimensionless ratio, α.

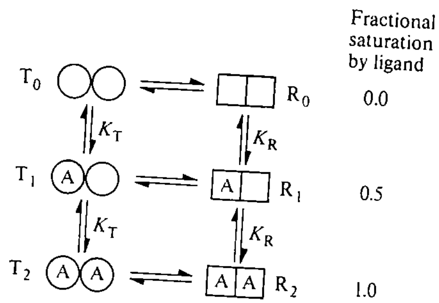


FIG. 6.11. Equilibria involved in binding of a ligand, A, to a dimeric protein according to the model of Monod, Wyman, and Changeux.[31] K_T and K_R are the intrinsic dissociation constants for ligand from the T and R states, respectively. Circles and squares represent the conformations of subunits in the T and R states, respectively.

constants L and c that could be adjusted to give the best fit to the experimental data. L is the ratio of the concentration of protein in the T state to that in the R state in the absence of ligand, i.e. $L=[T_0]/[R_0]$ in Fig. 6.11, and c is the ratio of the intrinsic dissociation constants for the ligand from the R and T states, respectively, i.e. $c=K_R/K_T$ in Fig. 6.11. The expressions derived[31] were

$$\bar{Y}=\frac{Lc\alpha(1+c\alpha)^{n-1}+\alpha(1+\alpha)^{n-1}}{L(1+c\alpha)^n+(1+\alpha)^n}, \tag{6.7}$$

$$\bar{R}=\frac{(1+\alpha)^n}{L(1+c\alpha)^n+(1+\alpha)^n}. \tag{6.8}$$

It is important to appreciate that the successive intrinsic dissociation constants for the T state are the same ($=K_T$) and that the successive intrinsic dissociation constants for the R state are the same ($=K_R$). *According to the model of Monod, Wyman, and Changeux, cooperativity in ligand binding arises because the protein is initially predominantly in one state (say T), but the ligand binds preferentially to the other state (R). As more ligand is added to the system, the protein is gradually swung over into the tighter-binding R state.* For the haemoglobin–oxygen system, the Monod, Wyman, and Changeux model gives a good fit to the experimental saturation curve with $L=9050$ and $c=0.014$,[31] i.e. the protein is initially more than 99.9 per cent in the T state, but oxygen binds over 70 times more tightly to the R state. An analysis of eqn (6.7) shows that cooperativity in ligand binding will be more pronounced when L is large and c is small.

Up to this point we have considered the Monod, Wyman, and Changeux model in terms of ligand saturation functions, which are readily applicable to a binding protein such as haemoglobin. However, in order to analyse kinetic data for an enzyme, where the observed velocity reflects both a Michaelis constant and a maximum velocity, some modifications to the basic equations are necessary. These modifications, in which dissociation constants are replaced by Michaelis constants and provision is made for the case that the two forms of the enzyme have different maximum velocities, are described by Dalziel.[33]

Monod, Wyman, and Changeux[31] proceeded to discuss the effects of regulator molecules (effectors) on the kinetic properties of allosteric enzymes in terms of the effects of these molecules on the $R \rightleftharpoons T$ equilibrium. They divided allosteric enzymes into two categories, i.e. K systems and V systems. In a K system, it was envisaged that both the substrate and the effector have different affinities for the T and R states of the enzyme. The presence of the effector modifies the affinity of the enzyme for the substrate, i.e. it affects K_m. Assuming that the substrate binds more tightly to the R state, it would follow that an effector that binds more tightly to the T state would displace the $R \rightleftharpoons T$ equilibrium towards the T state, thereby raising the K_m and decreasing the velocity at a given concentration of substrate. By contrast, an effector that bound more tightly to the R state would increase the velocity at a given concentration of substrate. Using these arguments, Monod, Wyman, and Changeux[31] were able to account for the behaviour of a K system such as aspartate carbamoyltransferase in the presence of activators (e.g. ATP) and inhibitors (e.g. CTP) (Fig. 6.12). (It should be noted that Monod, Wyman, and Changeux treated quantitatively only the case of exclusive binding, in which, for example, an inhibitor binds *only* to the T state. The more general case of non-exclusive binding, in which an inhibitor binds *preferentially* to the T state, is discussed by Rubin and Changeux[34]).

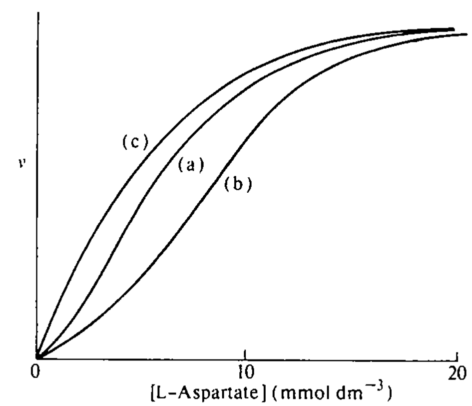
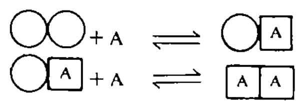


FIG. 6.12. Kinetic behaviour of aspartate carbamoyltransferase from *E. coli*. Curve (a) was obtained in the absence of effectors, curve (b) in the presence of 0.5 mmol dm^{-3} CTP, and curve (c) in the presence of 2 mmol dm^{-3} ATP.

In a V system, the substrate has the same affinity for the R and T states of the enzyme. However, the two states differ in their catalytic activities, so that the effector, by displacing the $R \rightleftharpoons T$ equilibrium, affects V_{max} rather than K_m. An example of a V system is phosphorylase b, which is activated by AMP.

(iv) The model of Koshland, Némethy, and Filmer

The model proposed in 1966 by Koshland, Némethy, and Filmer[35] to account for the ligand-binding properties of oligomeric proteins and enzymes can be seen as an extension of Koshland's 'induced-fit' hypothesis (see Chapter 5, Section 5.6) and as an attempt to define relationships between the various constants in the Adair equation (eqn (6.6)). In the 'induced-fit' hypothesis, it is postulated that the binding of substrate to an enzyme induces the appropriate conformation of the enzyme to allow catalysis to occur. Similarly, the binding of a ligand to one subunit in an oligomeric protein changes the conformation of that subunit, thereby altering the interaction of that subunit with its neighbours. In diagramatic terms the binding of a ligand A to a dimeric protein can be represented by the following equilibria:

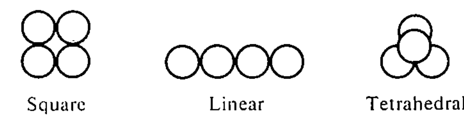


Features of the model are as follows.

(a) In the absence of ligand, the protein exists in one conformational state rather than as an equilibrium mixture of two conformational states as in the model of Monod, Wyman, and Changeux.
(b) The conformational change in the protein is *sequential*, i.e. the subunits change conformation sequentially as ligands bind, rather than in a concerted fashion as in the model of Monod, Wyman, and Changeux.
(c) The interactions between the subunits can be of a positive or a negative type, so that the binding of the second (and later) molecule of ligand can be more favourable (positive cooperativity) or less favourable (negative cooperativity) than the binding of the first molecule. By contrast, the model of Monod, Wyman, and Changeux allows only positive cooperativity in ligand binding, since the binding of ligand brings about a concerted transition of all the subunits to the form that has higher affinity for the ligand.

An expression for the fractional saturation of sites by ligand ($\bar{Y}$) is derived by considering the various equilibria with their characteristic dissociation constants, as was done for the Adair equation (eqn (6.6)). The model of Koshland, Némethy, and Filmer can then be used to derive relationships between the various dissociation constants in the Adair equation. Each dissociation constant is considered[35] to have three components:

(a) a dissociation constant characterizing the binding of ligand by the more favoured subunit conformation (□ in this scheme);
(b) an equilibrium constant for the conformational change of the subunit (i.e. ○ ⇌ □). This constant will be related to the difference in free energy between the two conformations;
(c) one or more interaction constants that depend on the degree of stability of the complex between the subunits relative to the stability of the isolated subunits. These constants depend on the geometrical arrangement of the oligomer, since this will determine the number and strength of the interactions between the subunits. For a tetrameric protein we could have the following arrangements:



and the form of the interaction constants would be different in each case.

Assuming a particular arrangement of subunits in the oligomer, an equation can be derived for the $\bar{Y}$ in terms of the free ligand concentration and the various constants referred to above. The equations derived from the model of Koshland, Némethy, and Filmer are more complex than those derived from the model of Monod, Wyman, and Changeux and elaborate curve-fitting procedures are usually necessary to derive the values of the various constants that give the best fit to the experimental data. For detailed accounts of the model of Koshland, Némethy, and Filmer, references 30 and 35–37 should be consulted.

(v) Relationships between the model of Monod, Wyman, and Changeux and the model of Koshland, Némethy, and Filmer

The concerted model of Monod, Wyman, and Changeux and the sequential model of Koshland, Némethy, and Filmer can be viewed as limiting cases of a more general model involving all possible conformational and liganded states of an oligomeric protein[38] (Fig. 6.13). These more general models have been analysed (see, for instance, references 39 and 40), but the resulting equations are extremely complex and for our purposes it is more useful to discuss only the limiting cases (i.e. the concerted and the sequential models).

A great deal of effort has been expended in attempts to show which model provides a more appropriate description of the behaviour of a particular protein or enzyme. It is true that in a number of cases, such as the binding of oxygen to haemoglobin, both models can give a reasonably satisfactory fit to the experimental data. However, there are at least three types of observation in which a distinction between the models can be made.

(a) Negative cooperativity in ligand binding. As mentioned in (iv) above the model of Monod, Wyman, and Changeux cannot account for negative cooperativity in ligand binding, whereas the model of Koshland, Némethy, Filmer can account for either positive or negative cooperativity. Thus, the observation of negative cooperativity in ligand binding can be used to exclude the use of the concerted model in a description of that particular system. Of the known examples of negative cooperativity,[41] the best-investigated is that of the binding of NAD^+ to the tetrameric enzyme glyceraldehyde-3-phosphate dehydrogenase from rabbit muscle, where the successive dissociation constants have been

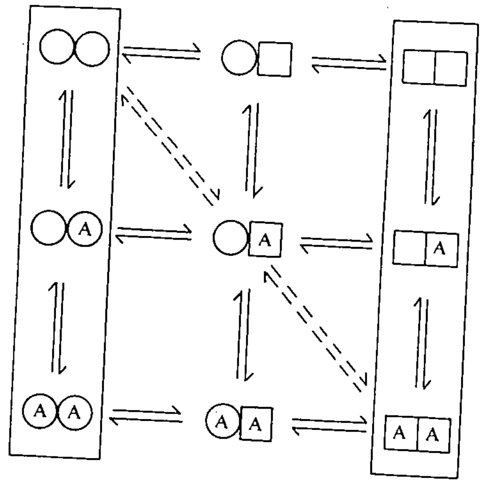


FIG. 6.13. A general model for the equilibria involved in the binding of a ligand, A, to a dimeric protein. The two vertical columns outlined show the species considered in the model of Monod, Wyman, and Changeux.[31] The dashed lines show the species considered in the model of Koshland, Némethy, and Filmer.[35]

determined as:

$$K_1 = 10^{-11}\ \text{mol dm}^{-3},\quad K_2 = 10^{-9}\ \text{mol dm}^{-3},\quad K_3 = 10^{-6}\ \text{mol dm}^{-3},$$
$$K_4 = 2 \times 10^{-5}\ \text{mol dm}^{-3}.$$

It should be pointed out that it is very difficult to distinguish between negative cooperativity and non-identical binding sites. Thus, if in a dimeric protein there were one tight-binding site and one weak-binding site for a ligand, we would appear to observe negative cooperativity in the binding of the ligand. Presumably, to prove that the binding showed negative cooperativity we would need to establish that the sites were identical in the absence of ligands—a not inconsiderable task.

In its most extreme form, negative cooperativity can be manifested as 'half-of-the-sites reactivity', where, for instance, only half of the active sites of an oligomeric enzyme are able to catalyse the reaction at the same time.[41,42] Tyrosyl-tRNA synthetase from *Bacillus stearothermophilus* shows this property (see Chapter 5, Section 5.5.5.2).

(b) Measurements of conformational changes and ligand binding. The conformational changes in enzymes and proteins that occur on binding of ligands can be monitored by a variety of means such as spectroscopic probes, changes in sedimentation coefficient, or changes in the reactivity of amino-acid side-chains.[43] Using such methods, the value of $\bar{R}$ (i.e. the fraction of protein in the R state) can be evaluated as increasing concentrations of ligand are added to the protein. The values of $\bar{Y}$ (i.e. the fraction of sites saturated) can be measured by

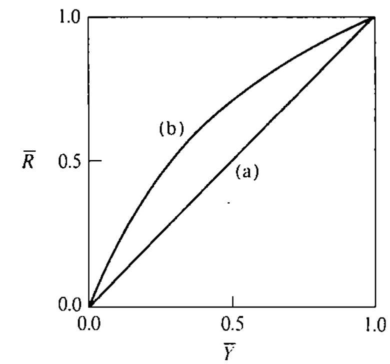


FIG. 6.14. Plots of $\bar{R}$ (the fraction of protein in the R state) against $\bar{Y}$ (the fraction of sites saturated). (a) Plot for a protein that conforms to the sequential model of Koshland, Némethy, and Filmer. (b) Typical plot for a protein conforming to the model of Monod, Wyman, and Changeux.

direct-binding studies.[44] The shape of the plot of $\bar{R}$ against $\bar{Y}$ can be used to distinguish between the concerted and sequential models (Fig. 6.14).

In the sequential model of Koshland, Némethy, and Filmer the conformational change in a subunit occurs only when a ligand binds to that subunit, and hence we would expect the plot of $\bar{R}$ against $\bar{Y}$ to be linear* (see (a) in Fig. 6.14). However, in the concerted model of Monod, Wyman, and Changeux, we would expect a non-linear plot of $\bar{R}$ against $\bar{Y}$ (see, for example, (b) in Fig. 6.14) because of the concerted nature of the $R \rightleftharpoons T$ transition. By examining the experimental data for $\bar{R}$ and $\bar{Y}$, it has been possible to classify the binding of AMP to phosphorylase *a* as being of the concerted type[45] and the binding of NAD^+ to glyceraldehyde-3-phosphate dehydrogenase from rabbit muscle as being of the sequential type.[46]

(c) Demonstration of isomerization reactions. The binding of NAD^+ to glyceraldehyde-3-phosphate dehydrogenase from yeast has been studied in great detail by the stopped-flow[47] and temperature-jump[48] methods (see Chapter 4, Section 4.4). The results were found to be in accord with predictions of the concerted model, but not with those of the sequential model, since the latter model cannot account for the observed isomerization reaction that occurs in the absence of ligand. (This isomerization reaction is denoted by the $T_0 \rightleftharpoons R_0$ equilibrium in the concerted model, see Fig. 6.11.)

(vi) Usefulness of the models

The various models described in (i)–(iv) can give only a limited insight into the nature of subunit interactions and cooperativity in ligand binding in regulatory

* It should be noted that a linear plot would only be expected in the simplest form of the sequential model, in which the ligand-induced conformational change is restricted to one subunit. If the conformational change extended across a subunit interface, deviations from linearity would be expected.

enzymes. It is necessary to have X-ray crystallographic data to understand the structural basis for these properties. At present, structural information is available on ligand-induced conformational changes in a number of enzymes, including phosphorylase *a*,[12] lactate dehydrogenase (see Chapter 5, Section 5.5.4), and glyceraldehyde-3-phosphate dehydrogenase,[49] but the best understood example is the haemoglobin–oxygen system. The elegant studies of Perutz[50] have shown how the binding of oxygen at the haem group on one subunit of deoxyhaemoglobin leads to a small movement ($\approx$0.07 nm) of the iron into the plane of the haem; this movement triggers a series of structural alterations in the protein that can be sensed at the other haem groups in the molecule that are between 2.5 and 3.7 nm distant.

In spite of these qualifying comments, it remains true that the models, especially the concerted model of Monod, Wyman, and Changeux and the sequential model of Koshland, Némethy, and Filmer, have provided a relatively simple conceptual basis for understanding the cooperative binding of ligands to proteins. The models have served to focus attention on regulatory enzymes and have stimulated a large number of elegant and detailed studies on these enzymes. Some of the experimental approaches used to study regulatory enzymes are summarized in Table 6.3.

TABLE 6.3
Some methods of study of regulatory enzymes

Type of information sought	Methods of study
Is cooperative behaviour observed in enzyme kinetic studies?	Perform assays of enzyme activity over a wide range of substrate concentrations. Examine the kinetic plots (see Chapter 4, Fig. 4.3) for non-linearity. Use the Hill plot (Fig. 6.10) to give measure of cooperativity
Is cooperative behaviour observed in ligand binding?	Use direct-binding methods,[44] e.g. equilibrium dialysis or ultracentrifugation (these methods separate free ligand from bound ligand). Test for cooperativity using, e.g. the Hill plot (Fig. 6.10) or the Scatchard plot (Fig. 6.15)
Do regulator molecules bind to a site distinct from the active site?	Test effects of regulator molecules on the kinetic plots. Are these effects consistent with binding to distinct sites? (see for example Chapter 4, Section 4.3.2.3)
	A distinction between substrate and effector sites can be made if the enzyme can be desensitized to the effects of the regulator molecule while retaining activity (Section 6.2.2.1)
Do conformational changes occur on binding of ligand?	X-ray crystallographic data gives the most detailed information, but less-direct methods, involving spectroscopic probes or measurements of the reactivity of amino-acid side-chains, can also be useful[43]
Does the enzyme consist of subunits?	The number, type, and arrangement of subunits in an enzyme can be determined from determinations of M_r, symmetry and cross-linking patterns. These methods are discussed in Chapter 3, Section 3.6

6.2.2.3 The significance of allosteric and cooperative behaviour in enzymes

Allosteric interactions between regulator molecules and substrates provide a versatile means of regulation, since they allow the activity of an enzyme to be controlled by changes in the concentrations of species other than the substrate and product of that enzyme-catalysed reaction. We have already mentioned this point in connection with the feedback inhibition of early enzymes in biosynthetic pathways (Section 6.2.2.1) and shall return to it in Section 6.3.1.

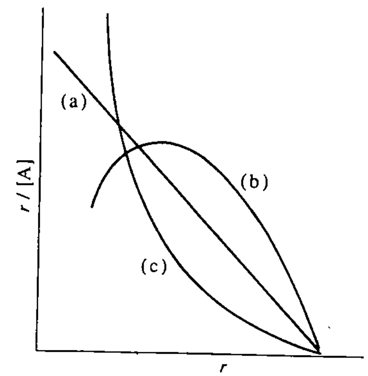


FIG. 6.15. Scatchard plots of ligand-binding data (see Appendix 6.1): r represents the number of molecules of A bound per molecule of P and [A] equals the concentration of free ligand. Case (a) corresponds to binding sites that are equivalent and independent; case (b) corresponds to positive cooperativity between ligand binding sites; case (c) corresponds to negative cooperativity between ligand binding sites (or possibly non-identical binding sites).

Enzymes that show cooperativity in kinetics have significantly different responses to changes in the concentrations of substrates from those that exhibit hyperbolic kinetics. The difference in response can be expressed in quantitative terms by calculating the change in substrate concentration required to increase the velocity from 10 per cent of V_{max} to 90 per cent of V_{max} as a function of the Hill coefficient, h, in eqn (6.9), which is analogous to the Hill equation (eqn 6.3):

$$v = \frac{V_{max}[S]^h}{K_m + [S]^h} \tag{6.9}$$

Value of h in eqn (6.9)	Required change in [S] to increase velocity from 10 per cent of V_{max} to 90 per cent of V_{max}
1	81-fold
2	9-fold
3	4.33-fold
4	3-fold
0.5	6561-fold

For an enzyme displaying hyperbolic kinetics ($h = 1$ in eqn (6.9)) an 81-fold change in substrate concentration is required; this change is reduced to just over 4-fold for an enzyme showing cooperativity in kinetics with a Hill coefficient of 3. The enhanced response could well be important in enabling an enzyme to adjust to changes in conditions, but it should be stressed that large variations in the concentrations of substrates are unlikely to occur in cells. Additional means of amplifying the response of enzymes to changes in conditions are discussed in Section 6.3.2 in connection with the regulation of metabolic pathways.

It is also evident that an enzyme that shows negatively cooperative kinetics ($h < 1$ in eqn (6.9)) has a diminished response to changes in substrate concentration (over certain ranges of concentration) compared with an enzyme that displays hyperbolic kinetics. Negative cooperativity could thus serve to insulate an enzyme from the effects of changes in substrate concentration.[41] In at least one case (the binding of NAD^+ to glyceraldehyde-3-phosphate dehydrogenase from yeast) it appears that an enzyme can show a mixture of positive and negative cooperativity in ligand binding,[41, 51] permitting further variations in the response of the enzyme to changes in substrate concentration.

The best understood example of cooperativity in ligand binding is the haemoglobin–oxygen system. In this case the physiological significance of the sigmoidal saturation curve (Fig. 6.7) is clear.[52] Haemoglobin is almost fully saturated with oxygen in the lungs, where the partial pressure of oxygen is about 0.13 atm, and yet unloads the oxygen in the tissue capillaries, where the partial pressure of oxygen is about 0.01 atm, much more completely than if the binding curve were hyperbolic with the four binding sites acting independently. The equilibrium between oxyhaemoglobin and deoxyhaemoglobin is also influenced by other factors, including the concentration of 2,3-bisphospho-D-glycerate, which acts as an allosteric regulator.[52] The significance of cooperative and allosteric behaviour in regulatory enzymes is not so well understood, although it can be appreciated in qualitative terms (see the example of 6-phosphofructokinase in Section 6.4.1.1).

A final word should be added about the rate at which ligand-induced conformational changes occur in regulatory enzymes. Several lines of evidence have indicated that proteins in solution can have highly flexible structures (see Chapter 3, Section 3.5.2). The time scale of conformational changes in enzymes appears to be generally in the range 10^{-2} to 10^{-4} s,[53] which is roughly equivalent to the time scale of catalytic events (see Chapter 4, Section 4.4). However, a number of cases of slow conformational changes in enzymes are now known.[30, 53–55] An example is the inhibition of hexokinase isoenzyme II from ascites tumour cells by D-glucose 6-phosphate, which has a half-time of over 10 s.[56] The physiological function of slow conformational changes is not well understood; some possibilities have been discussed.[54, 57]

6.3 Control of metabolic pathways

In Section 6.2 we were largely concerned with the more important ways in which the activities of single enzymes can be controlled. We now turn to considering the regulation of metabolic pathways that consist of sequences of enzyme-catalysed reactions. It is convenient to distinguish between two types of control that can be exerted on metabolism.[58] *Intrinsic* (or *internal*) *control* refers to the regulation of enzyme activity by the concentrations of metabolites, e.g. substrates, products, end-products of pathways, or adenine nucleotides (see the example of 6-phosphofructokinase in Section 6.4.1.1). This type of control is predominant in unicellular organisms. In multicellular organisms, the metabolism within any given type of cell must be related to the requirements of the whole organism. In such cases an extra level of control, *extrinsic* (or *external*) *control* is brought about by means such as hormones or nervous stimulation (see Section 6.4.2). These signals generally act via *second messengers* to affect the activities of target enzymes. Examples of second messengers include 3′:5′-cyclic AMP (cAMP), Ca^{2+}, inositol 1,4,5-trisphosphate, and diacylglycerol (Fig. 6.16). An excellent discussion of these regulatory mechanisms is given in reference 59.

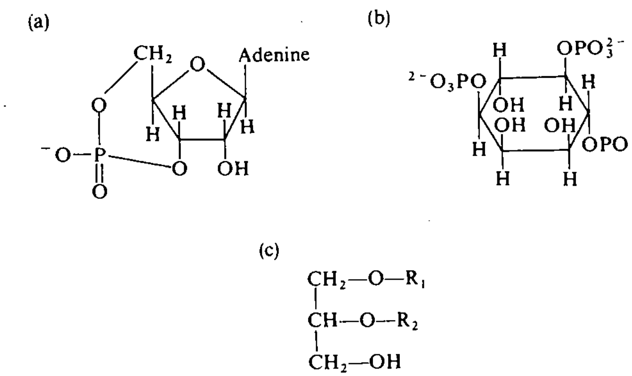


FIG. 6.16. Structures of some 'second messengers': (a) 3′:5′ cyclic AMP (cAMP); (b) inositol 1,4,5-trisphosphate; (c) diacylglycerol, where R_1 and R_2 represent fatty-acid chains, commonly stearic and arachidonic acids, respectively. Details of the action of these and other messengers can be found in reference 59.

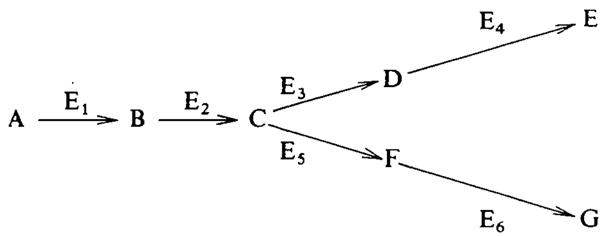
6.3.1 General considerations

We shall consider a metabolic pathway in which A is converted to F in a sequence of steps catalysed by enzymes E_1 to E_5:

$$A \xrightarrow{E_1} B \xrightarrow{E_2} C \xrightarrow{E_3} D \xrightarrow{E_4} E \xrightarrow{E_5} F.$$

It would be most economical to regulate the flux through the pathway, i.e. the rate at which F is formed from A, by regulating the catalytic activity of an early enzyme in the pathway such as E_1. This would then avoid the wasteful accumulation of intermediates. If E_1 were subject to feedback inhibition (Section 6.2.2.1) by F, then if F accumulates or is supplied exogenously, the production of

F would be shut off. On the other hand, if the concentration of F falls, the production of F would be resumed.



It might also be expected that if a pathway were branched, there would be a control point near the branch point. If an intermediate C is required for the synthesis of two products E and G, then it would seem reasonable that there should be some control mechanism to allow the rate of production of C to be altered to deal with various situations, for example, one in which E might be in excess but G not in excess. Such control could be achieved by a variety of interlocking controls near the branch point (for a review see reference 60). A good example is provided by the enzyme carbamoyl-phosphate synthase in *E. coli*. This enzyme catalyses the reaction:

Glutamine + 2ATP + bicarbonate → carbamoyl phosphate + 2ADP + orthophosphate + glutamate.

Carbamoyl phosphate is used in the production of pyrimidines and of arginine. As shown in Fig. 6.17, there is a variety of control mechanisms to ensure that an appropriate rate of production of carbamoyl phosphate is maintained in a variety of circumstances. Carbamoyl-phosphate synthase is subject to strong feedback inhibition by pyrimidine nucleotides but not by arginine. Thus, when pyrimidine nucleotides are in excess, the synthesis of carbamoyl phosphate is inhibited and may become too slow to support adequate synthesis of arginine. Under these circumstances, ornithine will accumulate and, at sufficiently high concentrations, will antagonize the effects of pyrimidine nucleotides and thus restore the activity of carbamoyl-phosphate synthase. When this happens the carbamoyl phosphate produced will be used exclusively for the production of arginine since the enzyme catalysing the production of *N*-carbamoyl-L-aspartate (aspartate carbamoyltransferase) is feedback-inhibited by the high concentrations of pyrimidine nucleotides. If arginine is present in excess, the enzyme catalysing the production of *N*-acetyl-L-glutamate (amino-acid acetyltransferase) will be feedback-inhibited (Fig. 6.17).

Other elaborate control mechanisms that have been demonstrated to exist include sequential inhibition, cumulative inhibition, and enzyme multiplicity.[60] An example of enzyme multiplicity is found in the biosynthesis of the aromatic amino acids in *E. coli*[61]. The first step in the pathway, i.e. the production of 7-phospho-2-keto-3-deoxy-D-*arabino*-heptonate from phosphoenolpyruvate

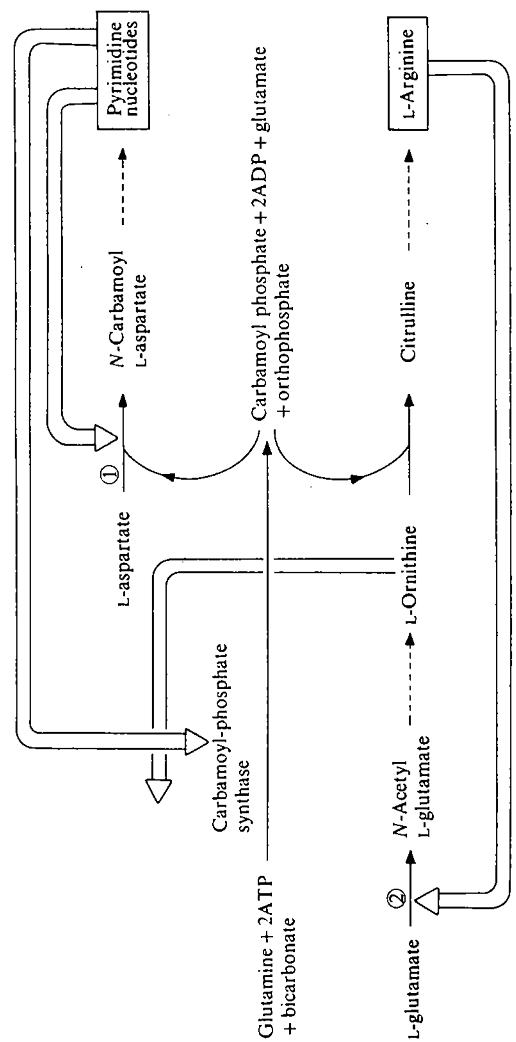


FIG. 6.17. Control of carbamoyl phosphate synthesis in *E. coli*.[60] Reaction ① is catalysed by aspartate carbamoyltransferase. Reaction ② is catalysed by amino-acid acetyltransferase.

and D-erythrose 4-phosphate is catalysed by three distinct enzymes, each of which is subject to control by a different end product. The first enzyme is subject to feedback inhibition by phenylalanine, the second to feedback inhibition by tyrosine, and synthesis of the third enzyme is repressed by tryptophan.

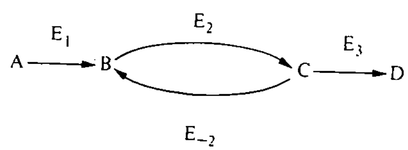
A final point that should be made is that within a cell there could be compartmentation of metabolites and enzymes. Our proposed mechanisms of control must therefore take into account any membrane-permeability barriers that may exist (see Chapter 8, Sections 8.2 and 8.3).

6.3.2 Amplification of signals

One of the features of a metabolic pathway (when compared with a single enzyme) is that considerable amplification of an initial signal is possible. The initial signal is usually a change in the concentration of a substrate or other ligand. Two particular mechanisms are considered to be important in signal amplification: these are substrate cycles (Section 6.3.2.1) and interconvertible enzyme cycles (Section 6.3.2.2).

6.3.2.1 *Substrate cycles*

Consider the following segment of a metabolic pathway:



in which there are separate enzymes to catalyse the conversion of B to C (E_2) and of C to B (E_{-2}). If E_2 and E_{-2} are active simultaneously, a *substrate cycle* will occur in which B and C are interconverted. An example of a substrate cycle occurs in the glycolytic pathway (Fig. 6.18).[62] Reaction (1) is catalysed by 6-phosphofructokinase, reaction (2) by fructose-bisphosphatase. If both enzymes are active simultaneously, the net result is the hydrolysis of ATP to yield ADP and orthophosphate. The importance of the substrate cycle in this case is that the activities of the two enzymes involved can be separately controlled (Section 6.4.1.1) and hence the net flux through this step, i.e. the rate of reaction (1) minus the rate of reaction (2), can be controlled much more precisely than if only a single enzyme were involved. The activities of the two enzymes are controlled in a reciprocal fashion, e.g. AMP activates 6-phosphofructokinase but inhibits fructose-bisphosphatase, so the effect of small changes in the concentrations of AMP on the net flux can be greatly amplified (Section 6.4.1.1).

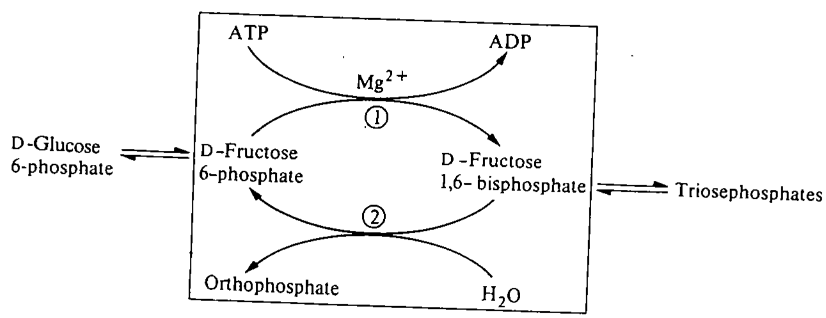
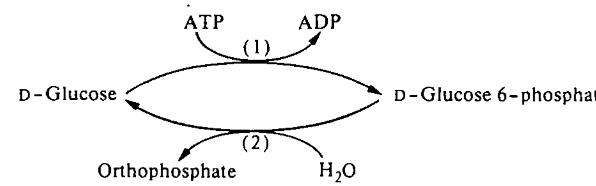


FIG. 6.18. The substrate cycle between D-fructose 6-phosphate and D-fructose 1,6-bisphosphate. Reaction (1) is catalysed by 6-phosphofructokinase and reaction (2) by fructose-bisphosphatase. Note the anomeric specificities of these enzymes (see Fig. 6.1).

A substrate cycle may also operate between D-glucose and D-glucose 6-phosphate in the liver.[63, 64]



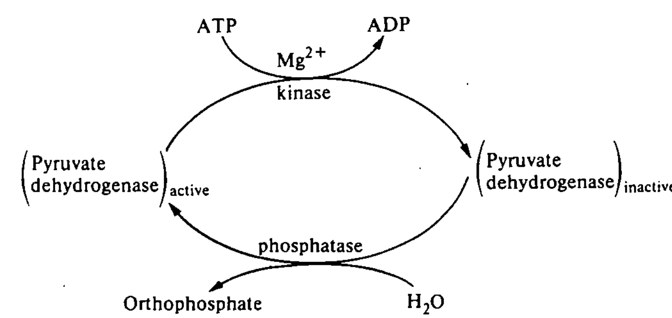
Reaction (1) is catalysed by hexokinase type IV (glucokinase) and reaction (2) by glucose 6-phosphatase. There is evidence that this substrate cycle may be important in the regulation of glucose metabolism in the liver.[63, 64]

Substrate cycles confer on the cell the possibility of more sensitive control of metabolic pathways. The price paid for this is consumption of energy in terms of ATP hydrolysis (Section 6.2.1.2). We shall discuss the evidence that substrate cycling does occur *in vivo* in Section 6.4.1.1.

6.3.2.2 *Interconvertible enzyme cycles*

In Table 6.2 we listed some examples of enzymes whose activity can be controlled by reversible covalent modification. An interconvertible enzyme cycle can provide a large amplification of an initial signal provided that the activities of the enzymes catalysing the interconversions are carefully controlled.[13, 18]

For example, the activity of the pyruvate dehydrogenase multienzyme complex from mammalian sources can be altered by phosphorylation (see Chapter 7, Section 7.7.4):



The kinase is active when the ratios of [NADH]/[NAD$^+$] and/or [acetyl-CoA]/[CoA] are high, so that when the concentration of NADH and/or acetyl-CoA is high, pyruvate dehydrogenase will be inactivated and the entry of pyruvate into the tricarboxylic acid cycle will be blocked.[14,65] Some reciprocal effects are observed on the phosphatase; for example, the phosphatase is inhibited by a high ratio of [NADH]/[NAD$^+$], so that the balance between the active and inactive forms of pyruvate dehydrogenase can be very sensitive to changes in the concentrations of these metabolites.[14]

In the case of phosphorylase, the conversion of the inactive *b* form to the active *a* form is catalysed by phosphorylase kinase, which is itself subject to enzyme-catalysed activation and inactivation. A sequence of events can be traced back to the binding of a hormone to a receptor on the cell surface (Fig. 6.19).

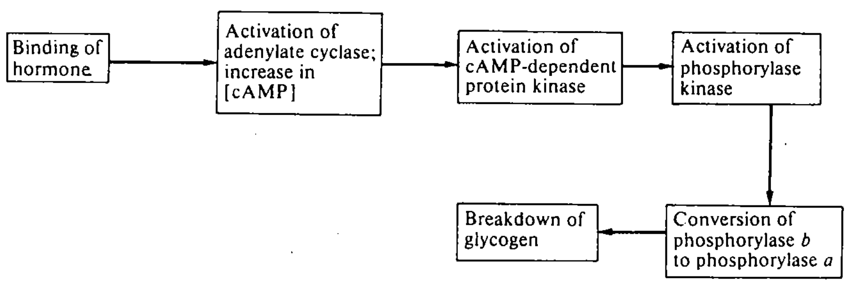


FIG. 6.19. Sequence of events involved in initiation of the breakdown of glycogen following binding of a hormone to a receptor on the cell surface.

The enzyme-catalysed activation of phosphorylase and of phosphorylase kinase can be reversed by the action of phosphatases that are themselves subject to regulation. There is also an elaborate mechanism to ensure that when phosphorylase is activated, the enzyme catalysing the synthesis of glycogen (glycogen synthase) is inactivated (Section 6.4.2.5).

For the present we should note that the sequence shown in Fig. 6.19 provides an excellent example of amplification. Experimentally it has been shown that there is only a small change in the concentration of cAMP on administration of a hormone such as adrenalin;[66] however, each molecule of protein kinase that is activated can rapidly activate many molecules of phosphorylase kinase and hence a great number of molecules of phosphorylase, thus leading to rapid breakdown of glycogen. It has been estimated that 50 per cent of the phosphorylase *b* could be converted to the active (*a*) form with only a 1 per cent increase in the concentration of cAMP,[67] the process occurring within about 2 s of the administration of hormone.[68]

6.3.3 Formulation of theories for the control of metabolic pathways

The development of a theory for the control of a particular metabolic pathway usually involves the following steps:

(1) detection of the major control points in the pathway;
(2) a study of the properties of the control enzymes;
(3) formulation of an initial theory;
(4) testing the correctness of the theory by obtaining and assessing experimental data (preferably from *in vivo* experiments); making any necessary modifications to the original theory.

We shall discuss these various steps in turn (Sections 6.3.3.1 to 6.3.3.4) and then illustrate them by discussing the present-day theories for the control of glycolysis (Section 6.4.1) and of glycogen metabolism (Section 6.4.2) in muscle.

6.3.3.1 Detection of major control points

Up to about 15 years ago it was general practice to divide the enzymes catalysing the various steps in a metabolic pathway into controlling and non-controlling types. Thus, the activity of a controlling enzyme would be changed by the regulatory signal, e.g. a change in the concentration of some metabolite, whereas the activity of a non-controlling enzyme would only change as a consequence of a change in the activity of a controlling enzyme. However, more recently it has become appreciated that the activities of all the enzymes in a pathway may have some influence on the overall flux through the pathway. In order to define metabolic control more precisely, and to provide experimentally testable models of control, a number of detailed mathematical analyses have been undertaken.[69–72] One of the best-known is that due to Kacser and Burns.[69,70] Each enzyme in the pathway is assigned a sensitivity coefficient, Z, which is a measure of the effect of a change in the activity of that enzyme on the flux through the pathway.* (The symbol Z has now been replaced by C^F_E, the flux control coefficient[73].) An enzyme with a high value of Z will be more important in controlling the flux through a metabolic pathway than one with a low value of Z.†

The different formulations and concepts of the various mathematical models of metabolic control continue to give rise to considerable debate.[75–78] However, the various formulations agree that in a metabolic pathway under given conditions, one (or very few) enzymes will dominate the regulation of the

* Mathematically, the sensitivity coefficient (Z_1) for an enzyme E_1 is defined as

$$Z_1 = \frac{\dfrac{dF}{F}}{\dfrac{dE_1}{E_1}} = \frac{d\ln F}{d\ln E_1},$$

where F is the flux through the pathway and E_1 is the catalytic activity of enzyme E_1

† In the glycolytic pathway, hexokinase and 6-phosphofructokinase have large values of Z (0.7 and 0.3, respectively, in erythrocytes; 0.47 and 0.53, respectively, in *S. cerevisiae*[74]). The other enzymes in the pathway have very low Z values and thus make a negligible contribution to the control of the flux.

pathway, though under different conditions the controlling reaction(s) may change. For our present purposes, therefore, we can effectively speak of major controlling and minor controlling enzymes. A major controlling enzyme should catalyse the slowest (or rate-limiting) step in the pathway. A reaction that is not at equilibrium *in vivo* is likely to be a rate-limiting step, since the reason that equilibrium is not achieved is presumably that there is insufficient active enzyme present to allow equilibration to occur. The initial identification of controlling enzymes therefore involves a search for those enzymes that (i) are present at low activities and (ii) catalyse reactions that are not at equilibrium. Additional methods involve searches for cross-over points and mathematical modelling techniques (see (iii) and (iv) below, respectively).

(i) The activity of each enzyme in a metabolic pathway is measured *in vitro* by performing an appropriate assay (see Chapter 4, Section 4.2) in a homogenate of the tissue concerned. It is important to measure the maximum velocity of each enzyme under conditions of pH, ionic strength, etc. that are similar to those existing in the cell. This may involve a partial purification of the homogenate to remove an inhibitor, for example. The results of the assays usually show that a reasonably clear distinction can be drawn between enzymes present in the cell at 'high activities' and those present at 'low activities'. In the glycolytic pathway (Fig. 6.20) hexokinase and 6-phosphofructokinase are 'low-activity' enzymes, whereas triosephosphate isomerase and pyruvate kinase are 'high-activity' enzymes.[5, 79]

(ii) In order to decide whether a given reaction in a pathway is at equilibrium *in vivo*, we must be able to measure the intracellular concentrations of the particular metabolites involved. The equilibrium constant for the reaction under conditions of pH, ionic strength, etc. that are similar to those that are thought to obtain in the cell can be determined by separate experiments using the isolated enzyme.

The usual method for determining the concentrations of metabolites in a given tissue involves the technique of 'freeze clamping'.[62] A sample of tissue is very rapidly frozen by compression between aluminium plates that are cooled to the temperature of liquid nitrogen, i.e. 77 K (−196°C). This rapid freezing effectively stops any processes that might tend to change the concentrations of metabolites. Before analysis of the tissue, it is essential to inhibit any enzyme-catalysed reactions; this is achieved by treating the frozen tissue with frozen perchloric acid, which will inactivate enzymes as the preparation is allowed to thaw. After centrifugation to remove protein and other precipitated material, the supernatant is neutralized and analysed for the metabolites of interest by appropriate chemical or enzymatic methods (see Chapter 10, Section 10.7). The content of a metabolite, determined as micromoles per gram of tissue can be expressed as a concentration, provided that the intracellular water content is known*

* The total water content of a tissue can be determined from the decrease in weight after drying. The contribution of extracellular water to this total can be assessed by measuring the concentration of some added compound (e.g. sorbitol) which is known not to penetrate the cells. The balance is the intracellular water.[80]

Results obtained on a rat heart perfused with glucose were:[81]

Metabolite	Intracellular concentration
ATP	11.5 mmol dm^{-3}
ADP	1.3 mmol dm^{-3}
D-Fructose 6-phosphate (F6P)	0.09 mmol dm^{-3}
D-Fructose 1,6-bisphosphate (FBP)	0.02 mmol dm^{-3}
AMP	0.17 mmol dm^{-3}

Thus, for the reaction catalysed by 6-phosphofructokinase,

$$\mathrm{F6P} + \mathrm{ATP} \xrightleftharpoons{\mathrm{Mg}^{2+}} \mathrm{FBP} + \mathrm{ADP},$$

the ratio of intracellular concentrations, known as the mass action ratio, Γ, is given by

$$\Gamma = \frac{[\mathrm{FBP}]\,[\mathrm{ADP}]}{[\mathrm{F6P}]\,[\mathrm{ATP}]} = 0.025.$$

Since the equilibrium constant for this reaction is about 1200, it is clear that this reaction is far removed from equilibrium *in vivo*.

By contrast, for the reaction catalysed by adenylate kinase,

$$\mathrm{ATP} + \mathrm{AMP} \overset{\mathrm{Mg}^{2+}}{\rightleftharpoons} 2\mathrm{ADP},$$

the mass action ratio is 0.85 which is close to the equilibrium constant for this reaction (0.44). Therefore, adenylate kinase catalyses a reaction that is close to equilibrium *in vivo*. Rolleston[82] suggests that if the ratio of Γ/K (K is the equilibrium constant) for a reaction is less than 0.05, the reaction can be considered as being non-equilibrium. More recently, ^{31}P nuclear magnetic resonance (n.m.r.) has been demonstrated to have a great deal of potential for measurement of the intracellular concentrations of phosphorus-containing metabolites such as ATP, ADP, AMP, phosphocreatine, etc.[83–86] It is possible to measure the concentration of these metabolites in a sample of tissue, an organ, or even a whole organism inserted into the wide-bore magnet of the n.m.r. spectrometer, and to detect changes in the concentrations caused by treatments such as anoxia, or addition of metabolites or hormones. The results obtained have confirmed the general conclusions reached using the freeze-clamping method, but it is evident that the non-destructive n.m.r. technique will allow a more detailed insight into the balance of metabolites in cells than was previously possible.

One difficulty in this work is that only the total concentrations of metabolites are measured, and no account is taken of the possible effects of compartmentation within cells (see Chapter 8, Section 8.2). It appears, for instance, that a high proportion of the ADP in muscle is bound to the protein actin[87] and so the concentration of free ADP is likely to be much lower than the measured total concentration. For the purposes of calculating mass action ratios, the concentrations of free metabolites should be used. We should therefore make deductions about whether particular reactions are at equilibrium *in vivo* with a certain degree of caution.

(iii) A third method of locating control points in a metabolic pathway is based on 'cross-over' experiments.[82, 88] In this type of experiment, a system is perturbed, e.g. by addition of some inhibitor or activator or by anoxia, and the changes in the concentrations of the various metabolites in a metabolic pathway are measured. The simple 'cross-over' theorem states that 'following a perturbation of a metabolic steady state, the variations in the concentrations of the metabolites before and after a controlling enzyme have different signs'. For instance, in the case of mouse brain, it was found that anoxia causes a decrease in the concentrations of D-glucose 6-phosphate and D-fructose 6-phosphate, but an increase in the concentrations of D-fructose, 1,6-bisphosphate, dihydroxyacetone phosphate, and D-glyceraldehyde 3-phosphate.[89] There is thus a 'cross-over' between D-fructose 6-phosphate and D-fructose 1,6-bisphosphate and the perturbation (anoxia) has clearly activated 6-phosphofructokinase so that the concentrations of its substrates are lowered and of its products raised. This clearly indicates that 6-phosphofructokinase is a controlling enzyme whose activity can respond to the perturbation. For a detailed appraisal of the 'cross-over' theorem, reference 88 should be consulted.

(iv) As an additional indication of the points of control in a metabolic pathway, we can make use of mathematical-modelling techniques. In the glycolytic pathway (Fig. 6.20), the various reactions are divided into equilibrium and non-equilibrium categories (see (ii) above).[71] For each equilibrium enzyme (e.g. fructose-bisphosphate aldolase and triosephosphate isomerase), an equilibrium constant is defined, and for each non-equilibrium enzyme (e.g. hexokinase and 6-phosphofructokinase) the rate can be expressed in terms of the concentration of substrates and the appropriate maximum velocity (taking into account any product inhibition, etc.). It is then possible to write down expressions for the concentrations of metabolites and the flux through the pathway and to solve the resulting equations for various assumed conditions, such as a steady state.[71] This type of approach predicts the steady-state concentrations of glycolytic intermediates in erythrocytes reasonably well, and also shows the importance of the enzymes hexokinase and 6-phosphofructokinase in the control of the pathway.[90] A fuller description of mathematical-modelling techniques is beyond the scope of this text and the reader should consult references 70 and 91–93.

6.3.3.2 *Properties of the controlling enzymes*

From the various approaches described in Section 6.3.3.1, it is possible to locate the probable sites of control of a metabolic pathway. The next step is to study the catalytic properties of the controlling enzymes, especially those properties that might be involved in regulatory mechanisms. Of particular interest are the following questions.

1. What type of kinetic behaviour is observed (hyperbolic, sigmoidal, etc.)? How does the activity of the enzyme vary with changes in the concentration of substrate, particularly in the range of substrate (and enzyme) concentrations that occur *in vivo*?
2. Is the activity of the enzyme affected by any regulator molecules at concentrations that are known to occur in the cell? What is the nature of the regulation (allosteric, competitive inhibition, etc.)?
3. Is the activity of the enzyme subject to regulation by changes in its covalent structure, e.g. by phosphorylation or dephosphorylation? How are the enzymes catalysing these changes regulated?

6.3.3.3 *Formulation of an initial theory for the control of a metabolic pathway*

From the studies of the controlling enzymes (Section 6.3.3.2), it should be possible to decide which means of regulation is most likely to be involved in the control of a metabolic pathway. On the basis of the initial theory for the controls it is possible to make predictions that can be then tested.

6.3.3.4 *Test of the correctness of a theory for the control of a metabolic pathway*

In order to test the correctness of a theory, it is necessary to gather data about the activities of enzymes, the concentrations of substrates and effectors, and the state of covalent modification, etc. in a variety of metabolic circumstances. It is important to determine these parameters under conditions that reflect the *in vivo* situation as closely as possible; non-destructive techniques such as ^{31}P nuclear magnetic resonance[83–86] (Section 6.3.3.1) have great potential in such studies. If it emerges that, for instance, the variations in the concentrations of substrates and/or effectors are too small to account for the observed changes in enzyme activity or in the flux through the pathway, then the initial theory must be modified or discarded altogether and a new theory formulated and tested.

6.4 Examples of control of metabolic pathways

We shall now discuss how present-day ideas about the control of metabolic pathways have been developed by considering two particular pathways, namely glycolysis (Section 6.4.1) and glycogen metabolism (Section 6.4.2). These pathways illustrate most of the points made in Section 6.3.3. The experiments described refer mainly to the regulation of the pathways in muscle that have been particularly well studied because muscle is a highly specialized tissue in which the fluxes through the pathways vary dramatically in different metabolic circumstances. These metabolic pathways in muscle must therefore be under a high degree of control.

6.4.1 Regulation of glycolysis

The glycolytic pathway is the sequence of reactions in which glucose is converted to pyruvate, and is shown in Fig. 6.20. This pathway appears to be of almost universal importance in energy production.

Using the various approaches described in Section 6.3.3, it has been shown that the major controlling points for the pathway are the steps catalysed by 6-phosphofructokinase and by hexokinase,* since these enzymes are present at low activities and catalyse reactions which, *in vivo*, are far removed from equilibrium.[62, 79] It should also be mentioned that the reaction catalysed by pyruvate kinase is not at equilibrium *in vivo*, but since the activity of this enzyme is relatively high in muscle, its role in the control of the pathway is somewhat open to question.[62, 79]

The importance of 6-phosphofructokinase and hexokinase as controlling enzymes in the glycolytic pathway in a variety of tissues has also been confirmed by 'cross-over' experiments[89] and by mathematical-modelling studies.[74, 90]

6.4.1.1 *Regulation of the activity of 6-phosphofructokinase*

The kinetic behaviour *in vitro* of 6-phosphofructokinase is highly complex.[95–97] As shown schematically in Fig. 6.21, there are mutual interactions between the two substrates D-fructose 6-phosphate (F6P) and ATP. At low concentrations (in the physiological range) of F6P, it is found that high concentrations of ATP (in the physiological range) cause inhibition (Fig. 6.21(a)). This inhibition is due to binding of ATP to a site distinct from the active site (see Chapter 4, Section 4.3.1.4) as is shown by the fact that the enzyme can be desensitized (Section 6.2.2.1) to ATP inhibition by photo-oxidation in the presence of methylene blue.[98] The inhibition at high concentrations of ATP is accentuated by citrate but can be counteracted by various metabolites including AMP (see Fig. 6.21(a)), ADP, orthophosphate, and D-fructose, 1,6-bisphosphate. These latter compounds tend to accumulate when ATP is degraded, for example during anoxia. At high concentrations (in the physiological range) of ATP, the plot of velocity against concentration of F6P is markedly sigmoidal with a very low velocity at low concentrations of F6P (Fig. 6.21(b)).

An initial theory, based on the data shown in Fig. 6.21(a), is that the concentration of ATP controls the activity of 6-phosphofructokinase and hence the flux through the glycolytic pathway. Thus, when the concentration of ATP is high, as in resting muscle, 6-phosphofructokinase is inhibited and hence the rate of production of ATP falls. (Note that for each molecule of glucose converted to two molecules of pyruvate there is a net production of two molecules of ATP.) If this theory is correct, it would be expected that significant changes in the concentration of ATP would occur *in vivo* to account for the changes in the

* There is an additional point of control, namely the entry of glucose into the cell. This transport process is far removed from equilibrium and is regulated by insulin; an increase in the circulatory levels of this hormone will increase the rate of entry of glucose into the cell.[79, 94]

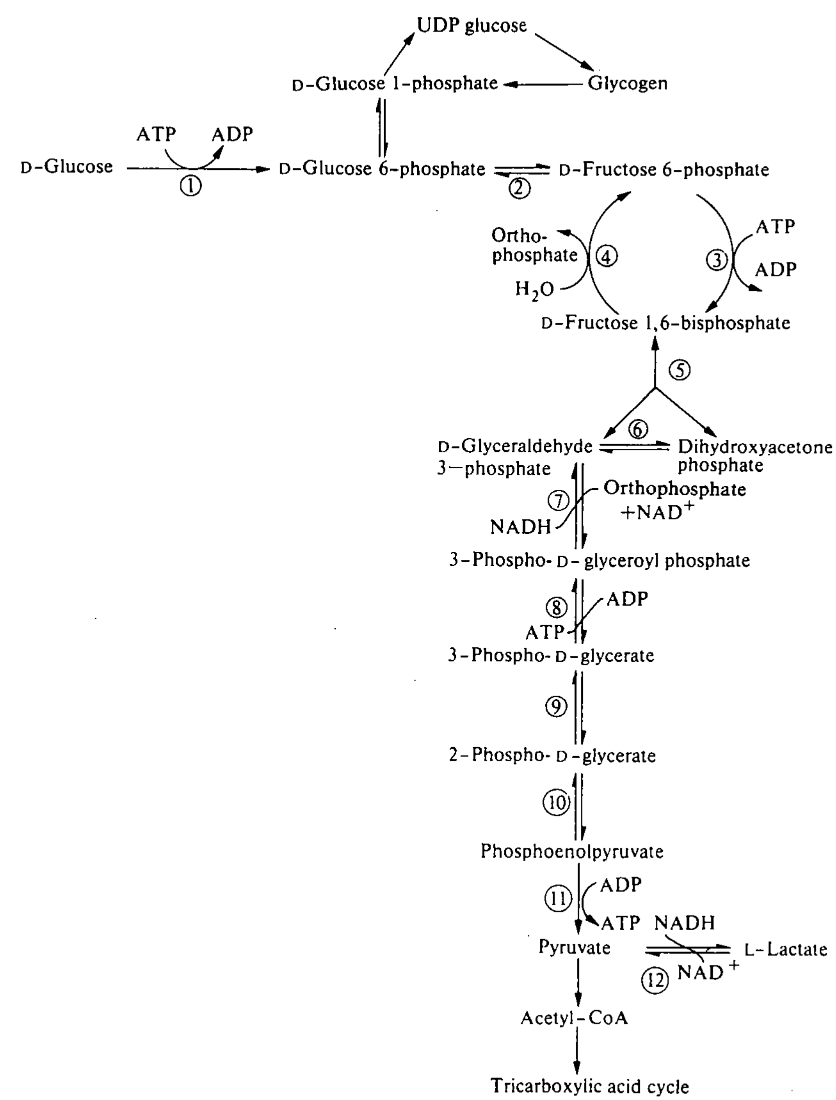


FIG. 6.20. The glycolytic pathway in muscle. The various steps are catalysed by the following enzymes: (1), hexokinase; (2), glucose-6-phosphate isomerase; (3), 6-phosphofructokinase; (4), fructose-bisphosphatase; (5), fructose-bisphosphate aldolase; (6), triose-phosphate isomerase; (7), glyceraldehyde-3-phosphate dehydrogenase; (8), phosphoglycerate kinase; (9), phosphoglycerate mutase; (10), enolase; (11), pyruvate kinase; (12), lactate dehydrogenase. The magnesium ions associated with the kinase-catalysed reactions have been omitted.

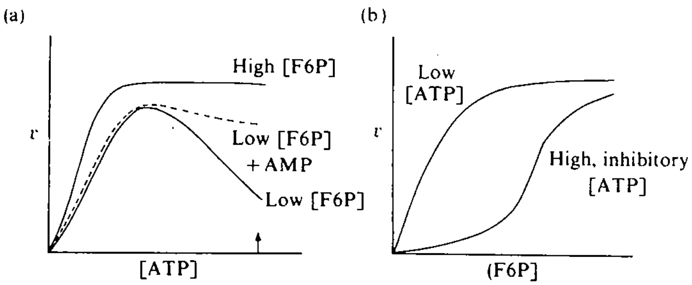


FIG. 6.21. Schematic representation of the kinetic behaviour of 6-phosphofructokinase.[95, 97] (a) Variation of velocity with concentration of ATP (the arrow on the abscissa indicates the prevailing physiological concentration of ATP). (b) Variation of velocity with concentration of F6P (D-fructose 6-phosphate).

activity of 6-phosphofructokinase (Fig. 6.21(a)). However, measurements on the flight muscles of blowflies showed that during flight there was only a small (10 per cent) decrease in the concentration of ATP compared with resting muscle, even though the flux through the glycolytic pathway had increased 100-fold.[99] Reference to the data shown in Fig. 6.21(a)), shows that this change in the concentration of ATP is nowhere near large enough to account for the change in the flux through the pathway.

It has been proposed that the effects of ATP on the activity of 6-phosphofructokinase can be amplified by other metabolites, especially ADP and AMP, which relieve the inhibition caused by high concentrations of ATP. The adenine nucleotides are linked in the reaction catalysed by adenylate kinase,

$$\mathrm{ATP+AMP} \overset{Mg^{2+}}{\rightleftharpoons} \mathrm{2ADP},$$

which is at equilibrium in muscle (Section 6.3.3.1). Since the concentrations of nucleotides in muscle are [ATP] > [ADP], [AMP], a relatively small change in the concentration of ATP brings about much larger changes in the concentrations of ADP and AMP.[62, 97] In the blowfly flight muscle, for instance, it was found that, during flight, the concentration of AMP increased by about 2.5-fold compared with resting muscle[99] (cf. a 10 per cent decrease in the concentration of ATP). The revised proposal is therefore that the activity of 6-phosphofructokinase is controlled by a combination of changes in the concentrations of ATP, AMP, ADP,* and other metabolites such as orthophosphate.

* The dependence of the activity of 6-phosphofructokinase on the concentrations of the adenine nucleotides has been explored by Atkinson[100, 101] who has introduced the term 'energy charge'. This is defined as 'one half of the phosphate anhydride bonds per adenosine' and is equal to ([ATP] + $\frac{1}{2}$[ADP])/([ATP] + [ADP] + [AMP]). The 'energy charge' varies from 0, when all the adenosine in a cell is present as AMP, to 1, when all the adenosine is present as ATP. Enzymes that are energy utilizing, such as ATP citrate (*pro*-3*S*)-lyase, have a different dependence on energy change from those that are energy-regenerating, such as 6-phosphofructokinase.

However, from detailed considerations of the kinetic behaviour of 6-phosphofructokinase (Fig. 6.21), it is considered unlikely that the combination of the changes in the concentrations of adenine nucleotides is large enough to account for the observed changes in the flux through the glycolytic pathway.[62, 97] Various suggestions have been made as to how the activity of 6-phosphofructokinase might be made even more sensitive to changes in the concentration of metabolites. One of these proposals[62, 102] is based on a substrate cycle between D-fructose 6-phosphate (F6P) and D-fructose 1,6-bisphosphate (FBP) (Fig. 6.18 and Section 6.3.2.1). Fructose-bisphosphatase, which is strongly inhibited by AMP, is found in significant amounts (approximately 10 per cent of the maximal activity of 6-phosphofructokinase) in various types of muscle where no significant gluconeogenesis occurs. The proposal has therefore been made that in resting muscle, in which the concentration of AMP is low, the low rate of conversion of F6P to FBP catalysed by 6-phosphofructokinase is opposed by conversion of FBP to F6P catalysed by fructose-bisphosphatase, giving a very low net flux through this step in the glycolytic pathway. However, when the muscle is made to do work, the concentration of ATP falls and that of AMP rises; 6-phosphofructokinase is now activated (i.e. no longer inhibited) and fructose-bisphosphatase is inhibited, giving a large net flux through this step. Calculations[102] suggest that the observed changes in net flux can be accounted for by changes in the concentration of AMP that are of the same order of magnitude as those detected by freeze-clamping measurements.[99] (Thus, a 250-fold change in net flux could be accounted for by a 2.5-fold change in the concentration of AMP.[102]) The hypothesis of substrate cycling accounts not only for the sensitivity of net flux to changes in the concentration of AMP, but also for the presence and distribution of fructose-bisphosphatase in various types of muscle. Fructose-bisphosphatase is not present in muscles such as heart muscle, where the energy demands are fairly constant and hence the need for regulation of glycolysis is much less than in the case of skeletal muscle. Lardy and his coworkers have also obtained direct evidence for substrate cycling in various tissues such as bumble bee flight muscle[103] and rat liver[104] using isotopically labelled phosphorylated monosaccharides (see Fig. 6.22).

However, the assumptions underlying the interpretation of the experimental data in these experiments have been questioned, particularly in the case of rat liver.[64] Also, mathematical modelling of the situation in rat liver suggests that under physiologically realistic conditions, when the compartmentation and the states of protonation of adenine nucleotides are taken into account, the rate of substrate cycling is negligible.[106]

Since about 1980, a good deal of attention has been focused on fructose 2,6-bisphosphate (F26BP) as a regulator of the activity of 6-phosphofructokinase.[107–109] This regulation appears to be of particular importance in the liver; indeed, F26BP is the most potent known activator of liver 6-phosphofructokinase, with effects being shown in the 0.1 μmol dm^{-3} concentration range. The activation of 6-phosphofructokinase is synergistic with AMP. F26BP is also a powerful inhibitor of liver fructose-bisphophatase, again acting synergistically

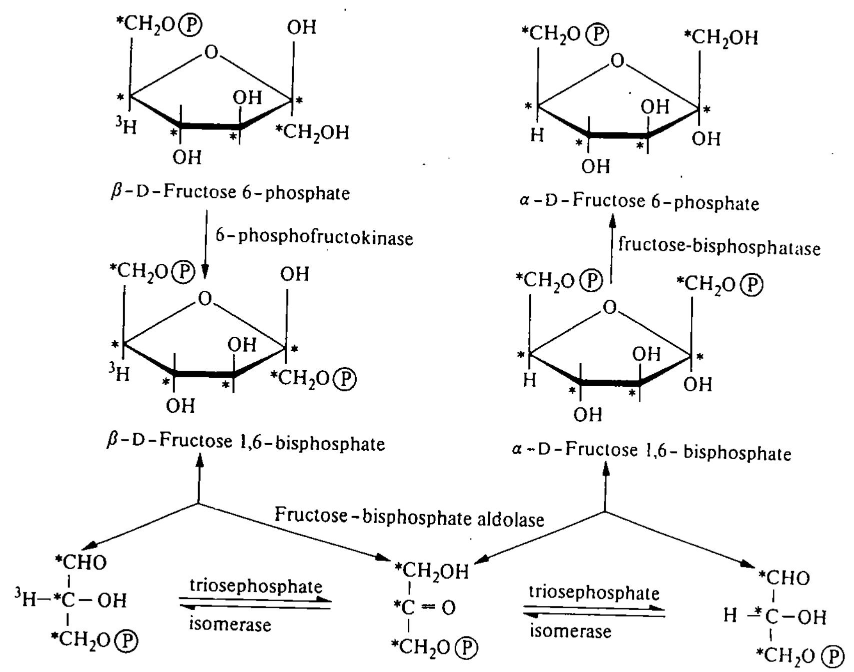


FIG. 6.22. System used to demonstrate substrate cycling. An asterisk represents ^{14}C and Ⓟ a phosphoryl group. On addition of [5-^3H, U-^{14}C]F6P to the system, the action of 6-phosphofructokinase and fructose-bisphosphate aldolase produces isotopically labelled D-glyceraldehyde 3-phosphate and dihydroxyacetone phosphate. The action of triosephosphate isomerase allows ^3H at C-2 of D-glyceraldehyde 3-phosphate to be exchanged for H from solvent water (see Chapter 5, Section 5.5.2.3). The action of fructose-bisphosphate aldolase and fructose-bisphosphatase then generates F6P in which ^3H is lost. If the remaining F6P is isolated at various times after the addition of [5-^3H, U-^{14}C]F6P and analysed for the content of ^3H and ^{14}C, the ratio of $^3H/^{14}C$ will fall if substrate cycling occurs. From the rate of decrease of this ratio, the rate of cycling can be deduced.[105]

with AMP. The synthesis and degradation of F26BP are catalysed by a kinase (6-phosphofructo-2-kinase) and phosphatase (fructose-2,6-bisphosphatase) whose activities are subject to control by reversible phosphorylation (Fig. 6.23). The latter kinase and phosphatase activities are located on an enzyme that consists of two identical polypeptide chains each of M_r 55 000.[107]

All the available evidence shows that, in liver, F26BP plays a major part in the hormonal and nutritional control of 6-phosphofructokinase and fructose-bisphosphatase activity and in determining the balance between glycolysis and gluconeogenesis.[109] Thus, the glycolytic flux in liver can be correlated with the level of F26BP but not with the levels of ATP, AMP, or citrate. The presence of F26BP is the signal that the level of glucose is high and that gluconeogenesis can

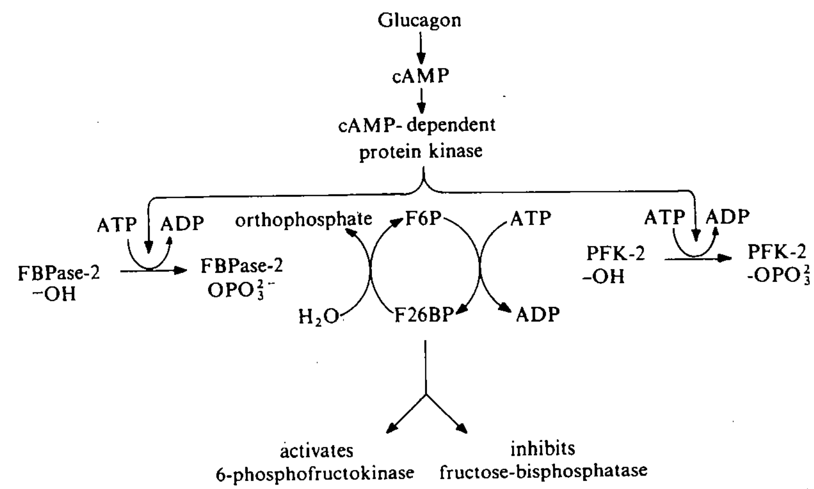


FIG. 6.23. Control of the synthesis and degradation of fructose 2,6-bisphosphate (F26BP).[106] In this scheme, FBPase-2 and PFK-2 represent fructose-2,6-bisphosphatase and 6-phosphofructo-2-kinase, respectively.

be switched off. The presence of glucagon, however, causes the level of F26BP to fall (see Fig. 6.23) and hence gluconeogenesis is favoured.

The importance of F26BP in the regulation of glycolysis in other tissues is less firmly established.[109–111] The concentration of F26BP in muscle is lower than that in the liver and this, coupled with the smaller effect of F26BP on muscle 6-phosphofructokinase compared with the liver enzyme, means that F26BP is unlikely to have a major effect on the glycolytic flux in muscle.

Among other factors that may be important in the regulation of the activity of 6-phosphofructokinase in muscle are the following.

1. 6-phosphofructokinase and fructose-bisphosphatase show different anomeric specificities (see Figs. 6.1 and 6.22), acting on the β-anomer of F6P and the α-anomer of FBP, and being activated by the α-anomer of FBP and the β-anomer of F6P, respectively. These specificities, taken together with the slow, spontaneous α→β anomerizations of F6P and FBP (rate constants approximately 1 s^{-1}, reference 6) create non-equilibrium intracellular distributions of the α- and β-anomers of F6P and FBP. These non-equilibrium distributions could then amplify the effects of AMP and other metabolites on the enzymes.[112]
2. The calcium-binding protein calmodulin has been shown to interact with 6-phosphofructokinase from muscle.[113, 114] Calmodulin acts as a Ca^{2+}-dependent inhibitor of the enzyme and appears to compete with ATP for binding to the regulatory site of the enzyme.

3. A number of reports have shown that 6-phosphofructokinase from muscle can be reversibly phosphorylated.[115] However, the phosphorylation appears to cause only minor changes in the regulatory properties of the enzyme.

It should be noted that the factors (1)–(3) have been studied *in vitro* and their possible importance *in vivo* has not been clarified.

In summary, it can be stated that although the importance of the step D-fructose 6-phosphate→D-fructose 1,6-bisphosphate as a control point in glycolysis is beyond dispute, the exact mechanism by which this step is regulated is still open to question and experimentation.

6.4.1.2 *Regulation of the activity of hexokinase*

Up to now we have concentrated attention on the regulation exerted at the D-fructose 6-phosphate→D-fructose 1,6-bisphosphate step in glycolysis. However, it is obvious that when 6-phosphofructokinase is inhibited, there would be an accumulation of the phosphorylated hexoses D-glucose 6-phosphate and D-fructose 6-phosphate unless some means existed to control the activity of hexokinase that catalyses the conversion of D-glucose to D-glucose 6-phosphate at the expense of ATP (see Fig. 6.20). It is found that hexokinases from a variety of mammalian tissues are inhibited by D-glucose 6-phosphate;* the inhibition is non-competitive (see Chapter 4, Section 4.3.2.2) with respect to D-glucose.[56] The inhibition of hexokinase would thus serve to regulate the rate of production of D-glucose 6-phosphate and hence D-fructose 6-phosphate.

There is, however, an additional factor to be borne in mind. As indicated in Fig. 6.20, hexokinase occurs near a branch point in the metabolism of monosaccharides, since D-glucose 6-phosphate is not only used to furnish D-fructose 6-phosphate for the glycolytic pathway but can also be used to produce D-glucose 1-phosphate via the action of phosphoglucomutase for the synthesis of glycogen. Under conditions in which the concentration of D-glucose is high, e.g. after an intake of carbohydrate in the food, it is desirable to build up reserves of glycogen. This would not be possible if hexokinase were inhibited by high prevailing concentrations of D-glucose 6-phosphate. The hormone insulin, secreted in response to high circulating concentrations of D-glucose, not only facilitates the transport of D-glucose into the cell, but also stimulates the synthesis of glycogen (Section 6.4.2.5). Thus when D-glucose concentrations are high, the excess D-glucose can be channelled via D-glucose 6-phosphate, D-glucose 1-phosphate, and UDP glucose to glycogen (Fig. 6.20). The dual control system afforded by regulation of the activities of hexokinase† and of 6-phosphofructokinase that occur before and after the branch point, respectively, thus allows the cell to respond to a variety of metabolic situations and provides more flexible regulation of the glycolytic pathway than could be achieved with a single control point.

* Hexokinase isoenzyme IV (also known as glucokinase) is not inhibited by D-glucose 6-phosphate. This isoenzyme, located in the liver, has a high K_m for D-glucose (see Chapter 4, Section 4.3.1.3). Thus phosphorylation of D-glucose can proceed in the liver at high prevailing concentrations of D-glucose 6-phosphate.

† As mentioned in Section 6.2.2.3, the inhibition of hexokinase isoenzyme II by D-glucose 6-phosphate from ascites tumour cells is rather slow.[56] This property allows the glycolytic intermediates D-glucose 6-phosphate and D-fructose 6-phosphate to rise to a new high level after addition of D-glucose to these cells before the hexokinase is inhibited.

6.4.2 Regulation of glycogen metabolism

The second example of control of a metabolic pathway that we shall discuss is the regulation of glycogen metabolism in muscle. Glycogen is a polysaccharide made up of D-glucopyranose units linked by α-1,4 glycosidic bonds with branches formed by 1,6 bonds about every 12–18 groups (Fig. 6.24), i.e. there is no unique molecular structure for glycogen. The M_r of glycogen is in the range from about 10^6 to 10^8 (consisting of between 6000 and 600 000 D-glucose units). In a variety of organisms, e.g. *E. coli*[116] and rat liver,[117] a protein molecule is involved as a 'primer' backbone in the biosynthesis of glycogen.

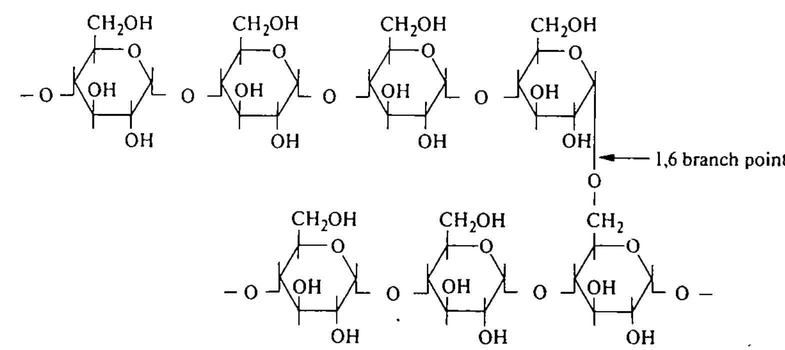


FIG. 6.24. The structure of glycogen showing the α-1,4 glycosidic bonds linking D-glucopyranose units and a 1,6 branch point.

The reactions involved in glycogen metabolism are relatively few in number (see Fig. 6.25) but the study of these reactions has revealed some very complex control mechanisms.

In both muscle and liver, glycogen has the function of being a reserve of D-glucose units. Liver glycogen can be used to help replenish D-glucose in the blood between meals and during exercise. In muscle, however, the enzyme glucose 6-phosphatase is lacking and hence the D-glucose 1-phosphate produced by the breakdown of glycogen is metabolized via the glycolytic pathway to provide ATP that acts as the fuel for muscular contraction. The amount of ATP that can be generated from the muscle glycogen is sufficient to sustain contraction for only a short time (of the order of one minute). Prolonged exercise requires the utilization of other sources of energy, e.g. triglycerides.[120]

Using the criteria for identification of controlling enzymes (Section 6.3.3.1), it has been established that the control points for glycogen breakdown and

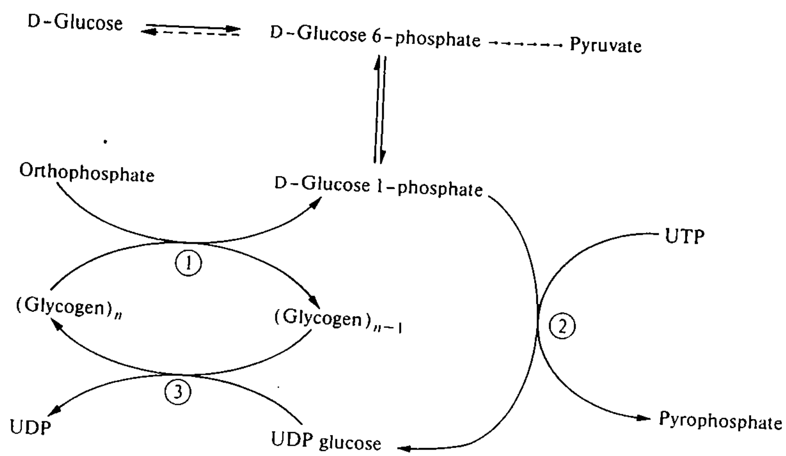


FIG. 6.25. Reactions involved in glycogen metabolism. The dashed line represents a reaction that occurs in liver but not in muscle. Reaction (1) is catalysed by phosphorylase plus debranching enzyme (which possesses two activities: amylo-1,6-glucosidase and 4-α-D-glucanotransferase[118]). Reaction (2) is catalysed by glucose 1-phosphate uridylyltransferase. Reaction (3) is catalysed by glycogen synthase plus branching enzyme (1,4-α-glucan branching enzyme[119]).

synthesis are the reactions catalysed by phosphorylase and glycogen synthase respectively.[62, 97, 121]

The main features of phosphorylase that implicate it as a controlling enzyme are that it is present at low activity (similar to the activity of 6-phosphofructokinase) and that it catalyses a reaction that, *in vivo*, is not at equilibrium.* A 'cross-over' type of experiment showed that when the breakdown of glycogen is stimulated by adrenalin, the catalytic activity of phosphorylase increases even though the concentration of its substrate decreases, i.e. phosphorylase is the controlling enzyme.[121] In the case of glycogen synthase, the main evidence that it is a controlling enzyme comes from the finding that it is present at low activity. 'Cross-over' experiments have helped to confirm this conclusion.[121]

* The equilibrium constant for the reaction catalysed by phosphorylase is given by

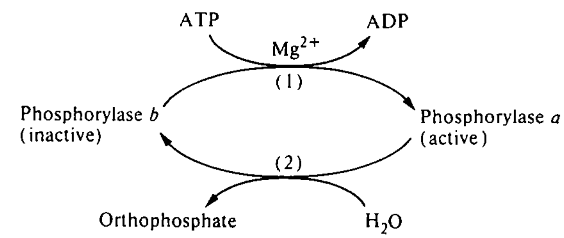
$$K = \frac{[(\text{Glycogen})_{n-1}]}{[(\text{Glycogen})_n]} \times \frac{[\text{G1P}]}{[\text{Orthophosphate}]},$$

but since it is impossible to determine the first term because of the variety of sizes of glycogen molecules, we write $K = [\text{G1P}]/[\text{orthophosphate}]$. At equilibrium this ratio is 0.3, but in muscle the ratio is approximately 100-fold lower. This is because in muscle the concentration of orthophosphate is high, and the concentration of G1P is low (the equilibrium between G1P and G6P lies 95 per cent to the side of G6P). Hence, in the muscle phosphorylase acts to break down glycogen. Glycogen synthesis proceeds via a different route, as shown in Fig. 6.25. Patients suffering from McArdle's disease lack phosphorylase but their ability to synthesise glycogen is not impaired.[122]

6.4.2.1 *Regulation of the activity of phosphorylase*

Phosphorylase has been studied intensively for over 35 years and a great deal is now known about its structure and function.[123, 124] In the last decade both the primary structure[125] and three-dimensional structure[12] of the enzyme have been elucidated, thus allowing many of the earlier observations to be interpreted in more detail.

Phosphorylase can be isolated from muscle in two distinct forms. One of these, phosphorylase *b*, is inactive in the absence of certain ligands, notably AMP (see Section 6.2.2.1), whereas the other, phosphorylase *a*, is active in the absence of these added ligands. Further work established that the interconversions of the two forms of phosphorylase were catalysed by a kinase and a phosphatase that could be isolated from muscle extracts.



(1) is catalysed by phosphorylase kinase
(2) is catalysed by phosphorylase phosphatase

As mentioned earlier (Section 6.2.1.2) the conversion of phosphorylase *b* to phosphorylase *a* involves phosphorylation of a single side-chain (Ser 14) in each subunit. The conversion is accompanied by a change in the quaternary structure of the enzyme (the *b* form is a dimer, the *a* form a tetramer) but this change is not an integral part of the activation process.[126]

Thus there appear to be two major means of regulating the activity of phosphorylase.

1. By ligand-induced conformational changes in the enzyme. AMP (or IMP) will activate phosphorylase *b*; this effect is counteracted by certain other ligands, notably ATP, ADP, UDPglucose and D-glucose 6-phosphate.[124] Hence, under conditions in which the concentration of AMP is high and the concentrations of ATP and D-glucose 6-phosphate, in particular, are low, phosphorylase *b* will be active. This provides a 'crude' control mechanism for the enzyme.
2. By a change in the covalent structure of the enzyme, i.e. conversion of phosphorylase *b* to phosphorylase *a*, catalysed by phosphorylase kinase. Phosphorylase kinase is itself subject to regulation and can be activated in two main ways: by phosphorylation catalysed by a cAMP*-dependent protein kinase and by Ca^{2+}-mediated stimulation.

* cAMP is used as an abbreviation for 3′:5′-cyclic AMP (see Fig. 6.16).

The sequence of events in which phosphorylase kinase is activated by cAMP-dependent protein kinase is shown in Fig. 6.26.

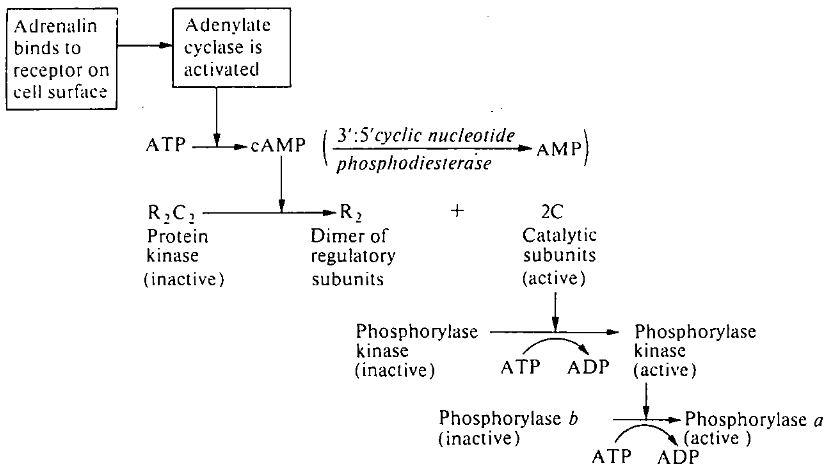
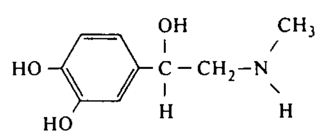
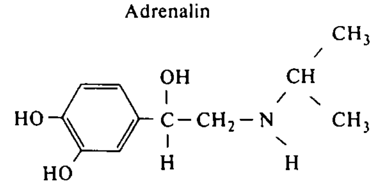


FIG. 6.26. The cascade mechanism for hormone-triggered activation of phosphorylase.

Binding of the hormone to the cell surface activates adenylate cyclase, which is a membrane-bound enzyme[127] (see Chapter 8, Section 8.4.5). The cAMP produced binds to the protein kinase,[128] causing dissociation of the inactive R_2C_2 complex to yield active catalytic subunits and a dimer of regulatory subunits. The active catalytic subunits catalyse the activation of phosphorylase kinase and hence of phosphorylase. This cascade mechanism allows for considerable amplification of the initial signal (Section 6.3.2.2). The administration of adrenalin to various tissues causes only small increases (less than fivefold) in the concentration of cAMP,[121] and indeed in the case of rabbit gracilis muscle it has been shown that the administration of 4 pmol of isoproterenol, an analogue of adrenalin, did not change the concentration of cAMP within experimental error, although the amount of phosphorylase *a* increased threefold.[129]



Adrenalin



Isoproterenol

Ca^{2+} ions in the concentration range of 0.1 to 1 μmol dm^{-3} can activate phosphorylase kinase in the presence of Mg^{2+} and ATP. This activation occurs by a process of autophosphorylation in which the enzyme catalyses its own phosphorylation,[130] at sites that are distinct from those that are phosphorylated by the action of cAMP-dependent protein kinase.[131,132] It has been known for many years that Ca^{2+} ions, in this concentration range, are released from the sarcoplasmic reticulum in muscle in response to a nerve impulse and can then act as a trigger for muscular contraction by allowing the components of the contractile apparatus (actin and myosin) to interact. Thus there is a clear link between the stimulation of muscular contraction, which requires ATP as a fuel, and activation of glycogen breakdown, which can furnish ATP via the glycolytic pathway (Fig. 6.20).

The mechanisms by which Ca^{2+} ions may affect the activity of phosphorylase kinase have been elucidated.[132] Phosphorylase kinase was originally thought to consist of three types of subunits with the quaternary structure $(\alpha\beta\gamma)_4$[133]. (The α and β subunits of M_r 145 000 and 128 000, respectively, are subject to phosphorylation and the γ subunit of M_r 45 000 contains the active site of the enzyme.) However, subsequent work[134] showed that the enzyme contains a fourth type of subunit (δ) of M_r 16 000 that is identical with the Ca^{2+}-binding protein, calmodulin, previously detected in a variety of other tissues such as brain and erythrocytes.[135] Activation of phosphorylase kinase by Ca^{2+} can proceed via Ca^{2+} binding to the δ subunit, which triggers a conformational change in the γ subunit converting it to an active form. A further degree of stimulation can be achieved by the binding of troponin-C (the Ca^{2+}-binding protein involved in the regulation of muscular contraction). Troponin-C interacts only with the β-subunit of phosphorylase kinase and only in the presence of Ca^{2+} ions.[132] Since the concentration of troponin-C in muscle is much higher than that of calmodulin, it may well be that activation of phosphorylase kinase via the binding of troponin-C is important in the synchronization of glycogen breakdown with muscular contraction.[132,136]

In summary, we have seen how the conversion of phosphorylase *b* to phosphorylase *a* can be brought about by the action of hormones such as adrenalin (acting via cAMP and cAMP-dependent protein kinase) or by a nerve impulse (acting via Ca^{2+} ions), thus providing a versatile amplified response to a variety of initial signals. It is now becoming increasingly recognized that the hormone-stimulated and nerve-impulse-stimulated pathways are interwoven.[132,137,138] For instance, Ca^{2+} ions acting via calmodulin are known to affect the activities of adenylate cyclase and 3′:5′-cyclic-nucleotide phosphodiesterase, which catalyse the synthesis and hydrolysis respectively, of cAMP. The Ca^{2+}-calmodulin system is also responsible for activating a multi-protein kinase of broad specificity that phosphorylates a number of proteins including glycogen synthase and tyrosine hydroxylase (which is involved in the synthesis of catecholamines).[137] In addition, cAMP-dependent protein kinase can catalyse the phosphorylation of troponin-I which may serve to regulate the interaction of actin and myosin and affect the force of muscular contraction.[139]

A number of experimental tests must be made in order to assess whether the various mechanisms for regulation of the activity of phosphorylase are operative *in vivo*. Two particular questions that need to be answered are: what are the concentrations *in vivo* of the various ligands such as AMP that affect the activity of phosphorylase *b*, and can the various events of the cascade mechanism depicted in Fig. 6.26 be demonstrated to occur *in vivo?* We shall deal with these questions in Sections 6.4.2.2 and 6.4.2.3, respectively.

6.4.2.2 *Concentrations of ligands* in vivo

The concentrations of a number of metabolites in muscle have been determined by the freeze-clamping technique (Section 6.3.3.1). These studies have shown that the concentration of AMP in resting muscle (approximately 100 μmol dm^{-3}) is greater than the concentration required to cause 50 per cent of the maximal activation of phosphorylase *b* (approximately 50 μmol dm^{-3}). The negligible activity of phosphorylase in resting muscle therefore implies that *in vivo* the effect of AMP is counteracted by high concentrations of ligands such as ATP, ADP, and D-glucose 6-phosphate; a conclusion confirmed by measurements of the concentrations of these ligands in resting muscle.[140] Ligand-induced activation of phosphorylase *b* can therefore occur *in vivo* only when the concentrations of activating ligands rise and/or those of counteracting ligands fall. Such a situation has been shown to arise in the perfused rat heart under anaerobic conditions,[141] and perhaps more convincingly in the muscles of I-strain mice that are deficient in phosphorylase kinase and cannot therefore activate phosphorylase by the $b \rightarrow a$ conversion.[142] A detailed study of the concentrations of various ligands in the muscles of I-strain mice before and after electrical stimulation of the muscle has led to the conclusion that, in this case, the increase in the activity of phosphorylase is largely due to a rise in the concentration of IMP rather than of AMP.[143]

Finally, mention should be made of the fact that *in vivo* a substantial proportion of the phosphorylase in muscle is probably bound to glycogen in the form of the 'glycogen particle' (see Chapter 7, Section 7.12). In this particle, the properties of the enzyme and the detailed effects of the various ligands may well be different from those observed in solution (see reference 124 for a discussion of this).

6.4.2.3 *Demonstration of the steps in the cascade mechanism (Fig. 6.26)*

The various steps in the cascade mechanism for the activation of phosphorylase have all been shown to occur *in vivo* in response to the administration of adrenalin.

(i) Conversion of phosphorylase b *to phosphorylase* a

The phosphorylation of phosphorylase *b* has been demonstrated *in vivo* in rabbit skeletal muscle.[144] On administration of adrenalin, up to 0.8 equivalents of ^{32}P were incorporated into phosphorylase *b* from ATP labelled at the γ-phosphoryl group with ^{32}P. In order to prevent the hydrolysis of phosphorylase *a* during the isolation procedure, it was necessary to homogenize the tissue in the presence of fluoride ions, which act as powerful inhibitors of phosphorylase phosphatase.

*(ii) Phosphorylation of phosphorylase kinase**

Careful work has shown that both the α- and the β-subunits of rabbit muscle phosphorylase kinase are phosphorylated in response to an intravenous injection of adrenalin.[145] In these experiments it was again necessary to homogenize the tissue in the presence of fluoride ions to prevent phosphatase action. By separation of the radioactively labelled peptides it was demonstrated that the sites of phosphorylation were the same as those that had previously been shown to be phosphorylated by the action of cAMP-dependent protein kinase *in vitro*.

(iii) Activation of cAMP-dependent protein kinase

It has proved more difficult to demonstrate convincingly that the cAMP-dependent protein kinase is activated *in vivo* in response to adrenalin. In theory it is possible to seek evidence for the cAMP-induced dissociation of the enzyme (Fig. 6.26) or for the action of the kinase. Both approaches involve a number of technical problems;[146] for instance in the second approach, account must be taken of the action of other protein kinases that are known to occur in tissues and that are not cAMP-dependent. Reasonably clear-cut demonstrations of the activation of cAMP-dependent protein kinase have been afforded by the work on phosphorylase kinase ((ii) above) and by the observation that, in rat liver, histone H1 could be phosphorylated in response to hormones at the site that is labelled by cAMP-dependent protein kinase *in vitro*.[147] A further discussion of the problems involved in this work are given in reference 146.

(iv) Increase in concentration of cAMP

We have already mentioned (Section 6.4.2.1) that administration of adrenalin and other hormones causes an increase in the concentration of cAMP, although the changes are relatively small. The reason that the activation of adenylate cyclase does not lead to large increases in the concentration of cAMP is that cAMP is hydrolysed to AMP in a reaction catalysed by 3':5'-cyclic-nucleotide phosphodiesterase (Fig. 6.26). The K_m for cAMP is higher than its prevailing concentration, so that as the concentration of cAMP is raised by activation of adenylate cyclase, the activity of the phosphodiesterase is also raised.[121]† It is therefore essential that the relatively small changes in the concentration of

* The phosphorylation of α- and β-subunits increases the activity of phosphorylase kinase some 15–20 fold, but this activity is still completely dependent on Ca^{2+}.[132] At the concentration of Ca^{2+} in resting muscle (0.1 μmol dm^{-3}), the activity is likely to be very low. The ability of adrenalin to bring about the conversion of phosphorylase *b* to phosphorylase *a* is thought to be due in large part to the inhibition of protein phosphatase-1 (see Section 6.4.2.4).[132]

† There may be as many as 6 or 7 phosphodiesterases in the cell with differing K_m values for cAMP, but the one generally believed to play a major role in controlling the level of cAMP has a high K_m.

cAMP lead to an amplified physiological response; this amplification is achieved by the system of interconvertible enzymes shown in Fig. 6.26.

Thus, in conclusion it can be seen that the principal events in the proposed cascade system (Fig. 6.26) have been demonstrated to occur not only *in vitro*, but also *in vivo*. Before ending this discussion of the regulation of glycogen metabolism, two further aspects should be mentioned: firstly, the regulation of the enzymes that reverse the actions of the enzymes in the cascade system (Fig. 6.26), and secondly, the regulation of glycogen synthase, the controlling enzyme in the synthesis of glycogen (Fig. 6.25). These aspects are discussed in Sections 6.4.2.4 and 6.4.2.5, respectively.

6.4.2.4 *How is phosphorylase switched off?*

It appears inevitable that an elaborate control system of the type shown in Fig. 6.26 should possess mechanisms able to switch off phosphorylase and so prevent unnecessary depletion of glycogen reserves. As described in Section 6.4.2.3, cAMP is hydrolysed to AMP in a reaction catalysed by 3′:5′-cyclic-nucleotide phosphodiesterase. The action of the kinases (cAMP-dependent protein kinase, Ca^{2+}-calmodulin multiprotein kinase, and phosphorylase kinase) could be reversed by the action of one or more phosphatases (Section 6.4.2.1), thereby allowing much finer control of enzyme activity (Section 6.3.2.2) at the expense of consumption of a small amount of energy. These phosphatases have now been characterized, and appear to be of two main types.[137,148] Type 1 phosphatases dephosphorylate the β-subunit of phosphorylase kinase preferentially, whereas the type 2 enzymes act preferentially on the α-subunit of this enzyme. The type 1 enzymes each possess the same catalytic subunit of M_r 37 000, whereas there are three distinct type 2 enzymes (termed 2A, 2B, and 2C) each with a different catalytic subunit. In addition, one or more small inhibitor proteins are known that inhibit type 1 phosphatases, but not type 2. Inhibitor-1 for instance is a remarkably stable* protein of M_r 20 000 that is present in muscle at concentrations (1.5 μmol dm^{-3}) comparable with the concentrations of type 1 phosphatase, phosphorylase kinase, and glycogen synthase. To make the picture even more intriguing, it has been found that inhibitor-1 inhibits the type 1 enzyme only after it has itself been phosphorylated at a threonine side-chain by the action of cAMP-dependent protein kinase[149] (Fig. 6.27). This phosphorylation has been shown to occur *in vivo* in response to adrenalin,[150] thus permitting coordinated control of kinase and phosphatase in response to the hormone.

Phosphatases 1, 2A, and 2C account for the vast majority of protein phosphatase activity in muscle and act on a variety of enzymes involved in major metabolic pathways including glycogen metabolism, glycolysis, fatty-acid synthesis, and amino-acid breakdown.[137] In skeletal muscle, type 1 phosphatase appears to be associated largely with glycogen particles (see Chapter 7,

* The purification of this protein involves heating partially purified muscle extracts at 363 K (90 °C) to precipitate unwanted proteins.

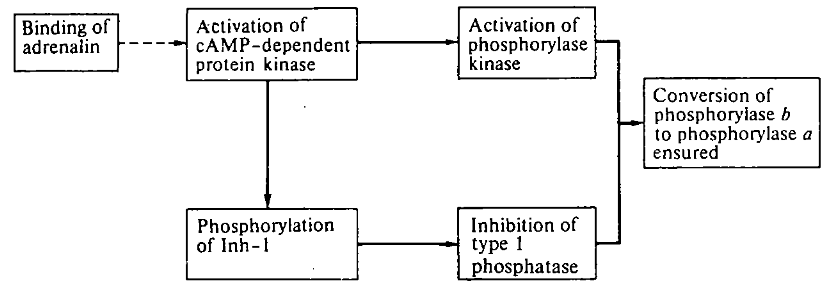


FIG. 6.27. Coordinated control of the activities of phosphorylase kinase and phosphorylase phosphatase following binding of adrenalin to the cell surface.

Section 7.12) via a glycogen-binding subunit of M_r 103 000. Type 2A and 2C phosphatases are confined to the cytosol.

Type 2B phosphatase is a Ca^{2+}-calmodulin-stimulated enzyme consisting of a catalytic subunit and a Ca^{2+}-binding subunit. It is of much narrower specificity than types 1, 2A, and 2C phosphatases. One of its most effective substrates is the phosphorylated form of inhibitor-1; thus, type 2B phosphatase could play a role in modulating the activity of type 1 phosphatase.[137]

6.4.2.5 *The regulation of glycogen synthase*

As mentioned in Section 6.4.2, glycogen synthase is the controlling enzyme in the synthesis of glycogen. Extensive work has shown that glycogen synthase is subject to a variety of controlling mechanisms that nicely complement those that exist for phosphorylase, thus permitting a coordinated control of glycogen metabolism.

Glycogen synthase can be converted from an active form (known as *a*, or I because its activity is Independent of D-glucose 6-phosphate) to an inactive form (known as *b* or D, because its activity is Dependent on D-glucose 6-phosphate) by the action of cAMP-dependent protein kinase.[151] Thus, as phosphorylase is switched on, glycogen synthase is switched off (Fig. 6.28).

The control of glycogen synthase is becoming better understood as more work is being performed on the purified enzyme (a tetramer of subunit M_r 86 000).[152] Two significant points emerge from recent work.

1. Glycogen synthase (like phosphorylase kinase; Section 6.4.2.3) can be phosphorylated at a number of sites (Fig. 6.29). It appears that there are at least nine distinct sites that can be phosphorylated *in vitro* (seven of which have been shown to be phosphorylated *in vivo*[152]). The cAMP-dependent protein kinase phosphorylates two serine side-chains near the C-terminus and a serine seven amino acids from the N-terminus; the latter can also be phosphorylated by phosphorylase kinase and by the Ca^{2+}-calmodulin multiprotein kinase (Section 6.4.2.1). At least three other kinases capable of phosphorylating glycogen synthase have been reported.[137,152] The various modifications produce forms of

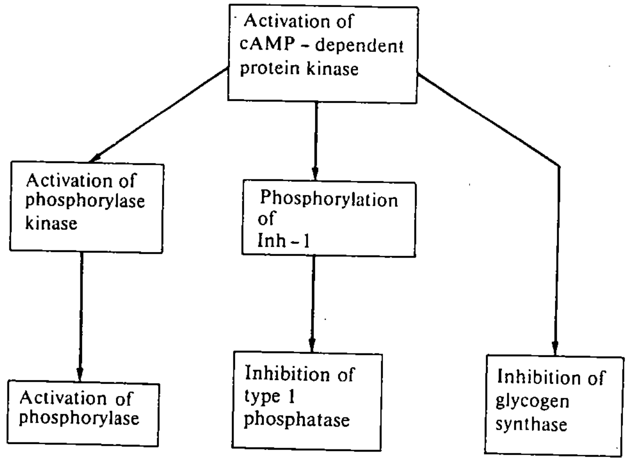


FIG. 6.28. Multiple effects of activation of cAMP-dependent protein kinase.

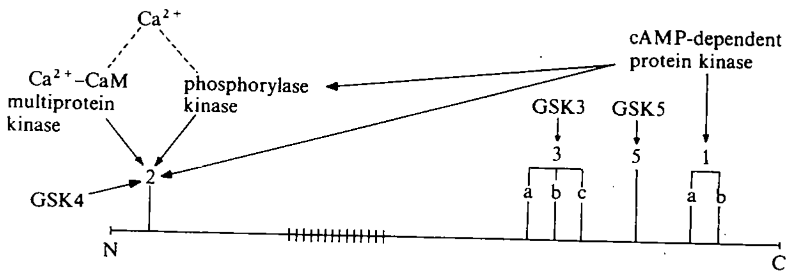


FIG. 6.29. The sites of phosphorylation of glycogen synthase.[137] GSK3, 4, and 5 are glycogen synthase kinases 3, 4, and 5. CaM is calmodulin. The polypeptide chain is not shown to scale.

glycogen synthase with distinctly different properties, thereby increasing the versatility of the response to signals such as cAMP and Ca^{2+}.

2. It has been known for many years[121] that insulin stimulates the activity of glycogen synthase, thus allowing D-glucose to be converted to glycogen under conditions in which glycolysis is inhibited (Section 6.4.1.2). The mechanism by which insulin exerts its effect is still not properly understood. The hormone has little effect on the concentration of cAMP or on the activity of cAMP-dependent protein kinase. Studies of the state of phosphorylation of the seven serine side-chains known to be phosphorylated in glycogen synthase *in vivo* have given some clues to the action of insulin.[153] In rabbits treated with propanolol (which blocks the action of adrenalin) plus insulin, there is a significant decrease in the degree of phosphorylation of sites 3a + 3b + 3c (Fig. 6.29) but no change in the degree of phosphorylation of the other sites (1a, 1b, 2, and 5). Dephosphorylation of sites 3a, 3b, and 3c is therefore responsible for the activation of glycogen synthase upon administration of insulin. The way this is brought about is not yet clear* since neither type 1 nor type 2A protein phosphatases, the major glycogen synthase phosphatase activities in skeletal muscle (Section 6.4.2.4), can bring about specific dephosphorylation of sites 3a, 3b, and 3c. It is possible that insulin has some specific inhibitory effect on the activity of glycogen synthase kinase 3 (see Fig. 6.29), but this has not been verified experimentally. There is currently intensive work on the mechanism of action of insulin in stimulating the synthesis of glycogen. The possibilities that polyamines such as spermine may act as second messengers and that the insulin receptor may act as a tyrosine kinase (catalysing the phosphorylation of tyrosine side-chains in proteins) are being explored.[137]

6.4.2.6 *Concluding remarks on the regulation of glycogen metabolism*

The preceding Sections (6.4.2.1 to 6.4.2.5) have illustrated the variety of mechanisms by which glycogen metabolism can be regulated, e.g. ligand-induced conformational changes in enzymes, hormonal action, and nerve impulses acting via Ca^{2+} ions. These mechanisms allow a variety of responses in different metabolic circumstances. It might appear that the recent work mentioned has only served to confuse the issue, since it has now been found that a bewildering variety of enzymes and other proteins are involved in regulation. However, some of the recent work has demonstrated that considerable economy is exercised on the part of an organism, since some of the enzymes and proteins have multiple effects. For instance, cAMP-dependent protein kinase and Ca^{2+}-calmodulin multiprotein kinase have effects on enzymes involved in a number of metabolic pathways† such as glycolysis, glycogen and lipid metabolism, and catecholamine synthesis.[137, 148] Protein kinases thus appear to play a central role in tramsmitting the effects of intracellular signals such as cAMP and Ca^{2+} to the target enzymes, which then bring about the metabolic changes in the cell. It also appears that the various phosphatases each have multiple activities (Section 6.4.2.4).

The regulation of glycogen metabolism is still under very active investigation and it is reasonable to expect that new features of the system will emerge; nevertheless, the account given here should give some impression of the present-day ideas on the subject.

6.5 Concluding remarks

In this chapter we have discussed the various means by which the activities of enzymes are controlled. The studies on single enzymes (Section 6.2) have

* Despite earlier reports to the contrary:[154] it now seems clear that insulin does not lead to any significant degree of dephosphorylation of inhibitor-1, which would activate type 1 protein phosphatase.

† The amino-acid sequences around the phosphorylated side-chain in the various substrates for cAMP-dependent protein kinase contain two (or more) basic amino acids on the N-terminal side and a hydrophobic amino acid on the C-terminal side.[137]

provided the conceptual basis on which theories of the control of metabolic pathways have been formulated (Section 6.3). At present, we are in the position that the control of a number of single enzymes can be understood in quantitative terms, and some progress has been made towards a quantitative understanding of the control of metabolic pathways. However, when the cellular level of organization is considered (see Chapter 8), the situation becomes even more complicated, since a number of new factors such as compartmentation are introduced. Much of the future work will be directed towards a detailed understanding of these new factors.

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Appendix 6.1

The Adair equation and the Scatchard equation for a protein containing multiple ligand-binding sites

Consider a system in which one molecule of protein, P, can bind up to n molecules of ligand, A.

	Sites occupied	Sites unoccupied
$P+A \rightleftharpoons PA$	1	$n-1$
$PA+A \rightleftharpoons PA_2$	2	$n-2$
$PA_2+A \rightleftharpoons PA_3$	3	$n-3$
$PA_{n-1}+A \rightleftharpoons PA_n$	n	0

Let the successive dissociation constants for PA, PA_2 ... be K_1, K_2, etc.

$$\text{i.e.}\quad K_1=\frac{[P][A]}{[PA]},\qquad K_2=\frac{[PA][A]}{[PA_2]},\quad \text{etc.}$$

A.6.1. The Adair equation

The fractional saturation of ligand sites (Y) is given by

$$Y=\frac{\text{total concentration of A bound}}{\text{total concentration of sites for A}}$$

$$=\frac{[PA]+2[PA_2]+3[PA_3]\ldots *}{n([P]+[PA]+[PA_2]+[PA_3]\ldots)}$$

$$=\frac{\dfrac{[P][A]}{K_1}+\dfrac{2[P][A]^2}{K_1K_2}+\dfrac{3[P][A]^3}{K_1K_2K_3}+\cdots}{n\left[[P]+\dfrac{[P][A]}{K_1}+\dfrac{[P][A]^2}{K_1K_2}+\dfrac{[P][A]^3}{K_1K_2K_3}+\cdots\right]},$$

* Note the factor n in the denominator. This arises because each molecule of P has n sites for A. The factors 1, 2, 3, etc. in the numerator arise because each molecule of PA_2 contains two molecules of A, and so on.

$$\therefore\quad Y=\frac{\dfrac{[A]}{K_1}+\dfrac{2[A]^2}{K_1K_2}+\dfrac{3[A]^3}{K_1K_2K_3}+\cdots}{n\left[1+\dfrac{[A]}{K_1}+\dfrac{[A]^2}{K_1K_2}+\dfrac{[A]^3}{K_1K_2K_3}+\cdots\right]}. \tag{A.6.1}$$

This is the Adair equation.

For the special case of a tetrameric protein with four sites for A, $n=4$. Then eqn (A.6.1) becomes

$$Y=\frac{\dfrac{[A]}{K_1}+\dfrac{2[A]^2}{K_1K_2}+\dfrac{3[A]^3}{K_1K_2K_3}+\dfrac{4[A]^4}{K_1K_2K_3K_4}}{4\left[1+\dfrac{[A]}{K_1}+\dfrac{[A]^2}{K_1K_2}+\dfrac{[A]^3}{K_1K_2K_3}+\dfrac{[A]^4}{K_1K_2K_3K_4}\right]}. \tag{A.6.2}$$

This is the Adair equation for a tetramer (see eqn (6.6)).

A.6.2 The Scatchard equation

We introduce a new parameter, r, which is the average number of molecules of A bound per molecule of P.

By definition, $r=nY$ where n equals the number of ligand-binding sites and Y equals the fractional saturation of ligand-binding sites. From eqn (A.6.1),

$$r=\frac{\dfrac{[A]}{K_1}+\dfrac{2[A]^2}{K_1K_2}+\dfrac{3[A]^3}{K_1K_2K_3}+\cdots}{1+\dfrac{[A]}{K_1}+\dfrac{[A]^2}{K_1K_2}+\dfrac{[A]^3}{K_1K_2K_3}+\cdots}.$$

In order to simplify this quite general expression, we have to derive a relationship between the successive K_s. Now if it is assumed that the sites are independent and equivalent (i.e. that the free energy of interaction of the ligand with each site is the same), then the K_s are related to each other by statistical factors, i.e. A can dissociate from the PA_2 complex in two ways, but A can associate with the $(n-1)$ vacant sites in the PA complex in $(n-1)$ ways. The general relationship is that the ith dissociation constant (K_i) is given by

$$K_i=\left[\frac{i}{n-i+1}\right]K,$$

where K is an intrinsic dissociation constant (i.e. one that takes into account these statistical factors).*

$$\text{Thus}\qquad K_1=\frac{K}{n},\qquad K_2=\frac{2K}{n-1},\qquad K_3=\frac{3K}{n-2},\quad \text{etc.}$$

Our expression for r now becomes

$$r=\frac{[A]\dfrac{(n)}{K}+\dfrac{2[A]^2(n)(n-1)}{2K^2}+\dfrac{3[A]^2(n)(n-1)(n-2)}{(2)(3)K^3}+\cdots}{1+\dfrac{[A](n)}{K}+\dfrac{[A]^2(n)(n-1)}{2K^2}+\dfrac{[A]^3(n)(n-1)(n-2)}{(2)(3)K^3}+\cdots}$$

* K is actually the geometric mean of all the dissociation constants, i.e. $K=(K_1K_2K_3\ldots K_n)^{1/n}$.

$$= \frac{\frac{[A](n)}{K}\left[1+\frac{[A](n-1)}{K}+\frac{[A]^2(n-1)(n-2)}{2K^2}+\cdots\right]}{1+\frac{[A](n)}{K}+\frac{[A]^2(n)(n-1)}{2K^2}+\frac{[A]^3(n)(n-1)(n-2)}{6K^3}+\cdots}.$$

The expressions in the numerator and denominator in this equation are both binomial expansions. Thus the expression can be simplified to give

$$r = \frac{\frac{[A](n)}{K}\left[1+\frac{[A]}{K}\right]^{n-1}}{\left[1+\frac{[A]}{K}\right]^{n}}$$

$$= \frac{\frac{[A](n)}{K}}{\left[1+\frac{[A]}{K}\right]}$$

$$\therefore \quad r = \frac{n[A]}{K+[A]}.$$

This equation can be rearranged to give the Scatchard equation:

$$\frac{r}{[A]} = \frac{n}{K} - \frac{r}{K}. \qquad \text{(A.6.3)}$$

Thus, a plot of $r/[A]$ against r gives a straight line of slope $-1/K$ and intercept on the x-axis of n, *provided that the sites are equivalent and independent* (see Fig. 6.15).

7
Enzymes in organized systems

7.1 Introduction

In the previous chapters we have discussed results obtained from the study of highly purified enzymes. By examining single purified enzymes it is possible to study the structure (Chapter 3), the kinetics (Chapter 4), and mechanism of action (Chapter 5) of a single enzyme in great detail and without interference from competing reactions, and this is essential for an understanding of the molecular basis of enzyme catalysis. However, in the intact cell, enzymes do not act as separated, isolated catalysts in dilute aqueous buffers with kinetics approximating to those described by the Michaelis and Menten equation (see Chapter 4, eqn 4.4). In many subcellular compartments the protein concentrations are high (e.g. in the mitochondrial matrix, 500 mg protein cm^{-3}),[1] and many enzymes compete with one another for substrates and effectors and they may be physically associated with other enzymes to varying extents. In the next two chapters we consider how enzymes may behave in the intact cell, firstly by examining the types of organized enzyme systems that exist in the cell, and secondly by examining the nature of the environment in which enzymes operate *in vivo* and how this differs from the conditions under which most enzyme assays are performed *in vitro*.

7.2 Organized enzyme systems

The organization of enzymes can be seen as a progression of increasing complexity extending from those enzymes that exist as single separate polypeptide chains, through to oligomeric enzymes, which may show allosteric interactions between the subunits, and ultimately to multienzyme proteins.* In the previous four chapters, we have already considered enzymes having single polypeptide chains and oligomeric enzymes and thus in the next sections we examine the following types of organization: (i) enzymes which although catalysing a single reaction, also interact in a specific manner with a macromolecular template and therefore require a more complex structure; (ii) multienzyme proteins with well-defined structures which have more than one distinct catalytic activity; and (iii) loose associations of enzymes in which the stoichiometry may be variable. RNA nucleotidyltransferase (DNA-dependent RNA polymerase) is the best example of the first type, since, in addition to catalysing nucleotide

* There is not yet a standard nomenclature for multienzymes. We use here that proposed by Karlson and Dixon.[2] Multienzyme proteins include all proteins with multiple catalytic domains, and they may be subdivided into those that are covalently linked (multienzyme polypeptides) and those that are non-covalently linked (multienzyme complexes).

transfers, the enzyme has to recognize specific regions on the DNA template in order to initiate and to terminate the synthesis of particular species of RNA. It is discussed in Section 7.3. There are several examples of the second type, e.g. pyruvate dehydrogenase (Section 7.7), tryptophan synthase (Section 7.9), and fatty-acid synthase (Section 7.11.2), in which more than one step is catalysed by the system concerned. The third type is represented by the glycogen particle (Section 7.12).

7.3 RNA nucleotidyltransferase from *E. coli*

RNA nucleotidyltransferase from *E. coli* represents an intermediate stage in the complexity of its organization between that of a typical allosteric enzyme and a large multienzyme protein. It comprises four different polypeptide chains, each having distinct functions in the complex but at the same time each helping to coordinate the synthesis of RNA on a DNA template. The RNA nucleotidyltransferase from *E. coli* is discussed specifically here for the following reasons: (i) it is the most highly purified and extensively studied RNA nucleotidyltransferase, (ii) all the evidence available suggests that it is responsible for the synthesis of all the RNA (mRNA, rRNA, and tRNA) present in uninfected *E. coli* cells, in contrast to the situation in eukaryotes where there is a multiplicity of RNA nucleotidyltransferases serving different functions. The enzyme catalyses the complete transcription cycle (Fig. 7.1; see also Chapter 9, Section 9.2 for an explanation of the process of transcription), which includes the steps of (i) recognition of the specific promoter regions on the genome, (ii) transcribing the appropriate gene or group of genes controlled by a particular operator, and (iii) terminating transcription at the appropriate region of the genome. The RNA nucleotidyltransferase is thus concerned with the synthesis of a large number of different mRNAs, each subject to different controls, and also with the synthesis of tRNA and rRNA. (For reviews on transcription and RNA nucleotidyltransferases, see references 3, 4, 5, and 6).

7.3.1 Structure of RNA nucleotidyltransferase

The enzyme from *E. coli* has been purified in several different laboratories and shown to be composed of four different polypeptide chains, $\alpha(M_r\ 36\,512)$, $\beta(M_r\ 150\,619)$, $\beta'(M_r\ 155\,162)$ and $\sigma(M_r\ 70\,236)$ and has the composition $\alpha_2\beta\beta'\sigma$. The genes for the polypeptides have been cloned and sequenced. The enzyme contains two Zn atoms, which are not removed by dialysis against chelating agents, although they are released when the enzyme is dissociated. Their role is at present unknown. Two models have been proposed for the quaternary structure (see Section 7.3.4). Genetic evidence from *E. coli* mutants suggests that these are the only proteins involved in transcription, but an additional protein, *nus* A protein is implicated in elongation.[7] The σ subunit can dissociate from the holoenzyme ($\alpha_2\beta\beta'\sigma$) to yield a core enzyme ($\alpha_2\beta\beta'$). Both the holoenzyme and the core enzyme will catalyse RNA synthesis on a DNA template *in vitro* (see

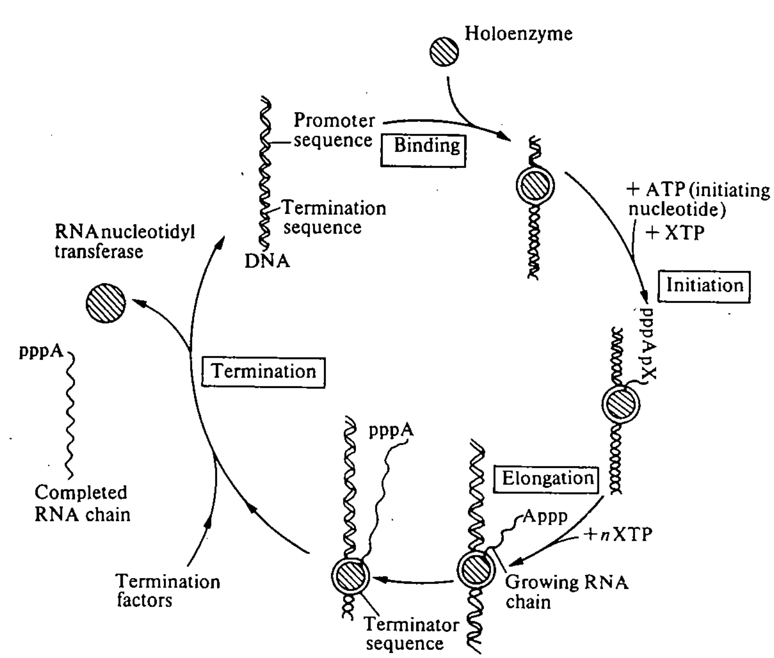


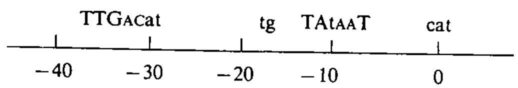
FIG. 7.1. The cycle of events catalysed by RNA nucleotidyltransferase (RNA polymerase) from *E. coli*. The release of the σ subunit following initiation is not shown.

Fig. 7.1 under binding, initiation, and elongation), but whereas the holoenzyme only initiates RNA synthesis on one strand of the DNA, commencing at the promoter sites, the core enzyme catalyses transcription on both strands starting at random positions. During normal transcription by the holoenzyme, the σ subunit is released soon after initiation. The σ subunit clearly has some role in normal initiation. The *nus* A protein is able to bind to the core enzyme in a similar fashion to the σ protein.[7]

7.3.2 Binding of RNA nucleotidyltransferase to DNA and initiation of transcription

We discuss next the present knowledge concerning the overall reaction mechanism and then subsequently the possible roles played by the individual subunits. The process of transcription is initiated by recognition of the promoter region on the DNA template (for a review of promoters see references 8, 9). It is estimated that the complete chromosome of *E. coli* contains in the region of 2000 promoters,[9] associated with different operons. The dissociation constants for the interaction between the core enzyme and DNA and between the holoenzyme and DNA are 0.5×10^{-11} mol dm^{-3} and 10^{-7} mol dm^{-3}, respectively. The σ subunit thus reduces the *general* binding affinity of RNA nucleotidyltransferase by about 20 000 times, and it is thought that this is important in enabling the holoenzyme to locate a particular region of DNA, namely the promoter site(s). The dissociation constant for the interaction of the holoenzyme with a promoter region is about 10^{-14} mol dm^{-3}. There is thus very tight binding between a promoter region and the holoenzyme and this has enabled the binary complex to be isolated on nitrocellulose filters. The different promoters will compete for RNA nucleotidyltransferase molecules and their effectiveness in competition will determine their frequency of usage. This is generally referred to as promoter strength, strong promoters being most effective.

The promoter regions are identified by nuclease digestion of the binary complex between DNA and the RNA nucleotidyltransferase, since the latter protects the promoter region. Over 100 promoter regions in *E. coli* have been sequenced[9] and by comparing these a consensus sequence can be formulated. There are two regions around positions -35 and -10 that are highly conserved. (The numbering indicates the number of nucleotides before the position at which transcription is initiated). In addition to the highly conserved regions (shown in large capitals), there are moderately conserved regions (small capitals) and weakly conserved regions (lower case letters). The contact points between the RNA nucleotidyltransferase and promoters are heavily concentrated in the consensus regions around -35 and -10.



The positioning of the holoenzyme at the start sequence for transcription is accurate to within $\pm$one nucleotide e.g. at the *lac* promoter in *E. coli*, transcription is initiated with the ribonucleotide sequence pppApApU, etc. or pppGpApApU, etc. Once the first phosphodiester bond has formed, the ternary complex (holoenzyme: DNA: oligoribonucleotide) is more stable and does not readily dissociate, although the σ subunit dissociates[10] from the holoenzyme after the first few phosphodiester bonds have formed, leaving the core enzyme: DNA: oligoribonucleotide complex to continue the process of elongation. Although certain prokaryotes have been shown to produce more than one σ factor, e.g. *Bacillus subtilis* uses at least five σ factors at different stages of its life cycle,[11, 12,134] for a long period only one σ factor had been isolated from uninfected *E. coli*. However, a second factor has more recently been discovered[13] as a result of studying proteins synthesized in response to heat shock.

An important question is how the holoenzyme locates the promoters within the double-stranded DNA of the chromosome. It is possible by using the values of estimated diffusion constants for DNA nucleotidyltransferase and the promoter to calculate the maximum rate at which the RNA nucleotidyltransferase could locate a promoter by a diffusion-controlled process.[14] However, when the calculated maximum rate is compared with experimentally determined rates for the strongest promoters, the latter are approximately 100-fold greater. In a detailed discussion Von Hippel *et al.*[14] suggest that the holoenzyme locates non-specific sites on the DNA initially, thus providing an 'expanded target', and then translocates along the DNA to the promoter region.

7.3.3 Elongation and termination of transcription

Elongation of the nascent RNA chain is catalysed by the core enzyme and the process continues, new nucleotides being added in a sequence complementary to that of one strand of the DNA. The mechanism is considered to be as shown in Fig. 7.2. It has been estimated that errors in transcription are between 1 in 2400 and 1 in 42 000 depending on conditions. The process of termination involves three steps: (i) termination in growth of the polynucleotide chain, (ii) release of nascent RNA, and (iii) release of the core enzyme from the DNA template. The mechanism of termination is not fully understood but is partly dependent on the nucleotide sequence and partly on the presence of a protein factor ρ.[15] The termination sequences of a number of genes have been examined and they show a number of common features, namely, a GC-rich region followed by several uracil residues and terminating with one or two adenine residues (Table 7.1); these terminate independently of protein factor ρ. The ρ factor is an oligomeric protein, the monomers having an M_r 46 000. It has been shown from *in vitro* experiments to promote termination and release of the RNA chain but not of the core enzyme. The termination sites that require ρ for termination appear to contain fewer conserved sequences.

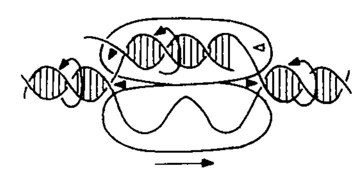


FIG. 7.2. Proposed mechanism of transcription. A limited region of the DNA is unwound and a DNA:RNA hybrid of about 12 base pairs is formed during transcription. The axis of the DNA template rotates relative to that of the enzyme. Unwinding of the DNA:RNA hybrid occurs during the rotation. The model is supported by photochemical cross-linking, chemical modification, and nuclease protection experiments.[14]

TABLE 7.1
Termination sequences in RNA recognized by RNA nucleotidyltransferase

λ 6S RNA	. . . A A C G G U U U C G G G G A U U U U U U A—3′—OH
λ 4S RNA	. . . U U G C C G C C G G G C G U U U U U U A—3′—OH
ϕ 80 M_3 RNA	. . . C C G C U U C A A G G G C U U U U U A—3′—OH

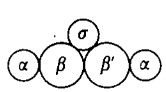
(λ and ϕ 80 M_3 are bacteriophages)

Thus, the holoenzyme is necessary for correct initiation of RNA synthesis, after which the σ factor is no longer necessary, and the additional proteins such as *nus* A and ρ may be required for elongation and termination. In the next section we consider the roles of the subunits in the process of transcription.

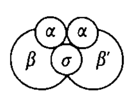
7.3.4 The roles of the subunits of RNA nucleotidyltransferase

The roles of the individual subunits of RNA nucleotidyltransferase have been studied in the following ways: (i) examining the properties of isolated subunits, (ii) partial and complete reconstitution of the holo- and core enzymes, and (iii) from a study of mutations affecting the subunits.[16] The individual subunits have all been isolated. The α subunit is present in solution as α_2, it carries a binding site for the β subunit and is involved in the specificity of interaction with a gene, possibly by binding to the promoter region. However, in spite of the fact that its complete amino-acid sequence is known, it is the subunit about which there is least evidence for the function. The β' subunit is the most basic of the subunits; it is able to bind directly to DNA and is undoubtedly involved in template binding. The β subunit participates in binding the substrates, is involved in initiation and elongation, and it also carries the catalytic site for phosphodiester bond formation. The substrate-binding site was detected by affinity labelling (see Chapter 5, Section 5.4.4.3) using uridine triphosphate analogues. RNA nucleotidyltransferase from *E. coli* is strongly inhibited by the antibiotics rifamycin and streptolydigin and this is due to interaction with the β subunit. Proof of this is that resistant strains of *E. coli* have a modified β subunit, and that if RNA nucleotidyltransferase is reconstituted using the β subunit from a resistant strain and the other subunits from a sensitive strain, then the resultant enzyme is insensitive to the antibiotics.

The details of the quaternary structure are still not completely clear. Alternative models have been proposed by Coggins *et al*,[17] and by Hillel and Wu[18] and they are discussed in detail in reference 5. The model of Coggins *et al.* is based on cross-linking studies using *N*,*N'*-bis-(2-carboximidoethyl)-tartaramide dimethyl ester dihydrochloride (see Chapter 3, Section 3.6.12) and that of Hillel and Wu[18] on low-angle neutron scattering and cross-linking with methyl 4-mercaptobutyrimidate.



Model of Coggins et al.[17]



Model of Hillel and Wu[18]

The main point of difference between the two is whether the α-subunits are separated.

Partially reconstituted complexes have been isolated containing $\alpha_2\beta$ and $\beta'\sigma$. The $\alpha_2\beta$ does not bind DNA but it does bind rifamycin. The existence of $\alpha_2\beta$ suggests that β contains at least one strong binding site for α. Similarly the existence of $\beta'\sigma$ suggests that β' contains the binding site for σ. The holoenzyme has been reconstituted *in vitro* by the following pathway and a similar route is thought to operate *in vivo*:[16]

$$2\alpha \rightleftharpoons \alpha_2$$
$$\alpha_2+\beta \rightleftharpoons \alpha_2\beta$$
$$\alpha_2\beta+\beta' \rightleftharpoons \alpha_2\beta\beta'$$
$$\alpha_2\beta\beta'+\sigma \rightleftharpoons \alpha_2\beta\beta'\sigma$$

This pathway is consistent with the arrangement of the subunits in the holoenzyme as determined by Hillel and Wu[18] and by Stöckel *et al.*[19] However, it is possible that, although the α subunits associate during the first stage of assembly, a subsequent rearrangement may occur. That this is likely is suggested by the observation that the initial $\alpha_2\beta\beta'\sigma$ complex formed is inactive, and thus may undergo rearrangement before becoming active.

7.3.5 RNA nucleotidyltransferases from bacteriophages

RNA nucleotidyltransferase from *E. coli* has to recognize a variety of promoter sites on the chromosome and transcription from these sites is subject to a variety of different controls and it is perhaps partly for this reason that the enzyme is a multisubunit protein. By contrast, the bacteriophage RNA nucleotidyltransferases have only to recognize a few promoters and this is reflected in a simple enzyme structure. For example, the RNA nucleotidyltransferases coded for by genes on the bacteriophages T_3 and T_7 comprise single polypeptide chains of M_r 110 000. These RNA nucleotidyltransferases are highly specific in their recognition of DNA. The T_7 enzyme recognizes only poorly the promoters of T_3 and vice versa. Thus, the complexity of any given RNA nucleotidyltransferases is partly a function of the range of species of RNA it is required to synthesize.

7.4 The occurrence and isolation of multienzyme proteins

In a metabolic pathway a number of enzymes catalyse a sequence of reactions linked so that the product of one enzyme-catalysed reaction becomes the substrate for the next enzyme. Such a sequence may be represented as

$$A \xrightarrow{E_1} B \xrightarrow{E_2} C \xrightarrow{E_3} D \xrightarrow{E_4} E \xrightarrow{E_5} F$$

in which A, B, C, D, and E are substrates for the enzymes E_1, E_2, E_3, E_4, and E_5. Both the efficiency of these reactions, measured as the yield of the end product F from the starting substrate A, and the overall rate of conversion of A to F will depend partly on the extent of coordination occurring between the five enzymes.

For some metabolic pathways it has been shown that certain of the enzymes are physically associated with one another to form multienzyme proteins, e.g. the fatty-acid synthase complex (Section 7.11.2) and enzymes of the tryptophan biosynthetic pathway (Sections 7.9 and 7.11.3).

The multienzymes that have been most studied are probably amongst the more stable ones. There may be many others in which the associations are weak and are thus more difficult to detect as multienzymes. The isolation and characterization of a multienzyme protein usually presents more technical difficulties than isolation of a single enzyme. This is well illustrated by two examples discussed in more detail later in the chapter, namely, the pyruvate dehydrogenase complex (Section 7.7) and the yeast fatty-acid synthase complex (Section 7.11.2). In the case of the pyruvate dehydrogenase complex from *E. coli*, although it is clear that the multienzyme complex contains three different enzyme activities, the precise number of each of the polypeptide chains present in the complex is still debated. This may be because (i) different isolation procedures yield complexes of slightly different composition, (ii) some dissociation may occur during isolation, or (iii) in the intact cell complexes of slightly variable composition do exist. In a multienzyme complex having a large number of subunits, e.g. 50–100, a small percentage error in the determination of the M_r of the complex will significantly affect the proposed number of subunits.

Yeast fatty-acid synthase posed a different type of problem. It proved difficult to separate the polypeptide chains corresponding to the seven catalytic activities of the fatty-acid synthase. It was then shown that the whole complex comprises two multifunctional polypeptide chains (see Section 7.11.2.1) and that the smaller fragments observed in earlier experiments were the result of limited proteolysis during isolation.

Evidence for the existence of a multienzyme complex usually appears when a component enzyme of the complex is being isolated and it is found to copurify with another enzyme from the same metabolic pathway. If the ratio of the enzyme activities remains constant during the isolation procedure, then this is further evidence for the existence of a definite multienzyme complex, e.g. carbamoyl phosphate synthase, aspartate carbamoyltransferase, dihydro-orotase (see Section 7.10).

Two enzyme activities that were initially shown to copurify from rat liver and were subsequently found to be two catalytic activities associated with a single polypeptide chain are fructose 2,6-bisphosphatase and 6-phosphofructo-2-kinase. The enzyme catalysing both these activities has been purified from a number of tissues and it has been shown that the two activities reside in separate parts of the polypeptide chain.[20, 21] The importance of this enzyme is discussed in Chapter 6, Section 6.4.1.1. Another example of phosphatase and kinase activities residing in the same polypeptide chain is that of isocitrate dehydrogenase kinase and isocitrate dehydrogenase phosphatase. Both activities copurify and are coded for by a single gene.[22, 23]

7.5 Phylogenetic distribution of multienzyme proteins

Although only a limited number of multienzyme complexes have so far been studied, it does seem that in general multienzyme complexes and multienzyme polypeptides are both more common and more complex in eukaryotes than in prokaryotes; this is borne out by the examples of pyruvate dehydrogenase, fatty-acid synthase, and carbamoyl phosphate synthase, aspartate carbamoyltransferase, dihydro-orotase and the enzymes of the tryptophan biosynthetic pathway discussed in this chapter. A typical prokaryote cell, e.g. *E. coli*, has a very much smaller volume than a typical eukaryote cell, e.g. liver parenchyma (1 μm^3 as compared to 6000 μm^3). Even making allowance for the intracellular compartmentation in eukaryotes brought about by the various intracellular membranes, the volume in which free diffusion of substrates and enzymes can occur is much greater than in a prokaryote cell. Thus, if the intracellular concentrations of substrates and catalytic centers are similar in prokaryotes and eukaryotes, then the diffusion time is more likely to be rate limiting in the latter. The comparative biochemical information on carbamoyl phosphate synthase, aspartate carbamoyltransferase, dihydro-orotase, the tryptophan biosynthetic pathway, and fatty-acid synthase suggests that evolution has resulted in multienzyme polypeptides and more tightly associated multienzyme complexes. In addition, genetic evidence suggests that this is accompanied by a clustering or even fusion of the genes coding for these proteins.

7.6 Properties of multienzyme proteins

What are the advantages of enzymes being physically associated with one another rather than occurring as separate entities within cells? The answers may be found by comparing the properties of multienzyme proteins with those of separated enzymes. Firstly, the transit time, i.e. the time required for a product of one enzyme to diffuse to the catalytic site of a second enzyme will be less in a physically associated multienzyme protein than when the enzymes are not associated. Whether the transit time becomes the rate-limiting step in a metabolic pathway will depend on the catalytic activities of the enzymes. Two instances in which the transit times may become important are (a) when an intermediate has a short half life and (b) when an intermediate has a high M_r and thus would diffuse more slowly. There is as yet no well-documented example of the first situation. Carbamoyl phosphate probably has a short half life in the cytosol, since there are enzymes capable of rapidly degrading it. The association of carbamoyl phosphate synthase with aspartate carbamoyl transferase, which channels carbamoyl phosphate within the multienzyme protein, probably prevents its degradation.[24] An example of the second situation is that of the pyruvate dehydrogenase system, in which the second step in the reaction sequence involves the two enzyme-bound intermediates hydroxyethyl-thiamin-diphosphate-pyruvate dehydrogenase and dihydrolipoamide acetyltransferase.

Secondly, in addition to the transit time being reduced in a multienzyme protein, so also would be the transient time, that is the time required to change from one steady-state rate to another. Thus in a system with two enzymes

$$A \xrightarrow{E_1} B \xrightarrow{E_2} C$$

in a steady-state in a cell, if the concentration of A is increased then the time required to reach the new steady-state rate (transient time) will be much less in the case of a multienzyme protein.[25]

A third possibility afforded by multienzyme proteins is that of channelling or compartmentation. Once an intermediate has been formed in a multienzyme protein, it is not generally available to be acted upon by other enzymes outside the complex. Examples of this occur with carbamoyl phosphate formed in the multienzyme complex for pyrimidine biosynthesis in *Neurospora crassa* (see Section 7.10) that is not available for the biosynthesis of arginine, and also with ammonia generated by glutaminase in liver mitochondria, which is preferentially used by carbamoyl phosphate synthase.[27]

There is also some evidence for coordinate effects of a single ligand on a whole multienzyme protein, e.g. in the *arom* complex discussed later (Section 7.11.3), where the substrate of the first enzyme is able to activate all five catalytic activities of the multienzyme protein.

The examples of multienzymes discussed in the subsequent sections of this chapter have been chosen to illustrate the various properties of multienzyme proteins that have been described in this section. The pyruvate dehydrogenase, oxoglutarate dehydrogenase, and tryptophan synthase systems are the multienzyme complexes about which most structural information is available. They also illustrate how the intermediate substrates remain attached to the multienzyme complexes. Fatty acid synthase and the *arom* complex are examples of multienzyme polypeptides. The multienzyme complex of carbamoyl phosphate synthase and aspartate carbamoyltransferase is used to illustrate the importance of channelling an intermediate (carbamoyl phosphate) into one of two competing pathways. The 'glycogen particle' is much less well defined than the other multienzyme proteins and presumably represents a looser association between glycogen and the glycolytic enzymes.

7.7 The pyruvate dehydrogenase system

The pyruvate dehydrogenase multienzyme complex is one of the most thoroughly investigated multienzyme proteins. The complex catalyses the conversion of pyruvate to acetyl-CoA, which is the step connecting the glycolytic sequence to the tricarboxylic acid cycle and is an important control point in intermediary metabolism. The reactions catalysed by the complex, which comprises three distinct enzymes, are decarboxylation, dehydrogenation, and the formation of a

thioester. Five cofactors are necessary for the complete reaction sequence, as shown below.

$$E_1\text{-TDP} + CH_3COCOO^- \rightleftharpoons E_1\text{-TDP}\text{—}CHOHCH_3 + CO_2 \quad (1)$$

$$E_1\text{-TDP}\text{—}CHOHCH_3 + E_2\text{-lip}\begin{smallmatrix} S \\ | \\ S \end{smallmatrix} \rightleftharpoons E_1\text{-TDP} + E_2\text{-lip}\begin{smallmatrix} SH \\ SCOCH_3 \end{smallmatrix}, \quad (2)$$

$$E_2\text{-lip}\begin{smallmatrix} SH \\ SCOCH_3 \end{smallmatrix} + CoASH \rightleftharpoons E_2\text{-lip}\begin{smallmatrix} SH \\ SH \end{smallmatrix} + CH_3COSCoA \quad (3)$$

$$E_2\text{-lip}\begin{smallmatrix} SH \\ SH \end{smallmatrix} + E_3\text{-FAD} \rightleftharpoons E_2\text{-lip}\begin{smallmatrix} S \\ | \\ S \end{smallmatrix} + E_3\text{-FAD reduced*} \quad (4)$$

$$E_3\text{-FAD reduced} + NAD^+ \rightleftharpoons E_3\text{-FAD} + NADH + H^+ \quad (5)$$

E_1 = pyruvate decarboxylase† TDP = thiamin diphosphate

E_2 = dihydrolipoamide acetyl transferase $\text{lip}\begin{smallmatrix} S \\ | \\ S \end{smallmatrix}$ = oxidized lipoamide

E_3 = dihydrolipoamide dehydrogenase $\text{lip}\begin{smallmatrix} SH \\ SH \end{smallmatrix}$ = reduced lipoamide

The complex that has been most extensively investigated is that from *E. coli*, although it has also been purified from a number of other bacteria[28–30] and mammalian sources.[31,32] The complex isolated from *E. coli* is discussed in Sections 7.7.1 to 7.7.3. The regulation of the complexes from mammalian sources differs from that of *E. coli* and these differences are discussed in Section 7.7.4.

7.7.1 Structure and isolation of the pyruvate dehydrogenase multienzyme complex from *E. coli*

Although it is clear that the multienzyme complex contains three distinct enzymes, pyruvate decarboxylase (E_1), dihydrolipoamide acetyltransferase (E_2), and dihydrolipoamide dehydrogenase (E_3), there is still uncertainty concerning the number of each of these enzymes in the complex. There is reasonable agreement about the M_r of the separate polypeptide chains at least of E_1 and E_3, and the M_r values of E_1 and E_2 have been inferred from the DNA sequence.[33,34] However, there is less agreement about the M_r of the intact complex (Table 7.2).

* E_3-FAD reduced. The precise structure of the reduced flavoprotein is not yet clear. There are two sulphydryl groups in close proximity to the FAD on E_3. It seems probable that a semiquinone intermediate FADH and the sulphydryl groups are both involved in the hydrogen transfer.[32]

† The Enzyme Commission recommended name for E_1 is pyruvate dehydrogenase (lipoamide). However, many workers in the field use the name pyruvate decarboxylase for E_1; this is both more informative and less confusing and is therefore adopted here.

Estimates of the M_r of the complex have ranged since 1970 as follows: 3×10^6 (1970),[35] 3.8×10^6 (1972),[36] 4.6×10^6 (1972),[37] 6.1×10^6 (1979)[38] and 5.36×10^6 (1985).[39] The values of M_r have been determined by different methods and using different preparations of the complex. It is difficult to determine the M_r of a large complex with sufficient accuracy to estimate its precise subunit composition. It is also possible either that some of the subunits may become detached during the isolation procedure or even that some variation or heterogeneity in the composition of the complex exists *in vivo*. Heterogeneity of composition is suggested when the M_r is determined by active enzyme centrifugation[40] (for details of this method, see Chapter 3, Section 3.2.4). Although knowledge of the precise subunit composition is necessary before the three-dimensional structure of the complex can be known with certainty, it is less important for understanding the mechanism of action of the multienzyme complex and will not be discussed further (for further details of the structure see references 31, 38 and 39).

TABLE 7.2
M_r values of the subunits of the pyruvate dehydrogenase multienzyme complex from E. coli

Pyruvate decarboxylase (E_1)	96 000–100 000
Dihydrolipoamide acetyltransferase (E_2)	60 500–91 000
Dihydrolipoamide dehydrogenase (E_3)	56 000
Pyruvate dehydrogenase multienzyme complex	3×10^6–6.1×10^6

References 40, 41, 42, and 43. For review, see reference 39.

The pyruvate dehydrogenase complex is isolated from *E. coli* by the steps indicated in Fig. 7.3. Precautions such as the inclusion of protease inhibitors have to be taken to prevent proteolysis occurring during isolation. The isolated multienzyme complex can be dissociated into its constituent subunits by use of $4\ \mathrm{mol\,dm^{-3}}$ urea, high pH, and calcium phosphate gel (Fig. 7.3). The multienzyme complex may also be reconstituted from the subunits and full enzyme activity restored. It is clear from both the dissociation and reconstitution studies that E_2 has binding domains for both E_1 and E_3, but that E_1 and E_3 do not bind in the absence of E_2. The separated acetyltransferase (E_2) has octahedral symmetry and this is consistent with the composition of the multienzyme complex having a total of 24 E_2 polypeptide chains and is also in agreement with the majority of M_r determinations. This arrangement of E_2 occurs not only in the *E. coli* pyruvate dehydrogenase complex, but also in the 2-oxoglutarate dehydrogenase complexes from *E. coli* and mammalian sources, and also in branched-chain oxo-acid dehydrogenases.[46] The 24 chains are thought to be arranged as shown in Fig. 7.4(a). The E_1 and E_3 enzymes are then considered to be associated with the E_2 chains. The stoichiometry favoured by most workers is

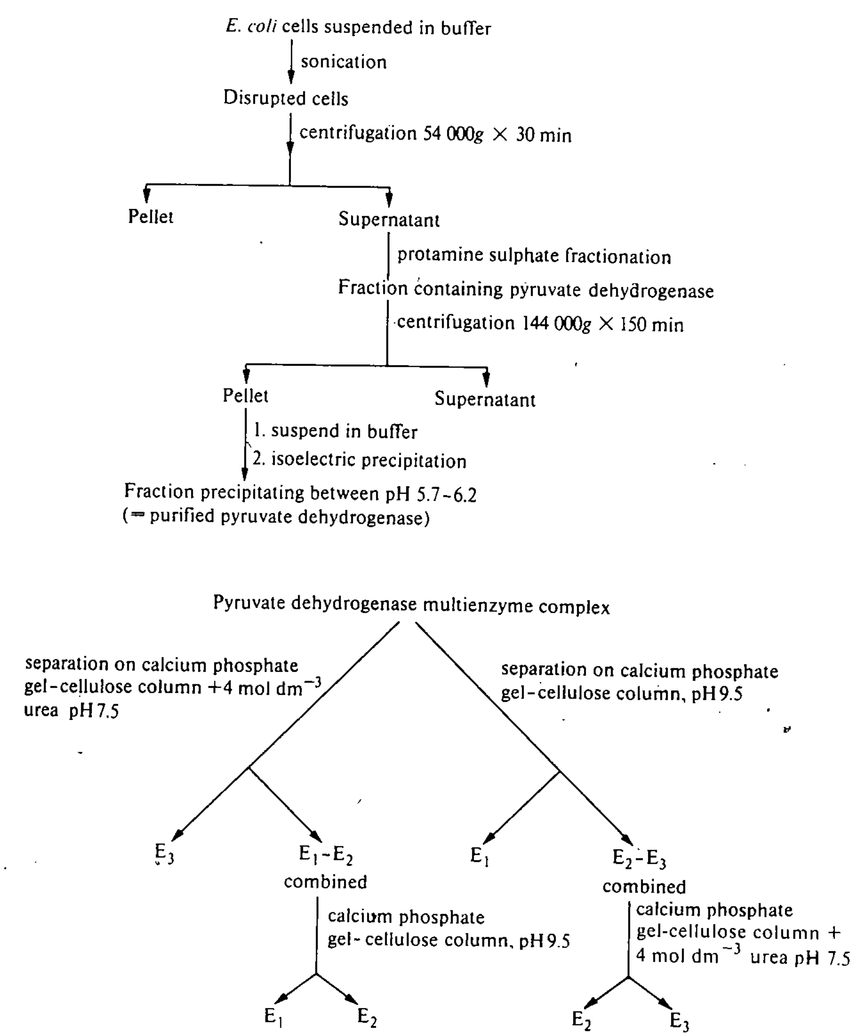


FIG. 7.3. Pyruvate dehydrogenase complex: isolation and separation of the subunits (see references 44 and 45).

24:24:12 for E_1:E_2:E_3[39], but Bates *et al.*[47] proposed a stoichiometry of 48:24:24.

Three of the cofactors thiamin diphosphate, lipoate, and FAD are firmly bound to the decarboxylase, the acetyltransferase, and the dehydrogenase, respectively, and do not dissociate during the reactions. Lipoate is covalently linked through an amide bond to a lysine residue on the acetyltransferase.

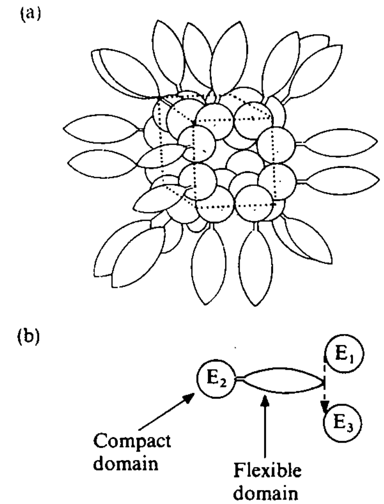
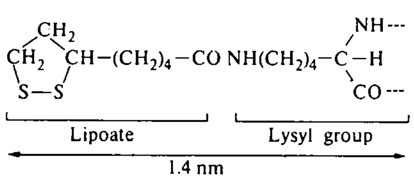


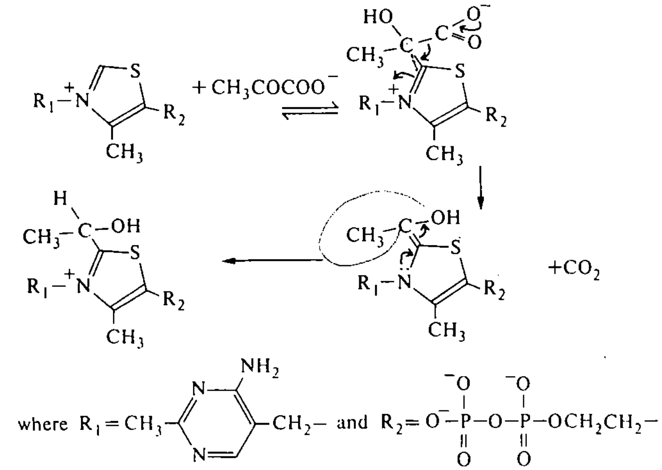
FIG. 7.4. Structure of pyruvate dehydrogenase multienzyme complex. (a) The probable arrangement of dihydrolipoamide acetyltransferase (E_2) forming the core of the pyruvate dehydrogenase multienzyme complex. Each E_2 polypeptide is thought to be made up of two domains, a central core containing the region of the active site and a flexible domain protruding out from the core. The latter contains the lipoate residues. The 24 E_2 polypeptide chains are arranged in the form of a cube, (a), and the E_1 and E_3 enzymes form a larger cube encasing the E_2 cube such that the flexible domain of E_2 can move between E_1 and E_3, as shown in (b). For details, see references 31 and 46.



Evidence shows that three lipoate residues are attached to each acetyltransferase molecule,[41,48] and the DNA sequence[49] shows three homologous regions in the N-terminal half of E_2, where the three lipoates are attached.[50] This arrangement seems to occur only in *E. coli* and *Azotobacter vinelandii.* All other pyruvate dehydrogenases so far studied have only one lipoate for each E_2.[46] Guest *et al.*[49] have shown that it is possible to delete one or two of the regions of the DNA coding for the lipoate attachments sites, and the E_2 that is expressed still shows full activity. Thus, the roles of the second and third lipoates are unclear.

7.7.2 Mechanism

The initial decarboxylation occurs when pyruvate combines with TDP on the decarboxylase probably by the mechanism shown below.



The second step involves the oxidative transfer of the hydroxyethyl group bound to E_1 on to the disulphide group of lipoamide on E_2, forming a thioester (reaction 2 in Section 7.7). For this step the two enzymes have to be in close contact. The acetyl group so formed is then transferred to CoA in the third step. The fourth and fifth steps are concerned with the regeneration of oxidized lipoamide by transfer of 2H from dihydrolipoamide on E_2 via FAD of the dehydrogenase (E_3) to NAD^+. The fourth step also involves the interaction between two enzyme-bound intermediates of E_2 and E_3 (reaction 4, Section 7.7).

It has been suggested[31] that the transfer of the acetyl group and of the two hydrogen atoms from dihydrolipoamide involves the flexible arm of lysyllipoamide on E_2 swinging between E_1 and E_3. This arm would be approximately 1.4 nm and therefore could swing through a diameter of 2.8 nm. Estimates have been made[51,52] of the distances separating E_1 and E_3 to test this hypothesis. By using an analogue of TDP, thiochrome diphosphate, bound at E_1, it is possible to estimate the distance to the FAD on the active site of E_3 by fluorescence energy transfer. The distance appears to be about 4.5 nm, which is too great for the flexible lysyllipoamide arm alone to be the explanation. However, the polypeptide chain of E_2 adjacent to the lysyllipoamide arm is also highly flexible, being rich in alanine, proline, and charged amino-acid residues;[50] this could increase the range of the arm.[136]

A question that arises with a multienzyme complex of this nature in which there are several copies of each of the enzymes is whether, in a given catalytic cycle, one molecule of pyruvate is oxidatively decarboxylated by a single specific cluster of E_1–E_2–E_3 within the complex. Several experiments have been aimed at answering this question. In the multienzyme complex, pyruvate decarboxylase (E_1) catalyses the rate-limiting step. This has been demonstrated by using an active-site-directed inhibitor of E_1, thiamin thiazolone diphosphate, which inhibits pyruvate dehydrogenase activity in parallel with the decrease in the rate of conversion of pyruvate to acetyl-CoA.[42] In contrast, the acetyltransferase (E_2) can be partially inhibited by reacting with *N*-ethylmaleimide before any reduction in the rate of conversion of pyruvate to acetyl-CoA occurs. Also, it is possible to remove nearly all the lipoyl moieties using the enzyme lipoamidase, without significant loss of activity.[53]

It is possible to reconstitute a pyruvate dehydrogenase complex that is deficient in pyruvate decarboxylase (E_1) and to use this to demonstrate that pyruvate decarboxylase (E_1) is capable of transferring hydroxyethyl groups to several acetyltransferases.[43] This is done by allowing reaction to occur in the absence of CoA and measuring the extent to which the acetyltransferase (E_2) becomes acetylated. From these results it seems probable that there is a communicating network of acetyltransferases (E_2) in which the acetyl groups may be transferred from one E_2 to another, thus enabling a transfer of acetyl groups and the associated reduction of lipoamide residues from an E_1 site to a distant E_3 site within the complex.[54] The possible advantages of this arrangement are discussed in reference 55.

7.7.3. Control of the pyruvate dehydrogenase system in *E. coli*

The pyruvate dehydrogenase complex occupies a key position linking the glycolytic pathway to the tricarboxylic acid cycle. There are a number of factors that control its activity in different organisms. The decarboxylation step is effectively irreversible under physiological conditions and thus controls the entry of pyruvate into the tricarboxylic acid cycle. Pyruvate dehydrogenase activity is inhibited by two of the products of its activity, namely, acetyl-CoA, which is a competitive inhibitor with respect to CoA, and NADH, which is competitive with respect to NAD^+ (Fig. 7.5). The combination of acetyl-CoA and NADH may also inhibit by acetylation of the bound lipoamide. The activity of the pyruvate dehydrogenase complex is stimulated by nucleoside monophosphates and inhibited by GTP. The latter effect is reversed by GDP.

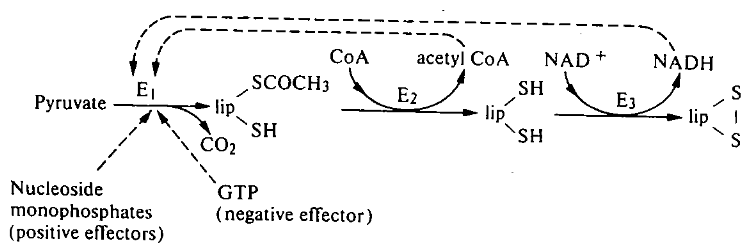


FIG. 7.5. Regulation of the pyruvate dehydrogenase complex in *E. coli.*

7.7.4 Organization and control of the pyruvate dehydrogenase system in mammalian tissues

The pyruvate dehydrogenase multienzyme complexes isolated from a number of mammalian tissues,[31] including pig heart, bovine kidney, and bovine heart, have been found to be larger than that of *E. coli*, and their regulation is subject to additional controls to those described in Section 7.7.3. The basic organization of the three components is similar, although the mammalian complexes show icosahedral rather than octahedral symmetry. The pyruvate decarboxylase (E_1) comprises two polypeptide chains α and β. There are 60 copies of each in the multienzyme complex.[46] It appears that in Gram-negative bacteria, e.g. *E. coli*, *Azotobacter vinelandii*, an octahedral arrangement occurs, whereas in Gram-positive bacteria, e.g. *Bacillus stearothermophilus*, fungi such as *Neurospora* and *Saccharomyces*, and higher eukaryotes, an icosahedral arrangement occurs.[56] The latter would mean that the components E_1, E_2, and E_3 are more likely to be present in multiples of 20, e.g. 60 rather than in multiples of eight, e.g. 24, which are expected for an octahedral arrangement.

The pyruvate dehydrogenase complexes from mammalian and avian sources, and from *Neurospora*, interact with two additional enzymes,[46] namely, [pyruvate dehydrogenase (lipoamide)] kinase and [pyruvate dehydrogenase (lipoamide)] phosphatase; the former is tightly bound to the complex, whereas the latter is very easily dissociated from the complex. The kinase is bound to the acetyltransferase (E_2) and sub-complexes containing E_2-kinase have been isolated. The substrate of the kinase is pyruvate-decarboxylase (E_1), which becomes phosphorylated on one of its serine* side-chains when the intracellular ratio of [ATP]/[ADP] is high; this results in inactivation of the multienzyme complex, specifically inhibiting the decarboxylation step (Fig. 7.6). The phosphatase catalyses the dephosphorylation of E_1, which converts the inactive multienzyme complex back into the active form (further details of this mechanism are given in references 57 and 58). Pyruvate inhibits the phosphorylation step, thus maintaining the complex in an active state. The ratio of phosphorylated to dephosphorylated pyruvate dehydrogenase multienzyme complex has been shown to vary with the physiological state of the mammal, e.g. whether it is well

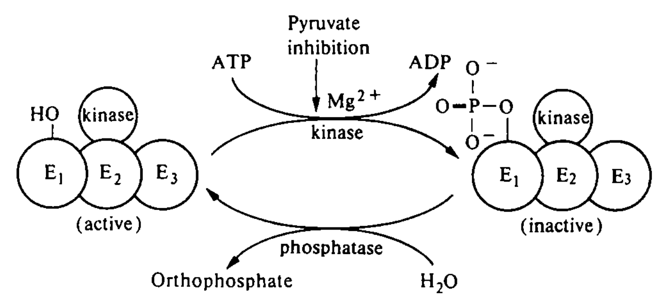


FIG. 7.6. Regulation of mammalian pyruvate dehydrogenase complexes by phosphorylation and dephosphorylation.

* There are three serine side-chains on E_1, which may become phosphorylated in succession, but phosphorylation of the first inactivates the complex. The significance of the three phosphorylation steps is not yet clear,[59–61] although it has been suggested that phosphorylation of the second and third site makes reactivation by the phosphatase slower.[62]

fed or starved (see also Chapter 6, Section 6.3.2.2). The pyruvate dehydrogenase multienzyme complexes from the lower eukaryotes *Neurospora crassa* and *Saccharomyces lactis* are also subject to similar covalent modification.[62]

In eukaryotes the pyruvate dehydrogenase multienzyme complex resides in the matrix within the inner mitochondrial membrane. The three enzyme components are coded for by nuclear genes. It has been shown that E_1, E_2, and E_3 are synthesized as cytosolic precursors each having a transit peptide to direct it into the mitochondrial matrix, where the peptide is removed by proteolysis and the multienzyme complex is assembled.[63]

7.8 Oxoglutarate dehydrogenase and branched chain 2-oxo acid dehydrogenase* multienzyme complexes

The oxoglutarate dehydrogenase multienzyme complex is not discussed here in detail because, in many respects, it is both structurally and functionally similar to that of pyruvate dehydrogenase. The complex has been isolated from *E. coli* and from mammalian sources. It consists of three distinct enzymes, oxoglutarate decarboxylase (E_1), dihydrolipoamide succinyltransferase (E_2), and dihydrolipoamide dehydrogenase (E_3) catalysing the conversion of 2-oxoglutarate to succinyl CoA, an integral step in the tricarboxylic acid cycle.

$$E_1\text{-TDP} + {}^-OOC.CO.CH_2.CH_2COO^- \rightleftharpoons E_1\text{-TDP—CHOH}.CH_2.CH_2.COO^- + CO_2 \qquad 1$$

$$E_1\text{-TDP—CHOH}.CH_2.CH_2.COO^- + E_2\text{-lip}\left\langle\begin{matrix}S\\|\\S\end{matrix}\right. \rightleftharpoons E_1\text{-TDP} + E_2\left\langle\begin{matrix}SH\\SCO.CH_2.CH_2.COO^-\end{matrix}\right. \qquad 2$$

$$E_2\text{-lip}\left\langle\begin{matrix}SH\\SCO.CH_2.CH_2.COO^-\end{matrix}\right. + CoASH \rightleftharpoons E_2\text{-lip}\left\langle\begin{matrix}SH\\SH\end{matrix}\right. + {}^-OOC.CH_2.CH_2.COSCoA \qquad 3$$

$$E_2\text{-lip}\left\langle\begin{matrix}SH\\SH\end{matrix}\right. + E_3\text{-FAD} \rightleftharpoons E_2\text{-lip}\left\langle\begin{matrix}S\\|\\S\end{matrix}\right. + E_2\text{-FAD reduced} \qquad 4$$

$$E_3\text{-FAD reduced} + NAD^+ \rightleftharpoons E_3\text{-FAD} + NADH + H^+ \qquad 5$$

TDP = thiamin diphosphate $\qquad \text{lip}\left\langle\begin{matrix}S\\|\\S\end{matrix}\right. =$ oxidised lipoamide

$\text{lip}\left\langle\begin{matrix}SH\\SH\end{matrix}\right. =$ reduced lipoamide

* These enzymes are sometimes referred to as α-ketoglutarate dehydrogenase and branched chain keto-acid dehydrogenase.

The arrangement of the enzymes within the multienzyme complex is similar to that in *E. coli* pyruvate dehydrogenase, based on octahedral symmetry.[46] The dihydrolipoamide succinyltransferase (E_2) binds both the decarboxylase (E_1) and the dehydrogenase (E_3). The amino-acid sequence of E_2 has been inferred from its DNA sequence.[64] Preparations of the isolated E_2 have a tetrad appearance similar to that of dihydrolipoamide acetyltransferase when examined by electron microscopy. The E_2 also comprises a compact and a flexible domain comparable with that of dihydrolipoamide acetyltransferase.[65] Dihydrolipoamide dehydrogenase isolated from pyruvate dehydrogenase and oxoglutarate dehydrogenase multienzyme complexes can be exchanged with one another in reconstitution experiments.

Like the pyruvate dehydrogenase complex, oxoglutarate dehydrogenase activity is regulated by the ratio of the concentrations of substrates and products ([succinyl CoA]/[CoA] and [NADH]/[NAD^+]), the products inhibiting activity. Activity is stimulated by AMP, which appears to promote tighter binding of the substrate, 2-oxoglutarate, and the cofactor TDP. ATP competes with AMP in reversing the stimulation. In the mammalian oxoglutarate dehydrogenase complexes there is no evidence for covalent modification by kinases and phosphatases. The branched chain 2-oxo acid dehydrogenase also has a similar structure.[132] It occurs inside the inner mitochondrial membrane and catalyses the overall reaction:

$$\left.\begin{matrix}R_1\\R_2\end{matrix}\right\rangle CHCO{\cdot}COO^- + NAD^+ + CoASH \rightleftharpoons \left.\begin{matrix}R_1\\R_2\end{matrix}\right\rangle CHCOSCoA + NADH + H^+ + CO_2$$

Its importance is in the oxidative decarboxylation of oxo-acids formed by transamination or deamination of valine, isoleucine and leucine. The multienzyme complex has E_1, E_2, and E_3 enzyme components and is also subject to regulation by a kinase and a phosphatase.[66]

7.9 The tryptophan synthase multienzyme complex from *E. coli*

The tryptophan synthase system is much smaller than either the pyruvate or oxoglutarate dehydrogenase systems, having an M_r of approximately 150 000. It is a striking example in which the interaction of two subunits greatly modifies their respective catalytic activities and this aspect is emphasized in the subsequent sections. The tryptophan synthase multienzyme complex catalyses the final steps in the biosynthesis of tryptophan (Fig. 7.7). The tryptophan synthase most studied is that from *E. coli* but it has also been investigated in a number of other bacteria, in fungi and in plants. Mammals lack tryptophan synthase and thus require a dietary source of tryptophan.

Tryptophan synthase from *E. coli* is coded for by two structural genes,* which, together with three other structural genes, form an operon† coding for the

* Structural genes are those genes that code for structural and enzyme proteins, as opposed to regulatory genes, which are concerned with the control of transcription or translation.

† An operon is a cluster of genes that are regulated as a unit and that code for metabolically related enzymes.

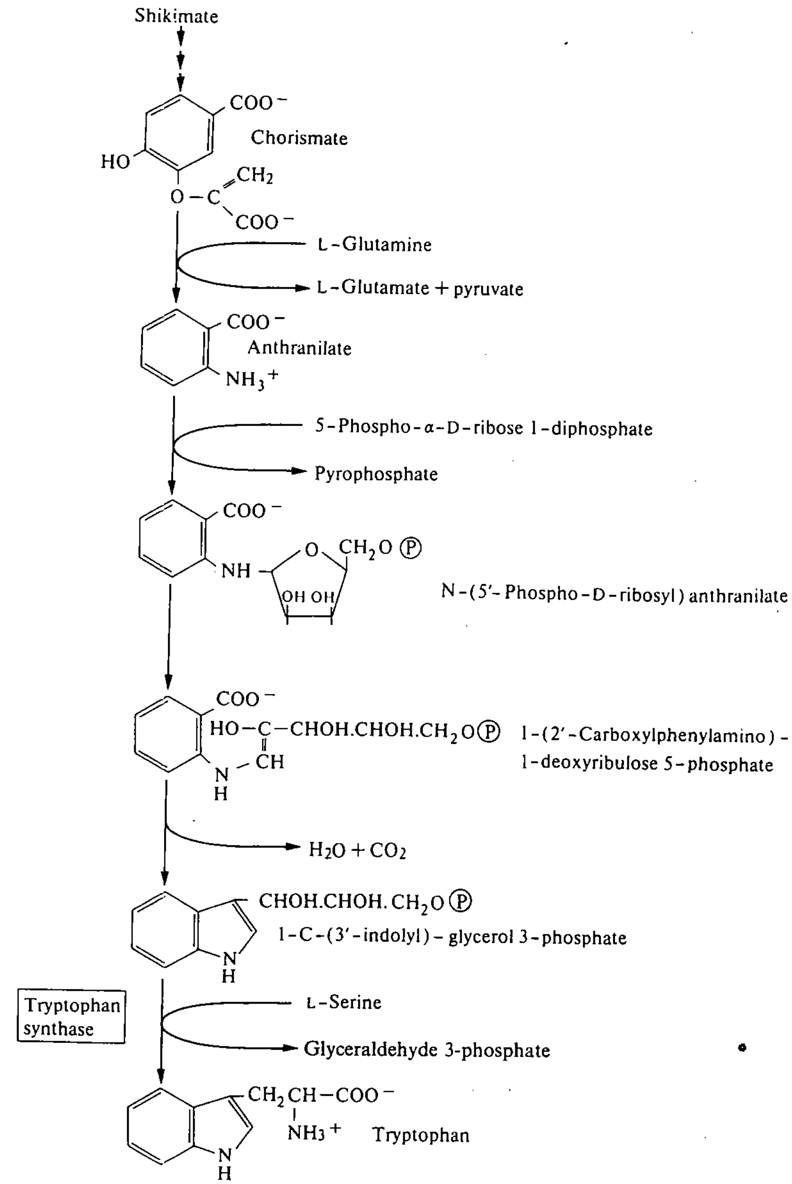


FIG. 7.7. The pathway for biosynthesis of tryptophan. Ⓟ = phosphoryl group.

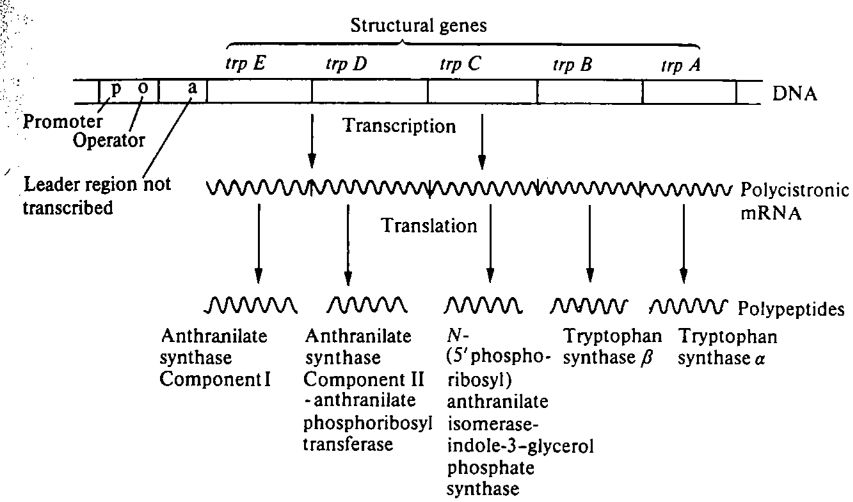
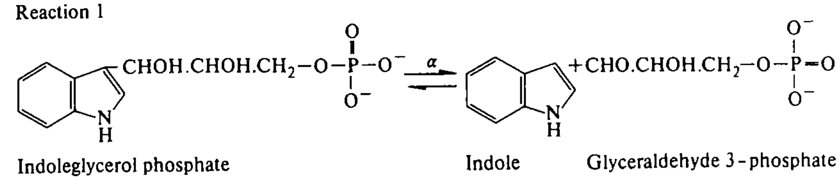


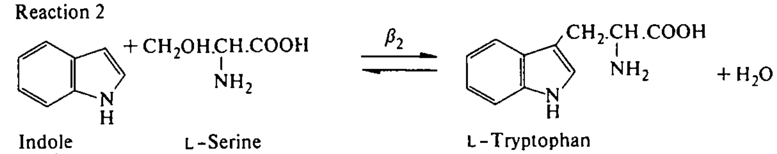
FIG. 7.8. The arrangement of the genes which constitute the tryptophan operon in *E. coli* (see reference 67).

enzymes required for the synthesis of tryptophan from chorismate (Fig. 7.8). It comprises two different polypeptide chains α and β and has the structure $\alpha_2\beta_2$. The separate polypeptide chains catalyse reactions 1 and 2, whereas the complex catalyses reaction 3, but will also catalyse reactions 1 and 2.

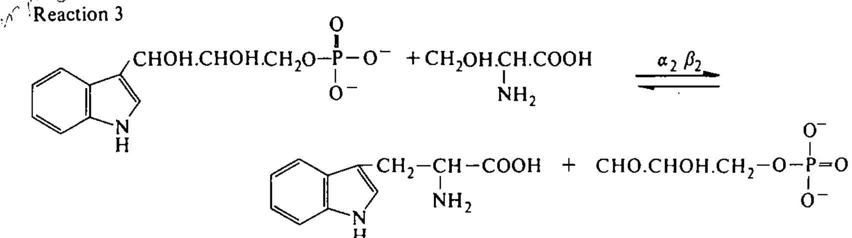
Reaction 1



Reaction 2



Reaction 3



7.9.1 Structure

The enzyme from *E. coli* can readily be dissociated into separate α and β_2 proteins, but more drastic conditions are required to separate β_2 into β (Fig. 7.9); this results in loss of activity. The activity is, however, restored upon reassociation. Each β polypeptide chain binds one molecule of pyridoxal phosphate. The complete sequences of the α subunit isolated from *E. coli*, *Salmonella typhimurium*, and *Enterobacter aerogenes* are known. Preliminary X-ray crystallographic data have been obtained for both subunits. Although the tryptophan synthase system in plants is also a two-component heteropolymer, the enzymes from *Euglena*, *Neurospora crassa*, and *Saccharomyces cerevisiae* are homopolymers (Table 7.3). The enzymes from *Neurospora crassa* and *Saccharomyces cerevisiae* were initially thought to be composed of separate α and β subunits but this was due to proteolysis during isolation and both have been shown to be homopolymers.[68–70] The homopolymeric nature of these enzymes is also supported by the genetic evidence of a single gene locus. This is believed to be an example where, during the course of evolution, gene fusion has occurred. The DNA sequence of the genes in both *E. coli* and *Saccharomyces cerevisiae* have been determined and they show a high degree of similarity.[71]

$$\underset{\substack{M_r\\146\,000}}{\alpha_2\beta_2} \xrightleftharpoons[]{\substack{\text{Reversible dissociation}\\\text{promoted by removal of}\\\text{pyridoxal phosphate*}}} \underset{\substack{M_r\\2\times 29\,000}}{2\alpha} + \beta_2 \xrightleftharpoons[]{\substack{\text{Reversibly dissociated}\\\text{by urea or guanidine}\\\text{hydrochloride}}} \underset{\substack{M_r\\2\times 44\,000}}{2\beta\ (\text{inactive})}$$

*Apparent association constant K_A (−pyridoxal phosphate) -4×10^6 (mol dm^{-3})$^{-1}$ and K_A (+ pyridoxal phosphate + serine) -2.6×10^9 (mol dm^{-3})$^{-1}$

FIG. 7.9. Dissociation of trytophan synthase (see reference 73).

TABLE 7.3

Structures of tryptophan synthases from different species

Source	Structure	M_r of complex	M_r of subunits	Comments
E. coli	$\alpha_2\beta_2$	146 000	2 × 29 000 2 × 44 000	Ready dissociation to $2\alpha + \beta_2$
Neurospora crassa	A_2	150 000	2 × 75 000	
Saccharomyces cerevisiae	A_2	154 000–167 000	2 × 80 000	
Euglena	A_2?	325 000	2 × 160 000	Complex contains four catalytic activities for E_9–E_{12} on Fig. 7.20

References 68–70 and 72.

7.9.2 Mechanism

The studies on the mechanism refer to the enzyme system from *E. coli*. The multienzyme complex, in addition to catalysing the complete reaction (reaction 3), will also catalyse the formation of indole (reaction 1) and the conversion of indole to tryptophan (reaction 2). The separate α and β_2 subunits will catalyse reactions 1 and 2, but at only 1/30th and 1/100th the rate catalysed by $\alpha_2\beta_2$. For this higher rate of reaction the α and β proteins have to be in physical contact. (They cannot, for example, be separated by a dialysis membrane.) Detailed kinetic studies, both rapid and steady state, have been made on reactions 1 and 2 catalysed by α and β_2, respectively, and these have been compared with similar studies using $\alpha_2\beta_2$.[74, 75] The general conclusion is that the mechanism is unchanged whether the single enzyme or the multienzyme complex is used, e.g. reaction 1 when studied in the direction of indole glycerol phosphate formation proceeds via an ordered ternary complex mechanism (see Chapter 4, Section 4.3.5.1) in which glyceraldehyde 3-phosphate binds first whether it is catalysed by α or by $\alpha_2\beta_2$. Although the mechanism appears the same, the rate constants are appreciably changed when a subunit is used compared to the multienzyme complex.

These increases in the rates of the partial reactions (reactions 1 and 2) that occur when either α or β_2 are combined may also be brought about by combining one defective protein with a catalytically active one. It is possible to obtain defective α or β_2 proteins either from suitable mutants of *E. coli* or by chemical modification of catalytically active subunits. These defective proteins can then be used to assemble hybrid $\alpha_2\beta_2$ proteins. The results of such experiments are summarized in Table 7.4. It can be seen that a subunit does not have to be catalytically active in order to effect a rate enhancement in the other reaction. It seems that so long as it can combine with the other subunit it will be effective.

TABLE 7.4

The catalytic activity of $\alpha_2\beta_2$ hybrids of tryptophan synthase formed by combining normal and defective subunits

Holoenzyme	Reaction 1	Reaction 2	Reaction 3
$\alpha_2\beta_2^d$	Rate enhanced with respect to α	Inactive	Inactive
$\alpha_2^d\beta_2$	Inactive	Rate enhanced with respect to β_2	Inactive
$\beta\beta^d$	Inactive	Half activity with respect to β_2	Inactive
$\alpha_2\beta\beta^d$	Active	Active	Full activity ($=\alpha_2\beta_2$)

Superscript 'd' indicates the defective subunit, i.e. one having no catalytic activity either isolated from a mutant, or produced by chemical modification.
References 76–78.

In addition to the increased rates of the partial reactions that occur when α and β_2 subunits are combined, another change occurs. Although the physiological role of the β_2 subunits is to catalyse reaction 2 it has been shown also to catalyse the following pyridoxal phosphate dependent reactions:

$$\text{L-serine} \overset{\beta_2}{\rightleftharpoons} \text{pyruvate} + NH_4^+.$$

$$\text{2-Mercaptoethanol} + \text{L-serine} \overset{\beta_2}{\rightleftharpoons} \text{S-hydroxyethyl-L-cysteine} + H_2O.$$

$$\text{2-Mercaptoethanol} + \text{L-serine} + \text{pyridoxal phosphate} \overset{\beta_2}{\rightleftharpoons}$$
$$\text{S-pyruvylmercaptoethanol} + \text{pyridoxamine phosphate} + H_2O.$$

When the subunits are combined to form $\alpha_2\beta_2$ not only is the catalytic efficiency of the partial reactions improved, but also the non-physiological side-reactions of the β_2 subunit are suppressed. Also, if the complete reaction (reaction 3) is carried out in the presence of [^{14}C]indole, it is found that no radioactivity is present in the tryptophan formed. This suggests that, when the complete reaction occurs, either that indole is not an intermediate or that indole formed in the first reaction remains enzyme bound, i.e. the indole is effectively channelled to the second catalytic site without equilibrating with indole in the surrounding medium. The kinetics have been studied in detail[79] and a number of schemes[75] have been suggested to explain how this might occur. At present the precise mechanism is not clear, especially since the three-dimensional structure of the complex, and hence the location of the active sites, has not been determined. A model that is consistent with kinetic and other evidence and that at present seems most favoured is shown in Fig. 7.10.

It is clear that the multienzyme complex must have two distinct active sites, since (i) the α and β_2 subunits are able to catalyse the partial reactions, and (ii) in *Neurospora*, where the enzyme is a single polypeptide chain, there is evidence for two separate sites. It is assumed that in the *E. coli* complex the two active sites must be closely juxtaposed so that the intermediate may be channelled from one catalytic centre to another. The distance of separation of the two catalytic sites has been estimated as 2 nm by measurement of fluorescence energy transfer between pyridoxal phosphate and fluorescent ligands attached to the α and β subunits.[82] The rate of the step in which indole is released in the partial reaction (reaction 1) becomes insignificant in the presence of L-serine. Thus the combination of the two subunits is assumed to induce certain conformational changes and also to bring the catalytic sites into close proximity; this increases the catalytic efficiency of the normal physiological reaction and also eliminates non-physiological side-reactions. For further details of the structure and mechanism, see references 68 and 73, and for the evolution of tryptophan biosynthetic enzymes, see reference 83.

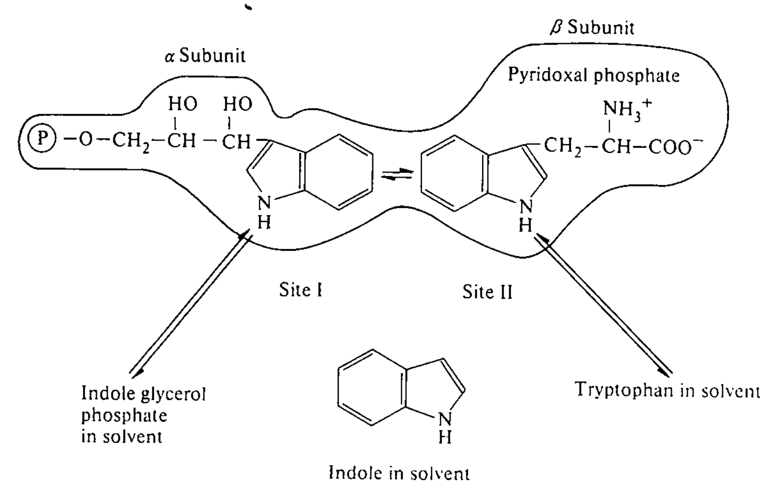


FIG. 7.10. Possible arrangement of the active sites of the α and β subunits of tryptophan synthase. Indole, present in the solvent medium, does not become incorporated into tryptophan when tryptophan synthase from *E. coli* is used to catalyse reaction 3 (see references 73, 80 and 81). Ⓟ = phosphoryl group.

7.10 Carbamoyl phosphate synthase and the associated enzymes of the pyrimidine and arginine biosynthetic pathways in *E. coli*, fungi, and mammalian cells

Carbamoyl phosphate is the metabolic precursor of both pyrimidines and arginine, and carbamoyl phosphate synthase catalyses a step that is at a branch point in the metabolic pathways (Fig. 7.11; and see also Chapter 6, Section 6.3.1). The carbamoyl moiety may become transferred either to aspartate or to ornithine. In certain tissues and organisms, carbamoyl phosphate synthase has been shown to be physically associated with aspartate carbamoyltransferase and in some cases also dihydro-orotase.

The multienzymes involved illustrate two important features, namely, (i) the channelling of a metabolite in a multienzyme protein and (ii) the evolutionary changes in the organization of a multienzyme protein.[84] We compare the organization of the pathways for the biosynthesis of pyrimidines and arginine in *E. coli*, *Saccharomyces cerevisiae*, *Neurospora crassa*, and mammalian systems. The main features of the systems in *E. coli*, *S. cerevisiae*, and *N. crassa* are illustrated in Fig. 7.12.

7.10.1 *E. coli*

It can be seen that in *E. coli* there is a single carbamoyl phosphate synthase that catalyses the formation of carbamoyl phosphate for both arginine and pyrimidine biosynthesis. Mutations in the carbamoyl phosphate synthase gene cause

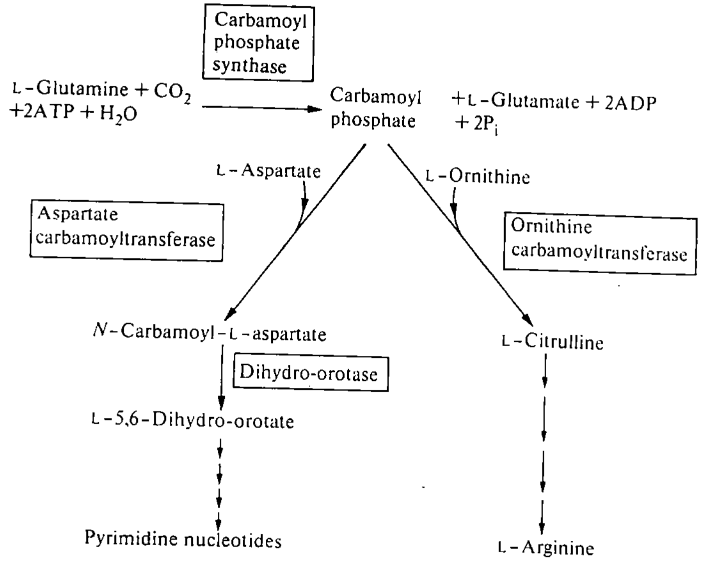
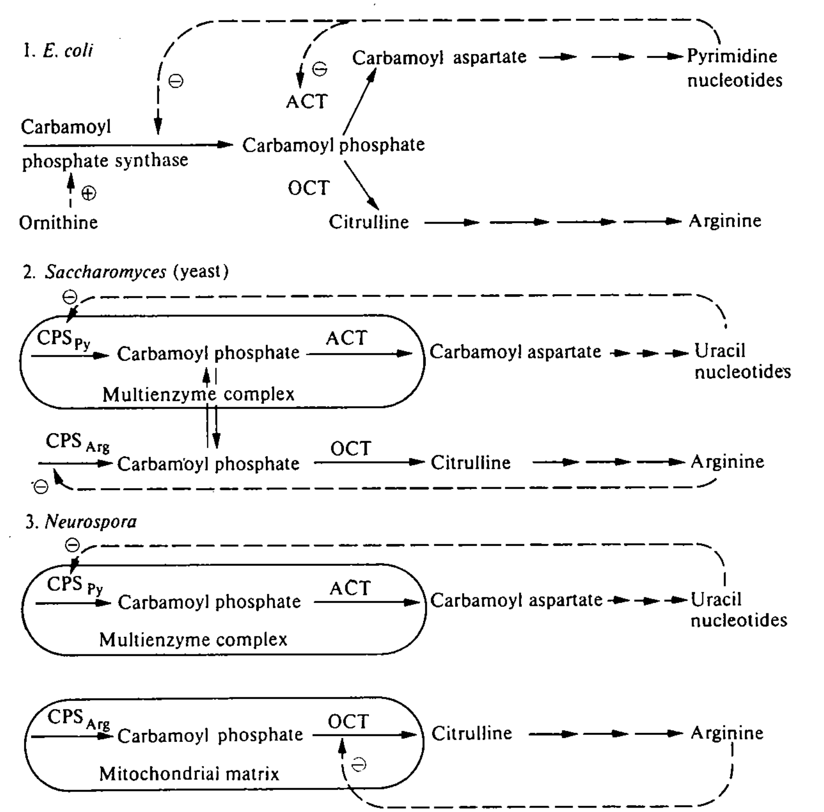


FIG. 7.11. The position of carbamoyl phosphate synthase on the metabolic pathways for arginine and pyrimidine biosynthesis. P_i = orthophosphate.

auxotrophy for both arginine and pyrimidines (i.e. both arginine and pyrimidine have to be supplied in the nutrient medium in order that growth can occur). Regulation of the balance in the supply of carbamoyl phosphate between pyrimidine and arginine biosynthesis is by the activator ornithine and the inhibitor UMP[85] (Fig. 7.12) and is more fully described in Chapter 6, Section 6.3.1. There is no evidence for any physical association of the enzymes in *E. coli*.

7.10.2 *Saccharomyces cerevisiae* (yeast)

In contrast to the situation in *E. coli*, the two fungi (*Saccharomyces*[86] and *Neurospora*[87, 88]) each have two distinct carbamoyl phosphate synthases, although there are differences between the two in the extent to which channelling of carbamoyl phosphate occurs. In *Saccharomyces* there are two distinct carbamoyl phosphate synthases (abbreviated to CPS_{Py} and CPS_{Arg}) and one of these, CPS_{Py}, is physically associated with aspartate carbamoyltransferase, but carbamoyl phosphate produced by either CPS_{Arg} or CPS_{Py} may be utilized in either biosynthetic pathway. The evidence for this is that mutants lacking either CPS_{Arg} or CPS_{Py} will grow on a minimal medium (i.e. without arginine or pyrimidine supplementation). However, mutants lacking CPS_{Py} will not grow on arginine-supplemented media and mutants lacking CPS_{Arg} will not grow on uracil-supplemented media. This is as expected from the feedback-inhibition patterns (Fig. 7.12), e.g. uracil inhibits CPS_{Py} and thus in a mutant lacking CPS_{Arg} no carbamoyl phosphate can be formed and growth is inhibited.



CPS = carbamoyl phosphate synthase
ACT = aspartate carbamoyltransferase
OCT = ornithine carbamoyltransferase.

FIG. 7.12. Comparison of arginine and pyrimidine biosynthesis in *E. coli*, *Saccharomyces*, and *Neurospora*. ⊕ = activation, ⊖ = inhibition.

The fact that mutants lacking either CPS_{Arg} or CPS_{Py} will grow on a minimal medium shows that carbamoyl phosphate formed may be exchanged between the pathways. However, the growth rate of these mutants on minimal medium is appreciably slower than that of the wild type, which suggests there is not a completely free exchange of carbamoyl phosphate between the pathways, i.e. some channelling occurs.

7.10.3 *Neurospora*

In *Neurospora*[87] there is efficient channelling of carbamoyl phosphate synthesized by the two enzymes. The channelling is achieved by means of a bifunctional enzyme protein containing CPS_{Py} and aspartate carbamoyltransferase and by different subcellular locations of the enzymes (Fig. 7.12). CPS_{Arg} and ornithine carbamoyltransferase are located in the mitochondria, whereas the CPS_{Py}: aspartate carbamoyltransferase multienzyme protein is present in the cytosol. Consistent with this is the finding that mutants lacking CPS_{Py} or CPS_{Arg} are both auxotrophs (i.e. they require supplementation with uracil or arginine, respectively, for growth to occur). Further evidence is that carbamoyl phosphate formed by CPS_{Py} *in vitro* does not exchange with added radioactively labelled carbamoyl phosphate. CPS_{Arg} in *Neurospora* is not feedback-inhibited by arginine as is the corresponding enzyme in *Saccharomyces*.[89]

7.10.4 Mammalian tissues

In the cytosol of mammalian tissues there is a multienzyme protein having three enzyme activities, namely, carbamoyl phosphate synthase, aspartate carbamoyltransferase, and dihydro-orotase (E_1, E_2, and E_3; Fig. 7.13).

HCO_3^- + Glutamine + 2 ATP + H_2O

↓ carbamoyl phosphate synthase (E_1)

Carbamoyl phosphate + 2ADP + Glutamate + Phosphate

↓ ← Aspartate; aspartate carbamoyltransferase (E_2)

Carbamoyl aspartate

↓ dihydro-orotase (E_3)

L-5,6-Dihydro-orotate

↓ orotate reductase (E_4)

Orotate

↓ ← 5-Phospho-α-D-ribose 1-diphosphate; orotate phosphoribosyltransferase (E_5)

Orotidine 5′-phosphate + Pyrophosphate

↓ orotidine 5′-phosphate decarboxylase (E_6)

UMP + CO_2

FIG. 7.13. Biosynthetic pathway for uridylate synthesis.

This complex has been studied in *Drosophila melanogaster*, bull frog eggs, rat liver, mouse spleen, and various cultured mammalian cells.[24, 90] It has been purified to homogeneity in mouse ascites hepatoma cells; the native complex has an M_r of about 800 000 and, after denaturation with sodium dodecylsulphate, a single size of polypeptide chain having an M_r of 200 000 is obtained.[91–94]

A method that has been used increasingly to study the separate domains of multienzyme polypeptides is limited proteolysis. In general the domains are compact but are separated by flexible regions and these latter are more sensitive to proteolysis (see also Sections 7.11.2.1 and 7.11.3). The CAD multienzyme polypeptide (carbamoyl phosphate synthase II, aspartate carbamoyltransferase, dihydro-orotase) has been studied in this way.[135] Using elastase and trypsin it is possible to remove the N-terminal domain having dihydro-orotase (DHO) activity, and the C-terminal domain having aspartate carbamoyl transferase (ATC), leaving a centre polypeptide with carbamoyl phosphate synthase II (CPSII) activity. (Carbamoyl phosphate synthase I is the other enzyme present in mitochondria functioning in the biosynthesis of citrulline.) This is shown diagramatically below.

N — Dihydro-orotase (M_r = 43 000) ↓ Carbamoyl phosphate synthase (M_r = 150 000) ↓ Aspartate carbamoyltransferase (M_r = 40 000) — C

There are two steps catalysed by CPSII, a glutaminase activity (1), followed by formation of carbamoyl phosphate (2);

$$(1)\ \text{glutamine} + H_2O \rightleftharpoons \text{glutamate} + NH_3$$

$$(2)\ NH_3 + HCO_3^- + 2ATP \rightleftharpoons H_2N\text{–}CO\cdot OPO_3^{2-} + 2ADP + P_i.$$

A model of the general structure of this multienzyme polypeptide is shown in Fig. 7.14, showing the exposed interdomain regions. It is found that when MgATP (substrate) and MgUTP (a negative effector) are bound to the enzyme, the rates of proteolysis by elastase and trypsin are retarded and this is believed to occur because a conformational change decreases the exposure of the interdomain regions.

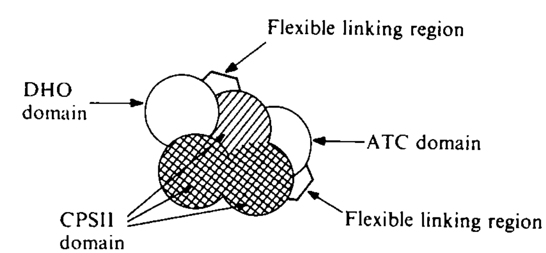


FIG. 7.14. A model of the domain structure of the multifunctional polypeptide CAD.[135]

It is interesting to compare the organization of the structural genes for the biosynthesis of UMP from carbon dioxide and glutamine or ammonia in different species. There are altogether six enzyme-catalysed reactions in the pathway, as shown in Fig. 7.13. In *E. coli* there are six gene loci corresponding to the six structural genes in the pathway. In *Saccharomyces* and *Neurospora* there are only five structural genes, since a single gene codes for the multienzyme polypeptide that contains E_1 and E_2 activities. In mammalian systems there appear to be only three structural genes coding for E_1–E_2–E_3, E_4, and E_5–E_6.[94] There is some evidence that E_5–E_6 in mammals exists as a multienzyme protein. Thus, during the course of evolution there is clustering on the chromosomes of genes that have related functions and in some cases there is gene fusion leading to the production of multienzyme polypeptides.

7.11 Multienzyme polypeptides: aspartate kinase–homoserine dehydrogenase, fatty-acid synthase, and the *arom* complex

In recent years it has become apparent that if enzymes are to be isolated with their complete polypeptide chains intact it is necessary to be particularly careful to avoid proteolysis occurring during isolation (see Chapter 2, Section 2.8.6). Many cells and tissues contain proteases that may be present in a latent form, e.g. within lysosomes, and these are often released during isolation. (For review of the problems associated with proteolysis during enzyme isolation see reference 95). On a simple statistical basis, proteolysis is more likely to cause a single break in a large polypeptide chain than in a small one. Thus a number of multienzyme proteins originally considered as multienzyme complexes have turned out to be multienzyme polypeptides. The examples of tryptophan synthase in *Neurospora* and *Saccharomyces* were mentioned in Section 7.9.1, but the most striking examples are the fatty-acid synthases and the complex of enzymes involved in aromatic amino-acid biosynthesis known as the *arom* complex; these are discussed in this section.

7.11.1 Aspartate kinase–homoserine dehydrogenase

The earliest reported example of a multienzyme polypeptide is aspartate kinase I—homoserine dehydrogenase I from *E. coli*.[96, 97] This multienzyme polypeptide is composed of four identical chains each having an M_r of 86 000. Each polypeptide chain has two active sites catalysing the reactions indicated below.

$$\xleftarrow{\qquad\qquad} M_r\ 86\,000 \xrightarrow{\qquad\qquad}$$

$$H_2N \underset{\text{Aspartate kinase}}{\text{———————}} \underset{\text{Homoserine dehydrogenase}}{\text{———————}} COOH$$

$$\text{ATP} + \text{L-Aspartate} \rightleftharpoons \text{ADP} + \text{4-Phospho-L-aspartate}$$

$$\text{L-Homoserine} + \text{NAD}^+ \rightleftharpoons \text{L-Aspartate } \beta\text{-semialdehyde} + \text{NADH}$$

These two reactions are the first and third steps in the biosynthesis of leucine, threonine, methionine, and lysine from aspartate.

The multienzyme polypeptide comprises 820 amino-acid residues. It has been shown that there is a region between residues 293 and 300 that is particularly sensitive to protease attack.[96] Two fragments that can be separated, and can be unfolded and refolded independently, contain the aspartate kinase and homoserine dehydrogenase activities, respectively. The C-terminal fragment, in addition

to containing homoserine dehydrogenase, also has the ability to self-associate. A model[97] has been proposed (Fig. 7.15) in which there are three compact domains linked by more flexible regions.

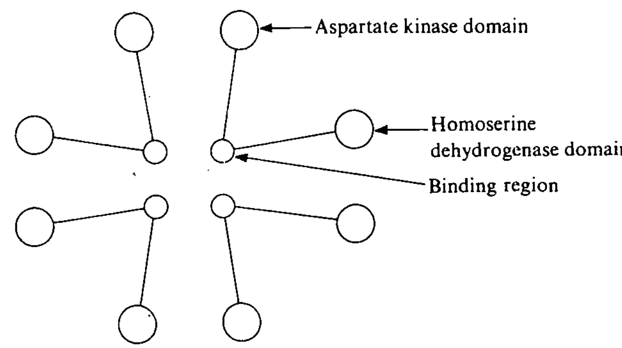


FIG. 7.15. Proposed tetrameric structure of aspartate kinase–homoserine dehydrogenase.[97]

7.11.2 Fatty-acid synthase

The ability to synthesize long-chain fatty acids is an attribute of all organisms and those in which fatty-acid synthesis has been examined in any detail show the same basic metabolic pathway, though with certain minor modifications, as illustrated in Fig. 7.16. Fatty-acids are synthesized from acetyl-CoA and malonyl-CoA and the overall process of extending an acyl chain by two methylene groups entails transacylations, two reductions, a dehydration, and a condensation and thus involves six catalytic activities as shown in Fig. 7.16. An additional enzyme acetyl-CoA carboxylase is necessary to catalyse the formation of malonyl-CoA but this is not part of the complex.

7.11.2.1 *Organization of the fatty-acid synthase system*

There are differences between organisms both in the predominant chain length of fatty acids they synthesize and in the mechanism of release of the long-chain acyl carboxylate, but the sequence of enzyme reactions is similar in all cases.[98] In spite of this similarity, the organization of the enzyme systems may differ. In *E. coli* six separate enzymes ([ACP] acetyltransferase, [ACP] malonyltransferase, 3-oxo-acyl-[ACP] synthase, 3-oxoacyl-[ACP] reductase, crotonyl [ACP] hydratase, and enoyl-[ACP] reductase), together with the acyl-carrier protein (ACP) have been isolated and purified. These enzymes together catalyse fatty-acid synthesis and there is no evidence that they are physically associated with one another. This non-aggregated type of fatty-acid synthase system (referred to as type II) has also been found in other bacteria, in the blue-green alga *Phormidium lunidum*, in *Euglena*, *Chlamydomonas*, Avocardo mesocarp, and in the chloroplasts from lettuce and spinach.

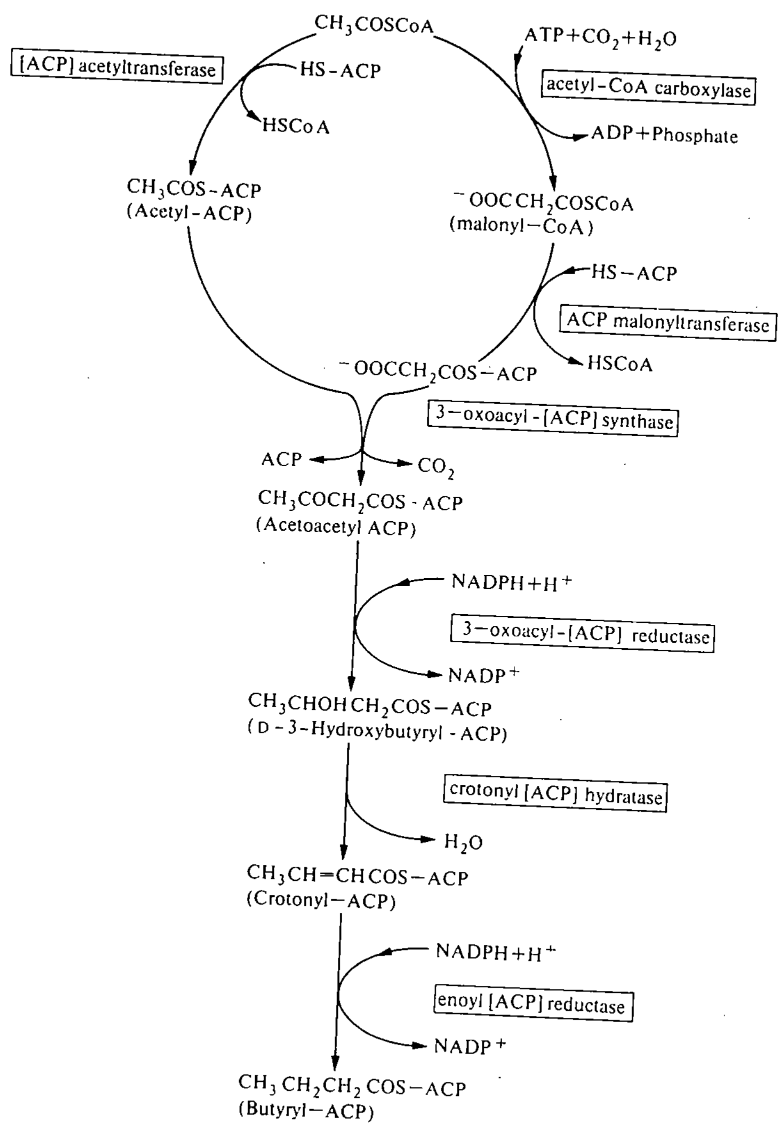


FIG. 7.16. The fatty-acid biosynthetic pathway. All reactions except the carboxylation of acetyl-CoA are catalysed by the fatty-acid synthase multienzyme system.

By contrast, in many animal tissues, avian tissues, yeast, *Neurospora*, and in some prokaryotes, the fatty-acid synthase system is found to exist as an aggregated multienzyme protein (type I). It was originally thought that these multienzyme proteins comprised several tightly associated polypeptide chains each having a single catalytic activity, but as a result of purifications in which rigorous measures have been taken to exclude proteolysis several of these have been shown to exist as multienzyme polypeptides. The type I fatty-acid synthases may be subdivided on the basis of their size into type IA, which have values of M_r of about 500 000 and subunit structures α_2, and type IB, showing a higher degree of aggregation and values of M_r 1–2.4 × 10^6 and subunit structures α_6–α_8, or $\alpha_6\beta_6$. Type IA are found in animal tissues, whereas Type IB occur in fungi, algae and higher bacteria such as *Mycobacteria*, *Corynebacteria*, and *Streptomyces*. The subunits of both types IA and IB have M_r values of between 185 000 and 250 000.

The yeast fatty-acid synthase system has been most thoroughly investigated both from a biochemical and genetic standpoint. By methods of genetic mapping[100] it has been shown that there are two unlinked genetic loci on the chromosome designated *fas 1* and *fas 2*, which code for the whole fatty acid synthase system. The two different subunits and the assignment of the catalytic activities to particular subunits is based on biochemical[101] and genetic[100] evidence. Fatty-acid synthase can be dissociated reversibly after modification of its free amino groups using maleic anhydride* (see Chapter 3, Section 3.4.2) and then the catalytic activities can be assigned to the separated polypeptide chains. The arrangement of catalytic centres on the α and β polypeptide chains is shown in Fig. 7.17, together with a possible model for the three-dimensional arrangement of the α and β subunits. The α chains are arranged in the form of a disc, and the β chains are looped across pairs of α chains above and below the plane of the disc.

In the smaller type IA fatty-acid synthases found in animal tissues, those that have been examined in detail have been found to be homodimers.[103–105] In the yeast fatty-acid synthase, the activities of [ACP]malonyltransferase, which is necessary to initiate fatty-acid chain growth (see Fig. 7.16), and of [ACP]palmityltransferase, which is necessary to terminate chain growth, are in identical regions of the polypeptide chain.[106, 107] On the other hand, in the mammalian multienzymes there are two different catalytic centres, making a total of seven different catalytic centres altogether. The fatty-acid synthases from chicken liver,[103] pigeon liver,[104] and rabbit mammary gland[105] each contain all seven catalytic activities within a single polypeptide chain and the multienzyme appears to be a homodimer arranged in head-to-tail fashion. Consistent with this is the finding of a single mRNA for all seven catalytic activities.[108]

The domain structure of the animal fatty-acid synthase has been studied by limited proteolysis using a variety of proteolytic enzymes. The presence of domains can readily be discerned by electron microscopy. By limited proteolysis the α chain is split into three domains of M_r 127 000, 107 000, and 33 000. The larger two can be further split. By locating the catalytic steps on particular domain fragments, it is possible to propose a model for the organization of the α_2 structure (Fig. 7.18). It can be seen that the growing chain is attached to the thiol

* Modification of free amino groups facilitates dissociation of the complex.[102]

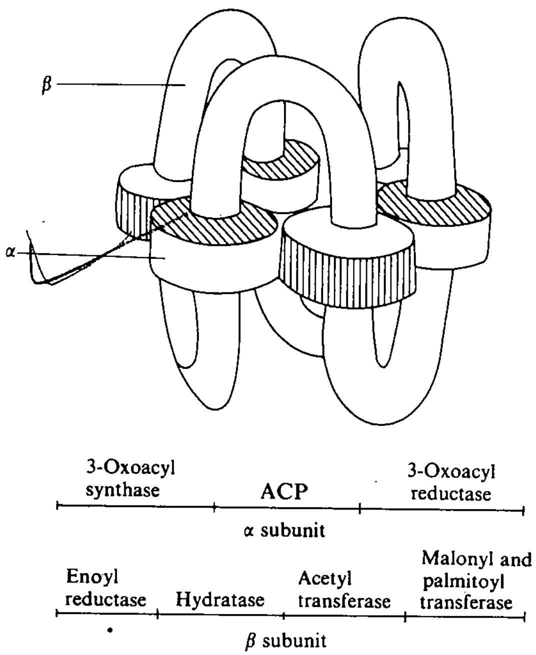


FIG. 7.17. A possible arrangement of the (α) and [β] subunits of yeast fatty-acid synthase ($\alpha_6\beta_6$), together with the location of enzyme activities on the polypeptide chains.

group on the end of the pantotheine moiety; it has to interact with the thiol on the synthase on the opposite α chain. The evidence for this stems from the observation that when the dimers are dissociated, $\alpha_2 \rightarrow 2\alpha$, the only activity lost is 3-oxoacyl-[ACP]synthase.[109]

A common feature of both type I and type II synthases is that the growing acyl chain is covalently attached to a protein through a phosphopantotheine

$$\text{Protein (Ser)OPO}_3^-\text{CH}_2.\text{C(CH}_3)_2.\text{CHOH.CONH(CH}_2)_2\text{CONH(CH}_2)_2\text{S}\underbrace{\text{COR}}_{\text{acyl group}}$$

prosthetic group. The acyl carrier protein has been isolated from several prokaryotes and sequenced. Its size varies between species, ranging from M_r values of 8600 to 16000.[110] In eukaryotes it is clear that the equivalent of an acyl-carrier protein exists, since phosphopantotheine becomes covalently bound to the fatty-acid synthase and thus the 'acyl-carrier protein' is a section of the multienzyme polypeptide.

7.11.2.2 *Control of the chain length of synthesized fatty-acids*

The process of fatty-acid synthesis is often described as a spiral because a sequence of reactions occurs that leads to the addition of $—CH_2CH_2—$ to a

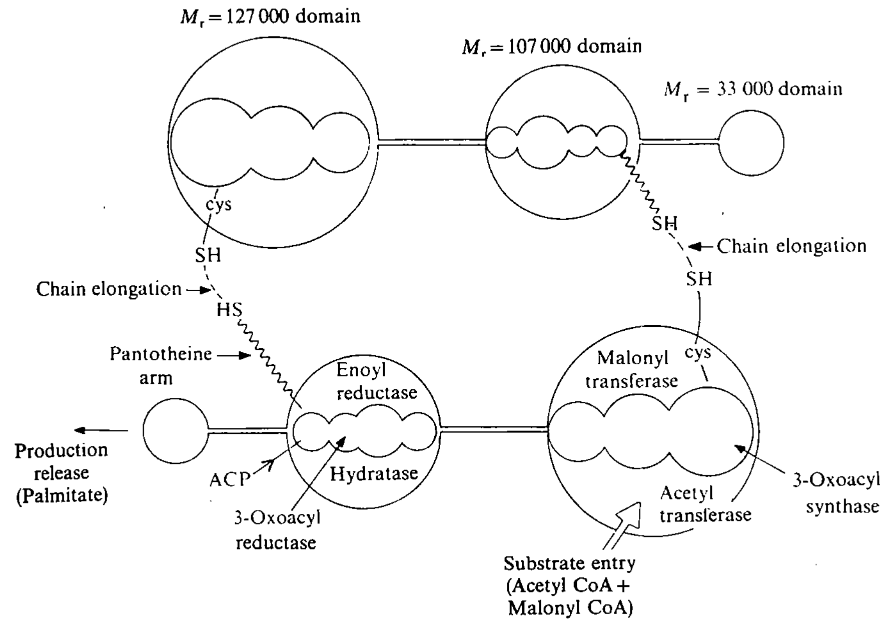


FIG. 7.18. The proposed domain structure of animal fatty-acid synthase, showing the two α subunits aligned head-to-tail.

growing acyl chain and then the sequence is repeated. During this growth the acyl group is attached to either the acyl-carrier protein or part of the multienzyme polypeptide. What determines the length of chain to be synthesized and how is it released from the enzyme? There is a fairly well-defined pattern for both the chain length and the degree of saturation of fatty acids synthesized by different species. In *E. coli* the predominant fatty acids synthesized are palmitic

Palmitic acid	$CH_3(CH_2)_{14}COOH$
Palmitoleic acid	$CH_3(CH_2)_5CH{=}CH(CH_2)_7COOH$
cis-Vaccenic acid	$CH_3(CH_2)_5CH{=}CH(CH_2)_9COOH$

acid, palmitoleic acid, and *cis*-vaccenic acid and the ACP derivatives of these acids transfer their acyl groups directly to glycerol-1-phosphate during phospholipid biosynthesis. The type of fatty acid synthesized in *E. coli* seems to be determined largely by the specificity of 3-oxoacyl[ACP] synthase, which catalyses the following two partial reactions:

$$\text{R CO-S-ACP} + \text{HS-enzyme} \rightleftharpoons \text{R-CO-S-enzyme} + \text{ACP—SH}$$

$$\text{R CO-S-enzyme} + \text{malonyl-S-ACP} \rightleftharpoons \text{R-CO-CH}_2\text{CO-S-ACP} + \text{HS-enzyme} + \text{CO}_2$$

where R = saturated or unsaturated alkyl group

The acyl group undergoing elongation is transferred to a cysteine side-chain on 3-oxoacyl[ACP] synthase.

When the maximum velocities of this synthase are compared using acyl-ACPs having different chain length of saturated hydrocarbon as substrates, there is very little difference comparing C_2 to C_{10} acyl ACPs, but the velocity decreases markedly for C_{14} acyl ACP and the enzyme is inactive towards C_{16} acyl ACP. When a similar comparison is made using substrates having unsaturated chains, the activity is low when hexadecenyl-ACP is used and inactive with *cis*-vaccenyl-ACP. Thus, the specificity of the synthase towards different acyl-ACPs sets an upper limit to the number of C_2 units that may be added to the growing chain.[111]

In yeast, the chain lengths of the fatty acids synthesized are predominantly C_{16} and C_{18} and control is probably similar to that in *E. coli*, namely, the specificity of 3-oxoacyl[ACP] synthase. The final step in the sequence is the transfer of palmityl-ACP or stearyl-ACP to coenzyme A catalysed by an acyltransferase.

$$\text{acyl-SACP} + \text{CoA} \rightleftharpoons \text{acyl-S-CoA} + \text{ACP}.$$

In yeast, palmityltransferase and malonyltransferase activities are associated with the same catalytic site on the multienzyme polypeptide, whereas in the mammalian fatty-acid synthases the two catalytic sites are distinct. A further difference is that in the mammalian system the sequence terminates with the release of free fatty acid, whereas, in yeast, acyl-CoA is released. This will greatly affect the rate of release of fatty-acid chains, since free fatty acids are readily released from the multienzyme complex, whereas acyl-CoAs, particularly of long-chain acids, are only slowly released. (For a review, see reference 112).

The sequence in yeast and in mammalian and avian systems[113] is as follows.

(i) *Yeast system*

1. E_1-Pantotheine-S-acyl + E_2-OH $\rightleftharpoons$ E_1-Pantotheine-SH + E_2-O-acyl.
2. E_2-O-acyl + CoASH $\rightleftharpoons$ Acyl-S-CoA + E_2-OH.

(ii) *Mammalian and avian systems*

1. E_1-Pantotheine-S-acyl + E_2-OH $\rightleftharpoons$ E_1-Pantotheine-SH + E_2-O-acyl
2. E_2-O-acyl + H_2O $\rightleftharpoons$ E_2-OH + acyl-OH

where E_1 and E_2 are different catalytic centres on the multienzyme polypeptides. Phosphopantotheine is covalently attached to E_1. The acyl group is transferred from the sulphydryl on pantotheine to a serine hydroxyl group on E_2.

7.11.2.3 *Control of fatty-acid synthesis in* Mycobacterium smegmatis

The control of fatty-acid synthesis in *Mycobacterium smegmatis* differs from that of *E. coli*, yeast, or mammalian systems and will be described here briefly. (For a detailed review see reference 114.) Fatty acids synthesized by *E. coli* are used primarily for incorporation into the phospholipids of the cell membrane. In mammals, they are, in addition, incorporated into triglycerides to be stored as an energy reserve. *Mycobacterium smegmatis*, which is a highly evolved prokaryote, synthesizes fatty acids both for incorporation into the cell membrane and into the cell wall. C_{16} and C_{18} fatty acids are used for membrane phospholipids, whereas C_{24} and C_{26} fatty acids are incorporated into the cell walls. Both C_{16}–C_{18} and C_{24}–C_{26} fatty acids are synthesized using a type I fatty acid synthase system. The fatty acids are synthesized whilst attached to the multienzyme protein via a phosphopantotheine residue and they are eventually transferred to CoA as in the yeast system (Fig. 7.19). Release of C_{16} and C_{18} acyl-CoAs from the enzyme is fairly slow (compare with yeast fatty-acid synthase, Section 7.11.2.2), but release of the C_{24} and C_{26} acyl-CoAs is very slow indeed, diffusion taking of the order of minutes.

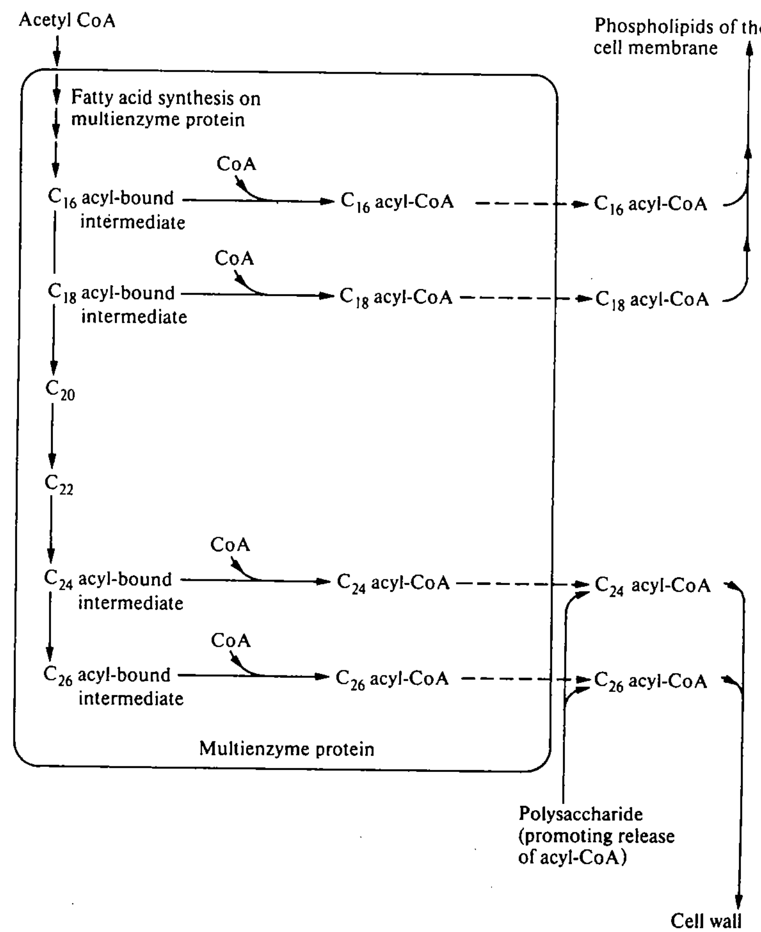


FIG. 7.19. Synthesis of fatty acids in *Mycobacterium smegmatis*.

Two factors regulate the balance between synthesis of C_{16} and C_{18} fatty acids and of C_{24} and C_{26} fatty acids: (i) high concentrations of acetyl-CoA stimulate the formation of C_{16}–C_{18} fatty acids; (ii) two classes of polysaccharides that contain decasaccharide segments of 6-*O*-methylglucose and 3-*O*-methylmannose residues promote the release of C_{24}–C_{26} acyl-CoAs from the fatty-acid synthase. Thus, in the presence of the polysaccharides, C_{24}–C_{26} fatty-acid synthesis is favoured, but when the polysaccharide concentration is low, slow release of C_{24}–C_{26} acyl-CoAs from the enzyme retards fatty-acid synthesis. This causes a build up of acetyl-CoA that in turn stimulates the synthesis of C_{16}–C_{18} acyl-CoAs.

7.11.3 Multienzyme proteins involved in the biosynthesis of aromatic amino acids

The aromatic amino acids tryptophan, tyrosine, and phenylalanine are synthesized *de novo* in most bacteria, fungi, and plants, whereas they are dietary requirements in many mammals. The biochemistry and genetics of the enzymes involved in this pathway have been studied in detail, particularly in certain prokaryotes and in fungi. The same pathway operates in all the organisms studied (as illustrated for *Neurospora crassa* in Fig. 7.20), but there is considerable variation between species in the organization of the genes, and hence for the enzymes for which they code.[115] When considering the conversion of chorismate to tryptophan, which is catalysed by enzymes E_8–E_{12} (Fig. 7.20), it is possible to identify seven different catalytic domains that can be completely separated in certain organisms. These are designated by the following letter code.

Domain E Catalyses the anthranilate synthase reaction with ammonia as amino group donor.
Domain G interacts with the E domain and provides the glutamine amidotransferase activity for the glutamine-dependent anthranilate synthase reaction.
Domain D is the catalytic site of E_9.
Domain F is the catalytic site of E_{10}.
Domain C is the catalytic site of E_{11}.
Domain A catalyses the conversion of indole glycerol phosphate to indole.
Domain B catalyses the conversion of indole to tryptophan.

The way in which these domains are linked shows considerable variation when different species are considered. Table 7.5 shows the linkage of polypeptide chains and subunit organisation for the best studied cases. Other fungi that have been studied, such as *Aspergillus nidulans*, *Coprinus lagopus*, and *Schizosaccharomyces pombe*, show the same pattern as *Neurospora crassa*.

In *Neurospora crassa* the biosynthetic pathway from erythrose 4-phosphate to tryptophan contains three multienzyme complexes (Fig. 7.20). There are also indications that weaker interactions may hold all three multienzyme proteins together with the remaining enzymes of the pathway in a 'super-multienzyme' protein.[116] In this section we discuss the properties of the *arom* complex, since certain features have been demonstrated with this multienzyme protein which have not been demonstrated in other multienzyme proteins.

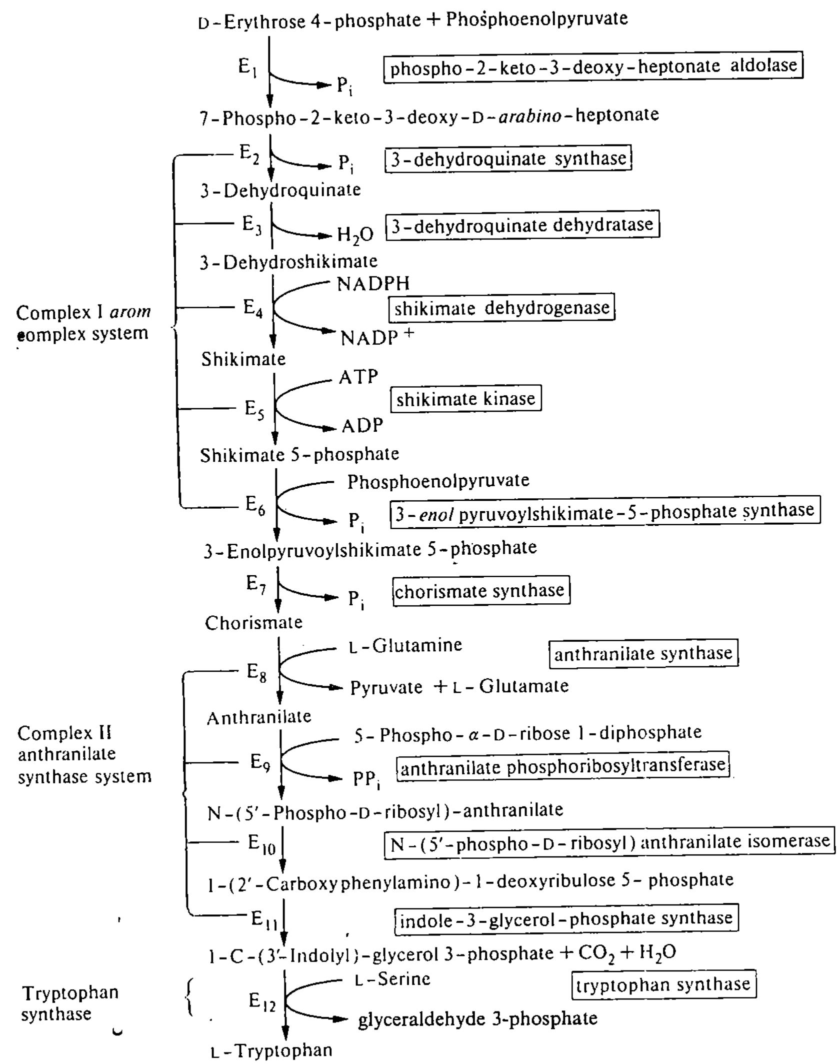


FIG. 7.20. Pathway for the biosynthesis of tryptophan in *Neurospora crassa*, showing the multienzyme proteins involved.

P_i = orthophosphate; PP_i = pyrophosphate

TABLE 7.5

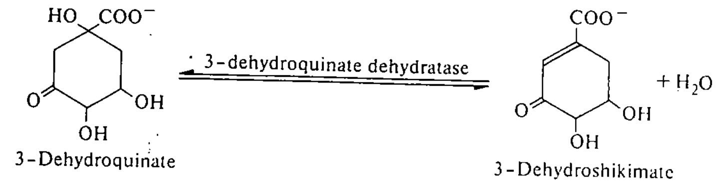
Organization of the domains for the conversion of chorismate to tryptophan

Organism	Domains					
Prokaryotes						
Escherichia coli	E–G–D		F–C		A, B	$(\alpha_2\beta_2)$
Pseudomonas putida	E, G $(\alpha\beta)$	D	F	C	A, B	$(\alpha_2\beta_2)$
Serratia marcescens	E, G $(\alpha\beta)$	D	F–C		A, B	$(\alpha_2\beta_2)$
Eukaryotes						
Neurospora crassa	E, G–F–C	$(\alpha_2\beta_2)$	D		A–B	(α_2)
Saccharomyces cerevisiae	E–G–C		D	F	A–B	(α_2)

'–' signifies a covalent link, ',' signifies a non-covalent link.

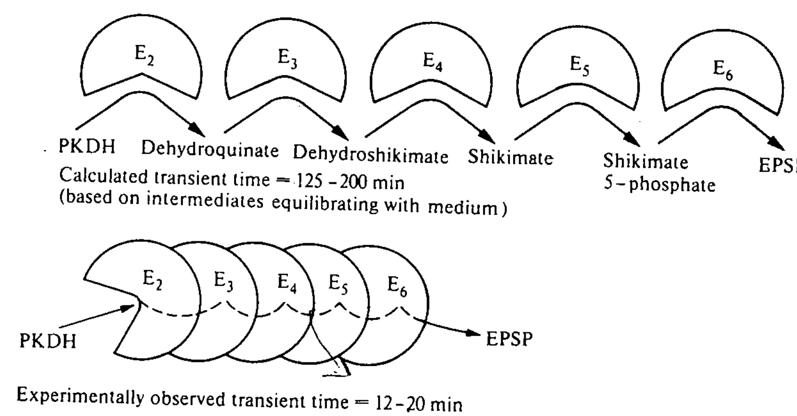
There are parallels between the history of the isolation and purification of the *arom** multienzyme protein and that of fatty-acid synthase type I. In 1972 the *arom* complex isolated from *Neurospora crassa*, having an M_r of 293 000, was shown to be separable into four distinct subunits having M_r of 54 000, 63 000, 84 000, and 95 000.[117] However, an isolation (Chapter 2, Section 2.8.6) in which rigorous attempts were made to inhibit proteolysis shows that the multienzyme protein comprises two polypeptide chains each having an M_r of 165 000.[118] From a study of the peptides obtained from hydrolysing the multienzyme protein, it is clear that the two chains of M_r 165 000 are identical or nearly identical[119] and each possesses the five catalytic activities (Fig. 7.20). The genes for all five catalytic activities are located in a cluster and may be transcribed as a polycistronic message, since the catalytic activities are transcribed and translated in an ordered sequence (compare with the tryptophan operon from *E. coli*, Fig. 7.8). It is probable that in the earlier isolation procedure some proteolysis occurred and that this resulted in rupturing the multienzyme polypeptide in the regions between the catalytic domains. It is noteworthy that in the early days of isolation of multienzymes, when the importance of exclusion of endogenous proteases was not fully realized, limited proteolysis occurred, preventing the isolation of the intact multienzyme polypeptide. Now that several multienzyme polypeptides have been isolated intact, limited proteolysis under carefully controlled conditions with exogenous proteases is being used as a method of exploring the domain structure.[120] In the case of the *arom* complex from *N. crassa*, limited proteolysis has been used to separate two fragments, one containing E_3/E_4 activity (Fig. 7.20) and the other E_6 activity. Different organizations occur in other species; in *E. coli* the catalytic activities appear to be on separate polypeptide chains, whereas, in plants, enzyme E_3 and E_4 copurify while E_2, E_5, and E_6 are separable (see reference 121).

A number of kinetic studies have been made to elucidate the working of the *arom* complex. It is thought that this complex may bring about a channelling of the intermediates so as effectively to separate the biosynthetic from the degradative pathways for aromatic amino acids. There is one enzyme activity—



3-dehydroquinate dehydratase, which is common to both biosynthetic and degradative pathways, and thus without a multienzyme polypeptide—which channels bound 3-dehydroshikimate on to the next catalytic domain there would be direct competition between the two pathways for these intermediates.

Calculations have been made to determine the expected transient time (see Section 7.6) for the reaction sequence from 7-phospho-2-keto-3-deoxy-D-*arabino*-heptonate to 3-enolpyruvoylshikimate 5-phosphate (Fig. 7.20) on the assumption that each step was catalysed by a separate enzyme and that the product of each step was released from each catalytic centre into the medium and was subsequently taken up by the next enzyme. When this calculated transient time is compared with the experimentally observed transient time for this multienzyme polypeptide, the latter is found to be approximately 1/10 to 1/15 of the former[122] (Fig. 7.21). Thus, when catalytic centres are in close proximity the metabolic rate can change much more rapidly to a new steady-state rate, and this



PKDH = 7-Phospho-2-keto-3-deoxy-D-*arabino*-heptonate
EPSP = 3-Enolpyruvoylshikimate 5-phosphate

FIG. 7.21. A comparison of the calculated and experimentally observed transient times for the enzyme-catalysed conversion of 7-phospho-2-keto-3-deoxy-D-*arabino*-heptonate to 3-enolpyruvoylshikimate 5-phosphate (see reference 122).

* The *arom* complex was so designated because the catalytic domains are coded for by a contiguous region of the chromosome named by geneticists as the *arom* cluster.

is probably due to the fact that the intermediates do not appreciably diffuse out into the medium but move directly from one catalytic centre to another.

Another related property of this multienzyme polypeptide is that of coordinate activation. It is found that the *arom* multienzyme polypeptide can convert 7-phospho-2-keto-3-deoxy-D-*arabino*-heptonate to 3-enolpyruvoylshikimate 5-phosphate (this requires E_2, E_3, E_4, E_5, and E_6; see Fig. 7.20) at approximately ten times the rate at which it can convert shikimate to 3-enolpyruvoylshikimate 5-phosphate (which requires E_5 and E_6 only). This is believed to be due to a process of coordinate activation of catalytic centres of the multienzyme polypeptide, with 7-phospho-2-keto-3-deoxy-D-*arabino*-heptonate possibly bringing about a conformational change in the whole multienzyme protein. There is evidence that 7-phospho-2-keto-3-deoxy-D-*arabino*-heptonate causes a reduction in K_m values of E_3, E_4, and E_6 for their respective substrates.[137] Also it is found that when 7-phospho-2-keto-3-deoxy-D-*arabino*-heptonate is bound to the *arom* complex it gives all five catalytic centres some protection against attack by proteases, suggesting possible conformational changes.

The organization of the anthranilate synthase complex (domains E, G–F–C) has been studied in *Neurospora crassa* using the technique of limited proteolysis to identify domains.[123] It comprises four polypeptide chains $\alpha_2\beta_2$. The α chain contains the catalytic site for anthranilate synthase capable of using ammonia as amino donor (domain E) and the β chain the catalytic site for phosphoribose anthranilate isomerase (domain F) and for indole glycerol phosphate synthase (domain C). When the multienzyme complex ($\alpha_2\beta_2$) is incubated with elastase, it cleaves the β chain between residues 237 and 238, splitting the multienzyme into two fragments. One of these fragments (M_r 98 000), comprising the complete α chain (M_r 70 000) and a fragment (M_r 29 000) of the β chain, contains the anthranilate synthase activity together with its glutamine-binding site (domain G). A model for the probable arrangement is shown below[123] (Fig. 7.22).

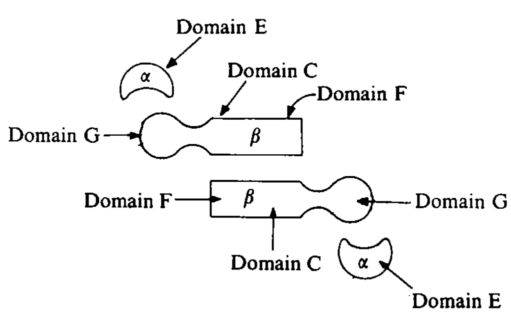


FIG. 7.22. A model for the arrangement of the anthranilite synthase complex ($\alpha_2\beta_2$) from *N. crassa*.

In *E. coli*, the phosphoribosyl anthranilate isomerase (domain F) and indole glycerol phosphate synthase (domain C) exist as a bifunctional monomeric enzyme. X-ray crystallographic studies have been carried out on this enzyme using an iodinated analogue of 1-(2′-carboxyphenylamino)-1-deoxyribulose 5-phosphate, the substrate of E_{11} (domain C) and the product of E_{10} (domain F).[124] This provided not only the three-dimensional structure of the enzyme but also a clear indication of the positions of the two catalytic sites. It is particularly interesting that the two catalytic sites are located back-to-back (Fig. 7.23) and this would appear to preclude the possibility of channelling of the reactants between the two catalytic sites. This is in contrast to another enzyme having two catalytic sites, namely, the DNA nucleotidyltransferase fragment known as the Klenow fragment[133] (see. Chapter 1, Section 1.3.2). This fragment contains both DNA nucleotidyltransferase activity and also 3′–5′ exonuclease activity. The latter serves an editing function (see Chapter 1, Section 1.3.2) in excising mismatched deoxyribonucleotides. In this case the two catalytic sites are juxtaposed so that the nascent chain from DNA nucleotidyltransferase activity passes directly through the second catalytic site for editing.[125]

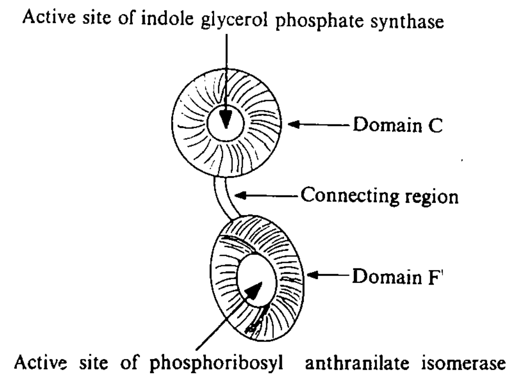


FIG. 7.23. Diagram showing the two catalytic sites in the bifunctional enzyme phosphoribosyl anthranilite isomerase and indole glycerol phosphate synthase from *E. coli*.

7.12 The glycogen particle

The multienzyme proteins or aggregates described in the last part of this chapter are those that are least well defined. The evidence for the existence of a subcellular particle containing glycogen together with the enzymes concerned with glycogen metabolism comes from studies of the isolation of glycogen from skeletal muscle and from liver. It has been known for many years from histological studies that glycogen is present in skeletal muscle in the form of particles (Fig. 7.24). If glycogen is isolated from skeletal muscle homogenates by procedures that do not cause protein denaturation, e.g. precipitation from the

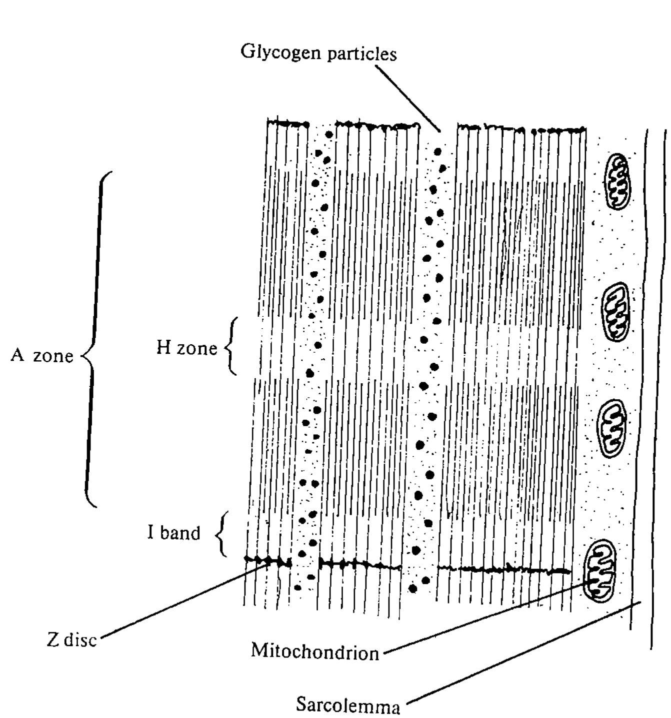


FIG. 7.24. The structure of a muscle fibre.

homogenate at pH 6.1 followed by high-speed centrifugation, then the particles containing glycogen also contain most of the glycolytic enzymes and they are of a similar size to those observed by histological studies. The evidence that these particles are not artefacts generated during the isolation procedure is supported by the following observations. (i) A high concentration of glycolytic enzymes is present after washing the particles. (ii) When three different methods are used to isolate the particles their composition is similar in each case.

The isolated glycogen particles have a reasonable well-defined composition, although not approaching the stoichiometric composition of the other multi-enzyme complexes. The enzymes present in glycogen particles are shown in Table 7.6. When the enzymes present in the particles are separated by polyacrylamide gel electrophoresis it is found that over 95 per cent of the total enzyme protein is accounted for by phosphorylase, glycogen synthase, debranching enzyme (see Chapter 6, Fig. 6.25), and phosphorylase kinase. Although phosphorylase phosphatase is tightly associated with the complex, it represents only 0.2 per cent of the protein of the complex.

Isolated washed glycogen particles are capable of catalysing the conversion of glycogen to lactate at a reasonable rate, e.g. 1.3 μmol per min per 100 mg protein compared to 40 μmol lactate per min per 100 mg protein in skeletal muscle *in vivo*. The glycogen particles contain many of the enzymes that control glycogen synthase and phosphorylase. They also contain AMP deaminase, which may control AMP levels. Much of the interest in the glycogen particle has been in the

$$\mathrm{AMP + H_2O \xrightleftharpoons[\text{deaminase}]{\text{AMP}} IMP + NH_3}$$

study of control of glycogen metabolism. In a number of situations the responses to ligands observed with the glycogen particle are similar to those observed in intact tissue. For example, the concentrations of Ca^{2+} and ATP required for phosphorylase kinase to phosphorylate phosphorylase, and the concentration of AMP required to inhibit purified phosphorylase phosphatase, in the glycogen particle are similar to those observed *in vivo*. Glycogen particles are polydisperse, varying in M_r from 6×10^6 to 1.6×10^9. Turnover of glucose moieties occurs more rapidly on the outside of the particles than on the inside. There is a different enzyme distribution in the particles depending on their size, e.g. phosphorylase activity is highest in small particles, whereas glycogen synthase activity is highest in the large particles.[127] Further study of the particle should help in understanding regulation of glycogen metabolism. The control of glycogen metabolism is discussed further in Chapter 6, Section 6.4.2 and also in references 126 and 128.

TABLE 7.6
Typical enzyme content of glycogen particles

Phosphorylase
Phosphorylase kinase
Phosphorylase phosphatase
Glycogen synthase
Protein kinase
Adenosinetriphosphatase
AMP deaminase
Adenylate kinase
Creatine kinase
Phosphoglucomutase

Reference 126.

7.13 DNA synthesis

There is also evidence that the enzymes required for DNA synthesis are organized in a complex that has been referred to as a replication apparatus.[129] Six of the enzymes required for catalysing the conversion of ribonucleoside diphosphates into DNA (DNA nucleotidyltransferase, thymidine kinase, cytidylate kinase, nucleosidediphosphate kinase, thymidylate synthase, and tetrahydrofolate dehydrogenase) have been found to cosediment during sucrose density-gradient centrifugation. The complex or replication apparatus can be extracted from the nuclei of S-phase* fibroblasts but not from G-phase fibroblasts, and a similar complex has been detected in yeast.[130] There is also evidence that the several of the aminoacyl t-RNA synthetases from eukaryotes exist as a multienzyme complex.[131]

7.14 Conclusions

In this chapter we have described a number of different multienzyme proteins, less well-defined aggregates of enzymes, and an enzyme in which the different polypeptide chains operate in a concerted manner to catalyse a complex enzyme reaction (RNA nucleotidyltransferase). Multienzyme proteins are generally regarded as providing a more efficient method of catalysing a sequence of reactions than separate enzymes. The increase in efficiency is thought to stem from (i) reduction in transit time, (ii) reduction in transient time, (iii) channelling of intermediates so as to reduce competition from other pathways, (iv) coordinate activation of a group of catalytic centres, (v) protection of unstable intermediates from non-enzymatic degradation. (These factors were discussed in Section 7.6.) Examples in support of points (i) to (iv) have been given throughout this chapter. As yet there appears to be no well-documented example on an unstable intermediate that would be degraded spontaneously if it were not efficiently transferred from one catalytic centre to the next.

However, although it is not difficult to see that multienzyme proteins may improve efficiency, it is not clear how important or essential this is *in vivo*. For example, it has been shown that the transient time for the *arom* complex may be reduced by 10–15-fold as a result of being organized as a multienzyme protein (see Section 7.11.3), and thus the system can respond more rapidly to a change in substrate concentrations. Whether or not this is important in the growth of the organism and its ability to respond to changing conditions has not yet been established. The organization of the fatty-acid synthase as a multienzyme protein in yeast and mammalian systems means that a growing acyl group on the enzyme complex is free from competition from other growing chains. Only one fatty acid can be synthesized at a time per fatty-acid synthase. There is no evidence to suggest that fatty-acid synthesis is less efficient, either in terms of the maximum rates achievable or of the efficiency with which the initial precursor is converted to the final products, in the organisms in which the enzymes appear to be completely separate, e.g. *E. coli* and plants, than in the multienzyme polypeptides from yeast or mammalian tissues.

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* The nuclear division cycle can be divided into the phases G_1 (presynthetic phase), S (synthetic phase during which DNA synthesis occurs), G_2 (postsynthetic phase), and the M phase (mitotic phase). For details, see reference 3.

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8
Enzymes in the cell

8.1. Introduction

In order to make measurements of enzyme activity, whether of single enzymes or a sequence of enzymes, the organism or tissue concerned is disrupted and the enzyme activity measured in crude homogenates or partially purified enzyme preparations. Disruption is necessary in order that the enzymes have free access to the added substrates. In an intact cell the membrane often either prevents the substrate from entering the cell or limits the rate at which it enters. In whole tissues, only the cells on the surface would have unrestricted access to the substrates. Much or all of the subcellular organization is lost in this process of disruption and the assays for enzyme activity are usually performed under conditions which differ considerably from those *in vivo*. In this chapter we consider how the process of cell disruption, and the changed environment *in vitro* compared with that *in vivo*, may effect single enzymes, and also their interactions with other enzymes and cell structures, since this is important in assessing how far results obtained from enzyme measurements *in vitro* (cf. Chapters 4–6) may be used to understand enzyme action *in vivo*.

The principal differences between enzyme reactions measured *in vitro* and those occurring *in vivo* are as follows.

(i) Loss of organization and compartmentation

In an intact cell, enzymes, substrates, cofactors, or effectors are not distributed uniformly throughout the cell. There are often considerable differences between the concentrations of any of these in subcellular organelles; there is even evidence that concentration gradients exist within the soluble phase of the cytoplasm i.e. the cytosol, (see Section 8.2.6), but the differences may not exist once a cell or its constituent organelles have been disrupted.

(ii) The dilution factor

During the course of preparation of either a crude homogenate or a purified enzyme, the suspending medium used causes dilution of the solutes present within the cell. Protein concentrations within cell compartments are at the lower end of the range of protein concentrations that exist in protein crystals.[1] For example, muscle cells have 23% protein by weight; red blood cells, 35 per cent, and most dividing cells 17–26 per cent, whereas protein crystals range between 20 and 90 per cent protein. In contrast, the protein concentration in an enzyme assay using a liver homogenate might be 5 mg cm^{-3}, and with a highly purified enzyme the protein concentration might be of the order of 5 μg cm^{-3}. The dilution of the cell constituents may affect a number of interactions between macromolecules and between macromolecules and small molecules and ions.

(iii) Relative concentrations of enzyme and substrate

In enzyme assays carried out under steady-state conditions (see Chapter 4, Section 4.1) there is a vast excess of substrate compared with that of enzyme. Typical concentrations might be in the region of 10^{-3} mol dm^{-3} for substrates and 10^{-6} mol dm^{-3} or below for enzymes. Measurements *in vivo* suggest that there are many instances in which enzymes are present in concentrations comparable with or greater than their substrates[2] and thus the kinetics observed under steady-state conditions may be somewhat different from those *in vivo*.

(iv) Closed and open systems

This aspect is related to the previous one. The enzyme assay carried out *in vitro* is essentially a closed system. The reactants are provided at the beginning of the assay and no replacement usually occurs during assay. The living cell, on the other hand, is an open system in which metabolites are continually entering and leaving. Thus, the concentrations of both the substrate and product of an enzyme reaction may be unchanged for a considerable period of time *in vivo* but nevertheless there may be considerable flux through the enzyme pathway (see Chapter 6, Section 6.3.1).

These four differences between enzymes in the cell and enzymes in the cuvette or test-tube will now be discussed in more detail. It will become apparent that certain aspects of this situation, e.g. compartmentation, have been explored extensively, whereas other areas, e.g. the high concentrations of proteins within organelles and their effects on reactions, have hardly been touched.

8.2 Intracellular compartmentation

Cells vary considerably in size, shape, and the amount of subcellular organization contained within them. A 'typical' cell illustrating subcellular organization is given in Fig. 8.1. Much of the subcellular organization can be discerned from light and electron microscopic observations, but in addition to this there are many examples of metabolic 'pools', where for a given metabolite there is kinetic evidence for compartmentation but for which no morphological counterpart can be discerned. A typical prokaryote cell has a volume in the range 0.5–500 μm^3 and well-defined subcellular structures have not been identified and isolated. A eukaryote cell is generally larger,[3] in the range 200 μm^3 to 15 000 μm^3 and many well-defined subcellular organelles have been characterized. The larger size probably necessitates a greater degree of subcellular compartmentation. In the following section we briefly describe the isolation and properties of the principal subcellular organelles of eukaryote cells. The main emphasis will be to give an indication of the size of each organelle, the selectivity of its surrounding membrane where appropriate, and the nature of the substances comprising the

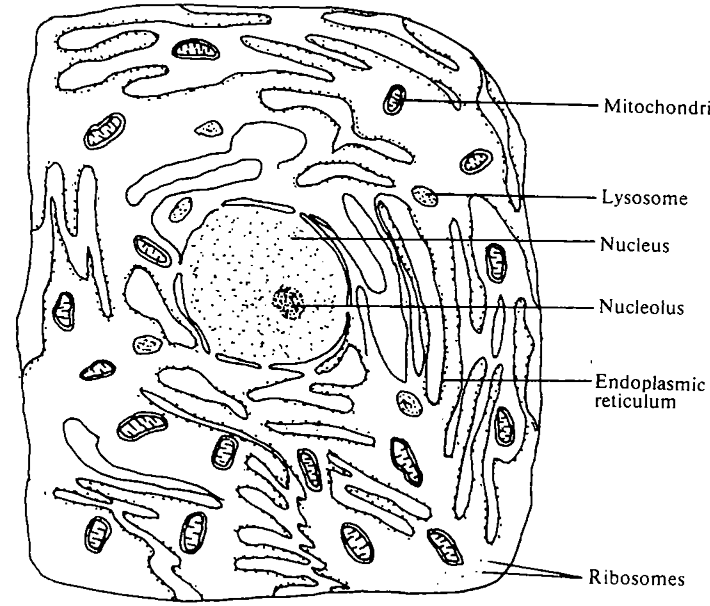


FIG. 8.1. A typical eukaryote cell showing the intracellular structure (subcellular organelles not drawn to scale).

matrix. Since the liver parenchyma cell is perhaps the most studied eukaryotic cell, the information given refers to the liver cell unless otherwise stated.

8.2.1 Isolation of subcellular organelles

The aim of subcellular fractionation procedures is to isolate subcellular organelles as free from contaminants as possible and with as little disruption as possible. The procedure involves two principal stages: (i) disruption of the tissue or cells to release the subcellular organelles and (ii) separation of the organelles from one another. It is necessary also to have methods for assessing the integrity of the organelles and the extent of contamination by other components. There is no single procedure suitable for all cells and tissues, and methods have to be adapted for the tissue concerned.

In the first stage, cells and tissues are disrupted by a variety of different methods according to the tissue involved (see Chapter 2, Section 2.5). The second step in the procedure is then to separate components from the homogenate usually by differential centrifugation. The rate of sedimentation in a centrifugal field increases with the size of a particle, the difference in density between the particle and the solvent, and the centrifugal field. There are small differences in the densities of the subcellular organelles, nuclei being denser than mitochondria, but the main factor influencing the separations are the differences in size of the organelles. The organelles are thus usually separated by centrifugation at increasing speeds, as indicated in Fig. 8.2. In order to obtain fairly pure preparations, the sedimented fractions are usually resuspended in buffer and then resedimented once or twice more. This procedure gives a fairly good separation of the main organelles: nuclei, mitochondria, and microsomes (disrupted endoplasmic reticulum). It is difficult to separate mitochondria from lysosomes because they have similar sizes and densities. An enriched lysosomal pellet may be prepared if a number of resuspensions and resedimentations are carried out, but this is unlikely even then to contain more than 10 per cent lysosomes. To obtain a complete separation of rat-liver lysosomes from mitochondria, the rats may be injected prior to sacrifice with Triton WR1339 (an anionic detergent, the sodium salt of an alkyl aryl polyether sulphate) or with colloidal gold. The Triton WR1339 or colloidal gold is phagocytosed by the

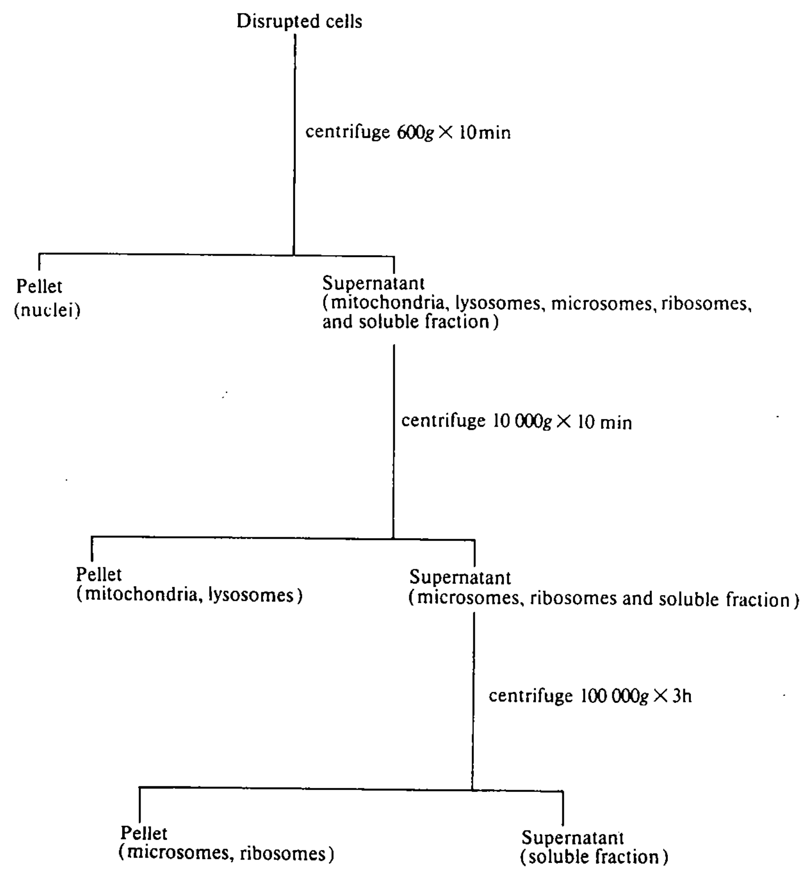


FIG. 8.2. Procedure for subcellular fractionation by differential centrifugation.

lysosomes, causing decreases or increases in their densities, respectively. They may then be separated from mitochondria.[4]

Further refinements to subcellular fractionations may be carried out; for example, smooth microsomes (disrupted endoplasmic reticulum lacking ribosomes) can be separated from rough microsomes (endoplasmic reticulum containing ribosomes) by centrifugation through layers of sucrose solution having different densities. The denser sucrose solutions retard the smooth microsomes, which are less dense than rough microsomes, because the former lack the more dense RNA.[5]

The purity of subcellular organelles can be assessed by biochemical and histological methods. Certain enzymes that have been found to be present either exclusively or predominantly in one subcellular organelle are used as marker enzymes to assess the extent of contamination of a particular subcellular fraction. For example, succinate dehydrogenase is a mitochondrial enzyme, and thus by measuring succinate dehydrogenase activity in a preparation of nuclei, the extent of contamination by mitochondria can be assessed. A list of marker enzymes for different subcellular fractions is given in Table 8.1. Besides the biochemical methods, it is useful to examine a preparation by light or electron microscopy, in order to check that the organelles are undamaged. Histochemical reagents can also be used. e.g. histochemical assay for acid phosphatase, or Feulgen's reagent for DNA (a reaction which depends upon the fact that partial hydrolysis products of DNA restore the colour of basic fuchsin that has been decolorized by sulphurous acid).

TABLE 8.1
Marker enzymes used in subcellular fractionation

Enzyme	Subcellular fraction
DNA nucleotidyltransferase	Nuclei
Nicotinamide-nucleotide adenylyltransferase	Nuclei
Succinate dehydrogenase	Mitochondria
Cytochrome *c* oxidase	Mitochondria
Glucose 6-phosphatase	Endoplasmic reticulum
Acid phosphatase	Lysosomes
Ribonuclease	Lysosomes
Catalase	Peroxisomes
Urate oxidase	Peroxisomes
5′-Nucleotidase	Plasma membrane
Glucose 6-phosphate dehydrogenase	Cytosol
Lactate dehydrogenase	Cytosol
6-Phosphofructokinase	Cytosol

8.2.2 The nucleus

The nucleus is the largest subcellular organelle. Its size does not vary substantially in a given type of cell, although there is a wide variation in the sizes of nuclei from different cell types and this variation is not related to the volume of the cytoplasm. For example, nuclei from certain fungi may be only 4 μm^3 in volume and occupy less than 1 per cent of the volume of the cytoplasm, whereas the nuclei of cells from the thymus gland are 300–400 μm^3 and occupy 70 per cent of the cell volume. The nuclei from liver cells have a volume of 300 μm^3 which is about 5 per cent of the total cell volume. The nucleus is bounded by two membranes which show some properties similar to those of the endoplasmic reticulum, e.g. all enzymes found on the nuclear membrane except cytochrome *c* oxidase are also found in the endoplasmic reticulum. However, they differ from the endoplasmic reticulum in buoyant density.[6, 7] There is a polymeric protein network lining the nucleoplasmic surface of the nuclear membrane.[8]

A characteristic feature of the nuclear membrane is the presence of pores. They vary in size and number according to the type of nucleus. In many mammalian nuclei the pores occupy 10–20 per cent of the surface area of the nucleus and the pore diameter is about 70 nm. Many small molecules and ions and also macromolecules are able to enter the nucleus by a diffusion process not requiring a source of metabolic energy, e.g. ATP. It seems likely that transport is through the nuclear pores. Larger proteins can enter through the pores by an active process that entails recognition by the pore of a signal sequence on the polypeptide chain.[9] There are differences in the concentrations of Na^+ and K^+ in the nucleus compared with outside the nucleus but these are probably due to different proportions of ions bound to macromolecules. It is thought that ribosomes (less than 50 nm wide from the outer edge of 40 S subunit to the outer edge of 60 S subunit) and RNA pass through the pores. The space within the nucleus into which sucrose and other small molecules cannot penetrate, is estimated at less than 10 per cent of the total volume.

The nucleus is the densest subcellular organelle (density approximately 1.35 g cm^{-3}) and this reflects the high concentrations of macromolecules within it (approximately 34 g protein, 9.5 g DNA, and 1.5 g RNA per 100 g nuclei).[10] The large amount of DNA in the nucleus necessitates a high degree of condensation from that of the double helix by as much as 50 000-fold in order to fit within the dimensions of the nucleus. The enzymes of the nucleus can be grouped according to their location (Table 8.2). The enzyme content of the soluble compartment of the nucleus is similar to that of the cytosol, especially in regard to that of the enzymes of the glycolytic pathway. The second and third groups (Table 8.2) of enzymes are bound to insoluble components of the nucleus which are not readily washed out of the preparations of isolated nuclei, but they may be extracted using concentrated salt solutions, e.g. 1 mol dm^{-3} NaCl. They comprise mainly the enzymes concerned with DNA and RNA synthesis and are attached to chromatin. The fourth group are only extracted by the use of detergents and are membrane-bound enzymes such as glucose 6-phosphatase and acid phosphatase. It is possible to isolate a nuclear matrix that comprises an insoluble skeletal framework connected to the nuclear membrane, and which has an appearance similar to that of the nucleus under the electron microscope. It comprises about 10% of the nuclear proteins, and processes such as replication and transcription are believed to occur when the enzymes concerned are associated with this framework.[13] The role of the nucleolus is principally the biogenesis of ribosomes and it develops and degenerates with each cell cycle.[14]

TABLE 8.2.
Enzymes present in the nucleus

1. Enzymes in the soluble space	**3. Enzymes concentrated in the nucleolus**
Glycolytic enzymes	RNA nucleotidyltransferase I
Pentose phosphate pathway enzymes	RNA methyltransferases
Lactate dehydrogenase	Ribonuclease
Malate dehydrogenase	
Isocitrate dehydrogenase	
Arginase	
2. Enzymes bound to chromatin	**4. Enzymes bound to membranes**
RNA nucleotidyltransferase II	Glucose 6-phosphatase
RNA nucleotidyltransferase III	Acid phosphatase
Nucleoside triphosphatase	
DNA nucleotidyltransferase	
Nicotinamide-nucleotide adenylyltransferase	

For further details see references 6, 7, 10–12.

8.2.3 The mitochondrion

The mitochondrion is typically a sausage-shaped organelle having dimensions of approximately 0.5 μm diameter, 4 μm length, and 1 μm^3 volume. Although it is a relatively small organelle, there are usually a large number of mitochondria per cell and collectively they occupy a significant proportion of the cell volume, e.g. liver parenchymal cells have 1000–1600 mitochondria per cell and they occupy 10–20 per cent of the volume of the cytoplasm. The mitochondrion has two membranes that have markedly different properties. The outer membrane is smooth in appearance whereas the inner membrane has a number of infoldings called cristae (Fig. 8.3). The inner mitochondrial membrane encloses the matrix, which is 60–70 per cent of the total mitochondrial volume. On the inner surface of the inner mitochondrial membrane are small mushroom-shaped particles containing mitochondrial adenosinetriphosphatase, which plays an important role in oxidative phosphorylation (see Chapter 2, Section 2.8.3). The number of cristae per mitochondrion varies with the tissue, being highest in those tissues having high potential metabolic rates, e.g. flight muscle (see Chapter 6, Section 6.4.1.1).

The mitochondrial matrix is granular in appearance and has a very high concentration of protein, usually about 500 mg cm^{-3}.[15] About 70 per cent of the mitochondrial protein is in the matrix, 20 per cent is part of the inner mitochondrial membrane and only 4 per cent is part of the outer membrane. The inner mitochondrial membrane has a ratio of protein:lipid of 75 per cent to 25 per cent, which is high compared to other cell membranes. It is estimated, using

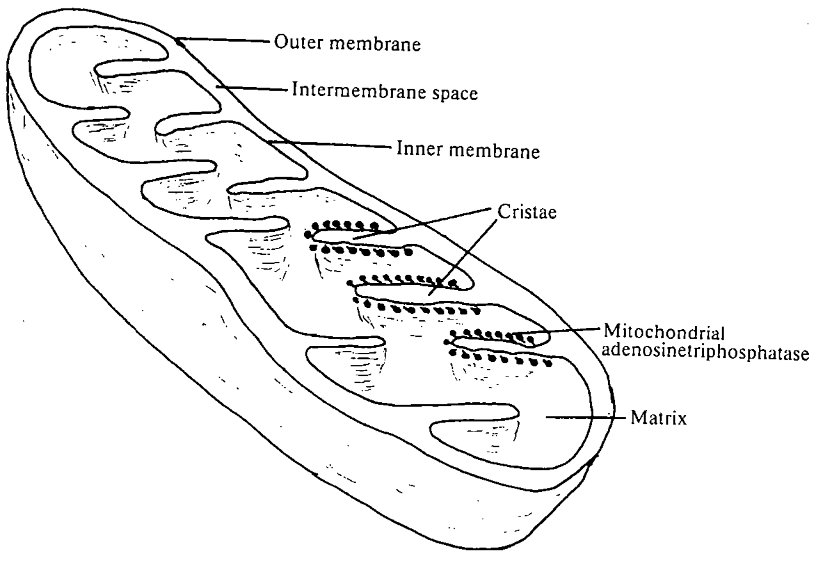


FIG. 8.3. Cross-section of a typical mitochondrion.

differential scanning calorimetry combined with freeze-fracture electron microscopy, that between one-third and one-half of the surface area is occupied by proteins,[16] most of which are involved in the electron-transport chain.

The chemical composition, permeability, and enzyme composition of the inner and outer membranes differ considerably. The outer membrane has a much higher proportion of lipid than the inner membrane, the latter having a very low content of cholesterol. Several of the enzymes present in the outer membrane are also present in the endoplasmic reticulum, e.g. cytochrome b_5 reductase and glycerophosphate acyltransferase, and it is often considered that the outer membrane may be derived from the endoplasmic reticulum. The outer membrane is generally permeable to small molecules having M_r up to 10 000, e.g. ATP, coenzyme A, and carnitine, whereas the inner membrane is generally permeable only to uncharged molecules having M_r up to 150. There are, however, a number of specific proteins in the inner membrane, called translocators, that are responsible for the transport of specific metabolites across the membrane. For example, translocators exist for inorganic phosphate, malate, succinate, citrate, isocitrate, 2-oxoglutarate, glutamate, aspartate, and adenine nucleotides (these are discussed further in Section 8.3.1.4). A list of the principal enzymes located in different regions of the mitochondrion is given in Table 8.3 and some are discussed later in the chapter. A number of the enzymes in the matrix have been shown to associate with one another and with proteins of the inner mitochondrial membranes. These are discussed in Section 8.5.3.2.

Although the concentration of protein is high in mitochondria, particularly in the matrix, the overall density of mitochondria is lower than that of the nucleus

TABLE 8.3

Principal enzymes or groups of enzymes present in the mitochondrion

1. Matrix Tricarboxylic acid cycle enzymes except succinate dehydrogenase Enzymes catalysing β-oxidation of fatty acids Pyruvate carboxylase Phosphoenolpyruvate carboxykinase (pigeon liver) Carbamoyl-phosphate synthase Ornithine carbamoyltransferase Glutamate dehydrogenase	**3. Intermembrane space** Adenylate kinase Nucleosidediphosphate kinase Nucleosidemonophosphate kinase L-Xylulose-reductase
2. Inner membrane Succinate dehydrogenase } + associated respiratory chain NADH dehydrogenase } Adenosinetriphosphatase 3-Hydroxybutyrate dehydrogenase Carnitine palmitoyltransferase Glycerol 3-phosphate dehydrogenase (FAD enzyme) 5-Aminolaevulinate synthase Hexokinase Cytochrome c oxidase	**4. Outer membrane** NADH dehydrogenase (rotenone insensitive) Cytochrome b_5 reductase Amine oxidase (flavin-containing) Kynureninase Acyl-CoA synthetase Glycerophosphate acyltransferase Cholinephosphotransferase Adenylate kinase Hexokinase Phospholipase A_2

For further details see reference 17

(1.1 g cm^{-3} compared with 1.35 g cm^{-3}), largely owing to the absence of high concentrations of nucleic acids.

8.2.4 The lysosome

Lysosomes are roughly spherical organelles or vacuoles slightly smaller than mitochondria and having diameters between 0.2 μm and 0.8 μm. They were first discovered by De Duve[18] as particles containing hydrolytic enzymes that sedimented with the mitochondrial fraction in subcellular fractionation. Not only their sizes but their densities (about 1.2 g cm^{-3}) are similar to those of the mitochondria. The characteristic feature of lysosomes is that they contain about 60 hydrolytic enzymes, most of which have acid pH optima (Table 8.4).

Lysosomes are bounded by a single membrane[19] enclosing a dense granular matrix. The membrane shows limited permeability to solutes, e.g. very little sucrose can penetrate the lysosomal membrane. Transport of many solutes is by carrier-mediated processes. In liver cells the lysosomes only occupy 0.3 per cent of the total cell volume and on average there are about 20 lysosomes per cell. About 35 per cent of the total protein is in the membranes and the remainder is in the matrix. The protein concentration in the matrix is about 200 mg cm^{-3} (calculated from reference 20 assuming 35 per cent of lysosomal protein is membrane protein) and the intralysosomal pH is about 1.5 units below that of

TABLE 8.4
Enzymes present in lysosomes

1. Proteolytic enzymes	**3. Hydrolysis of nucleic acids**
Cathepsins B, D, G, L	Deoxyribonuclease II
Elastase	Ribonuclease II
Collagenase	
2. Hydrolysis of glycosides	**4. Hydrolysis of lipids**
β-D-Glucuronidase	Phospholipases A_1 and A_2
β-*N*-Acetyl-D-hexosaminidase	Cholesterol esterase
Hyaluronoglucosaminidase	
Lysozyme	
Neuraminidase	
5. Others	
Acid phosphatase	
Aryl sulphatase	

For a more comprehensive list, see references 4 and 25.

the cytosol. Several of the proteins both of the membrane and matrix are glycoproteins and many are acidic. The low intralysosomal pH is thought to be maintained mainly by Donnan equilibrium[21] (an equilibrium that results in the asymmetric distribution of diffusible ions across a membrane, brought about by the presence of a larger ion that is unable to diffuse across the membrane), but there is also some evidence for a membrane-bound adenosinetriphosphatase that may act as a proton pump[22] (a process by which protons are transported across a membrane against the concentration gradient at the expense of metabolic energy). The proteins on the inner surface of the lysosomal membrane contain about 16 μg sialic acid per mg protein and this, together with other glycoproteins in the matrix, is thought to be responsible for the maintenance of the low pH.

Lysosomes are morphologically heterogeneous and this may reflect the way in which they function. Their mode of action is still uncertain, but De Duve and others have proposed the following theory.[23, 24] Primary lysosomes that have a full complement of hydrolytic enzymes but that have not participated in intracellular digestion are formed by pinching off portions of the smooth endoplasmic reticulum at the Golgi apparatus. The primary lysosomes fuse with other membranous structures containing substrates to be degraded and thereby form secondary lysosomes, i.e. lysosomes in which active digestion is occurring. Some of the low M_r products may diffuse out of the secondary lysosomes.

8.2.5 The endoplasmic reticulum

Although the endoplasmic reticulum is not generally regarded as an organelle it forms an extensive network throughout the cytoplasm and effectively separates the cytosol into two components; that which is enclosed within the endoplasmic reticulum (the lumen) and that which is outside it (Fig. 8.4). It is not possible to separate these two compartments by present biochemical techniques because when a cell is disrupted prior to carrying out subcellular fractionation, the endoplasmic reticulum is broken down and the membranous fragments form vesicles that constitute the 'microsome fraction' in subcellular fractionations. The amount of endoplasmic reticulum varies between different cell types but there is a good correlation between the density of reticulum and the extent to which the cell secretes proteins. The principal type of liver cell, the hepatocyte, and the cells of the exocrine pancreas (the cells concerned with its digestive functions rather than its hormonal functions) secrete a number of proteins and both have dense endoplasmic reticula (in the hepatocyte the endoplasmic reticulum occupies about 15 per cent of the total cell volume; 1 g of liver contains about 10 m^2 area of endoplasmic reticulum), whereas a non-secreting cell such as the erythrocyte is devoid of endoplasmic reticulum.

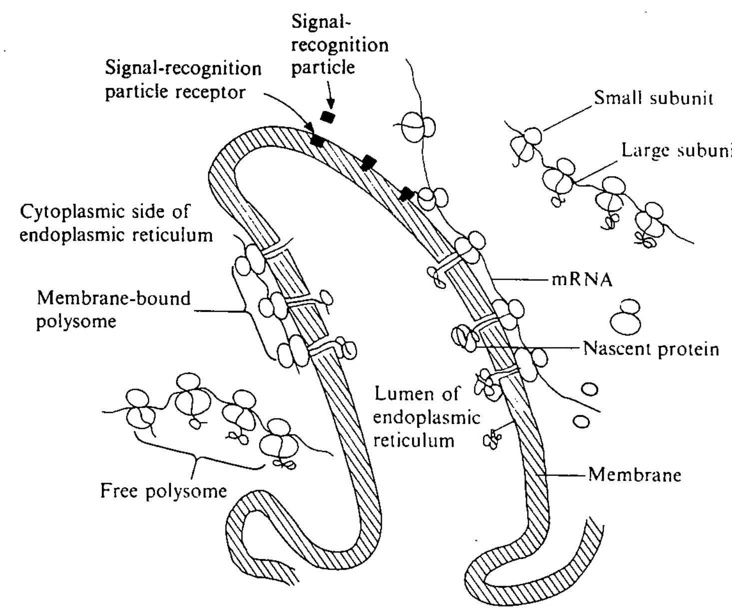


FIG. 8.4. The endoplasmic reticulum showing free and membrane-bound polysomes.

The endoplasmic reticulum, which is composed mainly of lipids and proteins, is organized into channels and it is assumed that proteins are collected and secreted into these channels. The proteins of the endoplasmic reticulum have both structural and catalytic roles. It is estimated that there are about 25–50 molecules of phospholipid per polypeptide chain in the endoplasmic reticulum. The membrane has limited permeability: uncharged molecules with M_r values up to about 600 can pass through freely, but charged species with M_r values greater than 90 and macromolecules, cannot pass freely. The amounts and distribution of many of the proteins have been estimated; cytochrome P450 is the most abundant, comprising 4 per cent of the total protein. The majority of the proteins appear to be on the cytoplasmic surface of the endoplasmic reticulum and those that are located on the luminal surface may be proteins of the cytosol that have been adsorbed on to the membrane (Table 8.5).

TABLE 8.5
Principal enzymes of the endoplasmic reticulum

Enzyme or group of enzymes	Location
Cholesterol biosynthetic enzymes	* Smooth endoplasmic reticulum
Steroid hydroxylation enzymes	Smooth endoplasmic reticulum
Carnitine acyltransferase	
Fatty acid elongating enzymes (C_{16}–C_{24})	
Glycerolphosphate acyltransferase	
Drug-metabolizing enzymes (aromatic ring hydroxylation, side-chain oxidation, deamination, dealkylation, dehalogenation)	Smooth endoplasmic reticulum
Protein synthesis	* Rough endoplasmic reticulum, cytoplasmic side
Glucose 6-phosphatase	Rough endoplasmic reticulum, luminal side
Adenosinetriphosphatase	Cytoplasmic side
Cytochrome b_5 reductase	Cytoplasmic side
NADPH-cytochrome reductase	Cytoplasmic side
GDPmannose α-D-mannosyltransferase	Cytoplasmic side
Nucleosidediphosphatase	Luminal side
β-D-Glucuronidase	Luminal side
UDP glucuronosyltransferase	
5′ Nucleotidase	Cytoplasmic side
Cholesterol acyltransferase	Cytoplasmic side

For further details see references 26, 27 and 28.
* Endoplasmic reticulum may be subdivided into rough endoplasmic reticulum and smooth endoplasmic reticulum. The areas of rough endoplasmic reticulum are those in which polysomes are present, whereas smooth endoplasmic reticulum lacks polysomes.

A large number of different processes occurs on the endoplasmic reticulum including protein synthesis and protein transport, biosynthesis of fatty acids, sterols, phospholipids, and the metabolism of foreign compounds, and this is reflected in the enzyme composition (Table 8.5). Protein synthesis occurs both on polysomes (Fig. 8.4), which are free in the cytosol, and on polysomes bound to the endoplasmic reticulum. The ribosomes that comprise the free and bound polysomes arise from the same ribosome pool. Whether a polysome remains free or bound is determined by the nature of mRNA that is being translated. Proteins that are destined either to become integral membrane proteins or to be transported across the endoplasmic reticulum on the lumenal side contain a signal sequence of between 16 and 30 amino acids, a preponderance of which are hydrophobic. The signal sequence is recognized by a signal-recognition particle which ensures that it becomes membrane bound. The growing polypeptide is extruded through the membrane eventually either becoming an integral membrane protein or being cleaved to remove the signal peptide and released into the lumen. These proteins may either become secreted out of the cell or be contained within vesicles or dispersed to the lysosomes. The endoplasmic reticulum is connected to the Golgi apparatus, which sorts these proteins in a manner dependent on their recognition regions for the lysosomes or for secretion. The enzymes destined for the lysosomes contain a recognition marker in the form of a mannose 6-phosphate residue that is recognized by a receptor (M_r 215 000) that transports the enzyme from the Golgi to the lysosomes (for details, see references 29 and 135).

8.2.6 The cytosol

The cytosol is the aqueous phase in which the cell organelles are suspended.* Operationally it is the soluble phase that remains as the supernatant after a tissue homogenate has been centrifuged for sufficient '$g \cdot$min' (Fig. 8.2) to sediment the microsome fraction and free ribosomes and is referred to as the 'high-speed' supernatant. It thus comprises most of the soluble phase from both the cytoplasmic and luminal sides of the endoplasmic reticulum. It will also contain soluble contaminants from any disrupted subcellular organelles. The 'high-speed' supernatant will also differ from the cytosol in that the former will be appreciably diluted with the buffer used for homogenization.

In recent years evidence has mounted that suggests that the cytosol is not simply a random mixture of proteins, nucleic acids, and other molecules. Certain groups of enzymes are known to form multienzyme complexes, e.g. fatty-acid synthase (see Chapter 7, Section 7.11.2), and in the case of glycolytic enzymes there is evidence for a type of association that is less well defined (see Section 8.5.3). It is possible that many other weak associations of macromolecules may occur that have not yet been detected either because they have not been sought or because the association may be so weak that the dilution of the cytosol during the isolation of a high-speed supernatant may be sufficient to cause dissociation.

In most tissues a substantial proportion of the protein of the cell is in the cytosol. The viscosity of the cytosol is approximately equivalent to that of a 15 per cent solution of sucrose. The cytosol may be regarded as a 'very crowded' protein solution. This causes the rate of diffusion of small molecules to be low (see Section 8.5.2). Measurements of the rates of diffusion of small molecules in the cytosol show that the rates are appreciably lower than in a protein solution of equivalent concentration. This has been interpreted as evidence that the characteristic macromolecules of the cytosol do form defined structures that retard the rates of diffusion of metabolites and other solutes.[30] The importance of diffusion rates in the cytosol is highlighted by evidence that ATP and O_2 gradients exist within the cytosol.[31] The presence of ATP gradients within the

* The term *cytoplasm* refers to the entire contents of the cell outside the nucleus, excluding the cell membrane. The cytoplasm therefore includes subcellular organelles such as mitochondria, lysosomes, etc.

cytosol has been demonstrated by showing that the catalytic activities of two ATP-requiring enzymes, one in the cytosol and one on the inside of the plasma membrane, differ in intact cells because their average distances from the sites of ATP generation (mitochondria) differ.

In addition to protein, the cytosol also contains most of the tRNA of the cell. The principal monovalent cations present are Na^+ and K^+ and these are thought to exist predominantly as hydrated ions in free solution, but the principal divalent cations, Mg^{2+} and Ca^{2+} are thought to be predominantly bound to nucleotides, nucleic acids, and acidic polysaccharides. A large number of enzymes are present in the cytosol, particularly those responsible for glycolysis, gluconeogenesis, fatty-acid synthesis, nucleotide biosynthesis, and aminoacyl-tRNA synthesis (Table 8.6).

TABLE 8.6
Enzymes or groups of enzymes present in the cytosol

1. Carbohydrate metabolism	**3. Amino-acid and protein metabolism**
Glycolytic enzymes including phosphorylase, phosphorylase kinase, and protein kinase	Aspartate aminotransferase
Glycogen synthase	Alanine aminotransferase
Fructose-bisphosphatase	Arginase
Phosphoenolpyruvate carboxykinase (rat liver)	Argininosuccinate lyase
Enzymes of the pentose phosphate pathway	Argininosuccinate synthase
Malate dehydrogenase	Aminoacyl-tRNA synthetases
Isocitrate dehydrogenase	**4. Nucleic-acid synthesis**
Lactate dehydrogenase	Nucleoside kinase
Citrate (*pro*-3*S*)-lyase	Nucleotide kinase
Malate dehydrogenase (oxaloacetate-decarboxylating) ($NADP^+$)	
Glucose-1-phosphate uridylyltransferase	
2. Lipid metabolism	
Acetyl-CoA carboxylase	
Fatty-acid synthase complex	
Glycerol-3-phosphate dehydrogenase (NAD^+)	

For further details see reference 36.

Other organelles that are not described here are discussed in the general reference 32, and in the following specific references: Chloroplasts, reference 33; Golgi apparatus, reference 34; Peroxisomes, reference 35; and Glycosomes, reference 137.

8.3 Compartmentation of metabolic pathways

In the intact cell the individual enzymes of a metabolic pathway function together and it is unusual for high concentrations of intermediates to build up. In addition, there are many connections between the main metabolic pathways, since some of the substrates, cofactors, and regulatory molecules, and sometimes the enzymes, are common to more than one pathway. These complex interactions can only be fully understood when, in addition to studying the isolated systems, some knowledge has been acquired concerning the intracellular locations and concentrations of the enzymes and substrates involved and any permeability barriers separating the components. The functions of some enzymes, e.g. malate dehydrogenase, isocitrate dehydrogenase, and ATP citrate (*pro*-3*S*)-lyase, cannot be understood fully without a knowledge of their location. We illustrate the importance of compartmentation using two examples: (i) the interrelationships between glycolysis, gluconeogenesis, fatty-acid oxidation and biosynthesis, and the tricarboxylic-acid cycle in the liver cell; and (ii) arginine and ornithine metabolism in the fungus *Neurospora*.

8.3.1 Compartmentation of carbohydrate and fatty-acid metabolism in liver

The processes of glycolysis, gluconeogenesis, β-oxidation of fatty acids, fatty-acid synthesis, and the tricarboxylic-acid cycle are of major importance in many cells. The enzymes concerned constitute a major portion of the protein of the cell. In this section we are primarily concerned with the intracellular location of the processes and their interconnections and we shall not consider the individual reactions in detail.

8.3.1.1 *Location of the enzymes catalysing the pathways*

The pathways discussed are illustrated in Fig. 8.5. The sequence of glycolytic enzymes from phosphorylase to pyruvate kinase and lactate dehydrogenase is located in the cytosol and in the soluble phase of the nucleus. It is not clear whether the soluble phase of the nucleus is continuous with that of the cytosol, but the concentrations of glycolytic enzymes are slightly higher in the nucleus.[10] In skeletal muscle there is evidence that the glycolytic enzymes are organized in a so-called 'glycogen particle' (see Chapter 7, Section 7.12) present in the cytosol. A similar structure may exist in liver and in other types of cell in which glycogen is stored, though they have not been investigated. Gluconeogenesis, the reversal of glycolysis, uses the same enzymes except for the three steps indicated below.

Step 1

$$\text{ATP} + \text{pyruvate} + CO_2 + H_2O \xrightleftharpoons{\text{pyruvate carboxylase}} \text{ADP} + \text{orthophosphate} + \text{oxaloacetate.}$$

$$\text{GTP} + \text{oxaloacetate} \xrightleftharpoons{\text{phosphoenolpyruvate carboxykinase}} \text{GDP} + \text{phosphoenolpyruvate} + CO_2.$$

Step 2

$$\text{D-Fructose 1,6-bisphosphate} + H_2O \xrightleftharpoons{\text{fructose bisphosphatase}} \text{D-fructose 6-phosphate} + \text{orthophosphate.}$$

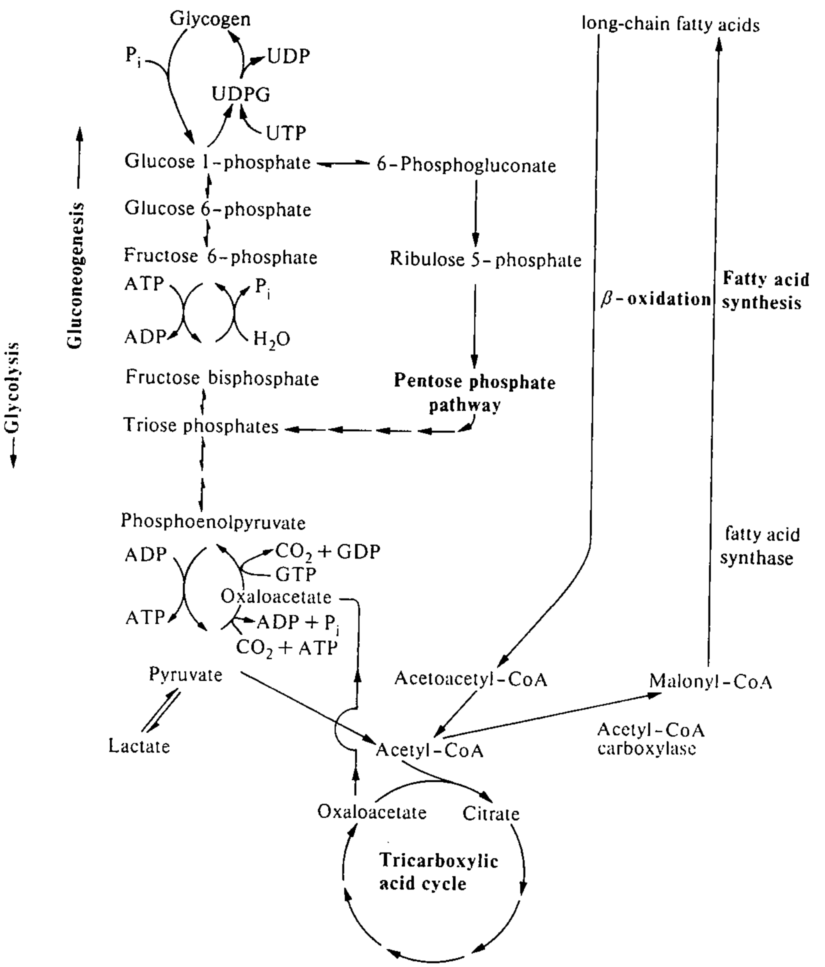


FIG. 8.5. Metabolic pathways for the biosynthesis and degradation of glycogen and of fatty acids. (P_i = orthophosphate.)

Step 3

$$\text{UTP} + \alpha\text{-D-glucose 1-phosphate} \xrightleftharpoons[]{\text{glucose 1-phosphate uridylyl transferase}} \text{UDPglucose} + \text{pyrophosphate.}$$

$$\text{UDPGlucose} + (1,4\text{-}\alpha\text{-D-glucosyl})_n \xrightleftharpoons[]{\text{glycogen synthase}} \text{UDP} + (1,4\text{-}\alpha\text{-D-glucosyl})_{n+1}.$$

Fructose-bisphosphatase and glycogen synthase are located in the cytosol, whereas pyruvate carboxylase is exclusively in the mitochondrial matrix. The

location of phosphoenolpyruvate carboxykinase varies with the species, being totally in the cytosol in rat, almost totally in the mitochondria in the pigeon, and distributed in both compartments in many other species.[37] The alternative pathway for conversion of glucose 6-phosphate (pentose-phosphate pathway) also exists in the cytosol.

The tricarboxylic-acid cycle and β-oxidation of fatty acids occur in the mitochondria, all enzymes being present in the mitochondrial matrix except succinate dehydrogenase, which is bound to the inner surface of the inner mitochondrial membrane. The electron-transport chain is located on the inner mitochondrial membrane. It now seems probable that other tricarboxylic-acid cycle enzymes form a loose association with the membrane. The fatty-acids are synthesized from acetyl-CoA using acetyl-CoA carboxylase and fatty-acid synthase present in the cytosol. Palmitic acid is the main product of mammalian fatty-acid synthases.[38] Palmitic acid may then act as the substrate for the malonyl-CoA-dependent elongase present on the endoplasmic reticulum, where the chain may be extended up to C_{24}. The desaturating enzymes that catalyse the following reaction also occur on the endoplasmic reticulum.[39]

$$CH_3.(CH_2)_x.CH_2.CH_2.(CH_2)_y.CO.SCoA + NADH + H^+ + O_2 \rightleftharpoons$$
$$CH_3.(CH_2)_x.CH{=}CH.(CH_2)_y.CO.SCoA + 2H_2O + NAD^+.$$

8.3.1.2 *Interconnections between pathways*

The principal connections between the pathways are through the redox cofactors NAD^+, NADH and $NADP^+$, NADPH; the adenine nucleotides AMP, ADP, ATP; and the metabolite acetyl-CoA. Metabolic intermediates of one pathway may also be regulators of other pathways, e.g. citrate, fructose bisphosphate, and acetyl-CoA. The degradative pathways generate NADH (glycolysis, β-oxidation, tricarboxylic-acid cycle) and NADPH (pentose-phosphate pathway) and ATP, whereas the biosynthetic pathways require NADPH (fatty-acid biosynthesis, cholesterol biosynthesis) or NADH (gluconeogenesis) and ATP and GTP. Acetyl-CoA is the metabolite that links the pathways. It can be seen from the locations of the pathways (Fig. 8.6) that acetyl-CoA will be generated in the mitochondria by either pyruvate dehydrogenase (mitochondrial enzyme) or by β-oxidation of fatty acids, and it will be utilized either by the tricarboxylic-acid cycle in the mitochondria or by fatty-acid synthesis in the cytosol. It can also be appreciated that ATP and NADH will be generated principally by oxidative reactions in the mitochondria whereas ATP, NADH and NADPH will be required for biosynthesis in the cytosol.

8.3.1.3 *Location of metabolites involved in the pathways*

Using methods described in Section 8.2.1 it is a relatively straightforward matter to determine the subcellular location of the enzymes catalysing metabolic pathways, especially when compared with the problems of determining the intracellular location and concentrations of metabolites and cofactors. The

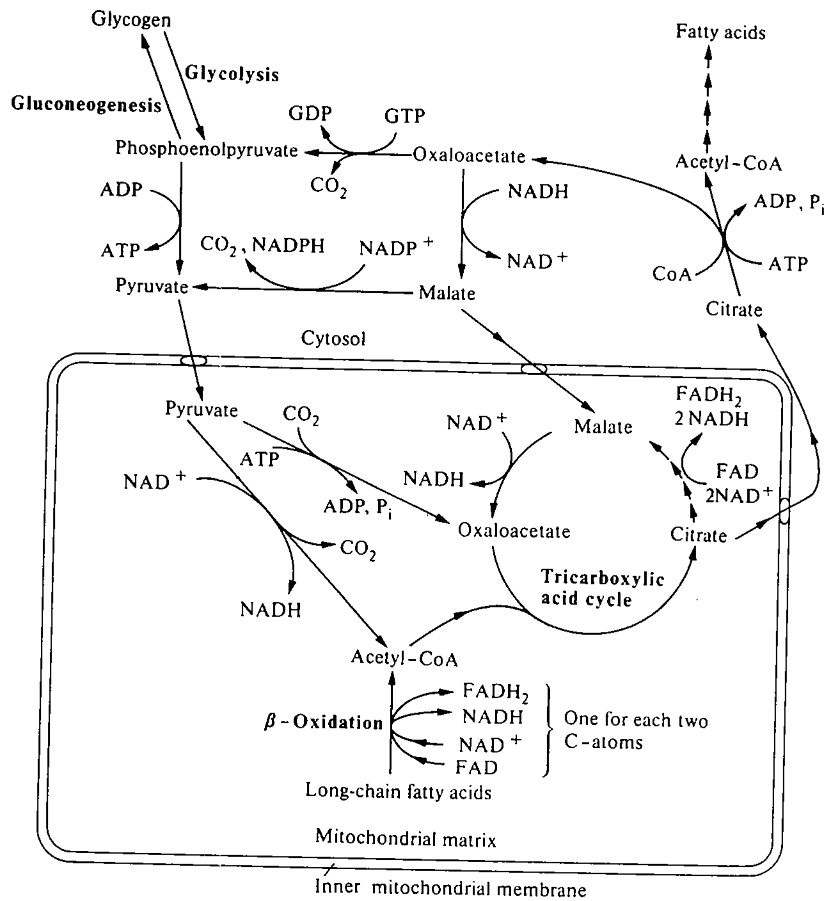


FIG. 8.6. The intracellular location of the principal metabolic pathways leading to the biosynthesis and degradation of carbohydrates and of fatty acids (P_i = orthophosphate).

direct determination of metabolite concentrations is difficult for two reasons: (i) the concentrations often change rapidly during the time required for isolation of subcellular organelles and (ii) substances having low values of M_r readily diffuse out of organelles and redistribute during isolation. Most attention has focused on the distribution of adenine nucleotides, the nicotinamide coenzymes, and the intermediates of the tricarboxylic-acid cycle in the mitochondrial compartments and in the cytosol. Little work has been carried out on other organelles. Four methods to determine these distributions have been generally used,[40] three direct and one indirect. The direct methods involve isolating mitochondria under conditions that minimize any change in the metabolite concentration or distribution, either by rapidly freeze-drying the tissue and then fractionating using organic solvents, or by disrupting the plasma membrane of isolated cells using digitonin and then rapidly sedimenting mitochondria within 30s. Using ^{31}P nuclear magnetic resonance it has been shown that it is possible to demonstrate compartmentation of metabolites such as orthophosphate and sugar phosphates in intact muscle.[41, 42] This non-destructive technique does not suffer from many of the limitations of the other methods; however, there are considerable technical problems in performing the experiments. The fourth method is an indirect method and makes use of reactions that are known to be near equilibrium in certain cell compartments (see Chapter 6, Section 6.3.3.1). The concentration of certain metabolites may then be determined from a knowledge of the concentrations of other metabolites that are common substrates of reactions at equilibrium (see reference 43).

All the methods have shortcomings. The method using non-polar solvents assumes that the metabolites are deposited during the freeze-drying process in the same location that they have *in vivo* and that they are not extracted by the non-polar solvent. The 'rapid' isolation method is still slow enough for certain changes to have taken place, e.g. it is estimated that a molecule of ATP can diffuse the length of a mitochondrion in < 0.3s.[44] ^{31}P nuclear magnetic resonance is restricted to studying phosphate esters. It is less sensitive than other methods and the equipment necessary is not widely available. The indirect method assumes certain reactions are at equilibrium, whereas they may only be close to equilibrium and some may be closer than others. Nevertheless, in spite of these shortcomings there is general qualitative agreement about the distribution of many of the metabolites in the mitochondria and cytosol, but in some cases there is substantial disagreement about the actual magnitudes of the concentration gradients.

8.3.1.4 *Location of nicotinamide cofactors, adenine nucleotides, and coenzyme A*

One of the most important groups of substances about which information is necessary is the nicotinamide cofactors, since these cofactors participate in a large number of reactions. It has been known since the 1950s that NAD^+, NADH, $NADP^+$, and NADPH cannot cross the inner mitochondrial membrane. This became apparent when it was found that intact mitochondria oxidized exogenously supplied NADH very slowly compared with the rate of oxidation of NADH by damaged mitochondria.[45] The concentrations of these coenzymes in the mitochondria and the cytosol have been determined by the indirect method (see Section 8.3.1.3). In this method certain enzymes that are present in high concentrations and whose subcellular distribution is known are assumed to catalyse reactions that are near to equilibrium, whereas reactions catalysed by other enzymes or transport processes are assumed to be not at equilibrium. A list of some of the reactions that are considered to be near equilibrium is given in Table 8.7. The determination of the free

TABLE 8.7

Enzymes catalysing near equilibrium reactions in vivo *which have been used to estimate metabolite compartmentation in rat liver*

Enzyme	Location	Equilibrium definition	K_{eq}	Activity (katal kg^{-1})
Lactate dehydrogenase	Cytosol	$K = \dfrac{[\sum \text{pyruvate}][\text{NADH}][\text{H}^+]}{[\sum \text{lactate}][\text{NAD}^+]}$	1.1×10^{-11} mol dm^{-3}	4.1×10^{-3}
Malate dehydrogenase	Cytosol	$K = \dfrac{[\sum \text{OAA}][\text{NADH}][\text{H}^+]}{[\sum \text{malate}][\text{NAD}^+]}$	2.9×10^{-2} mol dm^{-3}	6.6×10^{-3}
Glycerol-3-phosphate dehydrogenase	Cytosol	$K = \dfrac{[\sum \text{DHAP}][\text{NADH}][\text{H}^+]}{[\sum \text{G3P}][\text{NAD}^+]}$	1.4×10^{-11} mol dm^{-3}	
Isocitrate dehydrogenase	Cytosol	$K = \dfrac{[\sum \text{oxogl}][\text{CO}_2][\text{NADPH}]}{[\sum \text{isocitrate}][\text{NADP}^+]}$	1.2 mol dm^{-3}	3.7×10^{-4}
Phosphogluconate dehydrogenase	Cytosol	$K = \dfrac{[\sum \text{ribulose 5P}][\text{CO}_2][\text{NADPH}]}{[\sum \text{phosphogluconate}][\text{NADP}^+]}$	0.17 mol dm^{-3}	4.7×10^{-5}
3-Hydroxybutyrate dehydrogenase	Inner mitochondrial Membrane	$K = \dfrac{[\sum \text{acetoacetate}][\text{NADH}][\text{H}^+]}{[\sum \text{3HOB}][\text{NAD}^+]}$	4.9×10^{-9} mol dm^{-3}	
Glutamate dehydrogenase	Mitochondrial Matrix	$K = \dfrac{[\text{oxogl}][\text{NH}_4^+][\text{NADH}][\text{H}^+]}{[\text{glutamate}][\text{NAD}^+]}$	3.9×10^{-13} mol^2 dm^{-6}	1.97×10^{-3}

Values of K are defined at 311 K (38°C), ionic strength, 0.25, and measured *in vitro* near pH 7.0.
References 44,46
OAA = oxaloacetate; DHAP = dihydroxyacetone phosphate; G3P = glycerol 3-phosphate; oxogl = 2-oxoglutarate; and 3HOB = 3-hydroxybutyrate.

[NAD$^+$]/[NADH] ratio in the cytosol requires the following assumptions and measurements.

1. Lactate dehydrogenase is present exclusively in the cytosol and is present in sufficiently high concentration that the reaction is near equilibrium.

$$\text{Lactate} + \text{NAD}^+ \rightleftharpoons \text{pyruvate} + \text{NADH} + \text{H}^+$$

$$K = \frac{[\text{pyruvate}][\text{NADH}][\text{H}^+]}{[\text{lactate}][\text{NAD}^+]}.$$

2. The total concentrations of lactate and pyruvate in the tissue are determined after freeze clamping (see Chapter 6, Section 6.3.3.1).
3. It is assumed that the total concentration ratio of pyruvate/lactate is approximately the same as that in the cytosol. The mitochondria occupy about 20 per cent of the cell volume in liver and it is assumed that pyruvate and lactate freely diffuse into and out of the mitochondria.
4. It is also assumed that the ratio of the total concentrations of pyruvate and lactate is equal to the ratio of their free concentrations, i.e. no preferential binding occurs.

A further assumption is that the equilibrium constant measured *in vitro* with low enzyme concentrations is applicable *in vivo* (see discussion in Section 8.5.4).

By this method the [NAD$^+$]/[NADH] ratio has been determined in the cytosol. A similar ratio is obtained using the other enzymes catalysing reactions that are near equilibrium, namely, glycerol 3-phosphate dehydrogenase and malate dehydrogenase, and these findings give support to the validity of this approach.

The same procedure is used to determine the [NAD$^+$]/[NADH] ratio within the inner membrane of the mitochondria using reactions catalysed by the two enzymes 3-hydroxybutyrate dehydrogenase which is attached to the inner mitochondrial membrane, and glutamate dehydrogenase, which is in the mitochondrial matrix. It is necessary to measure the tissue contents of 3-hydroxybutyrate, acetoacetate, glutamate, 2-oxoglutarate, and ammonia. The procedure can be extended to determination of the ratio [NADP$^+$]/[NADPH] in the cytosol using glucose 6-phosphate dehydrogenase, isocitrate dehydrogenase, or the 'malic' enzyme (L-malate: NADP$^+$ oxidoreductase (oxaloacetate-decarboxylating), EC 1.1.1.40). The mitochondrial [NADP$^+$]/[NADPH] ratio is more difficult to determine because there is no NADP$^+$ -requiring enzyme that is exclusively mitochondrial. Calculations[46] have been made on the assumption that the mitochondrial glutamate dehydrogenase that reacts with either NAD$^+$ or NADP$^+$ *in vitro* (it is unusual for a dehydrogenase to use NAD$^+$ or NADP$^+$, interchangeably; see discussion in Chapter 5, Section 5.5.4.3) must react likewise *in vivo* and establish a near equilibrium as shown in the equations, but the results are still regarded as controversial.

(i) $NAD^+ + H_2O + \text{L-glutamate} \rightleftharpoons \text{2-oxoglutarate} + NH_3 + NADH$.

(ii) $NADPH + NH_3 + \text{2-oxoglutarate} \rightleftharpoons \text{L-glutamate} + H_2O + NADP^+$.

Sum $NAD^+ + NADPH \rightleftharpoons NADH + NADP^+$.

The results of determinations of the ratios $[NAD^+]/[NADH]$ and $[NADP^+]/[NADPH]$ are given in Table 8.8.

TABLE 8.8

The ratios of the concentrations of oxidized and reduced nicotinamide cofactors in rat liver determined by use of 'near equilibrium' enzymes

	Cytosol	Mitochondria
$[NAD^+]/[NADH]$	700–1000	7–8
$[NADP^+]/[NADPH]$	0.012–0.014	—

For further details see reference 43.

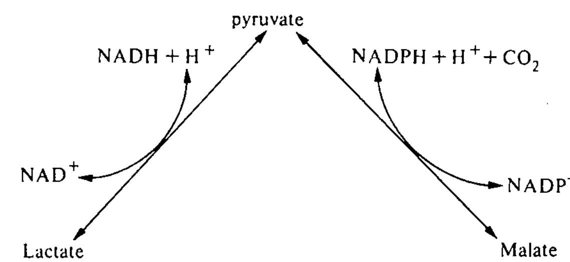
Although the exact values of these ratios vary with the method used and are sensitive to hormonal and nutritional changes, it is clear both that the ratio $[NAD^+]/[NADH]$ is much higher in the cytosol compared with the mitochondria and that in the cytosol the ratio of $[NADP^+]/[NADPH]$ is much lower than the ratio of $[NAD^+]/[NADH]$.

The principal significance of the different ratios $[NAD^+]/[NADH]$ and $[NADP^+]/[NADPH]$ is that it permits a further level of compartmentation within the cytosol. Spatial compartmentation occurs between pathways operating in the mitochondria and in the cytosol, but compartmentation based on enzyme specificity also occurs within the cytosol between NAD^+ and $NADP^+$-requiring dehydrogenases. The principal reactions generating NADPH are those of the pentose phosphate pathway,

$$\text{Glucose 6-phosphate} + NADP^+ \rightleftharpoons \text{glucono-}\delta\text{-lactone 6-phosphate} + NADPH,$$

$$\text{6-Phospho-D-gluconate} + NADP^+ \rightleftharpoons \text{D-ribulose 5-phosphate} + CO_2 + NADPH,$$

and the principal reactions utilizing NADPH are those of fatty-acid and cholesterol biosynthesis. There is a correlation between tissues synthesizing large amounts of fat, e.g. adipose and mammary tissue, and the activity of the pentose-phosphate pathway. Glycolysis and gluconeogenesis, on the other hand, require NAD^+ and NADH, respectively. Although the ratios of the NAD^+–NADH redox couple and of the $NADP^+$–NADPH redox couple are very different, they are nevertheless 'linked' to one another in the cytosol in such a way that changes in one couple will affect the other. The link between them is through enzymes that have common substrates, e.g. lactate dehydrogenase and the 'malic' enzyme, which are both present in the cytosol in large enough amounts to ensure equilibration.



Since the principal oxidative reactions, e.g. oxidation of NADH, occur in the mitochondria and nicotinamide cofactors do not cross the inner mitochondrial membrane, it is necessary for the reducing power generated by glycolysis to be transferred to the mitochondria. This is brought about by two shuttles, the malate-aspartate shuttle and the glycerol 3-phosphate shuttle (Fig. 8.7). The malate-aspartate shuttle uses two malate dehydrogenases, two aspartate aminotransferases, and two translocators.* NADH in the cytosol is oxidized in the reaction catalysed by malate dehydrogenase and the malate formed enters the mitochondria in exchange for 2-oxoglutarate. There malate is reoxidized to oxaloacetate, which is then transaminated to aspartate that leaves the mitochondria in exchange for glutamate entering. The cycle is completed by transamination of aspartate in the cytosol. The malate-aspartate shuttle seems to be the principal shuttle in liver and heart for the transport of reducing equivalents into the mitochondria. A second shuttle that has been studied principally in insect flight muscle is the glycerol 3-phosphate shuttle. As can be seen in Fig. 8.7, it differs from the malate-aspartate shuttle in that *sn*-glycerol 3-phosphate does not enter the mitochondrial matrix, since the flavin-linked glycerol 3-phosphate dehydrogenase present in the inner mitochondrial membrane appears to have its substrate-binding site on the outer face of the membrane but the reduced flavin is generated on the inner face of the membrane (see Section 8.4.3).

Equally important in linking the biosynthetic and degradative pathways are the adenine nucleotides. The principal site of generation of ATP is the mitochondrion and of ATP utilization is the cytosol. Adenine nucleotides do not cross the inner mitochondrial membrane freely but are controlled by the adenine nucleotide translocator (Fig. 8.8), which allows the passage of one molecule of ATP in one direction in exchange for one molecule ADP in the other direction (for a review see reference 47). AMP is not transported by the translocator and cannot

* Specific proteins on the inner mitochondrial membrane allowing selective transport of particular ions across the membrane. (See Section 8.2.3.)

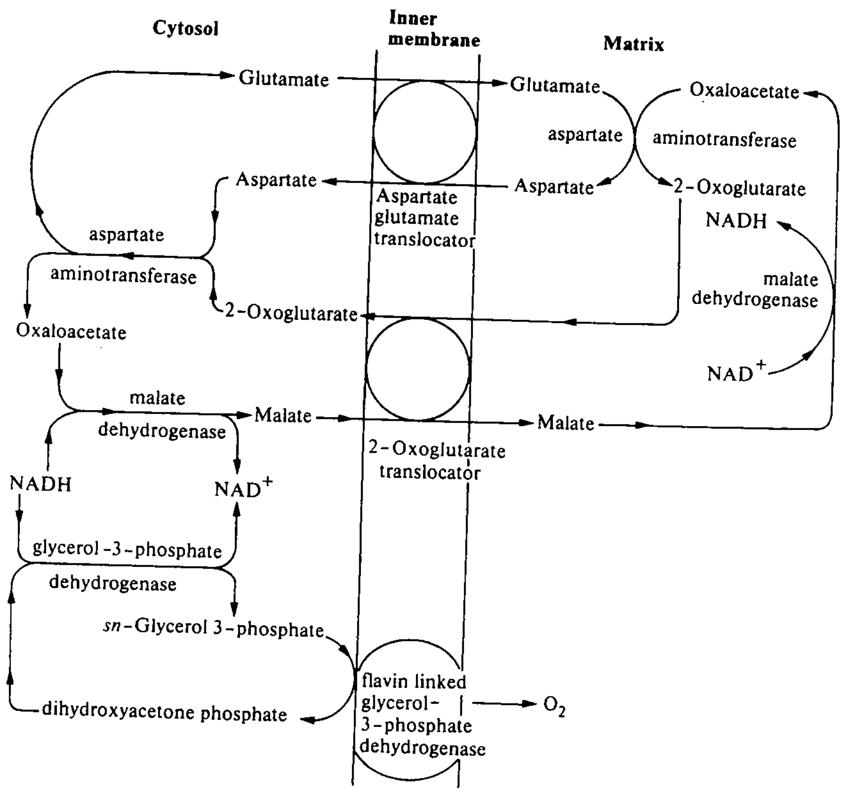


FIG. 8.7. The malate–aspartate shuttle and the glycerol 3-phosphate shuttle for the transport of reducing equivalents into the mitochondria.

cross the inner mitochondrial membrane without first being phosphorylated. Although the adenine nucleotide translocator constitutes as much as 6 per cent of the mitochondrial membrane protein, it appears to be the rate-limiting step in oxidative phosphorylation limiting the influx of ADP to the site of oxidative phosphorylation.[48] The ratio [ATP]/[ADP] in the cytosol is regulated by two enzymes present in large quantities and operating near equilibrium, namely glyceraldehyde-3-phosphate dehydrogenase and 3-phosphoglycerate kinase

(i) Glyceraldehyde 3-phosphate + NAD^+ + orthophosphate $\rightleftharpoons$

3-phosphoglyceroylphosphate + NADH + H^+.

(ii) 3-Phosphoglyceroylphosphate + ADP $\rightleftharpoons$ 3-phosphoglycerate + ATP.

Since these enzymes catalyse reactions near equilibrium, it can be seen that the ratio $[NAD^+]/[NADH]$ in the cytosol is linked to the phosphorylation state of the adenine nucleotides in the cytosol.

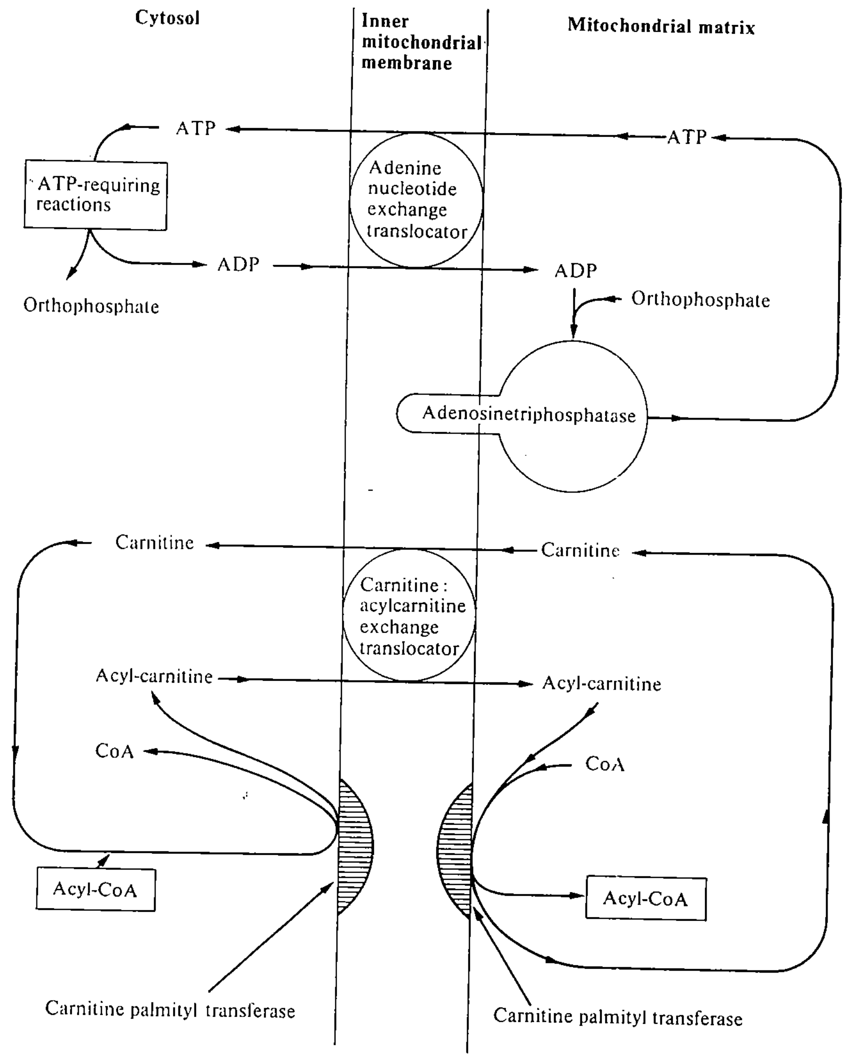


FIG. 8.8. Translocation of adenine nucleotides and of fatty acids across the inner mitochondrial membrane (see references 47, 49, and 138).

Besides nicotinamide cofactors and adenine nucleotides, the third important link between metabolism of fat and carbohydrate is acetyl-CoA, which can be either the end product of one pathway (β-oxidation or glycolysis) or the initial substrate of another (tricarboxylic-acid cycle or fatty-acid synthesis). The enzymes involved in acetyl-CoA metabolism, together with their intracellular locations, are shown in Fig. 8.9. Coenzyme A and its acylated derivatives are

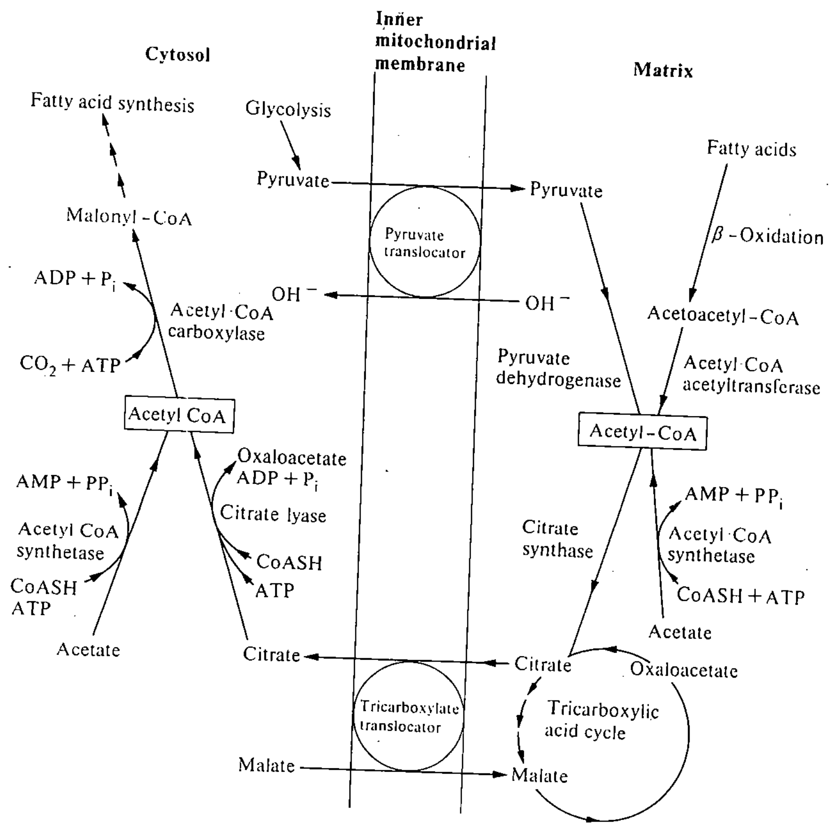


FIG. 8.9. The intracellular location of enzymes concerned with the metabolism of acetyl-coenzyme A. (P_i = orthophosphate; PP_i = pyrophosphate.)

unable to cross the inner mitochondrial membrane. Long-chain fatty acids are converted to their carnitine[50, 138] derivatives, which can then cross the membrane (Fig. 8.8)

$$\underset{\text{(Carnitine)}}{(CH_3)_3N^+.CH_2.CHOH.CH_2COO^-} + \text{Acyl-CoA} \rightleftharpoons (CH_3)_3N^+.CH_2.\underset{\substack{|\\ O\\ |\\ \text{Acyl}}}{CH}.CH_2.COO^- + \text{CoASH}.$$

As can be seen from Fig. 8.9, most of the important steps generating acetyl-CoA, i.e. pyruvate dehydrogenase and acetyl-CoA acetyltransferase, are present in the mitochondrial matrix. The principal enzymes using acetyl-CoA are citrate (*si*) synthase, which enables acetyl-CoA to enter the tricarboxylic acid cycle, and acetyl-CoA carboxylase, which is the first step in fatty-acid biosynthesis. Acetyl-CoA generated in the mitochondrial matrix is effectively transferred to the cytosol by conversion to citrate, the latter is transported using the tricarboxylate translocator and then cleaved back to acetyl-CoA by ATP citrate (*pro*-3*S*) lyase. Citrate synthase and ATP citrate (*pro*-3*S*) lyase are exclusive to the mitochondrial matrix and the cytosol, respectively. ATP citrate (*pro*-3*S*) lyase is present in high concentrations in the cytosol and catalyses a reaction operating close to equilibrium.

From the three groups of substances we have considered, i.e. nicotinamide cofactors, adenine nucleotides, and coenzyme A derivatives, it is clear that the inner mitochondrial membrane plays an important role in the compartmentation of these compounds, since none can freely cross the membrane. In addition to these coenzymes, several other metabolites are not uniformly distributed across the mitochondrial membrane and there are several translocators that control the movement of intermediates of the tricarboxylic-acid cycle and certain amino acids. It is necessary to know the distribution of these compounds in order to understand the regulation of intermediary metabolism so that calculations of reaction velocities can be made from knowledge of the kinetic properties measured *in vitro*.

It is difficult to measure the distribution of these metabolites directly, for the reasons discussed with regard to nicotinamide cofactors. Indirect measurements have been made using the estimated values for the $[NAD^+]/[NADH]$ ratio. For example, the concentrations of malate and oxaloacetate in the cytosol are estimated as follows.

(1) Total cell malate and oxaloacetate is measured.
(2) The ratio of free $[NAD^+]/[NADH]$ is calculated as described earlier in this section. Now, for the reaction catalysed by malate dehydrogenase (MDH):

$$\text{Malate} + NAD^+ \rightleftharpoons \text{oxaloacetate} + NADH + H^+.$$

$$K_{MDH} = \frac{[\text{oxaloacetate}][NADH][H^+]}{[\text{malate}][NAD^+]}.$$

Therefore

$$\frac{[\text{Malate}]_{cytosol}}{[\text{Oxaloacetate}]_{cytosol}} = \frac{[NADH][H^+]}{[NAD^+]K_{MDH}}.$$

If $[H^+]$ and the ratio $[NADH]/[NAD^+]$ are known for the cytosol and also K_{MDH}, the ratio [malate]/[oxaloacetate] in the cytosol can be calculated. A similar equation may be written for malate and oxaloacetate in the mitochondria.

Total cell malate = $\text{malate}_{cytosol} + \text{malate}_{mitochondria}$.
Total cell oxaloacetate = $\text{oxaloacetate}_{cytosol} + \text{oxaloacetate}_{mitochondria}$.

Thus, the distribution of both malate and oxaloacetate can be calculated. There is, however, less agreement between workers concerning the concentrations of dicarboxylic and tricarboxylic acids in the two compartments than about the [NAD^+]/[NADH] ratios. One reason for this is that the determinations are more indirect, relying on more than one reaction being at equilibrium, and thus discrepancies are likely to increase. There is also only qualitative agreement between results from the direct and indirect methods (see Section 8.3.1.3) of determining metabolite distribution. However, the data so far available do suggest that citrate, isocitrate, and possibly 2-oxoglutarate are more concentrated in the mitochondria, whereas glutamate, aspartate, and possibly oxaloacetate are more highly concentrated in the cytosol,[44, 51] but the actual concentration gradients obtained by different methods may differ by at least an order of magnitude and the results have to be treated with caution.

8.3.2 Compartmentation of arginine and ornithine metabolism in *Neurospora*

The metabolism of ornithine and arginine in *Neurospora* illustrates the importance of knowing not only the intracellular location of both enzymes and substrates but also the kinetic properties of the enzymes in order to understand their function *in vivo*. *Neurospora*, like many other fungi, is able to grow when supplied with glucose as the sole source of carbon. It is therefore readily able to synthesize both ornithine and arginine. *Neurospora* possesses all the urea-cycle enzymes (see Fig. 8.10) and also an active urease. When it is grown in a medium supplemented with arginine or ornithine it is readily able to assimilate these amino acids and use them in the biosynthesis of other amino acids. Table 8.9 shows some of the activities and K_m values of the principal enzymes present in *Neurospora* that utilize ornithine or arginine. Even when grown on minimal medium, *Neurospora* has substantial intracellular pools of ornithine and arginine, estimated at about 12 mmol dm^{-3} and 8 mmol dm^{-3}, respectively.

At first sight it would seem that with a high intracellular pool of arginine and such an active arginase, arginine would be continually degraded at a high rate. It would also appear that provided the supply of carbamoyl phosphate was not limiting, with the presence of all the urea-cycle enzymes (Fig. 8.10), there would be a continuous high rate of synthesis of arginine and hence a substrate cycle (see Chapter 6, Section 6.3.4.1), but this does not occur.

However, a number of additional factors have to be taken into consideration before extrapolations from *in vitro* to *in vivo* conditions can be made. Arginase activity is normally assayed at pH 9.5 after a preincubation with Mn^{2+}. Although this treatment results in optimum catalytic activity, it is non-physiological and if the preincubation is omitted and assays are carried out at pH 7.5 the activity is appreciably lower[52] and appears sigmoidal with respect to arginine (for the significance of sigmoidal kinetics see Chapter 6, Section 6.2.2.1). However, the intracellular location of the enzymes and substrates is perhaps the most important factor, as seen in Fig. 8.11. Part of the urea cycle is located in the mitochondria, but the steps from citrulline to urea are located in the cytosol,

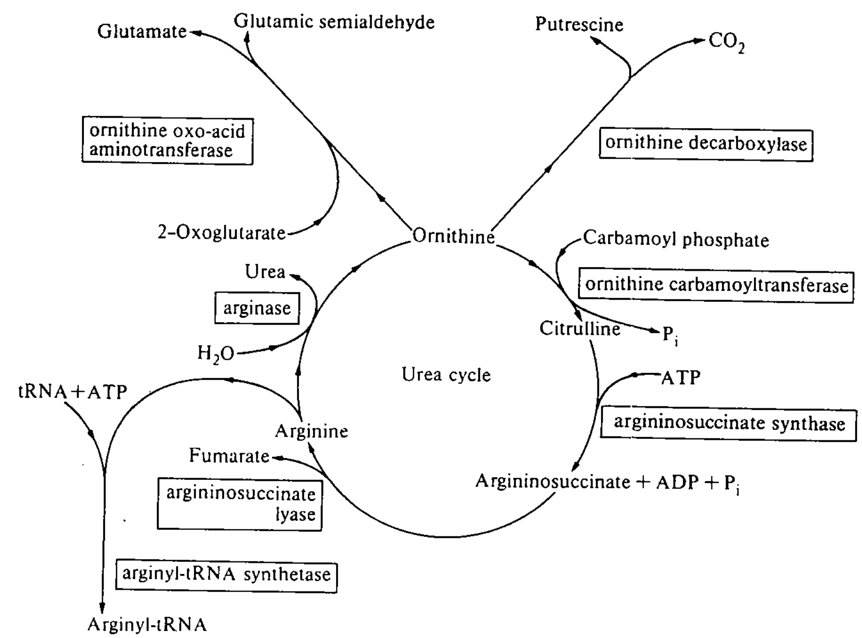


FIG. 8.10. The metabolic pathways for arginine and ornithine metabolism in the fungus *Neurospora crassa* (P_i = orthophosphate).

TABLE 8.9
The activities and Michaelis constants for ornithine- and arginine-metabolizing enzymes in Neurospora

Enzyme	K_m ornithine (mmol dm^{-3})	K_m arginine (mmol dm^{-3})	Activity (katal kg^{-1} protein)
Ornithine carbamoyltransferase	1.9		4.3
Ornithine oxyacid aminotransferase	2.0		0.23
Ornithine decarboxylase	0.2		8.3×10^{-4}
Arginase*		20	17.2
Arginyl-tRNA synthetase		0.02	1.3×10^{-5}
Argininosuccinate lyase		0.8	0.18
Argininosuccinate synthase			0.06

* Activity after Mn^{2+} activation.
For further details see references 55, 59–62.

where arginase and arginyl-tRNA synthetase are also located. It has been shown that approximately 98 per cent of the arginine and ornithine are located in a separate compartment known as a vacuole (Fig. 8.11) and thus the concentrations in the cytosol are much lower. This vacuole, which has been isolated and

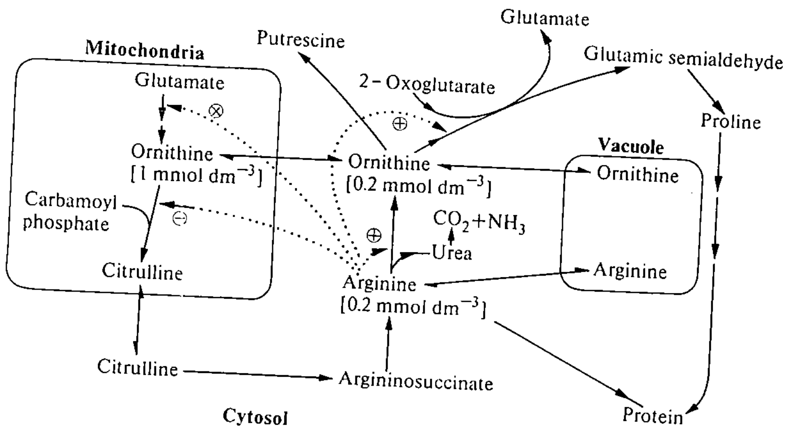


FIG. 8.11. The intracellular location of arginine and ornithine metabolism in *Neurospora crassa*: [ ] = concentrations in compartments indicated. Enzymes that are regulated by the intracellular concentrations of arginine are indicated ⊕ by induction, ⊖ by repression, and ⊗ by feedback inhibition (see references 53, 54 and 55).

partially characterized,[55] is capable of storing amino acids, particularly basic amino acids, and similar vacuoles have been found in other fungi and plants.[56]

The pathways are thus controlled to a large extent by the transport of substrates across both vacuole and mitochondrial membranes. In the intact organism the complete urea cycle does not operate to any appreciable extent, instead different halves of the cycle are operative under different physiological conditions. When a mutant deficient in urease is grown on a minimal medium and a small amount of [^{14}C]*guanidino*-arginine of high specific activity is added, most of the label is found in urea. The specific activities of the proteins labelled in short time periods show that the added [^{14}C]arginine does not equilibrate with the bulk of the arginine pool.[58] The low cytosolic concentration of arginine enables arginyl-tRNA synthetase to compete successfully with arginase (on the basis of their K_m values (Table 8.9) arginase will be operating at about 1 per cent maximum, whereas arginyl-tRNA synthetase will be at more than 90 per cent maximum provided $tRNA^{Arg}$ and ATP are not limiting). Thus, when grown on minimal medium the intracellular arginine, mainly in the vacuole, is synthesized using the urea-cycle enzymes to convert ornithine to arginine, whilst the arginase catalyses very little arginine degradation. On the other hand, when arginine is added to the growth medium it is the degradative part of the urea cycle that is active. The pool of arginine in the cytosol increases rapidly, e.g. from 0.2 mmol dm^{-3} to 15 mmol dm^{-3},[36] and, in addition, higher levels of ornithine-oxy-acid aminotransferase are induced and urea production is readily detectable. Also, the higher intracellular arginine inhibits further ornithine biosynthesis, probably by acting on acetylglutamate kinase, an enzyme on the ornithine biosynthetic pathway. The arginine taken into the cells is partly metabolized via arginase and ornithine-oxy-acid aminotransferase, leading to the formation of glutamate, proline, and other amino acids, and the remainder is taken up into the vesicles, where it largely displaces ornithine. The latter is transported into the cytosol where it is also metabolized by ornithine-oxy-acid aminotransferase. For reviews, see references 55 and 57.

8.4 Vectorial organization of enzymes associated with membranes

The intracellular membranes form the boundaries of subcellular compartments separating enzymes and metabolites in different regions of the cell, as has been discussed in the previous section. Associated with the intracellular membranes are a number of enzymes that are assymetrically arranged about the membranes. Several of these are concerned with the transport of metabolites and ions between compartments and their functioning can only be understood from studies of the intact membrane or organelles or in reconstituted systems where the enzyme, once isolated, has been reincorporated into membranes or membrane-like structures. In this section we discuss the methods used to study enzymes vectorially arranged in membranes and the progress that has been made in understanding how these enzymes function.

8.4.1 Enzymes in membranes

Much progress has been made in recent years in our understanding of the structures of membranes.[63] The 'fluid mosaic' model for membrane organization is generally accepted in which the phospholipids form a bilayer orientated with their polar head groups outside and the hydrocarbon chains towards the centre. The protein components of the membrane may be either on the surface of the lipid bilayer (peripheral proteins) or embedded in the lipid (integral proteins) as seen in Fig. 8.12. These two groups of proteins can be distinguished by the ease with which they are extracted from the membranes. Peripheral proteins form mainly electrostatic bonds with the charged phospholipid head groups and can be extracted by changes in ionic strength. The integral proteins bind to the hydrocarbon moiety of the phospholipids mainly by hydrophobic interactions and they are extracted from membranes using detergents or organic solvents. The extent to which an integral protein is embedded in the lipid bilayer varies from protein to protein, and this is reflected in the ease with which the protein is extracted. In addition, many of the integral proteins, once extracted, will aggregate once the detergent is removed because they contain a number of non-polar amino-acid side-chains that will take part in hydrophobic interactions. The range of interactions between proteins and the phospholipid bilayer is illustrated in Fig. 8.12.

Some proteins, such as fructose-bisphosphate aldolase and glyceraldehyde-3-phosphate dehydrogenase attached to the inner surface of the plasma membrane

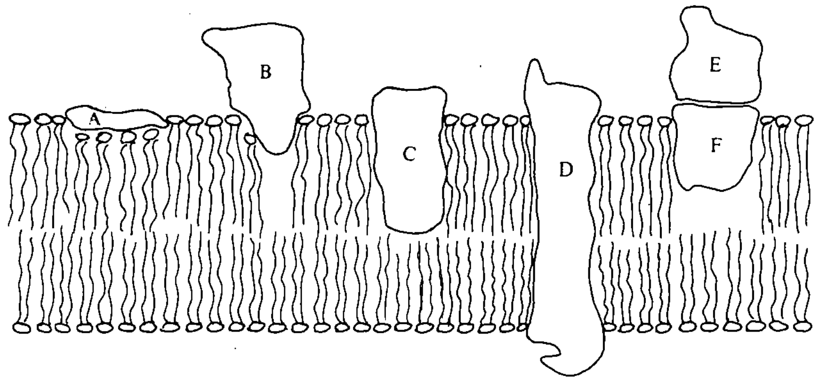


FIG. 8.12. Disposition of enzymes in a membrane.

A. A peripheral protein that can readily be extracted from the membrane by salt solutions, e.g. fructose-bisphosphate aldolase, glyceraldehyde-3-phosphate dehydrogenase in erythrocytes.
B. An integral protein with only a small portion of the polypeptide chain embedded in the bilayer, e.g. α-D-glucosidase and aminopeptidase in the brush border.
C. An integral protein with a large portion of the polypeptide chain embedded in the bilayer, e.g. acyl-CoA desaturase
D. An integral protein spanning the phospholipid bilayer, e.g. ion transporters such as adenosinetriphosphatase (Ca^{2+}-activated).
E. A protein attached to the membrane by a second protein (F) that is embedded in the bilayer, e.g. β-D-glucuronidase from the endoplasmic reticulum attached to a specific integral protein, egasin.

See references 39, 64–67.

of erythrocytes, can readily be extracted by 0.1 mol dm^{-3} EDTA or 1 mmol dm^{-3} ATP.[64] The integral proteins range from those in which only a small portion of the polypeptide chain is embedded in the bilayer, e.g. aminopeptidase from the intestinal brush border where a fragment of polypeptide chain of M_r weight 8000–10 000 is embedded within the membrane, whereas the remainder (M_r 280 000) is exposed in the lumen of the intestine.[65, 66, 68] to those in which the bulk of the polypeptide chain is presumed to lie in the bilayer, e.g. C_{55}-isoprenoid alcohol phosphokinase.[69] The latter enzyme from *Staphylococcus aureus* catalyses ATP-dependent phosphorylations of C_{55}-isoprenoid alcohols and functions in the biosynthesis of lipopolysaccharide. The enzyme has 58 per cent hydrophobic amino acids and when shaken with *n*-butanol–water mixtures it partitions in the butanol phase.

For some integral proteins the hydrophobic regions embedded in the bilayer merely act as an anchoring point on the enzyme. The enzymes may then be completely separated from phospholipid without any appreciable change in their kinetic properties, e.g. α-D-glucosidase and aminopeptidase[65, 66] from intestinal brush border, and nucleotide pyrophosphatase[70] from plasma membrane and

endoplasmic reticulum. Other integral proteins, however, require a minimum amount of phospholipid to retain activity, e.g. amine oxidase (flavin-containing)[71] from the outer mitochondrial membrane, cytochrome *c* oxidase[72] from the inner mitochondrial membrane, and adenosinetriphosphatase (Ca^{2+}-activated) of the sarcoplasmic reticulum.

8.4.2 The role of the membrane

The role the membrane plays in relation to the bound enzymes falls into one of four categories.

1. The membrane can act as an anchoring point. For example, the functions of aminopeptidase and α-D-glucosidase in the brush border are to hydrolyse peptides and maltose prior to their uptake across the brush-border membrane. Anchoring the enzymes presumably makes more economical use of the enzymes, since they are retained at the required location (see Section 8.4.3). This is analogous to the use of immobilized enzymes in industrial processes; Chapter 11, Section 11.5.

2. The substrates and products of an enzyme reaction may have only limited solubility in water and may be more soluble in the lipid bilayer, and thus the enzyme will operate more efficiently when lipid bound, e.g. enzymes involved in lipid and glycolipid biosynthesis, in redox reactions involving long-chain ubiquinones, or in glycosyl-transfer reactions involving polyisoprenyl carrier lipids.[73]

3. The phospholipid bilayer may act as the medium in which certain multienzyme complexes are organized. An example of this is the acyl-CoA desaturase complex,[39] which comprises at least three proteins—cytochrome b_5 reductase, cytochrome b_5, and acyl-CoA desaturase—which act together to desaturate fatty acids. All three proteins are localized on the cytoplasmic face of the endoplasmic reticulum (Fig. 8.13).

4. Membranes may act to separate substrates, products, and effectors in an enzyme reaction, or may allow controlled transport processes to occur. These separations and transport processes cannot readily be studied once the enzyme has been separated from the associated phospholipid membrane components and are usually studied using intact membranes or vesicles obtained from them, or using reconstituted systems. Three examples, each differing in the way in which the enzyme is disposed in relation to the membrane are considered in the next sections.

8.4.3 Hydrolases of the microvillar membrane

Enzymes of this group are good examples with which to illustrate the role of the membrane as an anchoring point for enzymes. Microvillar membranes occur on the lumenal surfaces of the epithelium of the small intestine and the proximal tubule of the kidney. The microvillar membranes are highly convoluted to increase their surface area, so as to facilitate efficient absorption—in the case of the small intestine, absorption of nutrients, and in the case of the proximal tubule, reabsorption of solutes filtered by the kidney. The role of hydrolases in

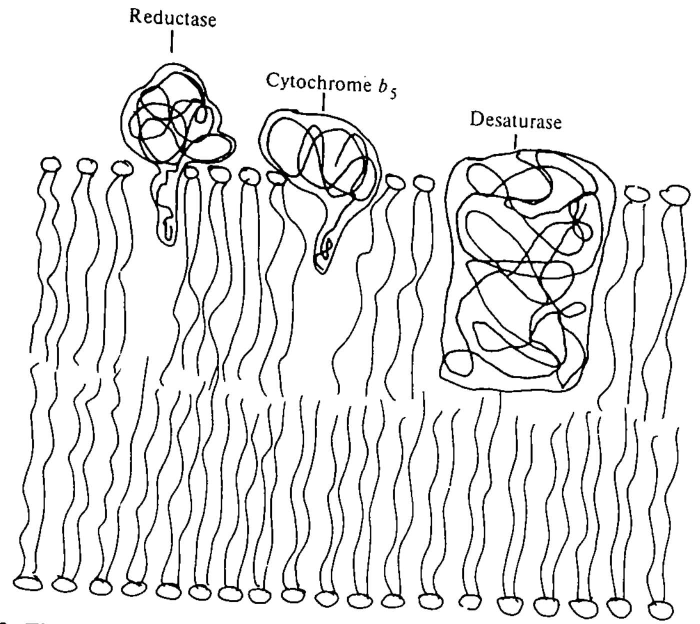
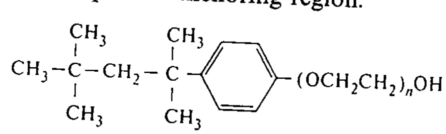


FIG. 8.13. The organization of the acyl-CoA desaturase complex in the membrane of the endoplasmic reticulum.

the small intestine is obviously to fulfil a digestive function. The proximal tubule contains a similar range of hydrolases but their function is not clear. The hydrolases form the major proportion of the integral membrane proteins of the microvillar membrane. A method that has been important in studying these enzymes is to compare the detergent extracted enzymes, usually using Triton X-100, with the enzymes extracted without detergent but after protease treatment to split off the hydrophobic anchoring region.



Triton X-100 (isooctyl phenoxy polyethylene oxide)

With all these hydrolytic enzymes, the protease-extracted enzymes show full activity when compared with the detergent-extracted enzyme. The size of the hydrophobic anchor can be determined from the difference in M_r of the detergent-extracted hydrolase and the protease-extracted hydrolase. In determining the M_r of the detergent-extracted form by gel filtration, allowance has to be made for the micelle of detergent attached to the hydrophobic region necessary for solublization. Most anchors range in M_r from 3000 to 10 000. The structural arrangement of the hydrolases so far studied falls into one of two types, either homodimers (α_2) or heterodimers ($\alpha\beta$). Aminopeptidase, alkaline phosphatase and dipeptidyl peptidase are homodimers, whereas sucrase-isomaltase and γ-glutamyl transferase are heterodimers. The probable structural arrangements are shown in Fig 8.14. All the hydrolases studied are glycoproteins having 13–15 per cent carbohydrate by weight, the carbohydrate moiety being attached to the region of the protein protruding from the membrane. Sucrase-isomaltase is one of the most thoroughly investigated. It is the most abundant protein of the microvilli of the epithelial cells of the small intestine, comprising 10 per cent of the integral membrane proteins. It accounts for all the sucrase activity of the microvilli, 90 per cent of the isomaltase activity, and 70–80 per cent of the maltase activity. It comprises two non-identical subunits, one of which contains the isomaltase catalytic site and is anchored directly to the membrane, the other contains the sucrase activity and also accounts for the maltase activity and is bound to the isomaltase polypeptide, but is not anchored directly to the membrane (Fig 8.14). The sucrase-isomaltase is synthesized as a single polypeptide precursor that is cleaved during post-translational processing. It is generally assumed that the role of the membrane with all these hydrolases is to ensure efficient hydrolysis coupled with absorption through the associated membrane. For further details, see references 65, 66, 74, and 75.

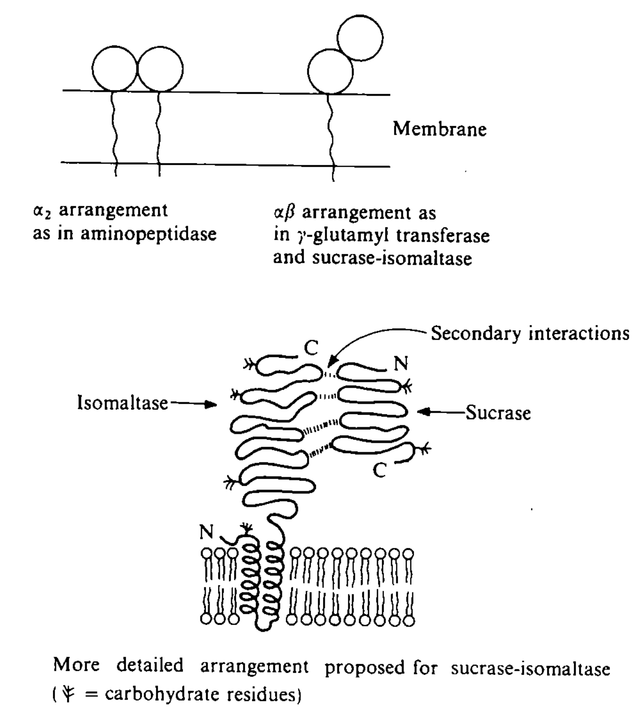


FIG. 8.14. The attachment of microvillar enzymes to the cell membrane.[74,75]

8.4.4 Adenosinetriphosphatase (Ca^{2+}-activated) of the sacroplasmic reticulum

In contrast with the enzymes described in the previous section, for which the membrane acts primarily as an anchoring point, adenosinetriphosphatases (ATPases) are good examples of transmembrane proteins involved in the movement of ions through the membrane. A number of different adenosinetriphosphatase enzymes are known in which the hydrolysis of ATP is coupled to the transport of ions across a membrane, e.g. adenosinetriphosphatase (Na^+, K^+-activated) of the plasma membrane, mitochondrial adenosinetriphosphatase associated with oxidative phosphorylation (linked to H^+-movement), and adenosinetriphosphatase (Ca^{2+}-activated) of the endoplasmic reticulum and sarcoplasmic reticulum. Of these, the adenosinetriphosphatase (Ca^{2+}-activated) is perhaps the one of the best understood in molecular terms and therefore is the example chosen here. The sarcoplasmic reticulum is the intracellular membrane surrounding each myofibril in the muscle fibres. Calcium ions are released from the sarcoplasmic reticulum in response to a nervous impulse. This triggers the process of muscle contraction. Relaxation occurs when Ca^{2+} is removed and this is achieved by adenosinetriphosphatase (Ca^{2+}-activated), which pumps Ca^{2+} back into the sarcoplasmic reticulum at the expense of ATP hydrolysis. Vesicles of the sarcoplasmic reticulum can be prepared and from these adenosinetriphosphatase (Ca^{2+}-activated) can be isolated. There are only four main protein components of this membrane and adenosinetriphosphatase makes us about 90 per cent of this protein. The abundance and relative ease of isolation of this protein has made it a good model system for studying ion transport.

The adenosinetriphosphatase (Ca^{2+}-activated) requires an annulus of 30 molecules of phospholipid for activity. Removal of the phospholipids causes aggregation of the adenosinetriphosphatase; however, the phospholipids can readily be exchanged with added phospholipids if they are mixed in the presence of detergent and the detergent is later removed. Alternatively, the phospholipids may be replaced by certain non-ionic detergents, e.g. dodecyl octaethylene glycol monoether, which provide the amphiphilic environment necessary for maintenance of activity.[76] The adenosinetriphosphatase can be incorporated into vesicles or liposomes (structures composed of multiple phospholipid bilayers completely enclosing an aqueous phase)[77] containing internal phosphate ions. These reconstituted systems catalyse ATP-dependent Ca^{2+} transport. The stoichiometry of a Ca^{2+} transport associated with ATP hydrolysis is well established but the detailed mechanism by which the Ca^{2+} is transported across the membrane is not yet clear.

The sum of the steps is probably as follows:

$$\underset{\text{outside}}{ATP + 2Ca^{2+}} + \underset{\text{inside}}{Mg^{2+} + 2K^+} \rightleftharpoons ADP + \text{orthophosphate} + \underset{\text{inside}}{2Ca^{2+}} + \underset{\text{outside}}{Mg^{2+} + 2K^+}.$$

Under suitable conditions it is possible to demonstrate the reversibility of this process, i.e. a Ca^{2+} concentration gradient across the membrane can drive the synthesis of ATP. During the process of Ca^{2+} transport, the adenosinetriphosphatase becomes phosphorylated by ATP and then dephosphorylated.[78]

The ATPase comprises a single polypeptide chain (1001 amino-acid residues), the sequence of which has been deduced from the sequence of the cDNA.[79] There are two genes that encode the Ca^{2+} ATPases of fast-twitch and slow-twitch muscles and the sequences show them to be highly conserved.[80] The secondary structure and topography of the ATPase has been predicted on the basis of the likely folding pattern, together with the charged residues involved in the Ca^{2+}-binding and transmembrane regions. In addition, electron micrographs show a globule attached to the membrane by a stalk and helices are observed on freeze-fracture as intramembranous particles. A probable topology of the enzyme is given in Fig. 8.15. As can be seen, there are three distinct domains on the cytoplasmic side of the membrane, eight helices running through the membrane, and five loops on the lumenal side of the membrane. The steps in the mechanism are considered as follows:

$$\begin{array}{ll} \text{Inside} & E + 2Ca^{2+} + ATP \rightarrow 2Ca{:}E.ATP \rightarrow 2Ca{:}E_1 \sim P + ADP \\ & \nwarrow \hspace{16em} \downarrow \\ \text{Outside} & E_2 \xleftarrow[\searrow P_i]{Mg^{2+}\text{-dependent}} E_2\text{–}P \leftarrow 2Ca{:}E_2 \sim P \end{array}$$

(The enzyme-bound calcium is shown without positive charge, since this is neutralized by the carboxyl groups on the enzyme.)

The initial step involves the binding of two Ca^{2+} and one ATP to the enzyme. The binding is cooperative and induces a conformational change (shown as

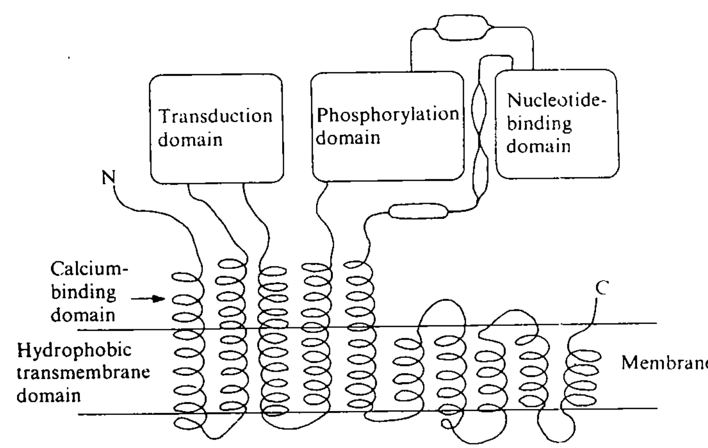


FIG. 8.15. The proposed structure of Ca^{2+}-ATPase.[80]

$E_1 \rightarrow E_2$) in the enzyme, which can be detected as a change in the fluorescence of tryptophan. The ATP is cleaved and phosphorylates Asp351 in the phosphorylation domain. The $2Ca:E_2{\sim}P$ has a lower affinity for Ca^{2+} than $2Ca:E_1ATP$ and is able to release Ca^{2+} on the lumenal side of the membrane, via the channel formed by the helices of the transmembrane region of the protein. The Ca^{2+} binding sites are believed to be the glutamic acid residues on the stalk of the helices close to the membrane.[80] It is not yet clear how many of the helices are involved in forming the channel, or how they are arranged in relation to one another.

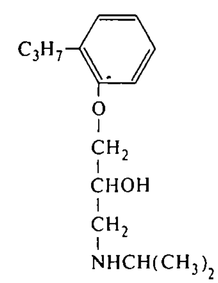
8.4.5 Adenylate cyclase

3′:5′-Cyclic AMP is perhaps the best-understood of the second messengers that are released inside cells in response to an external signal, generally a hormone (see Chapter 6, Section 6.4.2.6). Other intracellular messengers include Ca^{2+} ions, phosphoinositides, and diacylglycerol. Cyclic AMP is generated by the reaction catalysed by adenylate cyclase:

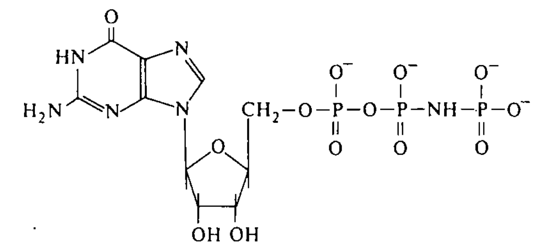
$$\text{ATP} \rightarrow 3':5'\text{cyclicAMP} + \text{pyrophosphate}.$$

The intracellular concentrations of cAMP range from 10^{-7} to 10^{-6} mol dm^{-3}. The rises are generally short-lived and the cAMP is degraded by a phosphodiesterase. In this section we focus on the mechanism by which the signal from the hormone is relayed across the membrane to stimulate adenylate cyclase activity. In the early studies of this system it was not clear whether there existed a single protein spanning the membrane and having both adenylate cyclase activity and a hormone-binding site, or whether more than one protein was involved. The question was resolved by some ingenious cell-fusion experiments.[81] Turkey erythrocytes, which possess an adenylate cyclase activity responsive to catecholamines, when treated with *N*-ethyl maleimide lose their catalytic activity. A second type of cell, a mouse erythroleukaemia cell, lacks catecholamine receptors but possesses adenylate cyclase activity. These two cell types were fused and the resultant hybrid cell could be stimulated by catecholamines to increase cAMP levels. It was possible to show that the coupling between the activities exhibited a temperature dependence that related closely to the fluidity of the cell membrane. Clearly, the receptor and the adenylate cyclase were distinct entities able to associate by lateral movement through the membrane.

The involvement of a third protein became evident as a result of showing that the efficient signalling was dependent on GTP, the latter being degraded to GDP and orthophosphate. A prolonged stimulation could be obtained by using a non-hydrolysable analogue of GTP, guanyl-5′-ylimidodiphosphate. Having demonstrated that three proteins were involved, in order to fully understand the mechanism it was necessary to isolate the components and to reconstitute the system and show that it functioned normally.



Dihydroalprenolol



Guanyl-5′ylimidodiphosphate

As a result of more recent work, it has become apparent that there are a number of examples of intracellular signalling that are triggered by binding of a hormone to the outside of a cell. They each comprise three proteins—a receptor, a transducer (G protein), and an effector. In the case of cAMP signals, the effector is adenylate cyclase, but with phosphoinositides and diacylglycerol the effector is phospholipase C. The most thoroughly studied system is that involving the β-adrenergic receptor and adenylate cyclase. There are four types[82] of catecholamine receptor, α_1, α_2, β_1 and β_2, which differ in their relative affinities towards agonists* and antagonists* and also in the effector to which they are coupled. In the case of the system involving the β_1 type, it has now been possible to isolate and reconstitute the components. These will be described briefly. All three proteins are present in very small amounts in the cell membrane, and since they are membrane proteins it is necessary to extract them from the lipid of the cell membrane. It is also necessary to have sensitive methods for detection. In the case of the receptor this requires a suitable binding assay to measure $H+R=HR$. Hormones show very high affinities for their receptors, since most circulating hormones are present in nanomolar concentrations. Binding must be measured at physiological concentrations of hormones, or equivalent concentrations of their agonists or antagonists, otherwise non-specific binding to other proteins will confuse the picture. This generally means using radioactively labelled agonists or antagonists having very high specific radioactivity. In the case of the β-receptor,[81] ^{3}H-labelled dihydroalprenolol was used; this has $K_d = 6 \times 10^{-9}$ mol dm^{-3}. The β-receptor has been purified from turkey erythrocytes. The receptors are first solublized using the detergent digitonin, since the receptor is an integral membrane protein, and it is then purified using a combination of ion-exchange chromatography and affinity chromatography. The latter is performed on alprenolol–agarose and elution is with alprenolol,[83]; a 12 000-fold purification can be achieved. The high purification factor is an indication of the small proportion of the total protein in the extract it represents. Since the β-receptor has been isolated, a number of other receptors have been isolated and

* Agonists are structural analogues that mimic the effect of the hormone; antagonists are structural analogues that block the effect of the hormone.

the genes for a number of them have been cloned and sequenced. It is now possible to compare the sequences of the β-receptor, cholinergic receptors, various drug receptors, and light receptors such as rhodopsin and bacteriorhodopsin.[82,84,85] All of these receptors have a number of features in common in their sequences, in particular seven stretches of between 20 and 28 hydrophobic amino-acid residues, which are consistent with the arrangement in the membrane shown in Fig 8.16.

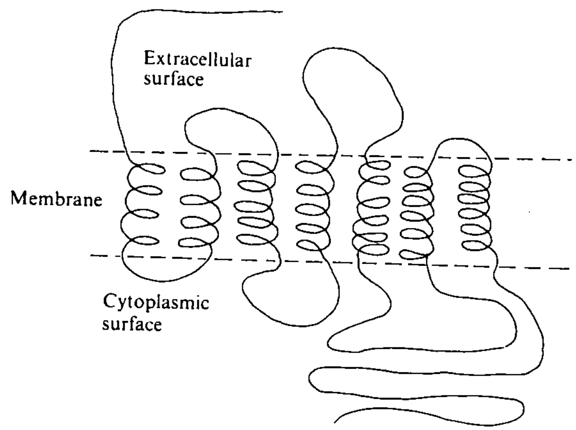


FIG. 8.16. Proposed model for receptor structure.

Although this arrangement has not been proven directly, it is analogous to that of the purple membrane in *Halobacterium*, which has been studied by high-resolution electron diffraction.[136] The first two cytoplasmic domains are the most conserved and may be involved in the interaction with the G-protein. The hormone binds into a pocket formed by the seven helices spanning the membrane.

The G-proteins have also been purified and there is a family of homologous transducing proteins.[86] They are peripheral membrane proteins and therefore more easily extracted. They are sometimes refered to as N-proteins (Nucleotide, as opposed to Guanine nucleotide binding proteins). Different types of G-proteins exist that may act positively (G_s, stimulating) on the effector protein or negatively (G_i, inhibitory). All the G-proteins that have been isolated have been found to be heterotrimers ($\alpha\beta\gamma$). Of the three polypeptide chains, the α chain differs most among members of the family, whereas the β and γ chains can be exchanged within a family. Two toxins have been useful in studying the mode of action of the G-proteins, namely cholera toxin and pertussis toxin. Both toxins are able to catalyse the ADP ribosylation of the α subunit of G-proteins. In the case of cholera toxin this leads to inhibition of GTP hydrolysis and thus to

prolonged activation of adenylate cyclase. Pertussis toxin causes inhibition of hormone-mediated activation of adenylate cyclase.

The third component, adenylate cyclase, has been purified from rabbit myocardial membranes[87] and from bovine brain.[88] In both cases there is an initial solubilization and the main purification step is affinity chromatography on forskolin–agarose. Forskolin is a diterpene isolated from the roots of the aromatic herb *Coleus forskohlii* that causes stimulation of adenylate cyclase.

Having purified the three components, it is then possible to reconstitute the system. This has been done in stages.[89] The receptor and G_s protein can be combined in phospholipid vesicles. Addition of an agonist then stimulates the binding of guanine nucleotides and causes the hydrolysis of GTP to GDP and orthophosphate. A complete reconstitution has been effected in which the proteins R, G, and C are combined in the proportions 1 : 10 : 1 and the phospholipid vesicles comprising phosphatidylethanolamine and phosphatidylserine in the ratio 3 : 2. It was then shown that isoproterenol, an agonist, was able to stimulate an increase in adenylate cyclase activity.[89] A number of possible mechanisms have been proposed; the mechanism most generally favoured[90] is shown in Fig. 8.17.

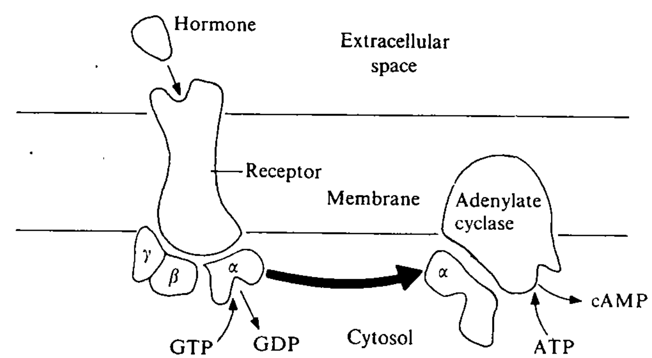


FIG. 8.17. Coupling between receptor, transducer, and adenylate cyclase.

8.5 The concentrations of enzymes and substrates *in vivo*

When cells or tissues are disrupted and extracts are prepared prior to assay, the suspension containing the enzymes is considerably diluted compared with that existing in the intact cell. This undoubtedly affects many aspects of an enzyme-catalysed reaction. Although many biochemists are aware of this problem when comparing measurements made *in vitro* and *in vivo*, there is little quantitative information that might enable a better extrapolation from *in vitro* to *in vivo* conditions. In this section we highlight some of these differences and discuss what progress has been made in understanding them.

8.5.1 The high concentrations of macromolecules within cells

The protein concentration in all of the intracellular compartments is high. In the mitochondrial matrix the concentration is estimated to be about 500 mg cm^{-3},[15] and in the cytosol concentrations may be in the region of 200 mg cm^{-3}. This has a number of effects on the properties of the medium in which enzymes act. The medium is much more viscous: a solution of albumin at a concentration of 200 mg cm^{-3} has 2–3 times the viscosity of water and one having a concentration of 500 mg cm^{-3} has 5–10 times the viscosity of water.[91] This will affect the rate of diffusion of solutes, particularly those of macromolecules (see Section 8.5.2). A protein concentration of 500 mg cm^{-3} means that individual protein molecules will be in close proximity to one another and this will tend to promote interactions between protein molecules that would not occur in dilute solutions. A good example of this is given in Section 8.5.3.3. It will also have a profound effect on the state of water molecules in the cell. The water molecules in close proximity to protein molecules differ in some of their properties from those of pure solvent. Apart from a lower freezing point, it can be shown by various physical methods, e.g. n.m.r. spectroscopy, i.r. spectroscopy,[92] that these water molecules do not rotate as freely as those in pure water. These differences are due to various types of secondary bonds that bind water molecules to protein either directly or indirectly. The water molecules in solutions of protein have been classified into three types[93] based on the differences in their physical properties, as summarized in Table 8.10. The actual contents of type II and III water will depend on the particular protein, since a higher proportion of polar amino-acid residues will lead to a higher proportion of bound water. The number of molecules of internal water (type III) for various proteins that have been studied by X-ray crystallography[94] is as follows: papain, 15; actinidin (a related protease), 15; lysozyme, 4; carboxypeptidase, 24; cytochrome *c*, 3; and penicillopepsin, 13. In the case of lysozyme, where there are four

TABLE 8.10

Physical properties of water in protein solutions

Type		Rotational correlation time[a]	Average number of grams of water per gram of protein
I	as in pure water	3×10^{-12} s	
II	loosely bound to protein; freezing point lowered	approx. 10^{-9} s	0.3–0.5
III	irrotationally bound; motion same as protein molecule, integral part of protein structure	10^{-5}–10^{-7} s	0.003

[a] Rotational correlation time can be considered as the time taken for a molecule to rotate through an angle of 360°. For a more detailed explanation see reference 95.
For review on water in biological systems, see reference 93.

molecules of internal water tightly bound, it has been shown that when lysozyme is gradually hydrated, enzyme activity is just detectable when these are between 150 and 250 water molecules per lysozyme. This corresponds approximately to the sum of the type II and type III water molecules. If the mitochondrial proteins behave in the same way as detailed in Table 8.10 then in the mitochondrial matrix with a concentration of 500 mg cm^{-3} protein, 25 per cent of water would be type II and 1.5 per cent type III. Thus, a high proportion of the water in the mitochondrial matrix will have properties different from those of pure water. The same will apply to other subcellular compartments, though in some cases to a lesser degree. Although the nucleus has a slightly lower protein content (approximately 34 g protein per 100 g nuclei)[10] than the mitochondrial matrix, it has a much higher content of nucleic acids, particularly DNA (9 g DNA per 100 g nuclei).[10] Isolated DNA has been shown to bind approximately ten water molecules per nucleotide residue (approximately 5.6 g water per g nucleotide)[96] and thus DNA binds proportionally more water than does protein. It is unlikely that DNA in the nucleus will bind this high proportion of water, since much of its surface will be covered with nucleoprotein. Nevertheless, it would appear that much of the water in the nucleus will be bound.

8.5.2 Diffusion rates within cells

When considering the flux through a metabolic pathway *in vivo*, it is important not only to know the activities of all the enzymes involved in order to determine the rate-limiting steps but also to know the 'transit times' of each step. The 'transit time' is the time required for the product of one enzyme-catalysed reaction to diffuse to the active site of the next enzyme. There is at present insufficient information available to make accurate estimates of the 'transit time' although a number of approximate estimates have been made.[97–99] The two principal parameters required for these calculations are (i) the distance separating two sequential enzymes in a metabolic pathway and (ii) the diffusion coefficient of the product of the first enzyme in the medium of which the cell is composed. The average distance separating two enzymes may be estimated if the number of molecules of each enzyme per cell is known, together with the volume of the relevant subcellular compartment. The diffusion coefficient of the enzyme, substrates, and products can easily be measured in aqueous solution, but it is clear that the diffusion coefficients are significantly lower in the cytosol compared with those in pure water. Various methods have been used to estimate the diffusion coefficients in the cytosol.[30] The ratio D_{H_2O}/D_{cells} varies from 2 to 5 for molecules of $M_r = 200$–400, to 200-fold for protein of $M_r = 40\,000$–$150\,000$.

From the calculations[97,98] made so far it seems that the 'transit time' for the steps in certain metabolic pathways could be rate limiting, particularly in the larger cell compartments, unless the enzymes are not randomly distributed but are organized. For example, the calculated 'transit time' for fructose 6-phosphate to diffuse to phosphofructokinase in the cytosol is too long in relation to the known rate at which glycolysis can occur. This suggests that the enzymes may not be randomly arranged in the cytosol but that some kind of ordering exists. There is also kinetic evidence that suggests that an enzyme (aspartate aminotransferase) and its substrate (aspartate) may not be uniformly distributed throughout the mitochondrial matrix, but that the concentration of aspartate may be highest in the centre and the aspartate aminotransferase highest at the periphery of the matrix.[100]

8.5.3 Enzyme associations *in vivo*

It seems probable that in living cells many weak interactions occur that are not easily detected once a cell has been disrupted and the components suspended in buffer solution. In the equilibrium

$$\text{Protein}_A + \text{protein}_B \rightleftharpoons \text{protein}_A \, . \, \text{protein}_B$$

dilution will favour the dissociation of the protein$_A$. protein$_B$ complex, and in many cases when cell disruption is carried out prior to subcellular fractionation, then the concentrations of proteins may be lowered by at least one order of magnitude. Thus, the multienzyme complexes discussed in Chapter 7 probably represent only the stronger protein interactions. In recent years attempts have been made to try to demonstrate other weaker interactions. Most attention has focused on the two major metabolic pathways, namely the glycolytic pathway occurring in the cytosol, and the tricarboxylic-acid cycle occurring in the mitochondrial matrix. The enzymes catalysing the steps in both these pathways are usually present in cells in high concentrations, because of their central importance in metabolism. Skeletal muscle is a particularly rich source of glycolytic enzymes since most of the ATP generation in white muscle is through glycolysis. It has been shown that a number of glycolytic enzymes bind to fibrous-actin,[101] one of the major proteins concerned in muscle contraction. The capacity of actin to bind glycolytic enzymes varies with the enzyme; the capacities being greatest with 6-phosphofructokinase, pyruvate kinase, fructose-bisphosphate aldolase, glyceraldehyde-3-phosphate dehydrogenase, glucose-6-phosphate isomerase, and lactate dehydrogenase. The kinetic properties of some of the enzymes change on binding and this, together with the fact that binding to actin is reversible, could be an additional means of regulation, but further work is required before the relevance of these observations to the situation in intact muscles can be assessed.

8.5.3.1 *Fructose-bisphosphate aldolase and glyceraldehyde-3-phosphate dehydrogenase*

Several attempts have also been made to demonstrate associations between individual enzymes of the glycolytic pathway. We have discussed in Chapter 7, Section 7.12, the evidence for the existence of a 'glycogen particle' that contains a number of glycolytic enzymes. Fructose-bisphosphate aldolase and

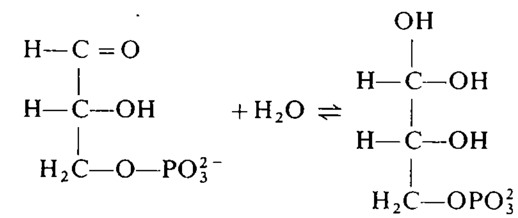
glyceraldehyde-3-phosphate dehydrogenase catalyse sequential steps in glycolysis:

Fructose-bisphosphate $\rightleftharpoons$ dihydroxyacetone phosphate + glyceraldehyde 3-phosphate.

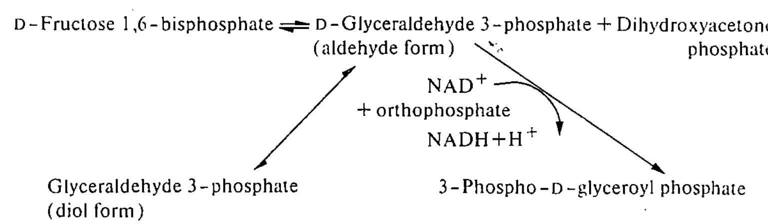
Glyceraldehyde 3-phosphate + NAD$^+$ + orthophosphate $\rightleftharpoons$ 3-phospho-D-glyceroyl phosphate + NADH.

(Triosephosphate isomerase converts dihydroxyacetone phosphate to glyceraldehyde 3-phosphate.)

So far direct methods to demonstrate association between the two enzymes, e.g. by ultracentrifugation or gel filtration, have yielded negative results; however, there is indirect evidence for some form of association.[102] When glyceraldehyde 3-phosphate is present in aqueous solution, there is normally an equilibrium between the aldehyde and the diol forms (see also Chapter 5, Section 5.5.2).



If the results of kinetic studies on glyceraldehyde-3-phosphate dehydrogenase alone are compared with those from fructose-bisphosphate aldolase coupled with glyceraldehyde-3-phosphate dehydrogenase, the rate of the coupled reaction is such as to suggest that the aldehyde form of glyceraldehyde 3-phosphate is used directly by the dehydrogenase and does not equilibrate with the diol. This in turn suggests a close association of the enzymes and is supported by measurements of polarization of fluorescence.

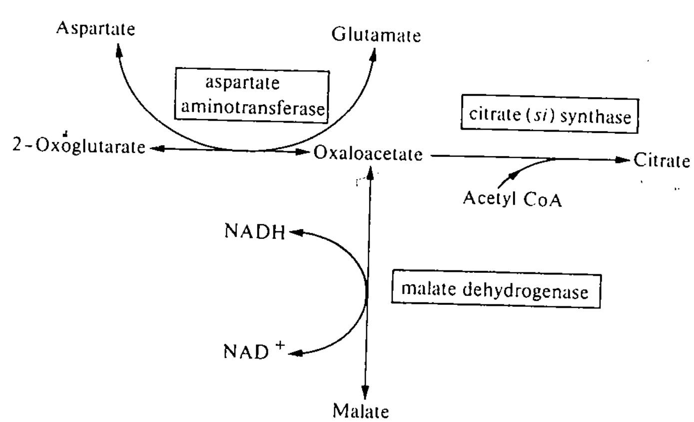


If fluorescein is coupled covalently to either fructose-bisphosphate aldolase or glyceraldehyde-3-phosphate dehydrogenase and the fluorescein-labelled enzyme mixed with varying amounts of the unlabelled form of the second enzyme, a change in fluorescence is observed, suggesting an interaction between the enzymes; the binding constant may be determined from the titration curve. This interaction between glyceraldehyde-3-phosphate dehydrogenase and fructose-bisphosphate aldolase has also been detected by active-enzyme centrifugation[104] (see Chapter 3, Section 3.2.4) and has a dissociation constant in the micromolar range[105]. There is also evidence for a weak association between fructose-bisphosphate aldolase and fructose-bisphosphatase isolated from rabbit muscle.[106]

8.5.3.2 *The tricarboxylic-acid cycle enzymes and related metabolic pathways*

The enzymes catalysing the tricarboxylic-acid cycle, together with aspartate aminotransferase and glutamate dehydrogenase, occur in the matrix of the mitochondria. Apart from succinate dehydrogenase that is covalently linked to inner mitochondrial membrane, all other enzymes are readily released when the inner mitochondrial membrane is disrupted. A number of investigations have been carried out to see whether there is any evidence for associations between the enzymes involved.

There is evidence for a weak interaction between mitochondrial malate dehydrogenase and citrate (*si*) synthase[107] and between malate dehydrogenase and aspartate aminotransferase.[108] During the purification of citrate (*si*) synthase it is difficult to remove the last traces of malate dehydrogenase activity, suggesting a type of interaction or similarity in properties.



In the mitochondrial matrix, because of the very high protein concentration, there is a limitation on free water and this could promote interaction between the two enzymes. In order to bring about water limitation *in vitro*, experiments have been carried out in which some of the water has been replaced by poly(ethylene glycol). It has been demonstrated that under these conditions citrate (*si*) synthase will co-precipitate with mitochondrial malate dehydrogenase but not with cytosolic malate dehydrogenase. In a similar set of experiments, the partitioning of malate dehydrogenase and aspartate aminotransferase in a biphasic system of water and dextran–trimethylaminopropyl poly(ethyleneglycol) was studied. If either malate dehydrogenase and aspartate aminotransferase (both mitochondrial), or if malate dehydrogenase and aspartate aminotransferase (both cytosolic) were used, the enzymes in each pair appeared in the same fraction after counter current distribution. If, on the other hand the pair of enzymes were not from the same subcellular fraction, e.g. mitochondrial malate dehydrogenase and cytosolic aspartate aminotransferase, then they appeared in different fractions, suggesting no interaction. Thus, the interaction between the enzymes seems to be specific. Further evidence for an association between these enzymes and other enzymes of the tricarboxylic-acid cycle comes from studies on immobilized enzymes.[109] When fumarate hydratase and mitochondrial malate dehydrogenase were immobilized on a gel filtration column, it was possible to demonstrate that the immobilized enzymes were able to bind enzymes catalysing related steps in metabolism, e.g. immobilized fumarate hydratase binds malate dehydrogenase and citrate synthase whereas immobilized malate dehydrogenase binds citrate synthase and immobilized aspartate aminotransferase binds malate dehydrogenase.

Carbamoyl phosphate synthase I is the most abundant liver mitochondrial protein,[110] being present in concentrations between 0.4 and 1.0 mmol dm^{-3}. In addition to playing a key role in the urea cycle, it appears to act as a linking enzyme by binding glutamate dehydrogenase and also enhancing the stability of the complex between glutamate dehydrogenase and aspartate aminotransferase. The association between glutamate dehydrogenase and carbamoyl phosphate synthase I has been demonstrated by cross-linking the two in the mitochondrial matrix.[110]

Although associations have been demonstrated only between pairs of tricarboxylic-acid cycle enzymes in mitochondria, there is evidence[111] for a multienzyme aggregate in bacteria containing several of the tricarboxylic-acid cycle enzymes. By preparing spheroplasts and then lysing them in suitable medium, it was possible to demonstrate by gel filtration that a multienzyme aggregate existed that contained citrate synthase, aconitase, isocitrate dehydrogenase, succinate-CoA ligase, and malate dehydrogenase, and was thus able to catalyse the steps from fumarate to oxoglutarate. It is possible that it may also contain oxoglutarate dehydrogenase. The processes of β-oxidation of fatty acids and the tricarboxylic-acid cycle involve a number of dehydrogenases that have to transfer their reducing equivalents to the electron-transport chain on the inner mitochondrial membrane. Several of these dehydrogenases, e.g. pyruvate dehydrogenase, oxoglutarate dehydrogenase, malate dehydrogenase, and hydroxyacyl-CoA dehydrogenase are able to bind to complex I, which catalyses the electron-transport chain from NAD^+ to coenzyme Q,[112]. Thus there is evidence for a physical link between the tricarboxylic-acid cycle and the electron-transport chain. The concentrations of the tricarboxylic-acid cycle enzymes in the mitochondrial matrix[113] are greater than 1 mmol dm^{-3} so that high concentrations of enzymes might be expected to be necessary to demonstrate interactions *in vitro*. These experiments illustrate the point that many weak interactions may occur *in vivo* but that special methods may have to be used to detect them: neither of these complexes has been detected in the analytical ultracentrifuge.

8.5.3.3 Arginase and ornithine carbamoyltransferase

Another interaction between two enzymes that is not easily demonstrated in dilute solutions is that between arginase and ornithine carbamoyltransferase from yeast.[114,115] The function of ornithine carbamoyltransferase is in the biosynthesis of arginine from ornithine (see Fig. 8.10). When arginine is added to the growth medium it acts as a good source of both carbon and nitrogen for the synthesis of many compounds, e.g. other amino acids, proteins, and carbohydrates and also is a preferred source of arginine compared to that synthesized endogenously. Two mechanisms operate to suppress ornithine carbamoyltransferase activity in this situation. One is the suppression of synthesis of ornithine carbamoyltransferase and is relatively slow acting. The second mechanism, which is fast acting, involves interaction between ornithine carbamoyltransferase and arginase. This latter mechanism is demonstrated in concentrated suspensions by comparing the activity of ornithine carbamoyltransferase in cell extracts. (Fig. 8.18, curve A) with that in permeabilized cells (curve B). In the latter, yeast cells are made permeable to low-M_r substances by use of nystatin (a polyene fungicide produced by *Streptomyces noursei*), but the proteins remain within the cells and thus the protein concentration is unchanged and therefore high.

In both situations A and B in Fig. 8.18 the synthesis of ornithine carbamoyltransferase is suppressed after arginine is added to the medium, but additionally in B, ornithine carbamoyltransferase becomes bound to arginase in

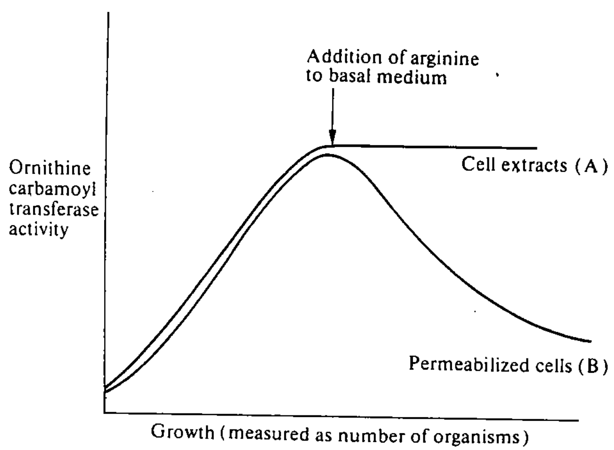


FIG. 8.18. The effects on ornithine carbamoyltransferase activity in yeast of adding arginine to the growth medium (see reference 114).

the presence of ornithine and arginine and the ornithine carbamoyltransferase activity is inhibited. Both arginase and ornithine carbamoyltransferase are trimeric, and the complex formed between them has been shown to have a stoichiometry $\alpha_3\beta_3$, and a dissociation constant of between 10^{-8} and 10^{-10} mol dm^{-3}:

$$\alpha_3\beta_3 = \alpha_3 + \beta_3.$$

The substrates ornithine and arginine promote the association of the complex $\alpha_3\beta_3$. It was originally proposed[114,115] that ornithine bound both to the catalytic site on ornithine carbamoyltransferase and also to a regulatory site. However, recent studies[116] using a bisubstrate analogue (transition-state analogue), δ-*N*-(phosphonacetyl)-L-ornithine,[117] have shown that a conformational change can be demonstrated on binding the catalytic site but give no evidence for a separate regulatory site. It seems probable that binding to the active sites of both enzymes induces the necessary conformational changes for their association.

8.5.3.4 Membrane-bound hexokinase

The binding of enzymes to other cell components may modify the properties of the enzyme concerned. In a number of mammalian tissues, hexokinase is distributed between the mitochondria and the cytosol. It has been suggested that an equilibrium exists between hexokinase that is free in the cytosol and that bound to the mitochondria and that this may act as a form of control mechanism. In one of the most studied tissues, the brain, up to 80 per cent of the hexokinase may be bound to the mitochondria and much of this may be released *in vitro* by increasing concentrations of glucose 6-phosphate. There is evidence that the kinetic properties of hexokinase are changed when it is released from the mitochondria. A higher proportion of hexokinase binds to mitochondria when ATP concentrations are diminished and this may enable hexokinase to compete more favourably with other ATP-requiring processes. There is also evidence that ATP released from the mitochondria is used by the hexokinase in preference to exogenous ATP. This in turn suggests that the initiation of the glycolytic pathway is directly coupled to ATP synthesis in mitochondria.[113]

8.5.4 The relative concentrations of enzymes and substrates

The constants, K_m and V_{max} (see Chapter 4, Section 4.3), determined by steady-state kinetic studies *in vitro*, are frequently used to estimate the rates at which reactions might occur *in vivo*; e.g. the functions of the various isoenzymes of hexokinase have been deduced from such kinetic data (see Chapter 4, Section 4.3.1.3). There can, however, be a major problem in such extrapolations, namely, that in steady-state kinetics it is assumed that $[S] \gg [E]$; this may not always be the case *in vivo*. Table 8.11 shows the intracellular concentrations of some of the glycolytic and tricarboxylic-acid cycle enzymes and their substrates. Although these are often quite wide variations in the calculated concentrations (compare

TABLE 8.11
Intracellular concentrations of enzymes and substrates in rat tissues

Enzyme	Enzyme concentration (μ mol dm^{-3})	Substrate concentrations (μmol dm^{-3})	References
1. Phosphoglucomutase	5	G6P = 450	120, 124
2. Fructose-bisphosphate aldolase (muscle)	66	FBP = 65, DHAP = 28, G3P = 3	120, 125
3. Glyceraldehyde-3-phosphate dehydrogenase (muscle)	35	G3P = 3, NAD$^+$ = 600, P$_i$ = 2800	120, 124, 125
4. Triosephosphate isomerase (muscle)	8	DHAP = 28, G3P = 3	120, 125
5. Pyruvate carboxylase (liver mitochondria)	10–40	Pyruvate = 50–1000 ATP = 1000–6000	120, 126, 128
6. Phosphoenolpyruvate carboxykinase (liver cytosol)	15	Oxaloacetate = 5, GTP = 100–600	37, 120
7. Citrate synthase (liver mitochondria)	35	Oxaloacetate = 0.01–0.1 Acetyl-CoA = 50–1000	120, 126, 128
8. Malate dehydrogenase (liver mitochondria)	70	Malate = 80–340 NAD$^+$ = 18 000	120, 129
9. Malate dehydrogenase (liver cytosol)	30	Malate = 300–450 NAD$^+$ = 500	120, 129
10. Aspartate aminotransferase (liver mitochondria)	50	Glutamate = 380–4800 Oxaloacetate = 0.01–0.1	120, 129
11. Aspartate aminotransferase (liver cytosol)	1.6	Glutamate = 2500–4400 Oxaloacetate = 5	120, 129

G6P = glucose 6-phosphate; FBP = fructose-bisphosphate; DHAP = dihydroxyacetone phosphate; G3P = glyceraldehyde 3-phosphate; and P$_i$ = orthophosphate.

Table 8.11 with for example reference 113), it can be seen that for many of these the enzyme concentration is of the same order of magnitude as that of the substrate concentration. Glycolytic and tricarboxylic-acid cycle enzymes are of major importance in liver and muscle and are therefore present in higher concentrations than the majority of other enzymes in these tissues, but other enzymes are known to be present in high concentrations, e.g. 0.78 mmol dm^{-3} cathepsin D, and 1.54 mmol dm^{-3} cathepsin B in the lysosomal matrix.[23]

One of the assumptions in steady-state kinetics is that, since $[S] \gg [E]$, the total substrate concentration is assumed to approximate to the free substrate concentration. Clearly this will not apply for enzymes present in high concentrations as an appreciable amount of the substrate will be enzyme bound. Various equations have been derived for the rate of an enzyme reaction when the substrate is not in great excess.[120–123] Perhaps the most important factor is that knowledge of free substrate concentration *in vivo* is required and that this cannot easily be determined.

It is not only in the case of enzymes that are present in high concentrations that it is unsafe to assume that the total substrate and free substrate approximate to the same value. An enzyme present in small amounts in a tissue may be in competition with an enzyme present in high concentrations for a common substrate that may be largely bound to the latter enzyme. Thus, the free substrate concentration may, in several cases, be appreciably lower than the total substrate concentration. This situation is well illustrated in the case of oxaloacetate in mitochondria. A number of reactions compete for oxaloacetate and the enzymes concerned are present in high concentrations (compared with substrate); thus a very high proportion of the total oxaloacetate in mitochondria must be enzyme-bound (Fig. 8.19). The kinetic properties (K_m and V_{max}) of citrate (*si*) synthase are such that one would expect any oxaloacetate formed to be converted to citrate, and not to phosphoenolpyruvate for gluconeogenesis; clearly this does not always happen. It has been suggested that there must be some type of macromolecular organization (multienzyme complex) that prevents the competition between the enzymes for oxaloacetate.

However, this overlooks another aspect of the difference between experiments carried out *in vitro* and the situation *in vivo*, namely that the latter behaves more

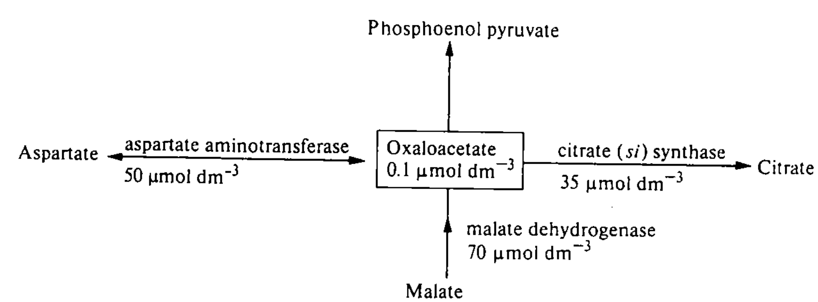


FIG. 8.19. The concentrations of enzymes metabolizing oxaloacetate in mitochondria.

like an open system rather than the closed system used *in vitro*. Although the intermediates of most metabolic pathways in cells are present in low concentrations, they are continuously being replaced by new source material and the end products are often removed from the cell. Thus, although the intracellular concentration of oxaloacetate is very low and undoubtedly much of it is enzyme bound, changes in the amounts of the enzymes utilizing oxaloacetate do not appreciably affect the intracellular concentration of free oxaloacetate in the 'near-open' system existing in the cell. Mathematical modelling with the aid of computers[130] has been made to determine the effect of a sharp increase in the amount of a protein that binds oxaloacetate (i.e. comparable with an oxaloacetate-metabolizing enzyme). This is found to cause only a transient fall in the intracellular concentration of free oxaloacetate. The reason is that oxaloacetate is rapidly replenished because it is linked metabolically to the other tricarboxylic-acid cycle intermediates that are present in much higher concentrations. In particular, the relatively large malate pool becomes slightly diminished to replenish oxaloacetate. Mathematical modelling techniques for studying the control of metabolic activity (see Chapter 6, Section 6.3.3.1) are still being developed and they require accurate kinetic data and knowledge of pool sizes to give reliable simulations, but they will play an increasing role in our understanding of metabolism in the whole cell, where the complexity of the situation makes computer simulations essential.

Another example in which the proportion of bound ligand is important *in vivo* is that of cAMP-dependent protein kinase.[131] This enzyme is present in muscle and when activated by a sufficiently high concentration of cAMP phosphorylates a number of proteins, amongst them phosphorylase kinase (as shown below).

$$R_2C_2(\text{inactive}) + 2\text{cAMP} \rightleftharpoons R_2(\text{cAMP})_2 + 2C\ (\text{active})$$

R_2C_2 = inactive cAMP dependent protein kinase (R = regulatory
C = catalytic)

$$\text{ATP} + \text{protein} \xrightleftharpoons{\text{active protein kinase}} \text{ADP} + \text{phosphoprotein}$$

(e.g. inactive phosphorylase kinase) (e.g. active phosphorylase kinase)

The role of phosphorylase kinase in the control of glycogen metabolism is discussed in Chapter 6 (Section 6.4.2.1). The kinetics of activation of cAMP-dependent protein kinase by cAMP are complex,[132] but the concentration of cAMP that gives 50 per cent activation (apparent K_a) of the enzyme is reported to be in the range 10^{-8} to 10^{-7} mol dm^{-3}. The cAMP content of most tissues is in the range 0.1 to 1.0 μmol kg^{-1} wet weight (i.e. if equally distributed throughout the cell would give 10^{-7} to 10^{-6} mol dm^{-3}) and this would appear to be a sufficient concentration to activate substantially the phosphorylase kinase. However, it is known that in tissues such as resting muscle the rate of glycogenolysis is very low. This led initially to the suggestion that cAMP must be

compartmentalized within the muscle so that the concentration to which the protein kinase is exposed is much lower. It is now clear that this explanation is unnecessary. The comparison between the *in vitro* experiment and the *in vivo* situation is false because the concentration of protein kinase in muscle is high (about 0.23 μmol dm^{-3})—approximately the same as that of cAMP. The apparent K_a will be dependent on the concentration of protein kinase, since high concentrations of the latter will favour association of the R and C subunits. Thus, at physiological concentrations of the protein kinase the apparent K_a is 1.5×10^{-6} mol dm^{-3}, i.e. higher concentrations of cAMP are required to effect activation. In addition, an inhibitor has been isolated from skeletal muscle that binds to the C subunit and inactivates it. Both these factors clearly have to be taken into account in explaining the activation *in vivo*.

Another factor to be considered, which arises from the high concentrations of certain enzymes in tissues, is the equilibrium position. When the enzymes in a reaction are present in only catalytic amounts, the equilibrium position for the reaction is unchanged, although in practice it may be difficult to study the equilibrium in the absence of the appropriate enzyme:

$$A + B \rightleftharpoons C + D$$

$$K_{eq} = \frac{[C][D]}{[A][B]}.$$

However, if the enzyme is present in concentrations similar to that of the substrates, so that appreciable amounts of the substrates and products are enzyme bound, then the equilibrium may be that between the enzyme–substrate complex and the enzyme–product complex. If the enzyme–catalysed reaction proceeds, for example by ternary complexes (Chapter 4, Section 4.3.5.1), the relevant equilibrium is

$$EAB \rightleftharpoons ECD$$

$$K'_{eq} = \frac{[ECD]}{[EAB]}.$$

A number of examples are given in Table 8.12. It can be seen that most of the values for K_{eq}, for the bound enzyme are near unity, and that this makes a very significant difference in a number of the examples cited. Thermodynamic arguments have been proposed (see reference 113) to account for the values of K'_{eq} near unity. These suggest that for the evolution of ideal catalysts the free-energy change should be near zero ($K_{eq} \approx 1$) (compare triose phosphate isomerase, Fig. 5.29); since this would give balanced rates of diffusion, chemical transformation, and desorption of steps in a metabolic sequence.

8.6 Conclusions

It can be seen from the preceding section that it has been possible to identify a number of factors that affect enzyme reactions and metabolic pathways *in vivo*,

TABLE 8.12
Changes in equilibrium by high concentrations of enzymes

Enzyme	$K^{\dagger}_{eq}$	K'_{eq} enzyme bound
1. Lactate dehydrogenase	0.37×10^{-4}	0.25
2. Alcohol dehydrogenase	0.5×10^{-4}	5–10
3. Pyruvate kinase	3×10^{-4}	1.0–2.0
4. Arginine kinase	0.1	1.2
5. Creatine kinase	0.1	≈ 1.0
6. Adenylate kinase	0.4	1.6
7. Phosphoglycerate kinase	8×10^{-4}	0.5–1.5
8. Hexokinase	2000	≈ 1.0
9. Adenosine triphosphatase	1.3×10^{5}	9
10. Triose phosphate isomerase	2.2	0.6
11. Phosphoglucomutase	1.7	0.4

References 113, 133, 134. †Equilibrium constant when catalytic amounts of enzymes are present, in contrast to when enzyme and substrate are present in comparable amounts.

but that are not always apparent in many of the better defined *in vitro* situations. Although many factors have been identified, in many cases it is not possible to be sure of the magnitude of these effects *in vivo*. In some cases this is because research interest in this area has only recently developed, in other cases it is a question of devising satisfactory techniques, for example there is still much controversy about the intercellular concentrations and distribution of certain metabolites because none of the methods used are ideal. Even for a metabolic pathway, such as glycolysis that has been actively studied since the turn of the century and in which the three-dimensional structures of many of the enzymes are known, there is still much to be learned about its organization in the cytosol and about its control *in vivo*.

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9
Enzyme turnover

9.1 Introduction

Since the early 1940s when Schoenheimer *et al.*[1] first used isotopically labelled precursors to study protein synthesis, it has been realized that most proteins in living cells are continually being replaced whether the organism is fully grown and in nitrogen balance or is growing and increasing its total protein (and hence nitrogen) content. However, it was not until the late 1960s that much information was gained about the rate at which individual enzymes are replaced or turned over in particular tissues and cells and the general importance of enzyme turnover as a control mechanism was appreciated (for reviews see references 2–10).

In any given tissue or cell type there is a wide range of turnover rates for different enzymes, e.g. in rat liver the half-lives of enzymes range from about 15 minutes to over 100 hours.[4,10] The underlying mechanisms that account for this wide range of turnover rates are not known. Mechanisms have been proposed to account for the degradation of certain enzymes or groups of enzymes, and some general correlations have been made between the properties of enzymes and their rates of turnover (see Sections 9.5 and 9.6), but more information is required before a general theory to account for the turnover of enzymes can be formulated.

At first sight the process of enzyme turnover might seem to be inefficient and wasteful of energy, since every peptide bond formed requires the hydrolysis of ATP and GTP, whereas proteolysis is not coupled to the generation of ATP. However, turnover is important if a cell is to be able to adapt to changes in its environment or if it becomes necessary to remove abnormal enzyme or protein molecules that may arise by mutation or by errors in gene expression. In general, the longer the life of an individual cell the more important is the process of intracellular enzyme turnover. For example, in a bacterium such as *Escherichia coli* growing under optimal conditions, cell division may occur every 20 min. Adaptation occurs largely by induction or repression of enzyme synthesis. If lactose is added to the growth medium, induction of β-D-galactosidase occurs. If lactose is then removed from the medium, existing enzyme molecules will rapidly be diluted out within the cells provided they continue to divide rapidly and β-D-galactosidase is no longer synthesized. Early experiments aimed at detecting turnover of proteins in exponentially growing bacteria suggested that the proteins were completely stable.[11] More recent experiments have shown that turnover does occur but that it is very limited. Increases in the rate of turnover occur when bacteria approach the stationary phase and the supply of nutrients becomes limiting. In contrast to that of rapidly dividing bacteria, the average life of an adult liver cell is between 160 and 400 days and many enzymes are completely replaced every few days. This turnover is necessary, for example, if a liver cell is to be able to adapt to changes in nutrients that it may encounter throughout its life.

When discussing enzyme turnover in this chapter we are referring to the intracellular processes by which individual enzyme molecules are degraded and replaced by synthesis of new enzyme molecules. The steady-state level of an enzyme in a cell depends on the rates of two opposing processes, namely the rate of enzyme synthesis and the rate of enzyme degradation. The general mechanism by which enzyme synthesis occurs is better understood than that of enzyme degradation, although, on the whole, enzyme degradation has a more important regulatory influence on enzyme turnover, particularly in eukaryotes. Therefore, we shall now consider in outline the mechanism of enzyme synthesis before considering the kinetics of enzyme turnover and possible mechanisms of enzyme degradation.

9.2 Mechanism of enzyme synthesis

The mechanism by which enzymes are synthesized is no different from that of protein synthesis in general. There are good accounts of this in many text-books[12–13] and monographs[10, 14, 15] and only the salient features are given here. The information that determines the primary sequence of an enzyme is contained in the order of the deoxyribonucleotide residues of DNA. This information is transferred to a sequence of ribonucleotide residues in mRNA in a process known as transcription; specificity is achieved by complementary base pairing between DNA and mRNA. In eukaryotes, the primary transcript generally undergoes processing to excise sections of the RNA before forming mRNA.[16] The sequence of ribonucleotide residues in mRNA is then expressed in the sequence of amino-acid residues in the enzyme in a process known as translation. The steps in these processes are shown in Fig. 9.1. The amino acids are activated by conversion to aminoacyl adenylates; these then react with species of tRNA to form aminoacyl-tRNAs (see Chapter 5, Section 5.5.5). For each different L-amino acid there exists at least one distinct tRNA. The tRNAs differ from one another in their nucleotide sequence and in particular in the trinucleotide sequence known as the anticodon, which recognizes a complementary sequence (by base pairing) on mRNA known as a codon. The conversion of amino acids to aminoacyl adenylates and thence to aminoacyl-tRNA is catalysed by a class of enzymes known as aminoacyl-tRNA synthetases (see Chapter 5, Section 5.5.5.) Each of these enzymes recognizes a specific amino acid and a specific tRNA.

L-Amino acid + ATP = L-aminoacyl-AMP + pyrophosphate.

L-Aminoacyl-AMP + tRNA = L-aminoacyl-tRNA + AMP.

Enzyme specificity is thus crucial in accurately relaying information contained in a ribonucleotide sequence to that in an amino-acid sequence during the process of translation. The ribosomes, together with a number of other specific proteins, constitute the machinery that in the presence of GTP enables aminoacyl-tRNAs, aligned on the mRNA, to react to form polypeptides and release the tRNAs.

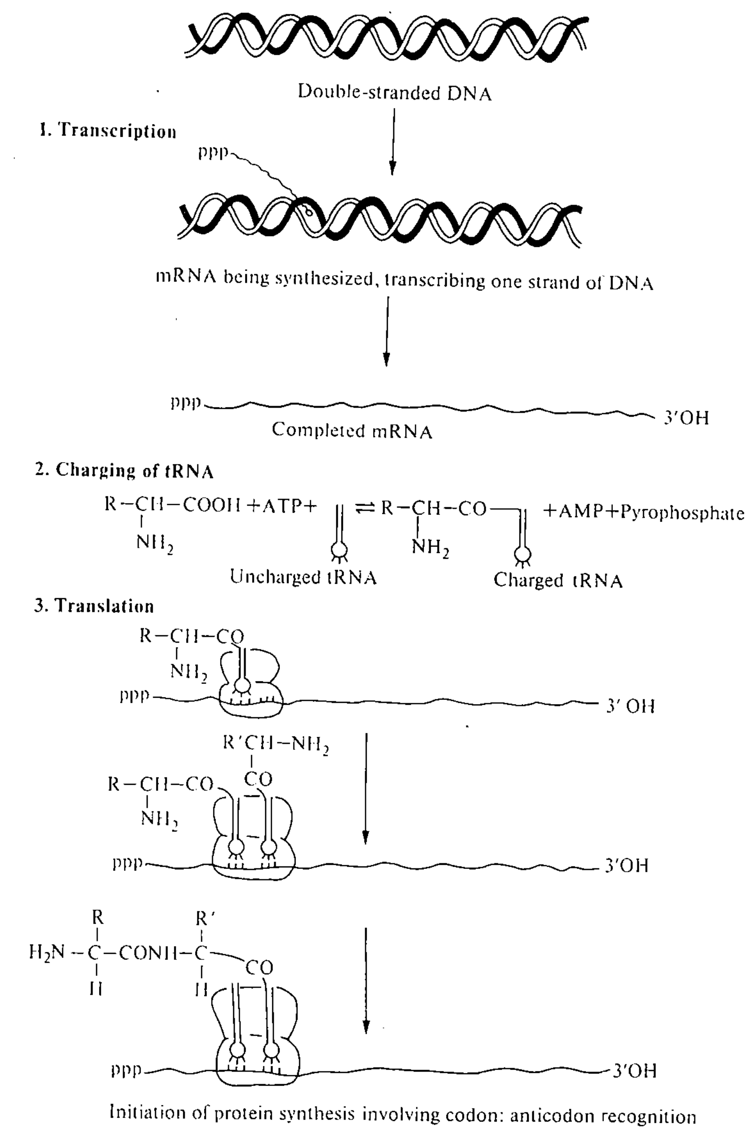

FIG. 9.1. Diagram to illustrate the principal features of the processes of mRNA and protein synthesis, as it occurs in prokaryotes. In eukaryotes the primary transcript is spliced to form mRNA.[16] (For details see references 14 and 15.)

The enzymatic steps that lead to the synthesis of an enzyme are thus well understood. However, the factors that regulate the rate of synthesis of the majority of enzymes are not clear. A limited number of inducible* and

* Inducible and repressible enzymes are those whose synthesis in cells is regulated by inducer and/or repressor molecules of low M_r.

repressible enzymes from *E. coli* have been studied in great depth (see reference 17), but the factors regulating the expression of the genes for the majority of enzymes, particularly those from eukaryotes, are not yet fully understood.

9.3 Kinetics of enzyme turnover

In order to understand the importance of enzyme turnover as a regulatory mechanism, it is necessary to make a kinetic analysis of the process. Equations that govern the steady-state levels of enzymes, and the rates of changes to new steady-state levels, have been derived by Schimke *et al.*[2, 10, 18] From the study of a number of enzymes in which the amounts of enzyme protein present in cells or tissues have been measured, it has been shown that the rate of degradation of an enzyme normally obeys first-order kinetics, i.e. the rate of enzyme degradation $=k[\mathrm{E}]$, where [E] is the enzyme concentration.* This means that the degradation of an enzyme is a random process; a newly synthesized enzyme molecule is as likely to be degraded as an old one. The process of enzyme synthesis has been shown generally to obey zero-order kinetics, i.e. the rate of synthesis of an enzyme is independent of the enzyme concentration within the cell. Thus, the rate of change of level of an enzyme in a cell ($\mathrm{d}[\mathrm{E}]/\mathrm{d}t$) is given by

$$\frac{\mathrm{d}[\mathrm{E}]}{\mathrm{d}t}=k_s-k_d[\mathrm{E}],$$

where k_s is the rate constant for enzyme synthesis and k_d the rate constant for enzyme degradation. In the steady-state, $\mathrm{d}[\mathrm{E}]/\mathrm{d}t=0$ and therefore $k_s=k_d[\mathrm{E}]$.

We now consider the expression for the rate of approach to a new steady-state level if either the rate of synthesis or the rate of degradation changes. If the rate constants for the new rates of synthesis and degradation are k_s' and k_d', respectively, $[\mathrm{E}_0]$ is the initial enzyme concentration, and $[\mathrm{E}_t]$ is the enzyme concentration at time t after the change in rates of synthesis or degradation, then the equation describing the time course of approach to a new steady-state level is given by

$$\frac{[\mathrm{E}_t]}{[\mathrm{E}_0]}=\frac{k_s'}{k_d'[\mathrm{E}_0]}-\left(\frac{k_s'}{k_d'[\mathrm{E}_0]}-1\right)\mathrm{e}^{-k_d't}.$$

For a derivation of this equation, see Appendix 9.1 and reference 18. This equation is useful for analysing changes in enzyme concentrations resulting from hormonal, nutritional, or other physiological changes. When the rate of degradation of an enzyme is measured, it is more common to express the result in terms of the half-life of the enzyme rather than in terms of the rate constant, k_d. For a first-order reaction the relationship is:

$$t_{\frac{1}{2}}=\frac{\ln 2}{k_d}=\frac{0.69}{k_d}.$$

Figure 9.2 illustrates the effect of differences in the rate of degradation of three enzymes on the rate of attainment of new steady states, when a tenfold increase in the rate of synthesis of the enzyme is induced by a stimulus. The effect of withdrawal of the stimulus 5 hours later is also shown. It can be seen that the enzyme having the shortest half-life, ornithine decarboxylase, responds very rapidly to the changed rate of synthesis and has almost reached the new steady state within 2–3 hours, similarly when the stimulus is removed the old steady-state level is reached within a further 2–3 hours. In contrast, the amount of the enzyme having the longest half-life, pyruvate kinase, is still increasing almost linearly after five hours and would require over 70 hours to achieve 90 per cent of the new steady-state level.

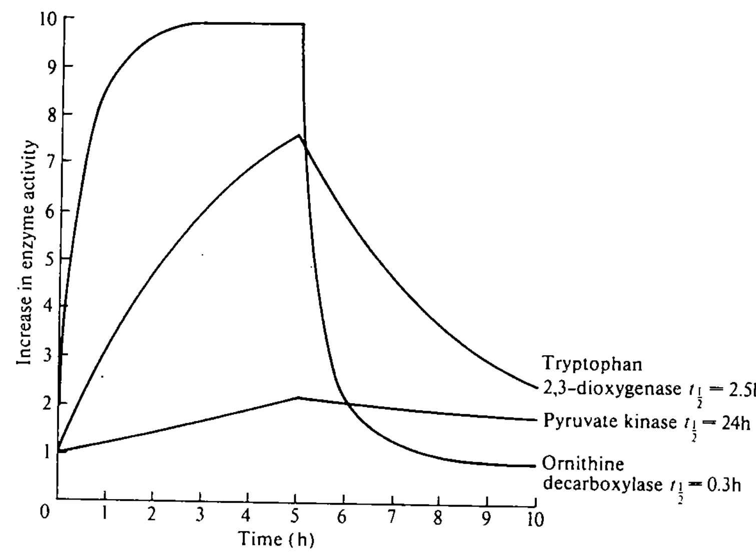

FIG. 9.2. The changes in enzyme activities resulting from a 10-fold increase in the rate of synthesis of each enzyme at zero time, and a 10-fold reduction in the rate of synthesis at 5 hours. Note how the differences in half-lives affect the rates of attainment of new steady states.

Thus, it is the rate of degradation of an enzyme that primarily determines how rapidly the amount of enzyme changes in response to a given stimulus. The shorter the half-life, the quicker the response, and thus the better the control of the level of enzyme. In deriving these equations it is assumed that the rate of

* There are, however, some exceptions; e.g. some proteins present in nerve and muscle show biphasic kinetics, that is, one fraction of the protein is degraded at a different rate from the remainder of the same protein in that tissue.[19]

growth of the cells is slow in comparison with the rate of synthesis of the enzymes so that changes in the total volume of the cells can be neglected.

9.4. Methods for measurement of rates of enzyme turnover

There are several methods that can be used to measure the rates of protein synthesis and protein degradation; the most appropriate will depend on the type of information required. In this section we outline the principal methods; there are several reviews and papers in which more detailed treatment is given.[3, 20–22] In order to understand how the concentration of a particular enzyme is maintained within a cell or tissue and how the concentration is changed in various physiological conditions, it is most useful to measure k_s and k_d. Knowledge of these rate constants will not give any insight as to the mechanism of synthesis or degradation, e.g. as to which are the rate-limiting steps in these processes, but they will indicate whether the change in concentration is due to a change in the rate of synthesis or of degradation, or of both.

A change in the enzyme content of a tissue or organism will usually first be observed from the measurement of enzyme activity. It is then necessary to establish whether this change represents a change in the amount of enzyme protein as opposed to some conformational change or covalent modification of the enzyme that alters its activity (see Chapter 6, Section 6.2). To distinguish these it is necessary to purify the enzyme and then measure the amount of enzyme protein. If previously purified enzyme is available it may be used to raise a specific antibody by injecting the enzyme into another species. The serum fractions containing the antibody may then be collected and used to titrate enzyme protein. Alternatively, monoclonal antibodies may be raised against the enzyme (for details see reference 13). The complex between the enzyme and the antibody precipitates and can be collected by centrifugation and then estimated. Isolation of the enzyme from the tissues at a number of different times, to prove that the amount of enzyme protein is changing, is a very time-consuming process and may require large amounts of tissue. The use of an antibody greatly simplifies the procedure and can be used with smaller amounts of tissue.

Once it has been established that a change in the amount of enzyme protein occurs, the rates of synthesis and degradation are measured. Isotopically labelled precursors are most frequently used for this, as outlined in the following sections.

9.4.1 Measurement of k_s

The rate of enzyme synthesis is usually measured either after giving a single pulse of a suitable radioactively labelled precursor, e.g. an amino acid, or by giving a constant infusion of the labelled precursor. In the first method (involving a single pulse), incorporation of labelled precursor into the enzyme is measured at short time intervals. The specific activity of the amino-acid pool in the tissue or cells concerned is also measured. If short time intervals are used, the

initial rate of incorporation is determined (i.e. when the specific activity of label in the enzyme is low and the loss of radioactivity by degradation is negligible). From these data k_s may be determined. One problem arises particularly in experiments with whole animals. The radioactive amino acids enter the plasma and thence pass into the cells, where they equilibrate with endogenous amino acids. The amino acids are then converted to aminoacyl-tRNAs, which are the ultimate precursors of the enzyme. Therefore, to calculate accurately the rates of synthesis, the specific radioactivity of the isolated aminoacyl-tRNAs should be used.[21] However, in rat liver, the tissue in which many measurements of k_s have been made, there is rapid equilibration between the free amino-acid pool within the cells and the aminoacyl-tRNAs and therefore in this particular tissue the specific radioactivity of the free amino acids may be used.

The second method is that of constant infusion of radioactively labelled amino acids. The amino acids are either infused intravenously or given in the diet until the specific radioactivity of the amino-acid pool within the cells has reached a plateau level. Measurements are then made of the incorporation of the label into the enzyme at the end of the infusion period and of the changes in specific radioactivity of the free amino acid within the tissue. From these measurements k_s may be determined; the mathematical relationships are complex and are described in references 20 and 21.

9.4.2 Measurement of k_d

The rate of degradation can also be measured by use of radioactivity labelled precursors. If a pulse of labelled precursor is given and the decay of the specific radioactivity of the labelled enzyme is measured at fixed time intervals, k_d can be calculated. Measurements are made for a longer period than those required for measurement of k_s. Ideally k_d is measured when the specific radioactivity of the enzyme is high, and the specific radioactivity of the amino-acid pool has declined (Fig. 9.3).

An assumption made in the calculation of k_d is that no significant re-utilization of the precursors released by protein degradation takes place. If re-utilization takes place, k_d may be considerably underestimated. This has been a major problem in the accurate determination of k_d, but re-utilization can often be minimized by a suitable choice of precursor; for example [*guanidino*-^{14}C]arginine is found to be a better label for this purpose than uniformly labelled arginine or other labelled amino acids when liver protein degradation is measured.[23] This is because the guanidino group is rapidly removed by arginase present in high concentrations in the liver. [^{14}C]carbonate has also been found to be a good precursor for measurement of k_d; much of the [^{14}C]carbonate is incorporated into the carboxyl groups of glutamate and aspartate.[24] The radioactivity of the latter amino acids declines rapidly because of decarboxylations that occur when glutamate and aspartate become transaminated and subsequently oxidized in the tricarboxylic-acid cycle. The $H^{14}CO_3^-$ produced in this way is diluted by the large intracellular pool of HCO_3^- and lost as $^{14}CO_2$.

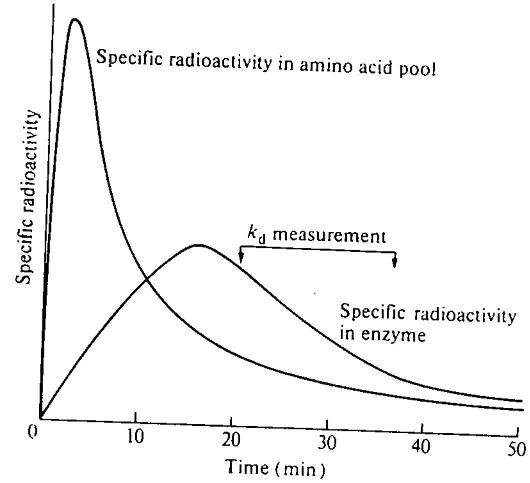

FIG. 9.3. Precursor-product relationship after a single pulse of radioactively labelled amino acid is given to a tissue or cells. Diagram shows the specific radioactivities in the amino-acid pool (precursor) and in the enzyme (product).

Another example in which reincorporation does not occur is that of 5-aminolaevulinic acid,[25] which is incorporated specifically into haem-containing proteins, e.g. cytochromes, catalase. Haem is degraded by a pathway that differs from that of its synthesis and thus reincorporation does not take place. The validity of this method depends on the assumption that the haem prosthetic group is degraded at the same rate as the protein moiety, which may or may not be the case.

The double isotope technique[20] is a convenient method of comparing the rates of turnover of proteins having the same cellular origin. It entails giving two successive pulses of a radioactively labelled amino acid, the first being ^{3}H-labelled and the second ^{14}C-labelled; these are shown as ^{3}H-A and ^{3}H-B, and ^{14}C-A and ^{14}C-B, respectively, in Fig. 9.4. where A and B are two different proteins. In the example in Fig. 9.4, the turnover of A is twice that of B. The proteins that are being studied are isolated at time *Y*. Typically the ^{3}H pulse would have been given 24 h before isolation of the proteins, whereas the ^{14}C pulse would have been given 30 min. before isolation (time *X*), although the precise times would depend on the proteins and tissue concerned. The times are chosen so that the specific activities of the first label (^{3}H) in the proteins being studied are declining, whereas the specific activities of the second label (^{14}C) are increasing (Fig. 9.4). The proteins are isolated at time *Y* and the ratio ^{14}C/^{3}H (b/a and d/c) in each is measured; these ratios are directly related to their turnover rates. The method has two advantages, namely, (i) that the proteins only have to be isolated at one time point, and (ii) that the differences in turnover rates between proteins are more readily apparent than when a single isotope is used.

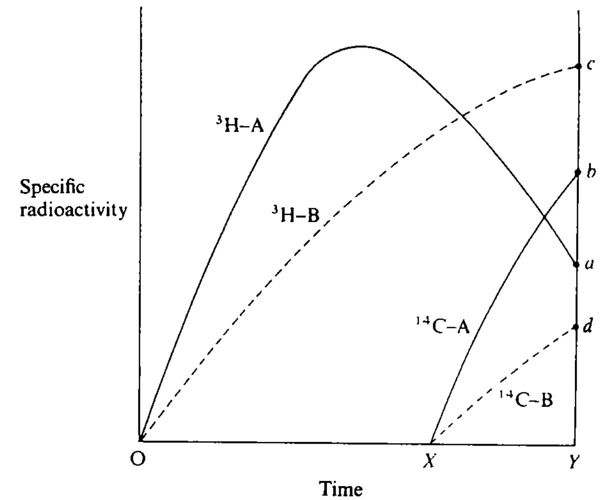

FIG. 9.4. Double isotope technique for measuring protein turnover.

A method that has been widely used to determine rates of degradation is to inhibit protein synthesis in the tissue concerned by administration of cycloheximide or puromycin and then to measure the subsequent decay of enzyme activity. This method has the advantage of simplicity in that it does not require the isolation of the enzyme, but it has the disadvantages (a) that it cannot distinguish enzyme inactivation from enzyme degradation and (b) that cycloheximide and puromycin, in addition to inhibiting protein synthesis, may also affect protease activity, thereby affecting the rate of enzyme degradation.[3] Evidence suggests that the latter complication is less serious when studying enzymes that have short half-lives.[27]

9.5 Results from measurements of rates of enzyme turnover

The rates of turnover of a number of enzymes have now been measured in a variety of tissues and organisms and it is possible to correlate these results with the structures and properties of the enzymes concerned. The most widely measured parameter is the enzyme's half-life. Some of the results obtained are given in Table 9.1. It is apparent that a wide range of turnover rates can occur within one tissue, e.g. the half-lives of liver proteins range from about 15 min (ornithine decarboxylase) to 7 days (6-phosphofructokinase). The half-life of a given enzyme may differ from one tissue to another (Table 9.2) or even between different compartments within the same cell, e.g. 5-aminolaevulinate synthase in the cytosol has a half-life of 20 min, whereas in the mitochondria the half-life is 60 min.[32] Although some results suggest that in certain organelles or membranes

TABLE 9.1
The half-lives of enzymes in rat liver

Enzyme	Half-life (hours)
Ornithine decarboxylase	0.3
5-Aminolaevulinate synthase (mitochondria)	1.2
RNA nucleotidyltransferase I	1.3
Tyrosine aminotransferase	1.5
Tryptophan 2,3-dioxygenase	2.0
Thymidine kinase	2.6
Hydroxymethylglutaryl-CoA reductase	4.0
Serine dehydratase	5.2
Phosphoenolpyruvate carboxykinase	8.0
Dehydro-orotase	12.0
RNA nucleotidyltransferase II	13.0
Glucose-6-phosphate dehydrogenase	24
Glucokinase	30
Catalase	33
Acetyl-CoA carboxylase	50
Glyceraldehyde-3-phosphate dehydrogenase	74
Pyruvate kinase	84
Arginase	108
Fructose-bisphosphate aldolase	117
Lactate dehydrogenase (LDH-5)	144
6-Phosphofructokinase	168

See references 2, 4, 28–31.

TABLE 9.2
The half-lives (in days) of enzymes in different tissues of the rat

Enzyme	Liver	Heart	Skeletal muscle	Kidney	Brain
Pyruvate kinase	3.5	4.2	21.6	—	—
Lactate dehydrogenase[a]	1.8–3.8	3.7–9.9	3.0–8.5	3.0–5.0	3.0–7.2
Ornithine oxo-acid aminotransferase	0.95	—	—	4.0	—
Fatty-acid synthase	6.4	—	—	—	2.8

See references 2, 28, 33, and 34.
[a] The half-life of lactate dehydrogenase varies with the particular isoenzyme.[35]

within an organelle a number of enzymes turnover at similar rates,[36] e.g. cytochrome *c* oxidase and adenosinetriphosphatase from the inner mitochondrial membrane of yeast turnover at similar rates during sporulation, and also ribosomal proteins turnover at similar rates, other data indicate that there is a considerable range of turnover rates of enzymes within a single organelle or membrane[38, 39] (Table 9.3) and it seems likely that in general organelles do not turnover as units. From the limited data available so far, it seems that the polypeptide chains within a multienzyme complex, e.g. pyruvate dehydrogenase,

TABLE 9.3
Variation in half-lives of enzymes within the same organelle of rat liver

	Half-life (hours)
1. Microsomal membrane	
Cytochrome b_5 (outside)	100
Cytochrome b_5 reductase (outside)	140
NADPH-cytochrome reductase (outside)	70
Nucleosidediphosphatase (inside)	30
Carboxylesterase (inside)	95
Hydroxymethylglutaryl-CoA reductase	4
NAD^+ nucleosidase	430
2. Mitochondria	
Carbamoyl phosphate synthase	185
Malate dehydrogenase	62
Glutamate dehydrogenase	24
Carnitine acetylase transferase	43
Ornithine oxo-acid aminotransferase	46
Cytochrome *c* oxidase	> 190
Amine oxidase (flavin-containing)	67
Pyruvate dehydrogenase system	
$E_1\alpha$	108
$E_1\beta$	120
E_2	134
E_3	120

See references 39–43.

and within an oligomeric enzyme, e.g cytochrome *c* oxidase and amine oxidase (flavin-containing), turnover at similar rates (Table 9.3).

9.6 Possible correlations between the rates of turnover and the structure and function of enzymes

A number of enzymes having short half-lives catalyse either the first or the rate-limiting steps in a metabolic pathway, e.g. ornithine decarboxylase, 5-aminolaevulinate synthase, tyrosine aminotransferase, tryptophan 2,3-dioxygenase, serine dehydratase, hydroxymethylglutaryl-CoA reductase, and phosphoenolpyruvate carboxykinase. This finding may be important, since the amounts of enzymes that turnover rapidly can be more precisely controlled than the amounts of those that turnover more slowly (see Section 9.3).

Although the list of characterized intercellular proteases* is growing, it seems highly improbable that there exists within each cell such a wide range of

* All proteolytic enzymes fall within the class of 'proteases'. These may be subdivided into 'endopeptidases' and 'exopeptidases'. The term 'proteinases' can be used interchangeably with endopeptidases.[44]

proteases that each is able to control the rate of degradation of either individual enzymes or small groups of enzymes; this would require an enormous number of proteases, the half-lives of which would in turn have to be regulated. What, therefore, are the characteristics of those enzymes that are degraded most rapidly? Do they have any common features? The outline of a pattern is beginning to emerge. The pattern may be sought from either a structural or a functional standpoint. From the structural standpoint, results obtained from a large number of proteins from the cytosol of liver suggest that there is a correlation between the subunit size of the protein and its rate of degradation, the larger subunits being degraded faster.[45,46] The explanation for this could simply be that larger polypeptides have more protease-sensitive sites and are thus more susceptible to proteolysis. It is not yet clear how general is this correlation, since exceptions have been noted in some mitochondrial[41] and lysosomal[47] enzymes.

As well as the size correlation, some data support a correlation between the turnover rate and hydrophobicity, the most hydrophobic enzymes being degraded fastest.[48] Enzymes with low p*I* are more labile.[49] However, there are exceptions to these general correlations.[50]

Interesting correlations can be made between the rate of degradation of enzymes and their ligand-binding properties. There are a number of examples in which the binding of a substrate, cofactor, or competitive inhibitor has been known to stabilize an enzyme against degradation (Table 9.4). The group of enzymes that has been most studied in this respect is the pyridoxal phosphate-requiring enzymes. A group of proteases were isolated that preferentially degraded the apoenzymes of pyridoxal phosphate-requiring enzymes. These proteases were believed to be responsible for the intracellular turnover of apoenzymes in liver and skeletal muscle.[57] However, later evidence throws doubt on this hypothesis, since the group-specific proteases in question were found to originate in the mast cells associated with skeletal muscle and liver.[58] Also, if rats were fed on a pyridoxal-(vitamin B_6) deficient diet so that they became pyridoxal deficient, no increase was observed in the rate of degradation of pyridoxal phosphate-requiring apoenzymes.[59] Thus, although there is good evidence for the destabilization of enzymes *in vitro* when depleted of ligands, it is doubtful whether this is an important mechanism *in vivo*.

The explanation for the wide variation in the rates at which many enzymes are turned over may be found by referring to the structures of the enzymes, i.e. those enzymes that have most protease-sensitive sites, that bind ligands least strongly, and that are denatured most readily are those most likely to be turned over most rapidly. One of the difficulties in making comparisons of structure with turnover is that the enzymes that are most interesting for studying turnover are the enzymes present in the cell in small amounts and about which least structural information is available. By contrast, the enzymes about which there is detailed structural information, including three-dimensional structure, are often those that are most stable *in vivo*. The situation has partly been remedied by Rechsteiner *et al.*,[60] who have developed a method for microinjection of purified

TABLE 9.4
Examples of ligands that stabilize enzymes against degradation

Enzyme	Ligand	Effect
Hexokinase	Glucose	Reduces rate of degradation by trypsin
Thymidine kinase	Thymidine	Stabilizes against degradation in cell extracts
Tryptophan 2,3-dioxygenase	Tryptophan or tryptophan analogues	Induces tryptophan 2,3-dioxygenase by stabilizing it against degradation
Serine dehydratase Tyrosine aminotransferase	Pyridoxal phosphate	Protects against group specific proteases that degrade pyridoxal phosphate-requiring apoenzymes
Aspartate carbamoyltransferase	Aspartate CTP or ATP	Increased susceptibility to trypsin Decreased susceptibility to trypsin
Tetrahydrofolate dehydrogenase	Methotrexate	Decreases rate of intracellular degradation
Ornithine decarboxylase	α-Methylornithine or diaminobutanone	Decreases rate of intracellular degradation, protects against chymotrypsin
Acetyl-CoA carboxylase	Palmityl-CoA citrate	Increased susceptibility to lysosomal protease Decreased susceptibility to lysosomal protease
Hydroxymethylglutaryl-CoA reductase	Mevinolin	Decreases intracellular degradation

See references 3, 51–56.

enzymes into cells. Using this method in a study of 35 proteins of known sequence and X-ray structure they found that the half-lives did not correlate with size, charge, or proportion of hydrophobic residues. More recently they have tried to relate the abundance of particular amino acids in a protein to turnover rate.[61] A correlation was found between the half-life and the abundance of four particular amino acids, namely proline, glutamate, serine, and threonine. All the proteins studied with a half-life less than 2 hr had regions rich in some of, or a combination of, these four amino acids.

In addition to making comparisons on purely structural grounds, it is useful to relate turnover to function. A number of workers have divided enzymes into long-lived and short-lived. This is a convenient functional division, since it appears to relate to the mode of degradation. Many long-lived enzymes are the so-called 'house-keeping' enzymes. These include enzymes catalysing non-rate-limiting steps in the major metabolic pathways. They are generally degraded by lysosomal proteases. The short-lived enzymes are present in much smaller amounts and are degraded by extralysosomal routes. Four categories of short-lived proteins can be distinguished at present. The first are enzymes catalysing regulatory steps in metabolic pathways; examples include all those cited at the beginning of this section. The second are enzymes required only at particular stages in the life of the cell, e.g. enzymes required for thymidine triphosphate synthesis, which is required only when the cell enters the presynthetic (G_1) phase of the cell cycle, when the total amount of DNA is doubled. The third type are the enzyme precursors that contain the signal peptide required for entry into an organelle. These are converted into the functional enzymes after entry to the organelle. The fourth are a rather different category, namely the abnormal or non-functional enzymes.

Abnormal proteins, e.g. mutant proteins or proteins containing chemically modified amino acids are degraded much faster than their normal counterparts. Mutations in β-D-galactosidase[62] from *E. coli* cause the enzyme to be degraded rapidly, and the incorporation of azetidine-2-carboxylate in place of proline into haemoglobin reduces its half-life from 120 days to 20 min.[63] The mechanism by which abnormal proteins are selectively degraded at a faster rate is not known.

9.7 The mechanisms of protein degradation

Enzymes degraded by lysosomal and non-lysosomal routes can often be distinguished by two methods. Chloroquine and ammonia are able to penetrate the lysosomal membranes and raise the internal pH,[64] thereby inhibiting lysosomal proteases and so affecting the breakdown of enzymes degraded by the lysosomal pathway. Secondly, the Q_{10} (see Chapter 4, Section 4.3.4.1) for degradation of short-lived proteins degraded by the non-lysosomal route is about 2,[65] whereas degradation via the lysosomes has a Q_{10} of about 4.[66] It is estimated that in the liver about 60 per cent of proteins are degraded in the cytosol and about 40 per cent in the lysosomes.[67]

The mechanisms by which enzymes are degraded are still only poorly understood. A few enzymes have been studied in detail and certain degrading systems have been examined, but how far one can generalize from these limited examples is not clear. It is generally assumed that enzymes are eventually degraded to small peptides or their constituent amino acids. This may involve, particularly in the case of short-lived enzymes, a highly selective initial step followed by further degradation by less specific proteases. The initial step is assumed to recognize specific features in the enzyme to be degraded, but the nature of these might be quite varied. The recognition might involve a highly specific protease that then partially degrades the enzyme; it might entail some kind of modification or tagging of the enzyme by addition of some covalent group; or it might involve binding to a membrane.

9.7.1 Intracellular proteases

In the past decade the number of intracellular proteases that have been characterized has increased considerably. Much of this has resulted from the use of highly specific substrates, e.g. peptides linked through an amide bond to a chromophoric group. The general method of detection of a protease would be that to incubate a cell extract with a protein such as casein and measure the degradation to peptides that are soluble in trichloracetic acid. Alternatively, a protein that has been chemically modified to contain a chromophore (e.g. azocoll, a chemically modified collagen or azocasein) is incubated with a cell extract, and the soluble, coloured peptides released are detected. These methods are satisfactory for detecting endoproteases of low or unknown specificity. The use of peptides modified with chromophores detects more highly specific proteases. The introduction of this method dramatically increased the number of proteases detected in yeast from 8 in 1979 to 29 in 1984.[68]

The lysosomes, which are the richest source of intracellular proteolytic activity in many tissues, contain a variety of exopeptidases and endopeptidases called cathepsins. The concentrations of cathepsins D and B within the lysosomes are estimated to be as high as 0.78 mmol dm^{-3} and 1.54 mmol dm^{-3}, respectively.[69] Lysosomal proteases are thought to be involved in the degradation of longer-lived proteins. It is now apparent that there are proteases present in all cell compartments.

Peptide-bond cleavage is an exergonic process and thus might not be expected to require energy coupling. However, it is now clear that in both eukaryotes and prokaryotes there exist ATP-dependent proteases.[44] Those found in the cytosol of eukaryotes are ATP stimulated, i.e. ATP is required for activity although it is not degraded. Those proteases found in bacteria and in mitochondria hydrolyse ATP during protein degradation. This ATP requirement may be the 'price' for specificity, in the same way that there is net degradation of ATP during substrate cycles (see Chapter 6, Section 6.2.1.2). The intracellular proteases known range in

M_r from 20 000 to 800 000. The sizes of these very large proteases may reflect the possession of a high specificity and complex regulatory mechanism.

9.7.2 How enzymes or proteins are selected for intracellular degradation

The key question in protein degradation is understanding the selection mechanism. Various methods of 'tagging' proteins for degradation have now been identified. These include the formation of a peptide bond between the protein and another protein called ubiquitin, the modification of the protein by phosphorylation or by oxidation, or the binding of the protein to a membrane. There may be other mechanisms yet to be discovered. The different methods will be considered in turn. The degradation of proteins in reticulocytes has been extensively studied and it has been shown by Herscho and co-workers[70, 71] that degradation involves tagging with ubiquitin (M_r 9000). Ubiquitin is so called because of its widespread distribution throughout eukaryotes, and it is possibly also present in prokaryotes. Its role in protein turnover has only been established in a few tissues. Although it is present in kidney and liver, it has not been shown to be involved in degradation in those tissues. Proteins become covalently attached to ubiquitin via their $-NH_2$ groups either the N-terminal α-NH_2 or ε-NH_2 of lysine. A protein may become multiply tagged. ATP is required for the conjugation, together with three other enzymes designated E_1, E_2, and E_3.

$$E_1\text{–SH} + \text{ATP} + \text{Ub} \rightarrow E_1\text{–S–Ub} + \text{AMP} + \text{PP}_i.$$

$$E_1\text{–S–Ub} + E_2\text{–SH} \rightarrow E_2\text{–S–Ub} + E_1\text{–SH}.$$

$$E_2\text{–S–Ub} + \text{protein} \rightarrow E_2\text{–SH} + \text{protein–Ub} \quad (E_3 \text{ catalysed}).$$

The ubiquitinated protein is thus tagged for degradation by proteolytic enzymes. Bachmair *et al.*[72] examined the specificity of the degrading system. They constructed a chimeric gene consisting of the genes for ubiquitin linked to β-galactosidase. When this gene is expressed it produced a conjugate protein with ubiquitin linked to the N-terminus of β-galactosidase via a peptide bond. The protein became de-ubiquitinated and degraded by the cell's proteolytic system. Site-directed mutagenesis (see Chapter 5, Section 5.4.5) was then used to replace the normal N-terminal amino acid of β-galactosidase (Met) in turn by other amino acids. It was then possible to study the degradation of the fusion products. The half-lives of the fusion products ranged from 2 min to over 20 h depending on the N-terminal amino acid as shown below:

$$\underbrace{H_2N\text{------------}}_{\text{ubiquitin}}\text{Gly-Met}\underbrace{\text{------------COOH}}_{\beta\text{-galactosidase}}$$

Replacement of Met by Ser, Ala, Gly, Thr, or Val has a stabilizing effect. Replacement of Met by Ile, Gln, Glu or Tyr has a destabilizing effect. Replacement of Met by Leu, Phe, Asp, Lys, or Arg has a strongly destabilizing effect. The N-terminal residue is therefore an important determinant in protein degradation through the ubiquitin pathway.

Another method of tagging an enzyme prior to its degradation is by phosphorylation. Two examples of this are fructose-bisphosphatase[73] from *Saccharomyces cerevisiae* and the NAD^+-requiring glutamate dehydrogenase from *S. cerevisae* and *Candida utilis*.[74, 95] When *S. cerevisiae* is grown with acetate as the source of carbon, it carries out gluconeogenesis to provide it with adequate hexoses and their derivatives. On the other hand, when it is supplied with glucose as the source of carbon, gluconeogenesis is suppressed. These differences are reflected in the stability of fructose-bisphosphatase, which catalyses a rate-limiting step in gluconeogenesis (see Chapter 6, Section 6.4.1). When grown on acetate, the k_d for fructose-bisphosphatase breakdown is 0.008 h^{-1}, whereas when grown on glucose it is 0.42 h^{-1}.[75] The role of phosphorylation is best seen when the acetate-grown cells are switched to a glucose medium. Within 2 min the activity of fructose-bisphosphatase drops very dramatically. This initial fall can be shown to be due to phosphorylation of a serine residue on fructose-bisphosphatase, although no proteolysis occurs. However by 30 min the fructose-bisphosphatase protein is degraded. Thus, it appears that the initial inactivation is through phosphorylation but that this then labels the protein for degradation. A similar phenomenon is observed with NAD^+-requiring glutamate dehydrogenase in *S. cerevisiae* and *C. utilis*. The NAD^+-requiring glutamate dehydrogenase in these organisms is principally concerned with oxidative deamination of glutamate:

$$\text{Glutamate} + NAD^+ \rightarrow \text{2-oxoglutarate} + NADH + NH_4^+.$$

When *S. cerevisiae* is grown with glutamate as the source of carbon and nitrogen, the NAD^+-requiring glutamate dehydrogenase activity is high. If the organism is switched to glucose and ammonia as the sources of carbon and nitrogen, then oxidative deamination of glutamate is not required and the NAD^+-requiring glutamate dehydrogenase activity rapidly declines. The mechanism is similar to that in the previous example, namely the glutamate dehydrogenase initially becomes phosphorylated and is then degraded by proteolysis. Two other enzymes that become more susceptible to proteolysis following phosphorylation are hydroxymethylglutaryl-CoA reductase[76] and pyruvate kinase.[77]

The method of tagging in the case of glutamine synthetase from bacteria is by oxidation.[78] The enzyme is oxidized by mixed-function oxidases and it has been shown that there is loss of a single histidine residue per subunit. The oxidized enzyme is then more susceptible to attack by proteases. Several other enzymes may also be tagged by oxidation, including glutamine phosphoribosyl pyrophosphate amidotransferase, phosphoenolpyruvate carboxykinase, tyrosine aminotransferase, fructose bisphosphate aldolase, and ornithine decarboxylase.[78]

There is some evidence that certain enzymes become associated with membranes prior to degradation.[5] The enzyme hydroxymethylglutaryl-CoA reductase (HMG-CoA reductase) contains two domains, a catalytic domain (548 residues) and an anchoring domain (339 residues). The latter anchors the enzyme to the endoplasmic reticulum. HMG-CoA reductase has a short half-life (Table 9.1). It has been shown[79] that if the catalytic domain is cleaved from the complete enzyme, then the rate of degradation of the truncated enzyme (catalytic domain only) is only one-fifth the rate of degradation of the complete enzyme. The truncated enzyme is no longer able to associate with the membrane, which is believed to be involved in initiating the breakdown.

Thus it can be seen that there are a number of possible methods of tagging enzymes to mark them for degradation, but it is not yet clear how generally each route is used.

9.8 The significance of enzyme turnover

The turnover of specific enzymes has been studied most extensively in mammalian tissues in response to nutritional and hormonal changes, and to a lesser extent during development. In microorganisms, enzyme turnover has been studied in response to changes in nutrients, particularly carbon and nitrogen sources. These aspects will be considered in turn.

9.8.1 Changes in enzyme-turnover rates in animals in response to changes in diet

Arginase and serine dehydratase present in the liver are concerned indirectly with protein breakdown. If an animal is fed on a high-protein diet, then the surplus protein is not stored as such; it becomes deaminated and the carbon skeletons of the amino acids are converted to precursors of lipid or carbohydrate before being stored in those forms. Non-essential dietary amino acids, of which serine is one, are degraded before the essential amino acids. Serine dehydratase catalyses the first step in the breakdown of serine:

$$CH_2OH.CH(NH_2)COOH \rightarrow CH_2{=}C(NH_2)COOH \rightarrow$$
$$CH_3C({=}NH)COOH \rightarrow CH_3CO.COOH + NH_3.$$

The ammonia resulting from deamination of the amino acids eventually enters the urea cycle to be converted to urea. Arginase catalyses the final step in urea formation. The levels of these two enzymes change in response to changing dietary intake. An increase in protein intake causes increases in the activities of both arginase and serine dehydratase in the liver. Both of these increases have been shown to be due to increases in total enzyme protein and not due to activation of pre-existing enzyme. A decrease in dietary protein causes a decrease in the amount of arginase. As can be seen from Table 9.5, changes in the levels of enzymes can be affected by changes in the rate of synthesis, in the rate of degradation, or of both. In the case of arginase a further adaptation has been observed by starving animals that had previously been fed on a low-protein diet. Under these conditions the arginase content of the liver increases in spite of there being no protein intake. Endogenous proteins are now degraded as a source of energy. In contrast to the increase in arginase activity when placed on a high-protein diet, under fasting conditions the increase in arginase activity is brought about in a more economical fashion by a decrease in k_d but no change in k_s. An enzyme that resides in the plasma membrane and is believed to be important in

TABLE 9.5

The effect of diet on the turnover rates of enzymes from rat liver

Enzyme	Change of diet	Effect on enzyme
Arginase	Low-protein → high-protein	Increased activity mainly due to increased k_s
	High-protein → low-protein	Decreased activity, k_s decreased, k_d increased
	Low-protein → fasting	Increased activity, k_d decreased
Serine dehydratase	Protein-free → high-amino-acid	Large increase in activity due to increased k_s
Ornithine oxoglutarate aminotransferase	Basal → high-protein	Increased activity, increase in k_s k_d unchanged
Acetyl-CoA carboxylase	Basal → fat-free diet	Increased activity, increased k_s
	Basal → fasting	Decreased activity, k_s decreased, k_d increased
	Basal → high-fat diet	Decreased activity, k_s decreased, k_d increased
Fatty acid synthase	Basal → fat-free diet	Increased activity, k_s increased k_d unchanged
	Basal → fasting	Decreased activity, k_s decreased, k_d increased
Pyruvate kinase	Basal → fasting	Decreased activity, k_s decreased, k_d slight increase
	Fasting → refeeding	Increased activity, k_s increased, k_d unchanged
Phosphoenol pyruvate carboxykinase	Fed → starved	Increased activity, k_s increased, k_d slightly decreased
	Starved → refed	Decreased activity, k_s decreased, k_d slightly increased

See references 2, 33, 42, 80–84.

amino-acid uptake into cells is γ-glutamyltransferase. The activity of this enzyme is much higher in hepatoma cells than in normal liver cells, and this probably reflects the need for a greater rate of uptake to sustain the more rapid growth of the tumour cells. The increase is brought about by a decrease in the rate of degradation of the enzyme. Its half-life in normal cells is 3 h compared to 24 h in hepatoma cells.[85]

Another important enzyme to be studied in response to dietary change is acetyl-CoA carboxylase. This enzyme, which catalyses the first (rate-limiting) step in fatty-acid synthesis, is sensitive to changes in dietary fat.

$$\text{Acetyl-CoA} + CO_2 + \text{ATP} = \text{malonyl-CoA} + \text{ADP} + \text{orthophosphate.}$$

High dietary fat lessens the need for endogenous fatty-acid synthesis and *vice versa*. Again, the changes in enzyme level can be brought about by changing either the rate of synthesis or the rate of degradation (Table 9.5). When animals are fasted, not only acetyl-CoA carboxylase but also fatty-acid synthase and L-malate-NADP$^+$ oxidoreductase (oxaloacetate-decarboxylating) are degraded more rapidly than in fed animals. L-Malate-NADP$^+$ oxidoreductase (oxaloacetate-decarboxylating) is one of the enzymes involved in lipogenesis, since it catalyses the formation of NADPH.

9.8.2 Changes in enzyme stability in microorganisms in response to changes in carbon- and nitrogen-containing nutrients

Changes in the carbon and nitrogen sources in the growth medium cause adaptation in microorganisms; this adaptation may involve synthesis of new enzymes and degradation or inactivation of existing enzymes. An enzyme that is no longer necessary for the growth of the microorganisms on a new medium may be lost either passively, i.e. diluted out as the organism continues to grow, or in an active fashion by inactivation or degradation. The active loss of an enzyme that is no longer required may involve reversible or irreversible steps. Reversible inactivation may occur by covalent modification (see Chapter 6, Section 6.2.1). Examples from microorganisms include the following.

(1) Phosphorylation of the pyruvate dehydrogenase complex in *Neurospora crassa*[86] which occurs in the presence of ATP and inactivates the complex.
(2) Phosphorylation of yeast NAD$^+$-dependent glutamate dehydrogenase to form a much less active enzyme.[74] This enzyme is responsible in yeast for glutamate breakdown and is phosphorylated when glutamate in the medium is exchanged for ammonia and glucose as sources of nitrogen and carbon.
(3) Inactivation of enzymes required for gluconeogenesis, i.e. phosphoenolpyruvate carboxykinase, malate dehydrogenase, and fructose-bisphosphatase in *S. cerevisiae* when the source of carbon is changed from acetate to glucose (see Section 9.7.2).
(4) Inactivation of ornithine carbamoyltransferase in yeast by binding arginase. This occurs when arginine is added to the growth medium and is described more fully in Chapter 8, Section 8.5.3.3.

9.8.3 Changes in enzyme turnover in response to hormonal stimuli

The systems most studied under this heading are various liver enzymes concerned with the catabolism of amino acids and their response to hormones, particularly the glucocorticoids. Some studies have been made with intact animals but others have used a strain of hepatoma cells grown in culture which responds very similarly to that of intact liver and which has the advantage that the hormone concentration can be more easily controlled. Glucocorticoids include cortisol, corticosterone, and cortisone and their principal effects are to raise blood glucose levels, mobilize amino acids, and stimulate gluconeogenesis in the liver. Trytophan 2,3-dioxygenase, the first enzyme in the tryptophan catabolic pathway, can be induced in liver by administration of either cortisol or of tryptophan or analogues of tryptophan. In each case the total amount of enzyme protein is increased, but in the case of cortisol administration it is due to an increase in the rate of enzyme synthesis, whereas in response to tryptophan or its analogues the rate of enzyme degradation is decreased, possibly as a result of stabilization of the enzyme in the presence of the substrate. The activity of tyrosine aminotransferase, the first step in tyrosine breakdown, is also induced by cortisol which increases the rate of its biosynthesis. Unlike tryptophan 2,3-dioxygenase, tyrosine aminotransferase is also induced by the pancreatic hormones insulin and glucagon.

Other liver enzymes that have been studied are alanine aminotransferase and ornithine oxo-acid aminotransferase. The former increases in response to glucocorticoids, which affect only the rate of synthesis, while the latter has been shown to increase in response to oestrogen, again by an increase in the rate of synthesis; there is no change in the rate of degradation. The effects are summarized in Table 9.6.

The mechanisms by which these hormones bring about increases in enzyme activity are still unclear. The steroid hormones have been shown to penetrate the cells of the target tissue, where they bind to receptors and ultimately bring about a change in gene expression.[87] In the case of certain enzymes, e.g. tryptophan 2,3-dioxygenase, there is evidence for increased production of mRNA, which is assumed to be due to changes in gene expression. Other hormones, such as glucagon and adrenalin, bind to receptors on the cell membrane and their effects are thought to be mediated through changes in the intracellular concentrations of cyclic nucleotides (see Chapter 6, Section 6.4.2.1).

9.8.4 Changes in enzyme turnover during development

Although changes in the levels of many enzymes have been studied in a number of different developmental systems and in some cases groups of enzymes have

TABLE 9.6

Changes in enzyme turnover in response to hormonal treatment

Tissue	Enzyme	Hormonal change	Effect on enzyme
Rat liver	Tryptophan 2,3-dioxygenase	Glucocorticoid administration	Increased activity, k_s increased, k_d unchanged
Rat liver	Tyrosine aminotransferase	Glucocorticoid, glucagon or insulin administration	Increased activity, increased k_s, k_d unchanged
Rat kidney	Ornithine oxo-acid aminotransferase	Oestrogen	Increased activity, k_s increased
Rat liver	Carbamoyl phosphate synthase	Thyroidectomy	Decreased k_d
Rat liver	Malate dehydrogenase	Thyroidectomy	Decreased k_d
Rat liver	Arginase	Glucocorticoid	Increased activity, k_s increased
Rat liver	Serine dehydratase	Glucagon	Increased activity, k_s increased

References 2, 34, 41.

been shown to appear rapidly at particular stages in development,[88] none of these changes has been analysed in as great detail as the dietary effects described in Section 9.8.1. The most widely used method of study is to measure enzyme activity during different stages of a particular developmental process, in some cases perturbing the system by the use of inhibitors. In more detailed studies, enzymes have been purified and the enzyme protein has been measured. These studies taken together allow us to build up an overall picture of gene expression during development. As yet very few studies[89] have been made to determine changes in k_s and k_d for particular enzyme systems during development. A number of simple systems have been used as models to study development, e.g. sporulation of yeast and bacteria and the life cycle of the cellular slime mould.

The sporulation of yeast illustrates the type of studies made. Certain strains of *Saccharomyces cerevisiae* can be induced to sporulate by starvation on a nitrogen-free medium. During the process of sporulation, modified cells or ascospores are formed within an ascus. Since this occurs during nitrogen starvation, all new proteins are made from existing nitrogen soruces. A large amount of protein turnover occurs during sporulation and the total protein of the culture remains fairly constant. Extensive degradation occurs to provide a source of amino acids for the synthesis of new proteins. By differential labelling of vegetative and sporulating yeast cells it has been shown that proteins in both the vegetative cells and the spores are degraded.[90] Two endoproteases, (A and B), one carboxypeptidase (Y) and two or more aminopeptidases are present in yeast cells. There are also specific inhibitors for some of the proteases but these are located in the cytosol, whereas the proteases are contained within vacuoles. The activities of proteases A and B increase several-fold during sporulation; the activity of proteases A appears essential for sporulation, since mutants lacking this enzyme are unable to sporulate. A number of enzymes have been shown to decrease their specific activities to less than 50 per cent of their initial value,[90] e.g. glutamate dehydrogenase ($NADP^+$-requiring), isocitrate lyase, malate dehydrogenase, cytochrome *c* oxidase, succinate dehydrogenase, fructose-bisphosphatase, aspartate aminotransferase, alcohol dehydrogenase, and hexokinase. It is not yet clear whether these changes are due to proteolysis, although this seems most probable. As can be seen, many enzymes that decrease in activity are those one might expect to accompany a decrease in metabolic activity.

Proteolysis occurs in bacteria during sporulation. This has been studied in *Bacillus subtilis*, where it has been shown that selective inactivation of certain enzymes occurs, e.g. aspartate carbamoyltransferase and amidophosphoribosyltransferase. This is due to inactivation followed by proteolysis.[91] (For review see reference 92.)

Large increases in proteolytic activity have been found associated with particular stages of development in many organisms, e.g. during germination and sporulation in the water mould,[93, 94] *Blastocladiella emersonii*, in the germination of macroconidia of *Microsporum gypseum*, and in the regression of the salivary glands of the insect larval instar of *Chironomus tentans*, but in the developmental systems studied up until now the rates of synthesis and degradation of individual enzymes have not yet been determined. Thus, more research is needed before the control of protein turnover during development is understood.

9.9 Conclusions

Enzyme turnover is an important method of regulation of enzyme activity and is controlled by regulation of either enzyme synthesis or enzyme inactivation/degradation, or in some cases by regulation of both steps. The general mechanism of enzyme synthesis is well understood, although its control has only been studied in depth in a limited number of prokaryote systems. The mechanism of enzyme inactivation is less well understood. A number of correlations with structure and function and with particular sequences of individual enzymes have been made. Many enzymes appear to become tagged before becoming susceptible to protease attack. The methods of tagging include conjugation with ubiquitin, phosphorylation, oxidation, or binding to membranes, but it is still not generally clear precisely which features of the enzymes are recognized by the tagging system, although some clues are beginning to emerge.

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Appendix 9.1.

Derivation of equation describing the time course of approach to a new steady-state level of enzyme after a change in its rate of synthesis and/or degradation has occurred.

The steady-state level of an enzyme in a cell or tissue is given by the equation:

$$\frac{d[E]}{dt} = k_s - k_d[E] \qquad \text{(see Section 9.3)}$$

where k_s is the rate constant for its synthesis (zero order with respect to [E]) and k_d is the rate constant for its degradation (first order with respect to [E]).

If there is a change in the rate of synthesis and/or degradation of the enzyme as a result of some hormonal, dietary, etc. influence then the enzyme concentration will change to a new steady-state level.

Let

k'_s be the new rate constant for enzyme synthesis,
k'_d be the new rate constant for enzyme degradation,
$[E_0]$ be the initial steady-state concentration,
$[E_t]$ be the enzyme concentration at time t after change in rate constants from k_s to k'_s and k_d to k'_d.

$$\frac{d[E]}{dt} = k'_s - k'_d[E] \tag{A.9.1}$$

$$\frac{d[E]}{(k'_s - k'_d[E])} = dt. \tag{A.9.2}$$

By integration of eqn (A.9.2),

$$\ln(k'_s - k'_d[E]) = -k'_d t + c. \tag{A.9.3}$$

At time $t = 0$, $[E] = [E_0]$, therefore $c = \ln(k'_s - k'_d[E_0])$. Thus at time t,

$$\ln(k'_s - k'_d[E_t]) - \ln(k'_s - k'_d[E_0]) = -k'_d t. \tag{A.9.4}$$

Taking antilogarithms of eqn (A.9.4):

$$\frac{k'_s - k'_d[E_t]}{k'_s - k'_d[E_0]} = e^{-k'_d t}. \tag{A.9.5}$$

This can be rearranged to give

$$\frac{[E_t]}{[E_0]} = \frac{k'_s}{k'_d[E_0]} - \left(\frac{k'_s}{k'_d[E_0]} - 1\right) e^{-k'_d t}.$$

See also reference 18.

10
Clinical aspects of enzymology

10.1 Introduction

Over the last thirty years there has been a considerable increase in both the measurement of enzyme activities and in the use of purified enzymes, in clinical practice. Before 1940 the only enzymes whose activities were measured for clinical diagnoses were the hydrolytic enzymes: lipase, amylase, phosphatases, trypsin, and pepsin and their measurement constituted less than 5 per cent of the analyses carried out in the average clinical chemistry laboratories. At the present time up to 25 per cent of the work of an average clinical chemistry laboratory may consist of enzyme assays for diagnoses and up to about 20 different enzymes are assayed routinely. The reason for the change in emphasis is largely due to increased understanding and awareness of the molecular details of metabolism and to the development of rapid and reliable methods of enzyme assay. The discovery in the 1950s of the relationship between the activities of serum aminotransferases and dehydrogenases and heart and liver diseases gave the impetus to search for other enzymes present in serum that could be of diagnostic value. In addition, during the last twenty-five years many enzymes have been isolated and purified (see Chapter 2) and this has made it possible to use enzymes to determine the concentrations of substrates of clinical importance. The availability of highly specific antibodies, often from single clones (monoclonal antibodies), in conjunction with purified enzymes to amplify detection has enabled a wide range of biological compounds to be measured by enzyme immunoassay. A further development arising from the increased availability of purified enzymes is the potential for enzyme therapy (see Section 10.8).

At the present time the major importance of enzymes in clinical practice is in the measurement of enzyme levels in serum as an aid to diagnosis and we shall therefore discuss this first (Sections 10.2–10.6). In Section 10.7 we shall describe the use of enzymes as a means of estimating concentrations of substrates, and finally in Section 10.8 we consider enzyme-replacement therapy. There are a number of books and reviews on clinical applications of enzymology in which more detailed information can be obtained.[1-7]

10.2 Determination of enzyme activities for clinical diagnosis

For the measurement of an enzyme activity to be useful as a routine diagnostic clinical method the following conditions should be fulfilled.

1. The enzyme should be present in blood, urine, or some readily available tissue fluid. Tissue biopsies should not be used routinely, although they may be used if the diagnostic value is sufficiently important.
2. The enzyme should be easy to assay and it is advantageous if the method can readily be automated.
3. The differences in the ranges of enzyme activities obtained from normal and diseased subjects should be diagnostically significant and there should be a good correlation between the level of enzyme activities and the pathological state.
4. It is also useful if the enzyme is sufficiently stable so that the sample may be stored at least for a limited time.

Serum is the fluid in which most measurements of enzyme activities are made. Urine can be used for a few enzymes that are cleared by the kidney. Red blood cells and white blood cells, although they are relatively accessible, have not so far been used extensively in diagnosis. The enzymes present in serum can be considered in one of two categories: (i) plasma-specific enzymes and (ii) non-plasma-specific enzymes. The former are enzymes whose normal function is in the plasma, e.g. enzymes concerned with blood coagulation,[8,42] complement activation,[9,43] and lipoprotein metabolism. The latter are enzymes that have no physiological function in the plasma, enzymes for which the cofactors or even substrates may be lacking. This category includes enzymes that are secreted by tissues, e.g. amylase, lipase, and phosphatases, and also enzymes associated with cellular metabolism, whose presence in normal serum at low levels may be due to turnover of cells within the tissue causing release of the enzyme content. The rationale of enzyme measurement for diagnosis is that if a tissue is broken down or if it is producing an abnormally high amount of intracellular enzyme that is then released, there is an elevation of enzyme activity in the serum. For several enzymes there is a very high concentration gradient (a factor of the order of 10^3 or greater) across the cell membrane to the extracellular fluid and thus a small amount of tissue damage may affect considerably the serum concentration of certain enzymes.[6]

In the diseased state a tissue may become inflamed or it may become necrotic.* If the latter occurs there is likely to be fairly complete release of the enzyme content from the dead cells. However, the pattern of enzymes detectable in the serum may not completely resemble that of the tissue from which it arises, since enzymes may be inactivated at different rates. Inflammation of a tissue may result in a change in the permeability of the cells, so that release of enzymes may occur from the cytosol but not from organelles,[10] e.g. glutamate dehydrogenase, which is present in high concentrations in the mitochondrial matrix, is released only when cell destruction is fairly complete.[11] The enzymes released into the serum seem to be removed at a fairly rapid rate. The mechanism of their removal has not been studied appreciably: it may be due to inactivation and degradation occurring in the serum or it may be due to clearance by the kidney. Some

* Necrosis is the death of a portion of tissue or organ.

experiments, involving intravenous injections of ^{125}I-labelled lactate dehydrogenase (LDH-5) into rabbits, have been carried out to study the fate of serum enzymes. The results suggest that LDH-5 undergoes denaturation in the plasma and that the products are excreted into the small intestine where they are further degraded and then reabsorbed, as amino acids and small peptides, back into the circulation.[12] The quite rapid removal of most released enzymes means that monitoring particular enzymes present in the serum of a diseased subject gives an up-to-date picture of the release of enzymes from diseased tissue. Monitoring enzymes in the serum is thus useful not only in the initial diagnosis but also in studying the course of the disease and the response to treatment.

Ideally it would be desirable to study tissue-specific enzymes for diagnostic purposes, since these would identify the tissue from which they arose. However, there are relatively few enzymes that are entirely tissue specific, although there are several that are much more abundant in one tissue than another, e.g. acid phosphatase in the prostate and acetylcholinesterase (EC 3.1.1.7) in erythrocytes. Although a particular enzyme activity may not be specific to one tissue, there may exist isoenzymes (see Chapter 1, Section 1.5.2) that show a different pattern in different tissues. The best-studied example is that of lactate dehydrogenase (further information on lactate dehydrogenase is given in Chapter 5, Section 5.5.4). Lactate dehydrogenase is composed of four subunits. There are two types of subunit, which combine in lactate dehydrogenase to give five possible combinations: α_4, $\alpha_3\beta$, $\alpha_2\beta_2$, $\alpha\beta_3$, and β_4. These five types may be separated by electrophoresis and it has been found that their distribution varies with the tissue (Fig. 10.1). Thus, although by measurements of lactate dehydrogenase activity in serum the tissue of origin could not be identified, identification might be possible

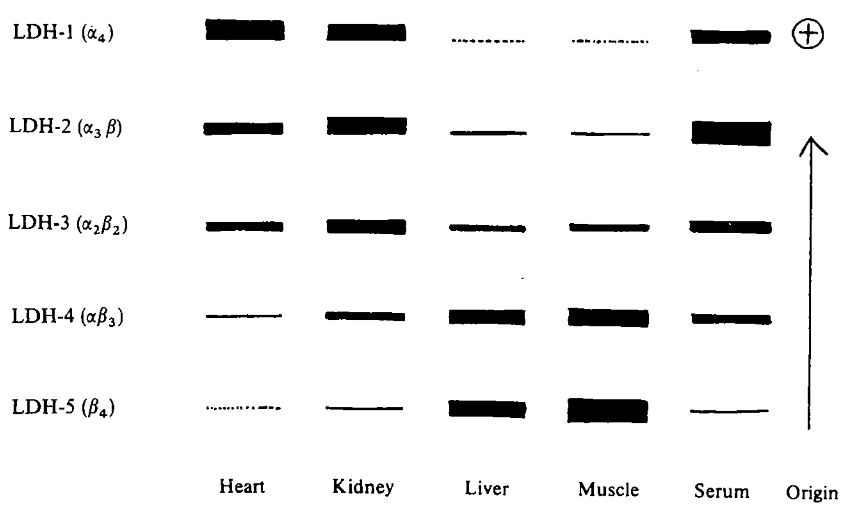


FIG. 10.1. Electrophoretic separation of isoenzymes of lactate dehydrogenase from different tissues.

if the isoenzyme distribution were determined by electrophoresis. Several enzymes that may be present in serum are known to exist in multiple forms, e.g. alkaline phosphatase, amylase, creatine kinase, ceruloplasmin, glucose 6-phosphate dehydrogenase, and aspartate aminotransferase, but these isoenzymes have not yet been as well characterized as those of lactate dehydrogenase. Some isoenzymes may be differentiated by criteria other than electrophoretic mobility, e.g. by substrate specificity, heat stability, or sensitivity to inhibitors (Table 10.1). It is now possible in some cases to distinguish isoenzymes by use of monoclonal antibodies. This method has been used to distinguish the different isoenzymes of human phosphofructokinase, and to identify which form is absent in inherited deficiencies of phosphofructokinase.[42] Monoclonal antibodies against specific isoenzymes may in future be used for the detection and quantitation of isoenzymes used as tumour markers, such as prostatic acid phosphatase, placental alkaline phosphatase, and terminal deoxynucleotide transferase.

In many cases where measurements of enzyme activity are used as an aid to diagnosis, more than one activity is measured in order to permit a greater degree of discrimination. Perhaps the two examples in which enzyme monitoring has been most useful and widely applied are those of liver and heart diseases. It must be emphasized that the measurement of enzyme activity itself is not sufficient evidence on which to make a diagnosis but must be taken in conjunction with the

TABLE 10.1
Methods used to determine the tissue of origin of serum enzymes

Enzyme	Differentiation by electrophoresis	Differentiation by other methods
Lactate dehydrogenase	LDH-1 predominant in heart, erythrocytes, brain and kidney	Differential heat stability, LDH-4 and LDH-5 show greater heat lability.
	LDH-3 predominant in leucocytes, adrenal and thyroid	LDH-1 is more sensitive to inhibition by excess pyruvate than LDH-5
	LDH-5 predominant in skeletal muscle and liver	
Acid phosphatase		Prostate enzyme inhibited by tartrate. Erythrocyte enzyme inhibited by formaldehyde. Differential activities towards substrates, phenyl phosphate, glycerol 2-phosphate and thymolphthalein phosphate
Alkaline phosphatase	Complex electrophoretic patterns. Liver enzyme most cationic	Placental enzyme shows greater heat stability. Enzymes from both placenta and intestine are sensitive to phenylalanine
Creatine kinase	Brain, lung and thyroid mostly CK-1 (or BB). Skeletal muscle almost exclusively CK-3. Cardiac tissue CK-3+ about 15% CK-2	

See references 1, 2.

clinical symptoms and other types of evidence, e.g. electrocardiographic evidence in the case of heart disease.

Increased activities of certain enzymes in serum are commonly found in malignant diseases, but no enzyme is specific for cancer and so far none has been very useful in the early detection of cancer.[13]

Besides the use of isoenzyme patterns to identify the tissue of origin, isoenzyme patterns are also used in forensic science. Since a number of enzymes in serum and red blood cells occur as different isoenzymes, the particular pattern in a blood sample may provide supporting evidence as to its origin. Isoenzyme patterns that are currently used by the Metropolitan Police Science Laboratory are adenosine deaminase, adenylate kinase, carbonate dehydratase, acid phosphatase, esterase, phosphoglucomutase, aminopeptidase and lactoylglutathione lyase.[14]

10.3 Clinical enzymology of liver disease

The enzymes most used in diagnosis of liver disease are serum aspartate aminotransferase (often referred to as serum glutamate oxaloacetate transaminase, SGOT), alanine aminotransferase (often referred to as serum glutamate pyruvate transaminase, SGPT), alkaline phosphatase, and γ-glutamyltransferase. Others that are also sometimes measured are lactate dehydrogenase and isocitrate dehydrogenase. Ornithine carbamoyltransferase is an enzyme that is exclusive to the liver, but it has not been widely used as a diagnostic aid because the assay is complex compared to those of other liver enzymes. The aminotransferases that catalyse the following reactions are usually assayed by coupling to the reactions catalysed by malate and lactate dehydrogenases so that the change in A_{340} due to oxidation of NADH can be measured continuously (see Chapter 4, Section 4.2).

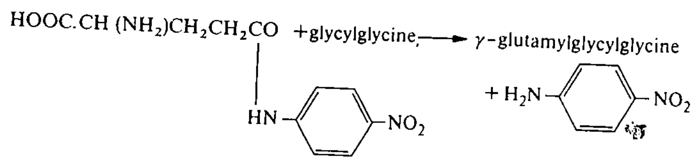
1. Aspartate + 2-oxoglutarate $\rightleftharpoons$ oxaloacetate + glutamate

   Oxaloacetate + NADH + H^+ $\rightleftharpoons$ malate + NAD^+.

2. Alanine + 2-oxoglutarate $\rightleftharpoons$ pyruvate + glutamate.

   Pyruvate + NADH + H^+ $\rightleftharpoons$ lactate + NAD^+.

Alkaline phosphatase is assayed by measuring the hydrolysis of p-nitrophenyl phosphate. γ-Glutamyltransferase is assayed by measuring the release of p-nitroaniline when γ-glutamyl-p-nitroanilide is used as one of the substrates. p-Nitroaniline absorbs at 400 nm.



The most commonly occurring liver diseases are viral or toxic hepatitis, cirrhosis, primary and secondary liver tumours, and obstruction of the bile-duct due to stones, tumours, or stricture. The conditions may be acute, intermittent, or chronic. The two aminotransferases are in the category of non-plasma-specific enzymes, i.e. they are enzymes concerned with intracellular metabolism and have no known function in the plasma. They are released from the liver when parenchymal cells (the principal cells of the liver) become necrotic as in, for example, viral or toxic hepatitis or cirrhosis; the increase in their activity in the plasma is related to the extent of cell breakdown. Aminotransferases in serum may also arise from cardiac tissue after myocardial infarction, but the proportions of asparate aminotransferase and alanine aminotransferase differ. This ratio was used to differentiate between various conditions, although it is only rarely used now. In cases of chronic hepatitis or cirrhosis, aminotransferases are often monitored over a period of serveral months to follow the progress of the disease and to provide an indication of the prognosis.

Measurements of serum alkaline phosphatase are also important in diagnosis of liver disease. Serum alkaline phosphatase arises from two main sources, namely, the liver and bone; the enzymes from these sources differ in their isoenzyme patterns. Liver alkaline phosphatase is a non-plasma-specific enzyme that is secreted from the sinusoidal surface (i.e. that adjacent to the blood space) of the liver cell and thus is present in the serum at low levels in the absence of liver cell damage. It is possible to differentiate obstructive jaundice from hepatitis by measurement of aminotransferases and alkaline phosphatase, since there is a proportionately greater increase in alkaline phosphatase activity in obstructive jaundice than in hepatitis, so that the ratios of activities of serum alkaline phosphatase/serum alanine aminotransferase one week after the onset of viral hepatitis might be 0.2–0.6 compared with 10–20 one week after the onset of obstructive jaundice.[4] Whether the bile-duct obstruction is intermittent or complete can be assessed by monitoring the alkaline phosphatase activity over a period of a few days (Fig. 10.2). Glutamate dehydrogenase can also be used in place of alkaline phosphatase to distinguish hepatic and biliary disease. The ratio of glutamate dehydrogenase/alanine aminotransferase is higher in cases of obstructive jaundice than, for example, in viral hepatitis.[11]

γ-Glutamyltransferase is now also being used extensively in the diagnosis of liver disease, since most of this enzyme activity detectable in the serum is of hepatic origin. The activity is greatly elevated in the serum in cases of chronic alcoholism, when it causes cirrhosis of the liver, and in cases of metastatic invasion of the liver. The increased serum level in cases of chronic alcoholism is due to both increased synthesis in the liver and increased release into the serum. γ-Glutamyltransferase activity follows a similar pattern to that of alkaline phosphatase in obstructive jaundice.

10.4 Clinical enzymology of heart disease

Enzymological methods have been particularly useful in providing supporting evidence for myocardial infarction (the process of formation of an area of dead heart muscle) and also monitoring the course of the infarct. In myocardial

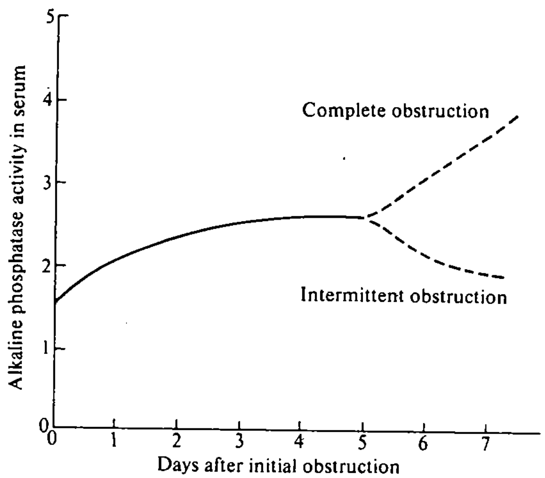


FIG. 10.2. Typical changes in the activity of serum alkaline phosphatase after complete and intermittent obstructive jaundice (see reference 4.)

infarction a coronary artery becomes obstructed and this leads to irreversible damage and necrosis of the heart tissue. Enzymes are released from the necrotic tissue into the plasma; the three enzymes most commonly assayed are creatine kinase, aspartate aminotransferase, and lactate dehydrogenase. Each enzyme shows a different time course for release into the plasma and subsequent disappearance. These differences depend on the concentration of each present in cardiac tissue, their rate of release, and their subsequent rate of clearance or degradation.[1,4] Their half-lives in the plasma are creatine kinase ≈ 15 h, asparate aminotransferase ≈ 17 h, and lactate dehydrogenase ≈ 110 h. Creatine kinase is the earliest to be detectable, rising 4–6 h after the onset of pain, reaching a peak at 24–36 h and then rapidly declining. Aspartate aminotransferase reaches a peak between 48 and 60 h and lactate dehydrogenase between 48 and 72 h; the latter declines more slowly than the former (Fig. 10.3). The prognosis and progress can be assessed by the increase in the activities of these enzymes. Creatine kinase, which does not occur in the liver unlike aspartate aminotransferase and lactate dehydrogenase, can therefore be used to differentiate heart disease from liver disease. However, since the level of creatine kinase rises and falls rapidly it is sometimes safer to measure aspartate aminotransferase or lactate dehydrogenase, but in other situations creatine kinase activity can be used in the early detection and prognosis of myocardial infarction.

Creatine kinase can be particularly useful if its estimation is accompanied by determination of the isoenzyme pattern. Creatine kinase derived from skeletal muscle, a common source of creatine kinase in serum, consists almost exclusively of the CK-3(MM) type, whereas creatine kinase derived from heart tissue,

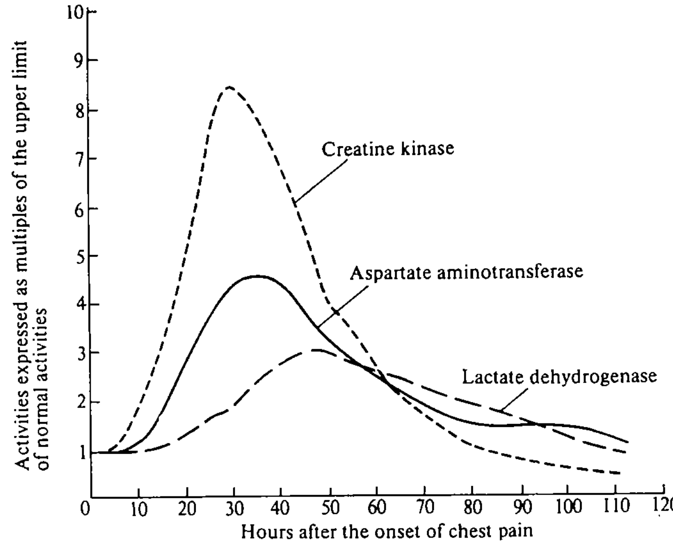


FIG. 10.3. Typical changes in the activities of serum creatine kinase, aspartate aminotransferase (SGOT), and lactate dehydrogenase following myocardial infarction. Activities are expressed as multiples of the upper limit of the normal level (see reference 1).

although predominantly CK-3(MM), also contains about 15 per cent CK-2(MB). Negligible CK-2 activity occurs outside heart tissue and thus it can be used as a specific indicator of damage to heart tissue in normal subjects.[15,16] An antibody against the M subunit is sometimes used to inhibit activity of the M subunit, and any remaining creatine kinase activity in serum will then be due to the B subunit of CK-2 and the CK-1(BB). Lactate dehydrogenases from liver and heart may also be distinguished by their isoenzyme patterns (Fig. 10.1).

Measurement of both serum aspartate aminotransferase and alanine aminotransferase can be used to differentiate between liver and heart disease. The ratio of activities of aspartate aminotransferase/alanine aminotransferase in heart tissue (20–25) is higher than the ratio in liver (3–5) and this is reflected in the ratios of enzymes released into the serum. However, this ratio of activities in the serum is not now widely used, because the measurement of, for example, creatine kinase or γ-glutamyltransferase is generally regarded as a more reliable method of distinguishing heart and liver diseases.

10.5 Other enzyme activities that become elevated in serum in disease

A number of other enzymes are assayed in serum and urine for diagnostic purposes; the more frequently measured ones are discussed below.

10.5.1 α-Amylase

α-Amylase is an endoamylase that hydrolyses the 1,4-α link in amylose and amylopectin. In humans the enzyme occurs in a variety of tissues, but the highest concentrations are in the pancreas and in the salivary glands. Low amylase activities can be detected in the serum and urine of normal healthy subjects. The most pronounced increases in serum and urinary amylase occur in acute pancreatitis, when the activities may rise to 20–30 times the normal levels. In other pancreatic diseases, e.g. chronic pancreatitis or carcinoma of the pancreas, serum and urinary amylase activities may be raised, but only slightly. Although the highest concentration of amylase is in the pancreas, the increased excretion of amylase may occur in a number of intra-abdominal disorders in which the tissue affected is in close proximity to the pancreas, e.g. gastric or duodenal ulcers, peritonitis, and mesenteric thrombosis. Amylase activity in urine may also be increased in mumps when the salivary glands become inflamed.

An unusual feature of amylase secretion compared with other enzymes is that it is readily detected in urine. Only enzymes having M_r values less than about 55 000 are cleared by the kidney and appear in urine. Most enzymes of importance in clinical diagnosis have M_r values greater than 70 000; two exceptions are uropepsinogen and α-amylase (49 000).[17]

During an attack of acute pancreatitis the level of α-amylase in the serum is usually raised after one day and falls on the second or third day; the level of the the enzyme in urine, on the other hand, may be raised earlier and persist longer.

10.5.2 Creatine kinase and fructose-bisphosphate aldolase

These two enzymes are useful as indicators of skeletal muscle disorders. Creatine kinase occurs predominantly in skeletal muscle, cardiac muscle, and brain, but in very low activities in all other tissues. The serum activities are raised in all forms of muscular dystrophy, the level depending on the mass of muscle that is becoming dystrophic. Although creatine kinase activity is also raised in myocardial infarction, the rise is short-lived and there would be no likelihood of confusing it with skeletal muscle defects; in addition, the isoenzyme patterns differ (see Section 10.4 and references 15, 16).

Fructose-bisphosphate aldolase is also present in high concentrations in skeletal muscle and thus could be used to support the conclusions reached from the assay of creatine kinase. However, fructose-bisphosphate aldolase is also present in many other tissues, though not at such high concentrations. Particularly important in this respect is the high fructose-bisphosphate aldolase content of red blood cells; it is therefore essential that a sample of serum to be assayed does not contain lysed red blood cells.

10.5.3 Alkaline phosphatase

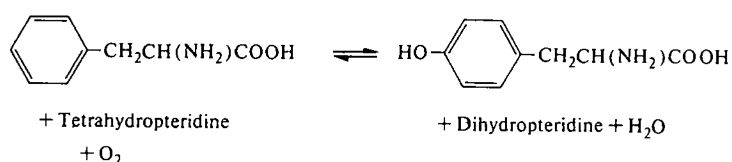
Alkaline phosphatase is a phosphatase having low specificity, hydrolysing a variety of organic phosphate esters, and having a pH optimum between 9 and 10. Its activity is detectable in a number of different tissues notably intestinal epithelium, kidney tubules, bone (osteoblasts), leucocytes, liver, and placenta. The true substrates of the enzyme are not known and it is usually assayed with p-nitrophenyl phosphate. Apart from its application in the diagnosis of obstructive jaundice, it is also very useful in the detection of bone disease. Its activity in serum is raised in those bone diseases in which there is increased activity of the bone-forming cells or osteoblasts as in, e.g. osteomalacia, rickets, Paget's disease, and hyperparathyroidism. The alkaline phosphatase present in normal serum is believed to be derived mainly from the liver cells. It is useful to be able to differentiate between increased activities in the serum due to secretion from liver, from bone, or from other tissues. The isoenzyme patterns of alkaline phosphatase from various tissues are quite complex, multiple bands being obtained from most tissues. For this reason differentiation by electrophoresis is not an easy method to use in a routine investigation. Other properties of the alkaline phosphatases from different tissues that have been useful are the effects of heat and various inhibitors (see Table 10.1). The results suggest that alkaline phosphatases are of three distinctive types, namely, placental, intestinal, and those from other tissues. There is an increase in alkaline phosphatase activity during the third trimester of pregnancy and this is due to release of placental enzyme. The latter can readily be differentiated from the bone and liver enzymes because it is quite stable to heating for 30 min at 329 K (56 °C) while the others are labile.[41]

10.5.4 Acid phosphatase

This enzyme, like alkaline phosphatase, is a phosphatase of low specificity and the natural substrates are unknown. The pH optimum is between 4.0 and 5.5. The enzyme occurs in a variety of tissues, e.g. liver, spleen, erythrocyte, and prostate. The highest concentration is present in the prostate and detection of prostatic carcinoma is the main purpose of clinical assays of this enzyme. Approximately one-third of the normal circulating acid phosphatase in adult males is derived from the prostate. The enzyme is very labile and must be assayed as soon as possible after collecting the serum. It is clearly important to have an early and unequivocal diagnosis of prostatic carcinoma; however, small deviations in the level of acid phosphatase in the serum could be due to release from other tissues or lysis of erythrocytes during serum collection. The erythrocyte and prostatic enzymes can be differentiated by their differing sensitivities towards formaldehyde and tartrate: 0.5 per cent formaldehyde almost completely inhibits the erythrocyte enzyme but hardly affects that of the prostatic enzyme, whereas 0.01 mol dm^{-3} tartrate has almost no effect on the erythrocyte enzyme whilst inhibiting the prostatic enzyme. Most recently, however, it has been found that the best substrate for the prostatic enzyme is thymolphthalein phosphate, which is only very poorly attacked by the erythrocyte enzyme and thus can be used to distinguish these enzymes.

10.6 The detection and significance of enzyme deficiencies

The previous section has been concerned with the detection of elevated enzyme activities in serum and urine and the diagnosis of the associated diseases. These represent the large majority of enzyme assays performed for clinical diagnosis. Much less frequently encountered are those subjects who suffer from enzyme deficiencies due to inborn errors in metabolism. More than 140 different inborn errors in metabolism are now known although none of them occur very frequently, e.g. phenylketonuria, 1 in 15 000 births (about 40 births per year in the UK);[18] cystinuria, 1 in 14 000; galactosaemia, 1 in 33 000; maple syrup disease, 1 in 330 000; alcaptonuria, 1 in 200 000.[19] In many cases the enzyme deficiency has been established (Table 10.2), although it is not generally known whether it is a mutation in the genome that has led to the production of a defective enzyme or failure to produce enzyme at all. The initial identification of the disease is rarely the result of an enzyme assay and in many cases the presence of a metabolite is used in detection of the disease; thus, phenylketonuria is usually detected by measurement of phenylpyruvate in urine or phenylalanine in blood. The ultimate determination of the enzyme defect may require a tissue biopsy. Some inborn errors in metabolism are relatively harmless, e.g. albinism, alcaptonuria, but others must be detected early if the defect is to be circumvented. Two important ones that fall in the latter category are phenylketonuria and galactosaemia. In phenylketonuria there is a lack of phenylalanine 4-monooxygenase, an enzyme required for the breakdown of phenylalanine:



Phenylalanine is being produced continuously in a normal subject as a result of protein turnover and dietary intake and is largely broken down via tyrosine to homogentisic acid. In the phenylketonuric subject, phenylalanine has to be degraded by a minor pathway via phenylpyruvic acid (Fig. 10.4).

Phenylketonuria is associated with mental retardation: it is not clear whether this is due to the build up of high concentrations of phenylalanine or its metabolites. The mental disturbance can be prevented if the subjects from birth to 7 years are fed a diet with a considerably reduced phenylalanine content, although there is evidence that the diet must be maintained continuously throughout adulthood to prevent the accumulation of deleterious effects.[21] Special diets with low-protein breads, cereals and vegetable fats have been compiled.[22] Early detection is thus very important and infants are now screened at birth in many countries for phenylketonuria.

TABLE 10.2

Some examples of inborn errors in metabolism due to enzyme deficiencies

Inborn error	Enzyme deficiency
Alcaptonuria	Homogentisate 1,2-dioxygenase
Phenylketonuria	Phenylalanine 4-monooxygenase
Cystinuria	(Renal and intestinal transport defect of cysteine, ornithine and lysine)
Maple syrup disease	Branched-chain oxo-acid oxidative decarboxylases
Galactosaemia	UDPglucose-hexose-1-phosphate uridylyltransferase
Glycogen storage diseases	Glucose-6-phosphatase, α-D-glucosidase, debranching enzyme, liver or muscle phosphorylase
Pentosuria	L-Xylulose reductase
Tay–Sachs disease	β-*N*-acetyl-D-hexosaminidase
Nieman–Pick disease	Phospholipase C
Gangliosidosis (generalized)	β-D-Galactosidase
Acatalasia	Catalase

For a more complete list see references 18–20.

In galactosaemia the deficient enzyme is galactose 1-phosphate uridylyltransferase. As can be seen from Fig. 10.5, this results in a block in the conversion of galactose 1-phosphate to glucose 1-phosphate. The deficiency results in a build up of high intracellular concentrations of galactose 1-phosphate, a high level of galactose in body fluids, and galactose excretion in urine. The pathological consequences, mental retardation, slow weight gain, enlarged liver, and cataracts are generally attributed to the high intracellular concentrations of galactose 1-phosphate. The harmful effects can be avoided by reducing the galactose content of the diet from birth. (Lactose present in milk is the principal source of galactose in infants.) Galactose 1-phosphate uridylyltransferase is normally present in many human tissues including erythrocytes. Thus, galactosaemia may be detected by assaying lysed erythrocytes for the enzyme.

The most frequently occurring hereditary disease known is glucose-6-phosphate dehydrogenase deficiency, in which the erythrocytes are affected. It is more common in Negros and in Sephardic Jews than in other races. The disease is manifested as haemolytic anaemia and occurs after the subjects have been treated with antimalarial drugs such as pamaquine or primaquine, or have received other drugs such as sulphonamides or aspirin, or have eaten fava beans. Erythrocytes metabolize glucose using the glycolytic and pentose phosphate pathways. The agents that cause haemolysis appear to inhibit glucose-6-phosphate dehydrogenase, which is present only in low levels in these subjects. This in turn causes reduction in the intracellular NADPH concentration and the latter is required for glutathione reductase. It is thought that haemolysis is

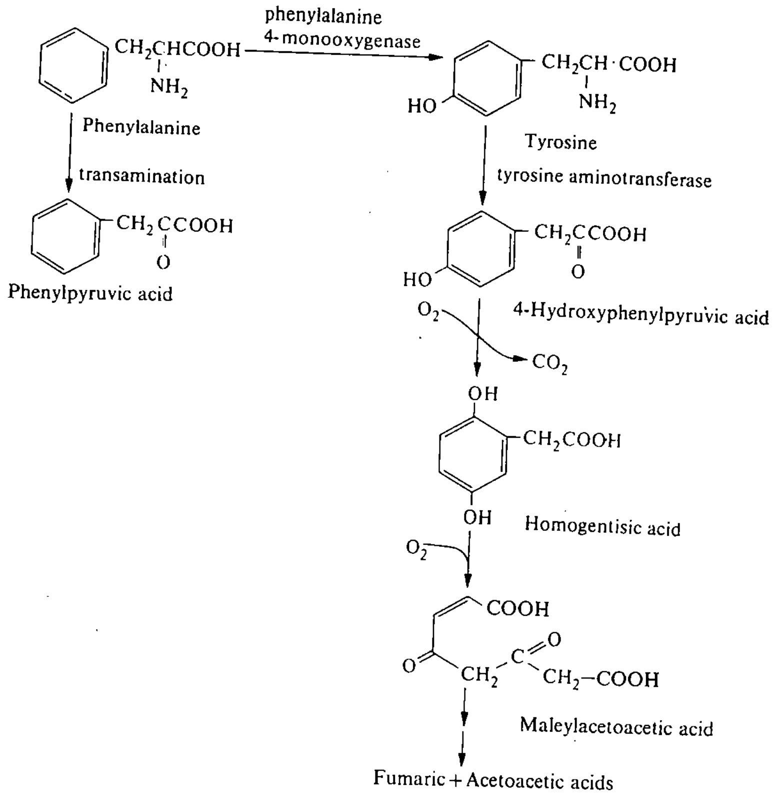


FIG. 10.4. Metabolic pathway for the degradation of phenylalanine in humans.

induced by lowering of the levels of glutathione:

$$\text{D-Glucose 6-phosphate} + NADP^+ \rightleftharpoons \text{D-glucono-}\delta\text{-lactone 6-phosphate} + NADPH + H^+.$$

$$H^+ + NADPH + \text{oxidized glutathione} \rightleftharpoons NADP^+ + 2\ \text{glutathione}.$$

Another example of drug sensitivity being associated with a genetically inherited deficiency occurs in the case of serum cholinesterase (EC 3.1.1.8) deficiency. Cholinesterase is normally present in serum at quite high levels. The enzyme is considered to be synthesized by the liver and lower levels of activity are detected in serum in certain liver diseases, e.g. acute hepatitis and advanced carcinoma of the liver, when the rate of synthesis of the enzyme is reduced. The enzyme is sometimes referred to as pseudocholinesterase since it hydrolyses a variety of esters of choline, in contrast to the acetylcholinesterase (EC 3.1.1.7) of nervous tissue and erythrocytes, which has a more restricted specificity. The

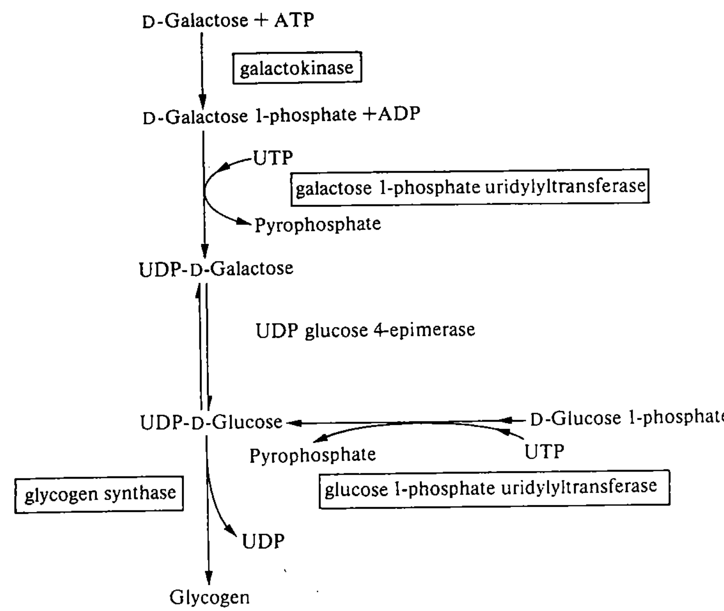


FIG. 10.5. The pathways of galactose metabolism.

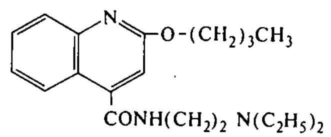
$$\text{Acylcholine} + H_2O \xrightleftharpoons{\text{cholinesterase}} \text{choline} + \text{carboxylate anion}$$

relative deficiency of cholinesterase in some individuals became apparent as a result of the use of suxamethonium (succinyl choline) as a muscle relaxant in surgery.

$$(CH_3)_3N^+(CH_2)_2O_2C(CH_2)_2CO_2(CH_2)_2N^+(CH_3)_3$$

Succinyl choline

In normal subjects the effects of suxamethonium are short-lived and this is due to its breakdown by serum cholinesterase. However, in certain individuals the effects of the drug were much more long-lasting; this was associated with low cholinesterase levels. When the cholinesterase from suxamethonium-sensitive individuals was further studied it became apparent that the enzyme differed from that of normal individuals in having a high K_m with all substrates tested. This enabled a simple screening test to be carried out on serum to determine whether a subject is likely to be sensitive to suxamethonium. The test involves determining the percentage inhibition of cholinesterase by a particular concentration of dibucaine measured under standard conditions. The cholinesterase from suxamethonium-sensitive individuals is inhibited by only about 20 per cent under conditions that reduce the normal enzyme activity by about 80 per cent.[18]



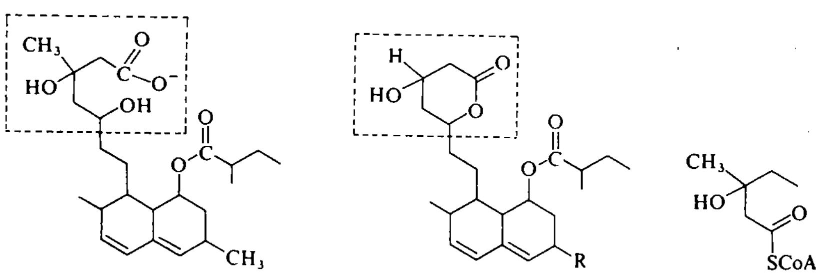
Dibucaine

The diseases described so far in this section are all genetically inherited diseases in which there is either a deficiency or a change in properties of a particular enzyme or protein. An important genetically inherited disease that involves a protein defect, though not an enzyme defect, is familial hypercholesterolaemia. It is included in this section because one of the successful treatments for the disease involves the use of a competitive inhibitor. Familial hypercholesterolaemia is characterized by high circulating cholesterol and cholesterol ester levels in blood. Subjects homozygous for the disease may have 4–6 times the normal circulating levels of cholesterol, and generally have symptoms of coronary disease in childhood or adolescence, with the deposition of arterial plaques. The manifestation of the disease is less severe in the more common heterozygous state, with cholesterol and cholesterol ester levels of about twice the normal range and increased frequency of coronary disease at middle age. The condition has been shown to arise from a deficiency in the low-density lipoprotein receptors that are responsible for the uptake of cholesterol and cholesterol esters by the tissues: hence the higher circulating levels of cholesterol and cholesterol esters.

It has been found that the circulating levels of cholesterol and cholesterol esters can be reduced, and therefore so can the tendency to form arterial plaques, if the subjects are given a competitive inhibitor of hydroxymethylglutaryl-CoA reductase (HMG CoA reductase). HMG CoA reductase catalyses the rate-limiting step in the formation of cholesterol. It is an enzyme having a short half-life (see Chapter 9, Section 9.6).

$$\text{HMG CoA} + 2\text{NADPH} + 2\text{H}^+ \rightarrow \text{mevalonate} + \text{CoASH} + 2\text{NADP}^+.$$

Three competitive inhibitors[23] of HMG CoA reductase have been found, namely, mevinolin, compactin, and monacol K, which can be used to inhibit the enzyme. All three are fungal products and have a structural resemblance to HMG CoA. Mevalonate is a precursor not only of steroids but also of a number of other important cell constituents such as ubiquinone, dolichol, and isopentenyl adenine (present as one of the modified bases in tRNA). However, sterol biosynthesis appears to be selectively inhibited at low concentrations.



Mevinolin | Compactin R = H | Monacol K R = CH$_3$ | HMG CoA

Competitive inhibitors of HMG CoA reductase; the regions resembling HMG CoA are outlined.

The last example in this section is rather different since it is a case where an enzyme assay is used to detect a vitamin deficiency. The enzyme transketolase, which catalyses the following reaction, requires thiamin diphosphate as a cofactor for activity:

$$\text{D-Ribose 5-phosphate} + \text{D-xylulose 5-phosphate} \rightleftharpoons$$
$$\text{sedoheptulose 7-phosphate} + \text{D-glyceraldehyde 3-phosphate}.$$

Normally thiamin deficiency rarely occurs in developed countries, where there is very little malnutrition. However, the deficiency is common in areas where there is severe malnutrition and it also occurs in the case of Wernicke's encephalopathy. This syndrome occurs in alcoholics, who become thiamin-deficient as a result of consuming little other than alcohol. The early stages of this syndrome may be detected as thiamin deficiency. Thiamin is normally converted into thiamin diphosphate, the cofactor for transketolase and also pyruvate and oxoglutarate dehydrogenases:

$$\text{Thiamin} + \text{ATP} \rightleftharpoons \text{Thiamin monophosphate} + \text{ADP}.$$

$$\text{Thiamin monophosphate} + \text{ATP} \rightleftharpoons \text{thiamin diphosphate} + \text{ADP}.$$

The erythrocytes are a good and readily available source of enzymes of the pentose-phosphate pathway such as transketolase. The transketolase activity is thus measured in the presence and absence of added thiamin diphosphate. In normal subjects the activity is not increased by addition of the cofactor, in contrast to the situation in thiamin-deficient subjects.

For reviews on genetically inherited enzyme deficiencies see references 18, 19, 24, 25.

10.7 The use of enzymes to determine the concentrations of metabolites of clinical importance

With the wider availability of purified enzymes it has become possible in recent years to measure a variety of metabolites by enzymatic methods. The principle is to use an enzyme to transform a metabolite into its product and then to estimate the amount transformed by utilizing the difference in the properties of the substrate and product. There are a number of advantages in using enzymatic methods to estimate metabolites in body fluids such as serum or urine. These methods can make use of the high specificity of the enzyme (see Chapter 1, Section 1.3.2) to estimate the metabolite in the presence of many other substances, thus avoiding purification steps that may be necessary prior to a chemical analysis. Enzymatic methods can sometimes be used to measure concentrations of labile substances that would be degraded by harsher chemical methods. There may also be certain disadvantages. The cost of purified enzymes may be too high for routine use. This will depend on the amount required to carry out each assay in a reasonably short time. In some cases it is now becoming possible to use immobilized enzymes that can then be reused (see Chapter 11, Section 11.5). Enzymes are susceptible to inactivation by a variety of agents. An inhibitor present in serum or urine might, for example, completely inhibit enzyme activity or it may only cause partial inhibition. In the latter case we might obtain a false result unless the time course of the reaction were studied to check that complete reaction had occurred.

A number of points have to be checked before an enzyme can be reliably used in the measurement of the concentration of a metabolite. (For a detailed discussion see reference 26.) The equilibrium position of the reaction is important and it is best, if possible to use a reaction in which the equilibrium is far to the right in order to obtain a quantitative conversion of the metabolite to the product of the reaction. If the equilibrium is not so favourable, this can sometimes be overcome either by removing the products by chemical trapping or by coupling to a second enzyme reaction. For example, the equilibrium constant for glucose-6-phosphate isomerase is close to unity,

$$K = \frac{[\text{fructose 6-phosphate}]}{[\text{glucose 6-phosphate}]} = 0.3,$$

but by coupling this to glucose-6-phosphate dehydrogenase the product of the isomerase is effectively removed.

1. D-Fructose 6-phosphate $\rightleftharpoons$ D-glucose 6-phosphate.
2. D-Glucose 6-phosphate + $NADP^+$ $\rightleftharpoons$ D-glucono-δ-lactone 6-phosphate + $NADPH + H^+$.
3. D-Glucono-δ-lactone 6-phosphate + H_2O $\rightleftharpoons$ 6-phospho-D-gluconate.

The third step, which is the spontaneous non-enzymic hydrolysis of D-glucono-δ-lactone 6-phosphate, ensures that reactions 1 and 2 proceed to virtual completion.

The equilibrium for the glutamate dehydrogenase reaction lies to the left, but glutamate can be estimated in terms of the NADH produced,

$$\text{L-Glutamate} + NAD^+ + H_2O \rightleftharpoons \text{2-oxoglutarate} + NADH + NH_4^+,$$

provided the 2-oxoglutarate is trapped by reaction with hydrazine:

$$\begin{array}{c} COO^- \\ | \\ C=O \\ | \\ CH_2 \\ | \\ CH_2 \\ | \\ COO^- \end{array} + H_2NNH_2 \rightarrow \begin{array}{c} COO^- \\ | \\ C=NNH_2 \\ | \\ CH_2 \\ | \\ CH_2 \\ | \\ COO^- \end{array} + H_2O$$

A sufficiently large amount of enzyme must be used so that the substrate is almost completely ($\geqslant$99 per cent) converted to the product in a short time (usually a few minutes). This will depend on the V_{max} for the enzyme and also on the K_m in relation to the amount of metabolite to be estimated. If the latter is, for example, approximately equal to the K_m, then when the substrate has become 99 per cent converted to the products the rate will only be 1 per cent of V_{max}.

The specificity of the enzyme should be known, since some enzymes that do not show absolute specificity will convert related substrates or homologues to their products, often at a reduced rate. If the enzyme preparation is not pure, then side-reactions will occur. Some examples of the use of enzymes to assay metabolites that are important in clinical chemistry are given below.

10.7.1 Blood glucose

The oral glucose tolerance test is still the most widely used diagnostic procedure in the investigation of glucose metabolism in man, e.g. in the investigation of diabetes. A subject is usually given an oral test dose of glucose and the blood glucose concentration is measured for the subsequent three to four hours. In a normal subject the blood glucose concentration rises during the initial 30–60 min and then falls rapidly back to the basal level due to the increased utilization by the tissues. In a diabetic subject the increase in the concentration of blood glucose is more pronounced and the decline to normal levels is much slower. The extent of the rise and the rate of the subsequent fall depends on the severity of diabetes. The test requires several measurements of blood glucose. Chemical methods that are used depend on the reducing properties of glucose and therefore are not specific for glucose. There are three enzymatic methods that are generally available: hexokinase coupled with glucose-6-phosphate dehydrogenase, glucose oxidase coupled with peroxidase, and glucose dehydrogenase.

1. $\begin{cases} \text{D-Glucose} + ATP \rightleftharpoons \text{D-glucose 6-phosphate} + ADP. \\ \text{D-Glucose 6-phosphate} + NADP^+ \rightleftharpoons \text{D-glucono-}\delta\text{-lactone 6-phosphate} + NADPH + H^+. \end{cases}$

2. $\begin{cases} \text{D-Glucose} + H_2O + O_2 \rightleftharpoons \text{D-gluconic acid} + H_2O_2. \\ H_2O_2 + \text{dye}_{reduced} \rightleftharpoons H_2O + \text{dye}_{oxidized}. \end{cases}$

3 D-Glucose + NAD^+ $\rightleftharpoons$ D-glucono-δ-lactone + $NADH + H^+$.

In the first method the NADPH produced would be measured by the change in A_{340}. Although hexokinase is not specific for glucose, glucose-6-phosphate dehydrogenase is highly specific for glucose 6-phosphate and thus the specificity is built into the second step. The second method using glucose oxidase is more frequently used because the enzymes and substrates are less expensive and the final absorption is in the visible range. Glucose oxidase specifically oxidizes β-D-glucose, and α-D-glucose is oxidized 150 times slower. Blood glucose contains an equilibrium mixture of both isomers. Most glucose oxidase preparations contain some aldose 1-epimerase as an impurity that enables rapid epimerization to occur and the total glucose to be estimated. The hydrogen peroxide produced in the first step oxidizes an acceptor, which then has a changed absorption spectrum. The most commonly used acceptor was *o*-dianisidine; however, on account of its mild carcinogenic properties it has been replaced by 2,2′-azino-di-(3-ethylbenzthiazoline)-6-sulphonate, which absorbs at 400–450 nm. This replacement also increases the sensitivity of the assay. The concentrations of glucose determined by glucose oxidase assays are usually lower than the true concentrations; this is due to the presence of substances in serum that interfere with the peroxidase reaction by acting as alternative acceptors.

The third method entails glucose dehydrogenase (EC 1.1.1.47) from *Bacillus cereus* and has the advantages that it does not require any additional coupling enzymes and also that the glucose dehydrogenase can be immobilized on nylon tubing and therefore can be reused several times (see reference 27; for further discussion on immobilized enzymes, see Chapter 11, Section 11.5.4).

10.7.2. Uric acid

Uric acid is the end product of purine metabolism in humans. Measurement of uric acid in blood and urine is important in the diagnosis of gout, when the levels of uric acid are raised. They are also raised in other conditions, e.g. in leukaemia and after ingestion of food rich in nucleoproteins. Uric acid may be estimated by either chemical or enzymatic methods. The chemical method involves the oxidation of uric acid to allantoin using phosphotungstic acid, whereas in the enzymatic method urate oxidase is used to catalyse a similar oxidation

Chemical

$$\text{Uric acid} + H_3PW_{12}O_{40} \rightarrow \text{allantoin} + CO_2 + \text{tungsten blue.}$$

Enzymatic

$$\text{Uric acid} + 2H_2O + O_2 \rightleftharpoons \text{allantoin} + CO_2 + H_2O_2.$$

The chemical method has the disadvantage of lack of specificity, whereas the enzymatic method has the advantage that urate oxidase shows absolute specificity but the possible disadvantage of requiring measurement of absorption in the ultraviolet region. The enzyme measurement uses the difference in absorbance at 293 nm between uric acid and allantoin.

The hydrogen peroxide formed could be coupled to peroxidase as in the glucose oxidase assay to enable measurements to be made in the visible part of the spectrum. The enzymatic method can be adapted for automated uric acid determination, whereas this is not easily possible with the chemical method.[28]

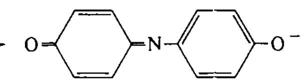
10.7.3 Urea

Determinations of the levels of urea in serum are frequently carried out as a test of renal function, especially in conjunction with urinary urea estimation, which

enables the glomerular filtration rate to be assessed. Urea is hydrolysed by urease to ammonium carbonate. The ammonia released is reacted with sodium hypochlorite and phenol under alkaline conditions to form indophenol that can be estimated spectrophotometrically.

$$\text{urea} + 2H_2O \xrightleftharpoons{\text{urease}} (NH_4)_2CO_3$$

$$NH_3 + NaOCl + 2C_6H_5OH \longrightarrow$$



Indophenol

10.7.4 Cholesterol, cholesterol esters, and triglycerides

The measurement of the concentrations of cholesterol and cholesterol esters in the serum is important in assessing the risk factors in arteriosclerosis and in myocardial infarction. High circulating levels of cholesterol and cholesterol esters are indicative of high risk. Cholesterol can be measured by a colorimetric method that is not highly specific and thus not very satisfactory. It also involves the use of concentrated sulphuric acid, which can be a nuisance in routine analysis. The introduction of straightforward enzymatic methods of high specificity has greatly simplified procedures. Cholesterol can be measured using a single enzyme, cholesterol oxidase, and cholesterol esters can be measured by using this enzyme together with cholesterol esterase. The product of cholesterol oxidase action, 4-cholesten-3-one, can be estimated by its absorption at 240 nm.

$$\text{Cholesterol ester} + H_2O \rightleftharpoons \text{cholesterol} + RCOO^-.$$

$$\text{Cholesterol} + O_2 \rightleftharpoons \text{4-cholesten-3-one} + H_2O_2.$$

The measurement of serum triglycerides in the plasma or serum is useful in following the course of diabetes mellitus, nephrosis, and biliary obstruction. It can also be measured enzymatically, usually after alkaline hydrolysis of the triglyceride. The glycerol formed can then be assayed using the following coupled assays:

$$\text{ATP} + \text{glycerol} \xrightleftharpoons{\text{glycerol kinase}} sn\text{-glycerol 3-phosphate} + \text{ADP}.$$

$$\text{ADP} + \text{phosphoenolpyruvate} \xrightleftharpoons{\text{pyruvate kinase}} \text{ATP} + \text{pyruvate}.$$

$$\text{Pyruvate} + \text{NADH} + H^+ \xrightleftharpoons{\text{lactate dehydrogenase}} \text{L-lactate} + \text{NAD}^+.$$

10.7.5 Other metabolites

A number of other metabolites that are measured less frequently can also be measured by enzymatic methods.

Creatine may be measured by the decrease in A_{340} using creatine kinase coupled to pyruvate kinase and lactate dehydrogenase:

$$\text{Creatine} + \text{ATP} \xrightleftharpoons{Mg^{2+}} \text{phosphocreatine} + \text{ADP}.$$

$$\text{ADP} + \text{phosphoenolpyruvate} \xrightleftharpoons{Mg^{2+}} \text{ATP} + \text{pyruvate}.$$

$$\text{Pyruvate} + \text{NADH} + \text{H}^+ \rightleftharpoons \text{L-lactate} + \text{NAD}^+.$$

The measurement of creatinine is important in the creatinine-clearance test, which is used to test glomerular filtration occurring in the kidney. The colorimetric method that has been used for many years is relatively non-specific and so the recent development of an enzymatic method using creatininase is a considerable improvement. It is used in conjunction with the coupling enzymes just mentioned for the assay of creatine:

$$\text{Creatinine} + \text{H}_2\text{O} \rightleftharpoons \text{creatine}.$$

Ammonia may be estimated using glutamate dehydrogenase:

$$\text{NH}_4^+ + \text{NADH} + \text{2-oxogluturate} \rightleftharpoons \text{L-glutamate} + \text{NAD}^+.$$

Ethanol may be estimated using alcohol dehydrogenase,

$$\text{CH}_3\text{CH}_2\text{OH} + \text{NAD}^+ \rightleftharpoons \text{CH}_3\text{CHO} + \text{NADH} + \text{H}^+,$$

and lactate in a similar way by using lactate dehydrogenase.

10.7.6 Enzyme immunoassay

In addition to the direct use of enzymes to measure the concentrations of their substrates in body fluids, there is also an indirect method in which enzymes are used as a type of amplification system in what is known as enzyme immunoassay. Since the first description of the method of radioimmunoassay by Yalow and Berson in 1960, a large number of methods have been developed for measurement of hormones, drugs, and a variety of small biological molecules that are based on antigen–antibody interactions. These methods rely on the specificity of antigen–antibody interactions for binding the species to be assayed, together with some method of increasing the sensitivity of detection by use of either a radioactive label or an enzyme label.

Small molecules such as hormones and drugs do not induce antibody formation when injected into a foreign species. However, if they are chemically combined to a protein such as albumin, and then are injected into an animal, they will induce antibody formation. The antibodies so formed will combine specifically with the antigen, forming an insoluble antigen–antibody complex. The antibodies will also combine with the original small molecule (known as a hapten) although they will not give rise to a precipitate. Immunoassays can thus be devised for any molecule (acting as a hapten) that, when coupled to a protein, is capable of inducing antibody formation. The method thus has potentially very wide application.

The second stage of the method is to detect the antigen–antibody or antibody–hapten complexes, preferably when they are present in low concentrations. Two of the methods used are either to incorporate a radioactive label on to the antigen or antibody (radioimmunoassay), or to incorporate an enzyme molecule on to the antigen or antibody (enzyme immunoassay). Radioimmunoassays do not in general entail the use of enzymes* and thus are outside the scope of this book (a good review of the method is given in reference 29). Radioimmunoassays are, however, being used in the detection of prostatic acid phosphatase and also the isoenzyme CK-2 of creatine kinase because they provide a more sensitive means of detection.[30]

The procedure of enzyme immunoassay is outlined in Fig. 10.6. The drug, hormone, etc. (designated compound L) to be assayed is chemically coupled to a protein and this conjugate is injected into a rabbit to raise the necessary antibodies. The rabbit serum is fractionated to obtain the fraction that is to be used as the reagent. A second conjugate is now made, this time between compound L and a purified enzyme. This conjugate must retain enzyme activity. A number of enzymes have been used in this type of coupling, e.g. alkaline phosphatase, β-galactosidase, horseradish peroxidase, lysozyme, malate dehydrogenase, and glucose-6-phosphate dehydrogenase. The solution containing an unknown concentration of compound L can now be assayed. The solution is mixed with a fixed concentration of the antibodies, containing an excess of antibody combining sites. A known concentration of enzyme-labelled compound L is then added and a portion of this combines with the remaining antibody-combining sites. It is then necessary to be able to measure the amount of free enzyme-labelled compound L. How this is done depends on whether the enzyme-labelled compound L–antibody complex shows enzyme activity (Fig. 10.6). In some cases the combination with the antibody blocks and the active site inhibits enzyme activity. The free and bound enzyme-labelled compound L can then be quantitated without prior separation. This is sometimes referred to as homogeneous enzyme immunoassay. In other cases, when the active site is not blocked, it is necessary to separate the free and bound enzyme-labelled compound L prior to measurement of enzyme activity (heterogeneous enzyme immunoassay).

There are a number of variants of this type of immunoassay. In one the enzyme is covalently linked to an antibody that is able to bind L (immuno-enzymetric assay). Another involves a sandwich technique in which an antibody–antigen complex involving the ligand is detected by a second antibody against the first. The second antibody has an enzyme covalently linked to it. The variants on the basic techniques are described in reference 31.

The procedure is thus very simple once a suitable antibody and enzyme-labelled compound L has been obtained. This method has been developed to

* Lactoperoxidase (iodide peroxidase EC 1.11.1.8) is sometimes used to catalyse the incorporation of ^{131}I or ^{125}I into the antigen.

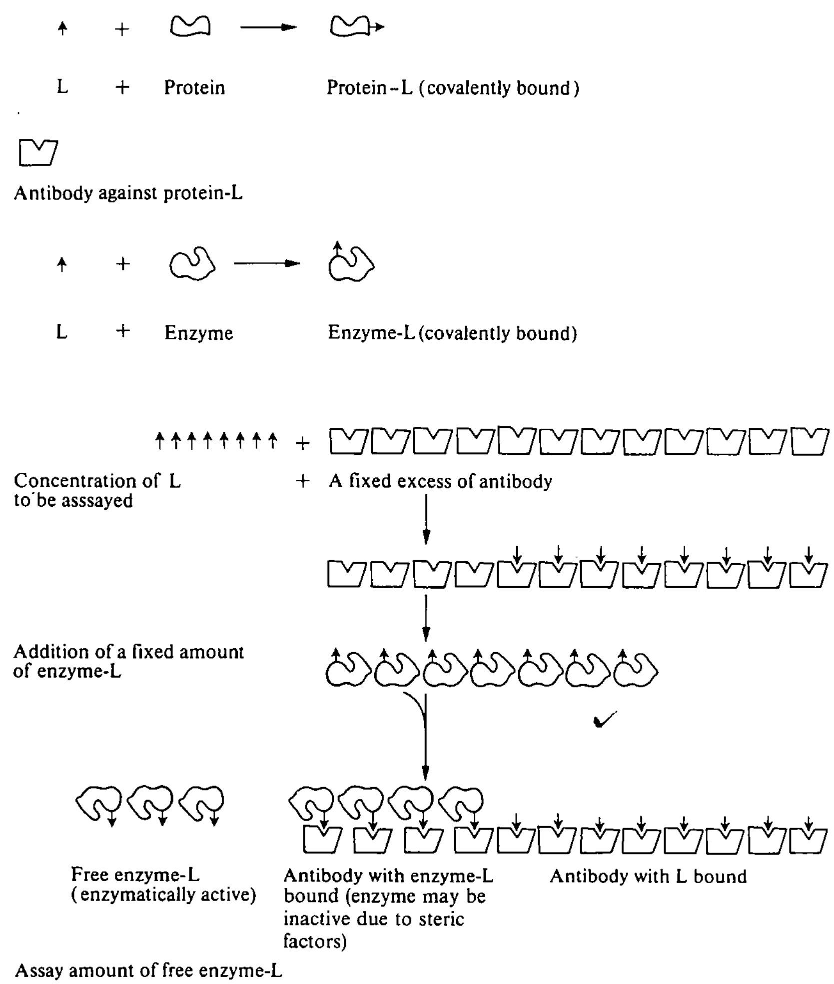


FIG. 10.6. The method of enzyme immunoassay. The diagram illustrates how this method is used to assay compound L after formation of a covalent derivative of compound L with a purified enzyme.

measure a variety of drugs, including opiates, barbiturates, amphetamine, cocaine, anticancer drugs, and hormones such as thyroxine, and to screen antigens from pathogenic organisms, and is being used increasingly in clinical laboratories. It has two advantages over radioimmunoassays in that the reagents are stable and it avoids the need to use radioactive materials. (For a review, see reference 32.) The method is likely to be improved in specificity by use of high-affinity monoclonal antibodies.

10.8 Enzyme therapy

Although the attempts at enzyme therapy in clinical trials have so far met with limited success, the technique of administering enzymes for therapeutic purposes seems sufficiently promising that is may become a feasible approach for treatment of certain diseases within the next decade. The principal areas in which work on enzyme therapy has so far concentrated are the removal of toxic substances from the blood, genetic deficiency diseases, and cancer. In addition, enzyme therapy may be used in the following areas:

(1) degradation of necrotic tissue by use of proteolytic enzymes such as trypsin, chymotrypsin, or bromelain;
(2) removal of blood clots using streptokinase or urokinase (see Chapter 11, Section 11.4.3.8); and
(3) treatment of pancreatic insufficiency by administration of suitably entrapped enzymes, e.g. proteolytic enzymes, as in cystic fibrosis.

How the enzyme is administered depends on which of the above objectives is being considered. If the enzyme is used to remove or metabolize a substance present in the blood, e.g. a toxic metabolite or a blood clot, then it is not necessary for the enzyme to enter the intracellular compartments; it is only necessary for the enzyme to be present in the blood. This type of application may be either intra- or extracorporeal. The latter, which involves the use of a bypass as in kidney dialysis, has the advantage that immobilized enzymes may be used and these can be retained outside the body and are not cleared by the kidney. Immobilized enzymes also have the advantage of being less immunogenic. There are several examples of this type of application,[33] such as removal of circulating urea by urease in cases of kidney failure, removal of bilirubin by bilirubin oxidase, removal of fibrin from blood clots by fibrolysin, and removal of asparagine by asparaginase (see Section 10.8.2). For intracellular enzyme therapy, it is necessary for the enzyme to be taken up by the appropriate target cells.

For more detailed accounts of enzyme therapy, see references 34–36.

10.8.1 Treatment of genetic deficiency diseases

There are now a large number of genetic deficiency diseases in which the enzyme deficiency has been identified (Table 10.2), although these only afflict small proportions of the population. The only treatment that is at present available for a number of these diseases is careful regulation of the diet to reduce the substrate loading and thereby minimize the harmful effects of the defect, e.g. for phenylketonuria, a diet low in phenylalanine; for galactosaemia, a diet low in galactose; for maple syrup disease, a diet low in leucine, isoleucine, and valine. The ultimate cure for these diseases would be to insert the gene that codes for the deficient enzyme into chromosomes. Although there are still many problems to be solved before this may be feasible, significant progress has been made. For example, phenylalanine hydroxylase, the enzyme deficient in phenylketonuria has been cloned and has been introduced into cultured human cells using a retrovirus.[21]

The introduction of the gene in somatic cells (somatic replacement), as opposed to germ cells, may thus become a feasible proposition. A more immediate prospect is the replacement of the deficient enzyme. Although purified enzymes are available for many of these deficiency diseases, there are formidable problems in delivering the enzyme to the required site in such conditions that it will remain stable and functional for a reasonable time.

When considering enzyme therapy to alleviate genetic deficiency disease, the prospects of a successful treatment are likely to be best in cases in which the enzyme can be most easily targeted to the tissue or organelle required, and also in cases in which an enzyme treatment for a limited period is likely to be of value. The latter is a consideration because any enzyme applied is likely to be turned over and eventually cleared from the system. These restrictions have made lysosomal storage diseases the prime candidates for enzyme therapy.[36] These are a group of diseases in which one of the normal lysosomal enzymes is deficient. Various substances are normally taken up into the lysosomes by the process of endocytosis, in which a portion of the membrane surrounds the substance and is then involuted and pinched off within the lysosome, and are subsequently degraded. However, in lysosomal storage diseases the substances are endocytosed but then accumulate. Because the lysosomes have the capability to endocytose foreign material, it would seem reasonable that they should take up added enzymes. There is also quite a good understanding of the signals that cause lysosomal proteins made in the Golgi apparatus to be taken up by the lysosomes. The presence of a mannose-6-phosphate residue on the protein targets it to the lysosomes. Work has been in progress since 1964 on attempts at replacement therapy in lysosomal storage diseases and a number of technical problems have been identified. Quite large amounts of the appropriate enzyme are required in a high state of purity and in a non-immunogenic form. Some of the early problems arose from immunological reactions, especially when the enzyme was obtained from a different organism. Many enzymes, when administered, are either inactivated or degraded fairly rapidly, although some experiments carried out using fibroblasts grown in culture have shown that enzymes can be taken up from the culture medium and will function within the cells. The lysosomal storage disease most studied is Gauchers disease. In this disease a glucocerebroside accumulates largely in macrophages. In this case a single treatment with a glucocerebrosidase might be expected to clear the lysosomes of glucocerebroside, but results so far show only marginal effects. The enzyme, although reaching the required site, has a very short half-life.

Perhaps the most promising developments are in the area of enzyme stabilization and in the vehicle used to supply the enzyme. In a number of cases it has been shown that enzymes may be stabilized against degradation either by their encapsulation within a matrix or by cross-linking to other substances; thus, cross-linking human α-galactosidase A to an anti-α-galactosidase antibody using hexamethyl isocyanate increased the half-life of the enzyme under physiological conditions from 9 min to 46 min.[37] A method that provides protection of the enzyme and may also improve its uptake is entrapment in erythrocyte ghosts or in liposomes.[38] The former may be prepared by lysing erythrocytes (these could be the subject's own erythrocytes) in dilute saline containing the enzyme and then resealing them in isotonic saline, thereby entrapping some of the enzyme. Liposomes containing enzyme are prepared from a mixture of cholesterol, phosphatidyl choline, and phosphatidic acid that is sonicated in aqueous suspension together with the enzyme. They have been prepared containing, for example, amyloglucosidase, and have been used in the treatment of a glycogen storage disease with encouraging results.[39]

10.8.2 Cancer therapy

The enzyme therapy that has been most tested as potential anticancer treatment has involved administration of asparaginase:

$$\text{L-Asparagine} + H_2O \rightleftharpoons \text{L-aspartate} + NH_3$$

it was observed that injection of normal guinea-pig serum caused complete regression of a lymphosarcoma in mice and delay in the appearance of other tumours in experimental animals. These effects occurred only when guinea-pig serum was used. Extensive analysis showed that the active factor present in guinea pig was asparaginase. This enzyme is widely distributed in plants, animals, and bacteria but it is absent from the serum of most common mammals. Other sources of asparaginase were tested and also found to cause regression of certain tumours. Bacteria represent one of the best sources of asparaginase and much subsequent experimentation has been carried out using asparaginase from *E. coli.* A number of different types of tumour, but particularly lymphosarcomas, have been found to be sensitive to asparaginase. The precise mechanism of action of asparaginase in causing tumour regression is not completely clear but it appears that the sensitive tumours, in contrast to normal host tissues, lack the ability to synthesize asparagine. Administration of asparaginase thus reduces the circulating level of asparagine. Asparaginase treatment has been given in a number of clinical trials and has been found most effective in relief of acute lymphocytic leukaemia. Unfortunately, there are a number of adverse side-effects such as liver damage and immune reactions, and in addition, repeated treatment can lead to the development of resistance. Attempts are being made to try to improve the treatment, i.e. to reduce the side-effects by administering the enzyme in microcapsular or other immobilized forms[33] or by giving asparaginase in combination with other chemotherapy. Other enzymes, e.g. glutaminase, serine dehydratase, arginase, phenylalanine ammonia-lyase, and leucine dehydrogenase, are also being tested as possible anticancer agents,[40] the rationale being that certain types of cancer cells may have lost the ability to synthesize amino acids and are thus dependent on uptake from the serum for supply. Normal cells can synthesize all but the eight essential amino acids. Thus, enzymes that degrade any of the non-essential amino acids from the serum might be effective therapeutic agents against a cancer cell having a particular amino-acid requirement.

10.9 Conclusions

Measurements of enzyme activity are being used increasingly in clinical diagnoses and enzyme preparations are being used increasingly to estimate metabolites and in trials for enzyme therapy. These trends are likely to continue to increase for several reasons. Sophisticated equipment is becoming more widely available in clinical laboratories and many assays can now be performed on automated equipment. A wider variety of pure enzymes is becoming commercially available and many of these have been successfully immobilized to support materials. This should enable the enzymes to be used more economically both in the assay of metabolites and in enzyme-replacement therapy.

In the field of clinical diagnosis, the assay of more than one enzyme or the combination of enzyme assay and isoenzyme distribution can give a reasonable degree of discrimination between different diseases. Many enzymes which are at present difficult to assay routinely may, in the future, be measured by radioimmunoassays or enzyme immunoassays, and this will greatly increase the range of possibilities. This, in turn, could widen the scope of enzyme diagnoses.

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11
Enzyme technology

11.1 Introduction

Enzymes have remarkable catalytic properties (see Chapter 1, Section 1.3.1), especially when compared with other types of catalysts. In recent years they have been exploited on an increasing scale in food, pharmaceutical, and chemical industries. The advantages of high catalytic activity, lack of undesirable side-reactions, and operation under mild conditions are often highly desirable. Also, with the advent of genetic engineering, a wider range of enzymes will become available on a larger scale and thus increase the scope of enzyme technology. Genetic engineering itself creates a demand for certain highly purified enzymes (see Section 11.5.5) and the whole area of enzyme technology is expanding rapidly.

The enzymes present in a number of bacteria and fungi have been used for many years by man, but it is only in recent years that isolated enzymes, as opposed to enzymes present in whole organisms, have been used in industrial processes. We first consider the main applications of reactions that are catalysed by enzymes in intact organisms, then consider the use of isolated enzymes, and follow by discussion of the recent and future applications of immobilized enzymes. This short chapter serves to illustrate some of the new and traditional technological applications of enzymology. There are many articles and books on enzyme technology (see references 1–5).

11.2 Use of microorganisms in brewing and cheesemaking

The oldest industries in which enzymes in living organisms have been used are those of brewing and cheesemaking. These processes were carried out centuries before there was any understanding of their biochemical basis. In fact, much of the emphasis in biochemistry at the end of the last century and in the early part of this century was aimed at understanding these fermentation processes.

As can be seen from Table 11.1, enzymes may be involved in either hydrolysis to monosaccharides or in glycolysis in the process of ethanol production. In the case of beer making, the starting material is the polysaccharide starch, whereas in winemaking the starting material is mainly mono- or disaccharides. The largest carbohydrates that most yeasts are able to ferment are trisaccharides. Therefore, when polysaccharides are used as the principal source of carbohydrates they must be degraded before fermentation can occur. The principal raw material used in brewing is the starch obtained from barley. Barley seeds are allowed to germinate and this leads to release of amylases that cause the breakdown of starch. The starch grains are contained within the cells of the endosperm, the

TABLE 11.1
The use of whole organisms as a source of enzymes in the food industry

Process	Organism used	Enzyme step or steps involved
Brewing and winemaking	Barley (*Hordeum*)	α-Amylase and β-amylase, endo-1,3-β-D-glucanase
	Saccharomyces spp	Limited proteolysis; oligosaccharidases (oligosaccharides → monosaccharides) Glycolytic enzymes (monosaccharides → ethanol)
Cheesemaking	*Streptococcus lactis* *Streptococcus cremoris*	Lactose → lactate + limited lipolysis and proteolysis
	Propionobacteria spp	Conversion of lactate → propionate + acetate $+ CO_2 + H_2O$
	Penicillium camemberti and *Penicillium roqueforti*	Limited lipolysis and proteolysis plus other undefined activities
	Calf stomach	Chymosin (rennin); caseinogen → casein
Vinegar production	*Acetobacter*	Ethanol to acetate

walls of which are surrounded by a number of polymers including 1,3-β-D-glucans. It is necessary for these glucans to be degraded so that the amylases can reach the starch granules (Fig. 11.1). Anaerobic glycolysis is the second stage in the process in which ethanol is the end-product. (For further details of the brewing process see references 6–9). New strains of yeasts that possess amylolytic

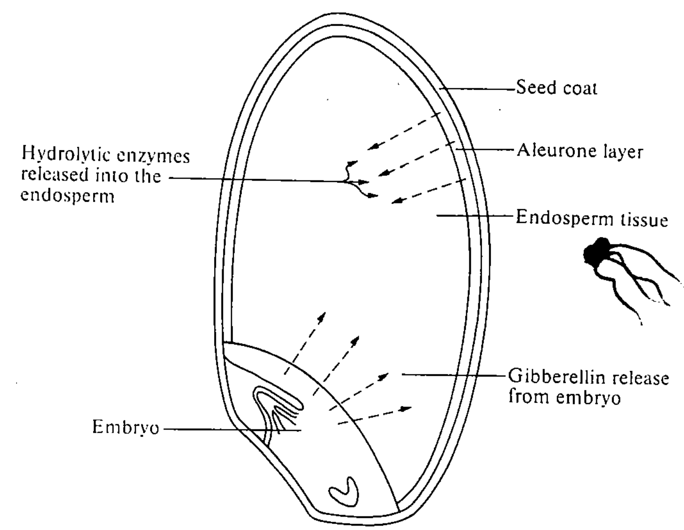


FIG. 11.1 The release of hydrolytic enzymes from a germinating barley seed.

activity are being constructed by recombinant DNA technology. These are able both to degrade starch to glucose and to ferment glucose and may replace traditional yeasts in the future.[10]

It should be remembered that ethanol is not only produced as a beverage but is an important solvent and fuel. There are considerable differences between the production of alcoholic beverages and of industrial ethanol. For the former it is important that the composition of the product is maintained constant and thus all components contributing to the flavour and appearance must be maintained. On the other hand for industrial ethanol, the main consideration is low-cost production. Much industrial alcohol is produced by catalytic hydration of ethene derived from crude oil. However, with the increased cost of petrochemicals there is a greater interest in the fermentation process, particularly if a cheap source of raw material is available. Thus, much effort is being expended on devising efficient methods of degrading the most abundant polysaccharide, cellulose. It is also important to use a strain of yeast that will allow maximum fermentation to occur. For example if sugar beet is used as the raw material, since it contains the sugar raffinose, it is important to use a raffinose-metabolizing strain of yeast. The possibility of production of ethanol using fermentative bacterium such as *Zymomonas* is being explored. It has the advantage of catalysing a faster fermentation[11].

For cheesemaking the main process is that of fermentation of lactose to lactic acid and this is most frequently carried out by *Streptococcus lactis* or *Streptococcus cremoris*. These organisms are used primarily because they have limited metabolic diversity and do not therefore produce large quantities of other products. However, it is the small quantities of other metabolities that are partially responsible for the characteristic flavours of cheeses. Other species of bacteria are used additionally in the case of certain cheeses that have gaseous cavities. These are the result of decarboxylations occurring late in the fermentation process,(usually brought about by *Streptococcus diacetylactis* and *Leuconostoc* spp.)

In Swiss cheese the flavour is partly due to propionic acid formation by *Propionobacteria* spp. The latter species only begins to thrive once appreciable lactic acid formation has occurred. This second fermentation, in which lactic acid is converted to propionic acid, does not result in a high energy yield compared to the energy yield in lactate formation and thus only develops in the later stages of fermentation. The ATP is generated in the decarboxylation of pyruvate.

$$\underset{\text{Lactate}}{3CH_3.CHOH.COO^-} \xrightarrow{-6[H]} \underset{\text{Pyruvate}}{3CH_3.CO.COO^-} \xrightarrow{+6[H]}$$

$$\rightarrow \underset{\text{Propionate}}{2CH_3.CH_2.COO^-} + CH_3COO^- + CO_2 + H_2O$$

The second stage in the production of(Camembert, Brie, or) blue cheeses is the ripening brought about by fungi *Penicillium camemberti* and *Penicillium roqueforti*. The *P. camemberti* grows on the surface of the ripening Camembert or Brie, whereas *P. roqueforti* has the ability to grow under fairly anaerobic conditions within the cheese. Although chemical analyses of the blue cheeses have shown a variety of carboxylic acids, alcohols, and esters present, very little is known about the enzymes involved.(A more detailed account of the organisms used in cheese making is given in references 9, 12 and 13.)

11.3 Use of microorganisms in the production of organic chemicals

Microorganisms are also used to produce a variety of complex organic compounds. In many cases the enzymes involved have not been characterized. The most widely used organisms for these processes are fungi, and several organic acids are synthesized by species of *Aspergillus*. The conditions of growth are arranged so as to produce the maximum amount of acids concerned, e.g. growth of *Aspergillus niger* on high concentrations of sugars and low concentrations of Mn^{2+} and Fe^{3+} gives a high yield of citric acid. In general it is not economical to produce simple organic compounds using microorganisms, since they can be produced more cheaply by chemical syntheses. However, this is not the case with more complex compounds such as antibiotics, for which microorganisms are used.(Table 11.2 summarizes the main processes in which the enzyme complements of certain microorganisms are used to synthesize organic compounds; further details are given in references 14 and 15.)

11.4 Use of isolated enzymes in industrial processes

Although several industrial processes are carried out using whole organisms as a source of enzymes, the efficiency of these processes can often be improved by using isolated enzymes. In addition, enzymes are now being used in a wider variety of processes. The advantages of using isolated enzymes compared to intact organisms are as follows: (i) it may be possible to obtain higher catalytic activity; (ii) undesirable side-reactions may be avoided; (iii) the processes may be able to be carried out more reproducibly. The principal disadvantage is likely to be the higher cost of using isolated enzymes and also that they may be more sensitive to inactivation than the enzymes within an intact organism. It is advantageous to use enzymes from thermophilic bacteria or fungi (when a suitable enzyme is available) so as to reduce the problem of heat inactivation.

11.4.1 Sources of isolated enzymes

The most important sources of isolated enzymes are bacteria and fungi, although enzymes from plant and animal tissues are also used. Fungi and bacteria have the advantage that they can be easily grown and it is usually not difficult to scale up a production process (for reviews, see references 16–18). In addition, the sources are not subject to seasonal or other factors, e.g. chymosin (rennin) production is

TABLE 11.2

The production of some organic compounds by use of microorganisms

Compound	Organism used for synthesis	Enzyme steps if known	Uses of compound
Citric acid	*Aspergillus niger*	Monosaccharides → citrate Glycolysis and tricarboxylic acid cycle involved	Soft drinks, jams, jellies, flavouring, blood transfusion, Sequestering agent in electroplating and leather tanning
Itaconic acid $H_2C{=}C(COOH){-}CH_2{-}COOH$	*Aspergillus* spp	As with citrate formation plus citrate ⇌ aconitate → CO_2 + itaconate	Copolymer in acrylic resins
Gluconic acid	*Aspergillus niger*	Glucose → gluconolactone → gluconate (glucose oxidase)	Calcium gluconate is used in Ca therapy, sequestering agent, food additive, pharmaceutical industry
Amphotericin B Chloramphenicol Cycloheximide Erythromycin Novobiocin Streptomycin	*Streptomyces* spp		Antibiotics
Griseofulvin Penicillin	*Penicillium* spp		Antibiotics
Bacitracin Gramicidin	*Bacillus subtilis* *Bacillus brevis*		Antibiotics
Carotenoid pigments	*Blakeslea trispora*	Synthesis from acetate	Colouring agent
Riboflavin	*Ashbya gossypi*	Synthesis from glycine, formate and CO_2	Nutrient

seasonal because it is dependent on the supply of calf stomachs. The majority of enzymes that have so far been used are hydrolytic enzymes and many of these are produced extracellularly by fungi. The possibility of producing larger quantities of enzymes from animal and plant sources by use of tissue culture methods is now being explored. Some proteins such as vaccines are already being produced in tissue culture.

With microbial enzymes it is often possible to increase the yields by changes in the growth conditions, addition of inducers, or strain selection, including increasing the number of gene copies by genetic engineering.[19–21]

With enzymes from animal and plant sources, the yields may be increased by the introduction of the appropriate genes and their promoter regions into the more rapidly growing microorganisms. However, there are often problems to be overcome before the gene is satisfactorily expressed and extracted from the microorganism as, for example in the case of chymosin in *E. coli*.[22, 41]

11.4.2 Isolation of enzymes

The methods available for isolating enzymes on laboratory scale (Chapter 2) or an industrial scale are the same. However, the criteria for selection of a particular method vary according to use. Enzymes required for food-processing and related industries and for detergents are generally required in large quantities at relatively low purity. Those enzymes required for clinical diagnosis and related areas are generally required in smaller quantities at a higher level of purity. Although a high state of purity is not generally required in food processing, it may be necessary to exclude certain contaminating enzymes; e.g. in biscuit manufacture, where proteases are used it is necessary that the extract containing protease is low in amylase activity. Many of the basic methods of enzyme purification require some adaptation for scaling up.[40]

As mentioned previously, many of the fungal enzymes used are extracellular enzymes and they can readily be separated from the mycelium by filtration. It is then usually necessary to concentrate the extract and this is done by spray drying, which is less costly than precipitation by ammonium sulphate or organic solvents. If the enzyme is intracellular, the mycelium has to be disrupted. Enzymatic methods of breaking the cell wall would be too costly and difficult to scale up, so high-pressure homogenization methods are usually used.

11.4.3 Enzymes isolated on an industrial scale and their applications

The principal isolated enzyme preparations that are used industrially are those concerned with carbohydrate metabolism (Table 11.3) and protein hydrolysis (Table 11.4) and these account for over three-quarters of current sales. Other enzymes that have been used less extensively include AMP deaminase, steroid 11β-isomerase, aminoacylase, streptokinase, penicillinase, hyaluronoglucosaminidase, hyaluronoglucuronidase, triacylglycerol lipase, catalase, keratinase, and phosphodiesterase I.

TABLE 11.3

Carbohydrate-metabolizing enzymes used in industry

Enzyme	Reaction or substrate	Sources of enzyme	Utilization
Amylase	Endohydrolysis of 1,4-α-D-glucosidic linkages in polysaccharides	*Bacillus subtilis* *Aspergillus niger* *Aspergillus oryzae*	Production of sugars and oligosaccharides from starch Enables starch to be fermented
Exo-1,4-α-D glucosidase (amyloglucosidase)	Hydrolysis of terminal 1,4-linked α-D-glucose residues successively from non-reducing end of polysaccharides	*Aspergillus niger* *Aspergillus oryzae* *Rhizopus* sp	Production of glucose from starch
Cellulase	Endohydrolysis of 1,4-β-D-glycosidic linkages in cellulose and β-D-glucans	*Trichoderma viride* *Aspergillus niger*	Cellulose → cellobiose for fermentation (cellulase preparations having very high activity have yet not been obtained)
Polygalacturonase	Random hydrolysis of 1,4-α-D-galactosiduronic linkages in pectin	*Mucor, Botrytis, Pencillium,* and *Aspergillus*	Extraction of fruit juice from pulp, clarification of wines and fruit juices
β-D-Galactosidase	Lactose → glucose + galactose	*Aspergillus niger* *Aspergillus oryzae*	Hydrolysis gives sweeter, more-soluble sugars
β-D-Fructofuranosidase	Sucrose → glucose + fructose	*Aspergillus oryzae* *Saccharomyces*	Hydrolysis gives sweeter, more-soluble sugars
Glucose oxidase	Glucose + O_2 → gluconolactone + H_2O_2	*Aspergillus niger* *Penicillum* spp	Analytical reagent for glucose; desugaring egg products, removing O_2 from mayonnaise and fruit juices susceptible to oxidation
Xylose isomerase (usually referred to as glucose isomerase)	Glucose ⇌ fructose	*Streptomyces* spp *Lactobacillus brevis*	Production of high-fructose syrup

TABLE 11.4

Protein-metabolizing enzymes used in industry

Enzyme	Reaction or substrate	Source of enzyme	Utilization
Papain	Peptide-bond cleavage	*Papaya* latex	Removal of turbidity (due to protein) in beer, meat tenderizing
Chymosin (rennin)	Clotting of milk	Calf stomach	production of cheese
Trypsin Chymotrypsin	Peptide-bond cleavage Peptide-bond cleavage	Animal pancreas Animal pancreas	Meat tenderizers, medical uses
Fungal proteases	Peptide-bond hydrolysis	*Aspergillus oryzae* *Aspergillus niger*	Meat tenderizers, breadmaking to improve viscoelastic properties of doughs, chill-proofing of beer (to prevent haze due to protein–tannin interaction)
		Mucor miehei *Edothia parasitica*	Used as a substitute for chymosin, since causes only limited proteolysis
Bacterial proteases		*Bacillus subtilis*	Detergents, removal of gelatin from films

In some instances isolated enzymes are replacing whole organisms in the processing of foods, but in many cases enzymes are being used in completely new processes. We consider briefly the principal uses of enzymes in industry.

11.4.3.1 *Alcoholic beverages*

In the traditional brewing process, where the starch from barley is hydrolysed prior to fermentation, the barley is allowed to germinate to an extent that ensures sufficient α- and β-amylases are produced for the hydrolytic steps not to be rate-limiting. The addition of hydrolytic enzymes prior to fermentation in brewing has not been extensively practised. Enzyme supplementation has been used more extensively in the production of spirits, where sorghum, potatoes, wheat, or rye may be used as a source of starch. The enzymes most used are amylases from *Bacillus subtilis, Aspergillus oryzae* or *A. niger*. The α-amylase from *B. subtilis* is more heat stable and is used when harsh conditions are required to liquefy particular cereals. However, the extract lacks α-1,6 glucosidase activity and therefore only degrades starch (amylopectin*) to oligosaccharides. The enzymes from *Aspergillus* are also able to cleave α-1,6 glucoside links and therefore degrade the starch more completely into fermentable sugars. When used in brewing, these enzymes result in the production of a low-calorie beer since there is very little oligosaccharide remaining (important for slimming drinkers!).

* Amylopectin is a polysaccharide containing D-glucose units linked α-1,4 and α-1,6.

11.4.3.2 *Breadmaking*

In breadmaking the carbohydrate for fermentation is also derived from the hydrolysis of starch. The aeration (CO_2 production) of bread during its manufacture will depend on adequate α- and β-amylase activity. In some countries, where harvesting conditions do not promote germination, amylase activity becomes a limiting factor. In this case, addition of the fungal amylases (see Section 11.4.3.1) is preferable on account of their greater heat lability. It is important that only limited starch hydrolysis occurs. The fungal enzymes are rapidly inactivated during baking. Fungal proteases may also be used in breadmaking to improve the viscoelastic properties of the dough.

11.4.3.3 *Cheesemaking*

The increased demand for cheese coupled with the reduced availability of chymosin (rennin), which is obtained from calf stomach, has led to a search for a substitute for chymosin. Chymosin causes the clotting of milk, a process which involves cleavage of a single peptide bond in casein κ between Phe 105 and Met 106, releasing the acidic C-terminal peptide. This release causes the micelles containing casein (which normally have a negative charge and so are mutually repelled) to aggregate and clot. (A detailed account is given in reference 23.) Proteolytic enzymes such as trypsin also cause clotting, but then further degrade casein. If this occurs in cheesemaking it leads to undesirable flavours. In the search for a substitute for chymosin, enzymes that cause clotting but only limited proteolysis are desirable. The fungal proteases from *Mucor pusillus*, *Mucor miehei*, and *Edothia parasitica* can substitute for chymosin, although they do not have quite such a good clotting/proteolysis ratio. They are now used on a commercial scale and account for more than one-third of the cheese production world wide.[24]

11.4.3.4 *Meat tenderizing*

Toughness in meat is due primarily to collagen and elastin and also actomyosin. Lower-quality cuts of meat are just as nutritious as prime cuts. Thus, to make maximum use of the carcass, efforts have been made to tenderize the lower-quality cuts. This is done by injecting proteolytic enzymes into the vascular system either before or after slaughter. The enzymes most used are papain, trypsin, chymotrypsin, and also microbial proteases from *Aspergillus*. Over 5 per cent of United States beef is subject to tenderizing procedures by packers.

11.4.3.5 *Sweeteners*

A quantitatively very important use of enzymes is in the production of glucose syrups to be used as sweetening agents in a variety of processed foods. A glucose–fructose mixture is sweeter than glucose alone (glucose is about 70 per cent as sweet as sucrose, whereas fructose is about 60 per cent sweeter than sucrose), and cheaper to produce than sucrose that is extracted from sugar cane or sugar beet. The glucose–fructose mixture is produced from starches, particularly corn starch. Two enzymes are used in the process: exo-1,4-α-D-glucosidase obtained from *Aspergillus niger* and xylose isomerase from *Streptomyces* spp. (The enzyme used for isomerization is xylose isomerase, EC 5.3.1.5, although it is usually referred to as glucose isomerase, since it will also isomerize glucose;[25] $K_{m(\text{D-glucose})} = 0.17$ mmol dm^{-3} and $K_{m(\text{D-xylose})} = 0.01$ mmol dm^{-3}.)

$$\text{Starch} \rightarrow \text{glucose} \rightleftharpoons \text{fructose.}$$

This process is discussed further in Section 11.5.1 on immobilized enzymes.

Another example of the use of an enzyme to cause an increase in sweetness is the use of β-D-galactosidase in ice-cream manufacture. Since the mixture of glucose and galactose is sweeter than lactose, this enzyme can be used to catalyse an increase in sweetness of a product.

11.4.3.6 *Clarification of beers, wines, and fruit juices*

Turbidity in beers, wines, and fruit juices is undesirable because turbid drinks are less acceptable to the consumer. Enzymes have been used to clarify a variety of beverages. In the case of fruit juices they are more easily prepared from extracted pulp if the viscosity is not too high. In the case of beer a haze or turbidity may develop on cooling and this appears to be due to a complex between tannin and protein. It can be successfully removed by addition of small amounts of protease. In the case of wines and fruit juices the turbidity may be due to starch or pectin and is removed by addition of amylases or polygalacturonases, which also cause a decrease in viscosity. (A detailed discussion of the enzymes used is given in reference 26.)

11.4.3.7 *Detergents*

Detergents containing enzymes were developed extensively during the 1960s, so that by 1969 about half of the detergents marketed contained enzymes. Following the indication of possible health hazards, the use of such detergents declined sharply in the early 1970s; since then methods have been developed for encapsulation of enzyme particles using non-ionic surfactants and this has reduced dust inhalation problems with powdered enzymes. The principal enzymes used in 'biological' detergents are mixtures of amylase and neutral and alkaline proteases that are active in the pH range 6.5 to 10 and at temperatures from 303 K (30°C) to 333 K (60°C). Detergents often contain oxidizing and chelating agents and so the enzymes must withstand these also. *Bacillus* neutral protease and *Bacillus* alkaline protease are the most suitable to date, but others from thermophilic organisms having high thermal stability are also being developed.[27] A mixture of carbohydrate-hydrolysing and protein-hydrolysing enzymes of low specificity is most useful, since polysaccharides and proteins require partial degradation before they are removed from garments. By contrast, lipids can be removed without any prior degradation by ionic detergents at alkaline pHs.

11.4.3.8 *Medical applications*

Isolated enzymes are used in medicine both as reagents in analysis and in therapy. We discussed the use of enzymes, e.g. glucose oxidase, in clinical analysis in Chapter 10, Section 10.7. As a therapeutic agent, trypsin has been used as an anti-inflamatory agent and as a wound cleanser. Hydrolytic enzymes have also been used in clinical trials to liquefy blood clots. Blood clots are liquefied under physiological conditions by the action of plasmin. Plasmin is produced slowly in the plasma by the action of fibrinokinase on plasminogen (plasmin precursor). Streptokinase (an enzyme from *Streptococcus haemolyticus*) or urokinase (an enzyme isolated from human urine) are effective activators of plasminogen and

$$\text{Plasminogen} \xrightarrow{\text{fibrinokinase}} \text{Plasmin}$$
$$\text{Insoluble fibrin} \xrightarrow{\text{Plasmin}} \text{Soluble peptides}$$

appear to be useful in relieving peripheral thrombosis. Plasminogen activators are also produced in other tissues. Attempts are being made to develop cell lines that produce larger quantities of these activators. The tissue plasminogen activators are often tissue specific.

Hyaluronidase* obtained from beef testes is a useful agent to aid diffusion when coinjected with other drugs such as antibiotics, adrenalin, heparin, and local anaesthetics. Hyaluronidase hydrolyses polyhyaluronic acid, the main polysaccharide component of connective tissue, and thus aids the diffusion of other substances by reducing the viscosity in the locality of the injection.

We have discussed the possible use of asparaginase in cancer chemotherapy in Chapter 10, Section 10.8.2. For further details of the medical applications of isolated enzymes, see references 28 and 29.

11.5 Immobilized enzymes

Attaching an enzyme to an insoluble support will permit its reuse and continuous use without a difficult recovery process. The attachment to the support may also stabilize the enzyme. If two or more enzymes catalysing a sequence of reactions are immobilized in close proximity to each other, an efficient immobilized multienzyme protein analogue can be produced. There are therefore potential advantages in immobilizing enzymes that are required on a large scale. Much research effort has been devoted during the last decade to producing satisfactory immobilized enzymes for large-scale application. For details of the methods used for immobilization, see references 30–33. A few immobilized enzymes are being used currently in industry (Table 11.5) but a large number are still at the research and development stage.

* The recommended names for the two enzyme activities commonly referred to as hyaluronidase are hyaluronoglucosaminidase and hyaluronoglucuronidase.

TABLE 11.5
Industrial applications of immobilized enzymes and whole cells

Enzyme or whole cells	Carrier	Methods of immobilization	Application
Aminoacylase	DEAE-Sephadex	Adsorption	Preparation of L-amino acids from a racemic DL-amino acid mixture
Xylose isomerase (glucose isomerase)	Duolite A7 (ceramic beads)	Adsorption	Preparation of fructose–glucose mixture
	Amberlite IRA904 exchange resin	Adsorption	
E. coli containing asparate ammonia-lyase	Polyacrylamide	Entrapment	Preparation of L-aspartate
Pseudomonas putida containing arginine deaminase	Polyacrylamide	Entrapment	Preparation of citrulline
Achromobacter liquidum containing histidine ammonia-lyase	Polyacrylamide	Entrapment	Preparation of uraconic acid
E. coli containing penicillin amidase	Polyacrylamide, cellulose	Entrapment	Preparation of 6-aminopenicillanic acid
Brevibacterium ammoniagenes containing fumarate hydratase	Polyacrylamide	Entrapment	Preparation of malate and fumarate
Lactase (β-galactosidase)	Silica particles	Adsorption	Preparation of lactose-free milk

There are a number of ways in which an enzyme may be immobilized, by adsorption, covalent linkage, matrix entrapment, or encapsulation, as summarized in Fig. 11.2. The uses of immobilized enzymes can be broadly subdivided into preparative and analytical. Examples of the former are large-scale isomerization of glucose and production of L-amino acids from racemic mixtures. Use of immobilized glucose oxidase is an example of the latter. We discuss in the following sections the current applications of immobilized enzymes.

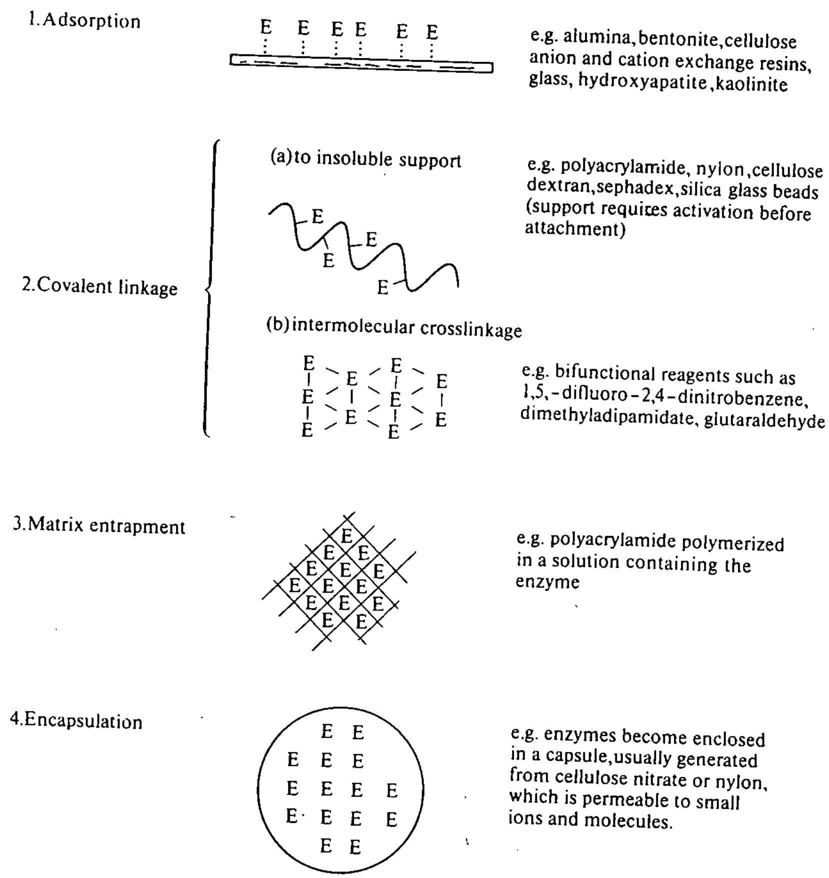


FIG. 11.2. Methods used for enzyme immobilization. (For details see references 31–33).

11.5.1 Production of syrups from corn starch

The production of syrups from starches was mentioned in Section 11.4.3.5. Syrups (sugar solutions) are used extensively as sweetening agents in a variety of manufactured foods. Several million tonnes of high-fructose corn syrup are now produced annually and this is displacing sucrose in traditional applications. It represents one of the biggest developments in food science since 1974. Fructose is more satisfactory than glucose because it is sweeter and does not crystallize so readily from concentrated solutions. The overall process requires two main enzyme-catalysed steps:

$$\text{Starch} \xrightarrow{\text{exo-1,4-}\alpha\text{-D-glucosidase}} \text{glucose} \xrightleftharpoons{\text{xylose isomerase}} \text{fructose.}$$

The first of these steps is catalysed by exo-1,4-α-D-glucosidase obtained from *Aspergillus niger*. This enzyme has been immobilized on porous silica. The second step is catalysed by xylose isomerase obtained from *Streptomyces* spp. This enzyme has been immobilized by attachment of DEAE-cellulose or to a porous ceramic carrier. The latter method has proved more successful in large-scale columns because of the non-compressible nature of the support. The equilibrium constant for the enzyme at 323 K (50°C), the operating temperature of the process, is approximately unity, so that approximately equal amounts of fructose and glucose are present in the product.

11.5.2 Production of L-amino acids from racemic mixtures

Over 5×10^7 kg of DL-amino acids are produced each year. Most of these are produced by synthetic methods and since most biological systems utilize only L-amino acids, a method for resolving the isomers is required for many purposes. Resolution of optically active acids by combining with optically active bases is time-consuming and expensive. Resolution by an enzymatic method that uses an immobilized enzyme is now carried out commercially in Japan. The procedure is shown in Fig. 11.3. The method is based on the specificity of aminoacylase for L-*N*-acetylated amino acids and the differences in solubility of free amino acids compared with their *N*-acetylated derivatives. The enzyme is immobilized by binding to DEAE-Sephadex columns. The columns can easily be recharged when necessary simply by passing more enzyme down the column.

11.5.3 Immobilized whole cells for use in amino-acid interconversions

Besides immobilization of single enzymes it is also possible to immobilize whole cells (for a review, see reference 31). This is often less costly because it does not entail enzyme isolation. The usual method is to entrap the cells by polymerizing acrylamide and *N,N'*-methylenebisacrylamide around them. This has been successfully used in Japan for the production of aspartate, citrulline, phenylalanine, tryptophan, serine, urocanic acid, and 6-aminopenicillanic acid by entrapment of strains of *E. coli* and *Pseudomonas putida* containing the appropriate enzymes.[34] Immobilized *E. coli* cells have been found to have an operational half-life of up to two years.[42] Aspartate and citrulline are used in the pharmaceutical industry, urocanic acid is used as an ultraviolet filter in suntan preparations, and 6-aminopenicillanic acid is used as a precursor of the

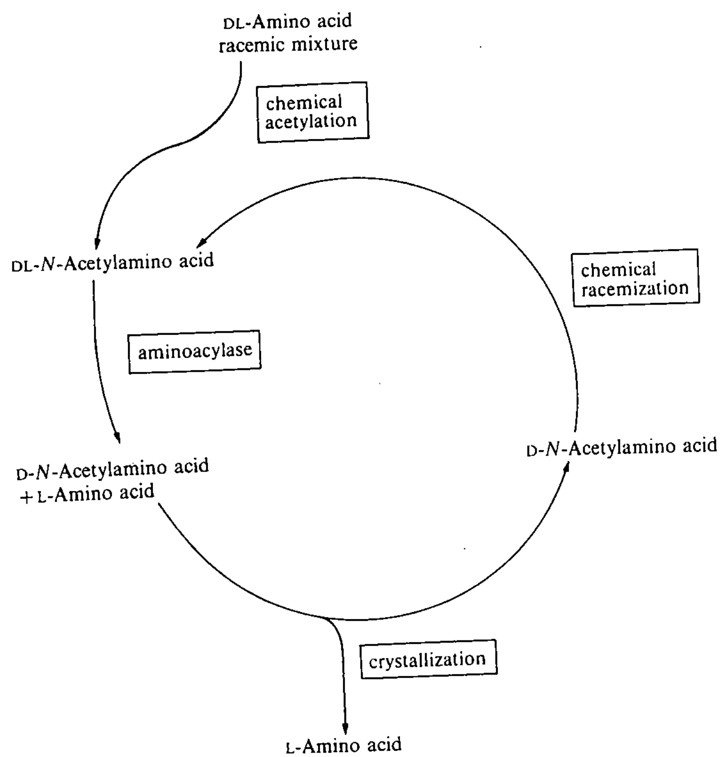
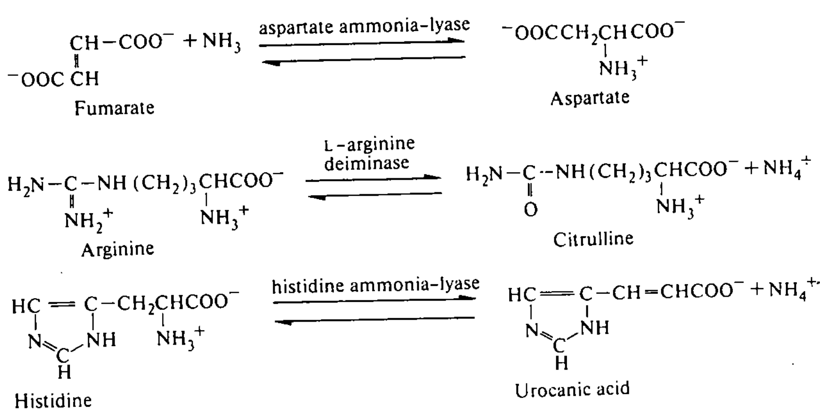
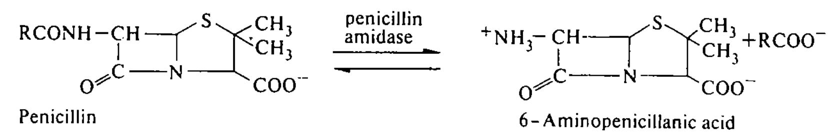


FIG. 11.3. Production of L-amino acids from racemic mixtures of amino acids by use of aminoacylase.

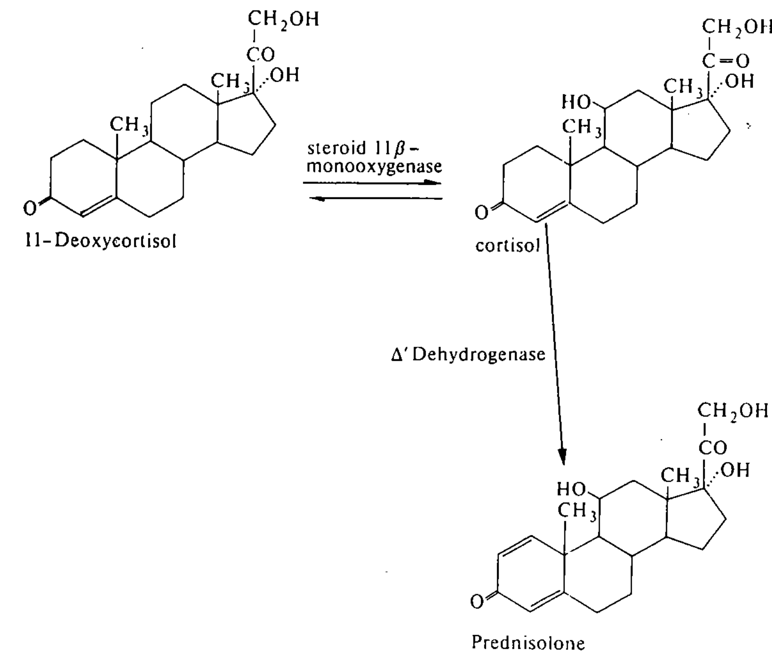
semisynthetic penicillins, e.g. methicillin, oxacillin, carbenicillin, and ampicillin. Aspartate, together with phenylalanine, is used in the sweetening agent Aspartame (*N*-L-α-aspartyl-L-phenylalanine methyl ester). The enzyme catalysed reactions for some of the steps are given below.





11.5.4 Other uses of immobilized enzymes

A number of other applications of immobilized enzymes are being developed. Lactose is a major byproduct of cheese manufacture, since much of the lactose occurs in the whey. A small-scale process in operation in Italy uses immobilized β-galactosidase to hydrolyse lactose to yield a mixture of glucose and galactose, the mixture being more useful as a sweetening agent. Cortisol is a useful drug for arthritis treatment and it can be made from the cheap precursor 11-deoxycortisol. Trials with immobilized steroid 11β-monooxygenase and a Δ'-dehydrogenase are being carried out to produce prednisolone, which is a superior therapeutic agent to cortisone.



The possibility of immobilizing urease to act as an artificial kidney in the removal of urea is being tested:

$$\text{Urea} + H_2O \xrightleftharpoons{\text{urease}} CO_2 + 2NH_3$$

This would be coupled to an absorbing device to remove the ammonia produced.

Immobilized enzymes can also be used in analytical procedures. The most developed application to date is in the analysis of blood glucose using immobilized glucose dehydrogenase. The glucose dehydrogenase from *Bacillus cereus* oxidizes only D-glucose and can be used specifically in the determination of this sugar:

$$\text{D-Glucose} + NAD^+ \rightleftharpoons \text{D-glucono-}\delta\text{-lactone} + NADH + H^+.$$

The NADH formed is measured spectrophotometrically or spectrofluorimetrically. The enzyme has been immobilized on to nylon tubing and thus can be reused, so that as little as 0.18 mg can be used for 1500 glucose determinations.

Another analytical development is that of an enzyme electrode that can be used for the continuous monitoring of specific substrates. It has the advantage over spectrophotometric methods that it can be used with fluids that are optically opaque, such as blood or culture media. An example of an enzyme electrode is that used to measure urea. Urease is immobilized within a gel layer in close proximity to cation-sensitive electrode that measures NH_4^+ concentration. As the urea is hydrolysed in close proximity to the electrode, the electrode detects the high local concentration of NH_4^+. A similar type of electrode can be used to monitor glutamate (Fig. 11.4). The arrangement is that NAD^+ is immobilized to

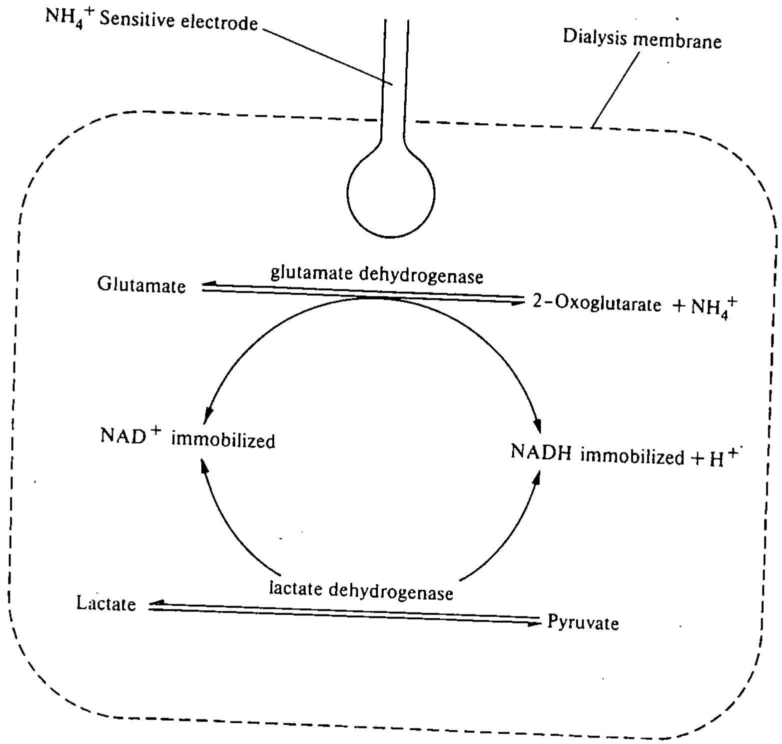


FIG. 11.4. Diagrammatic arrangement of an electrode used to monitor the concentration of glutamate (see reference 35).

an insoluble dextran, this and glutamate dehydrogenase and lactate dehydrogenase are all entrapped in a dialysis membrane surrounding the ammonium-ion-sensitive electrode. Again the NH_4^+ generated in close proximity to the electrode results in an electrode potential proportional to the logarithm of the glutamate concentration. Enzyme electrodes have been devised for measurement of a wide range of amino acids, lactic acid, acetic acid, formic acid, penicillin, and alcohols.[36]

11.5.5 Future developments

Several processes using immobilized enzymes have been tested at the pilot plant level, and can be expected to be developed on an industrial scale in the next few years. In addition, there are a number of areas where new developments are hoped for. These include immobilized enzymes for the oxidation of hydrocarbons, for the synthesis of steroids, vitamins, and antibiotics, for nitrogen fixation, for removal of pesticides from water supplies, and for the photolysis of water by a suitable photosynthetic cell.

The oxidation of hydrocarbons to specific alcohols and acids is expensive to carry out chemically. There are microbial cells capable of oxidizing carbohydrate, but the enzymes or the cells have not yet been successfully immobilized. We mentioned in Section 11.5.4 the production of prednisolone from 11-deoxycortisol. Similar procedures should permit the production of more steroids and antibiotics by the same type of method. An encapsulated cell capable of fixing nitrogen would enable some of the atmospheric nitrogen to be used as a souree of nitrogen to be incorporated into amino acids and proteins. Similarly, stabilization of a cell or components of a cell capable of carrying out photosynthesis would permit the utilization of solar energy for the photolysis of water to produce hydrogen as a fuel. There are still many problems associated with the production of a stable immobilized system that can photolyse water.[37]

There is still a great need for an enzymatic system capable of degrading cellulose: 40–60 per cent of waste disposed of by major cities is composed of cellulose or its derivatives. If cellulose and its derivatives could be efficiently degraded to monosaccharides a considerable saving of energy would result. There are two main difficulties here. Firstly, the cellulases that have been studied and isolated so far do not have very high specific activities. Secondly, cellulose is insoluble and occurs interwoven with other insoluble fibrous components of the cell. This greatly limits the area of contact between the enzyme and substrate and thus enzymatic degradation is likely to be very inefficient. There would be no advantages in this case in using an immobilized cellulase for the early stages of degradation, though immobilized cellulase might be used after the cellulose had been degraded to a soluble form. A number of cellulase genes have been isolated and cloned; the prospect of commercial preparations having high cellulase activity is now good.[38]

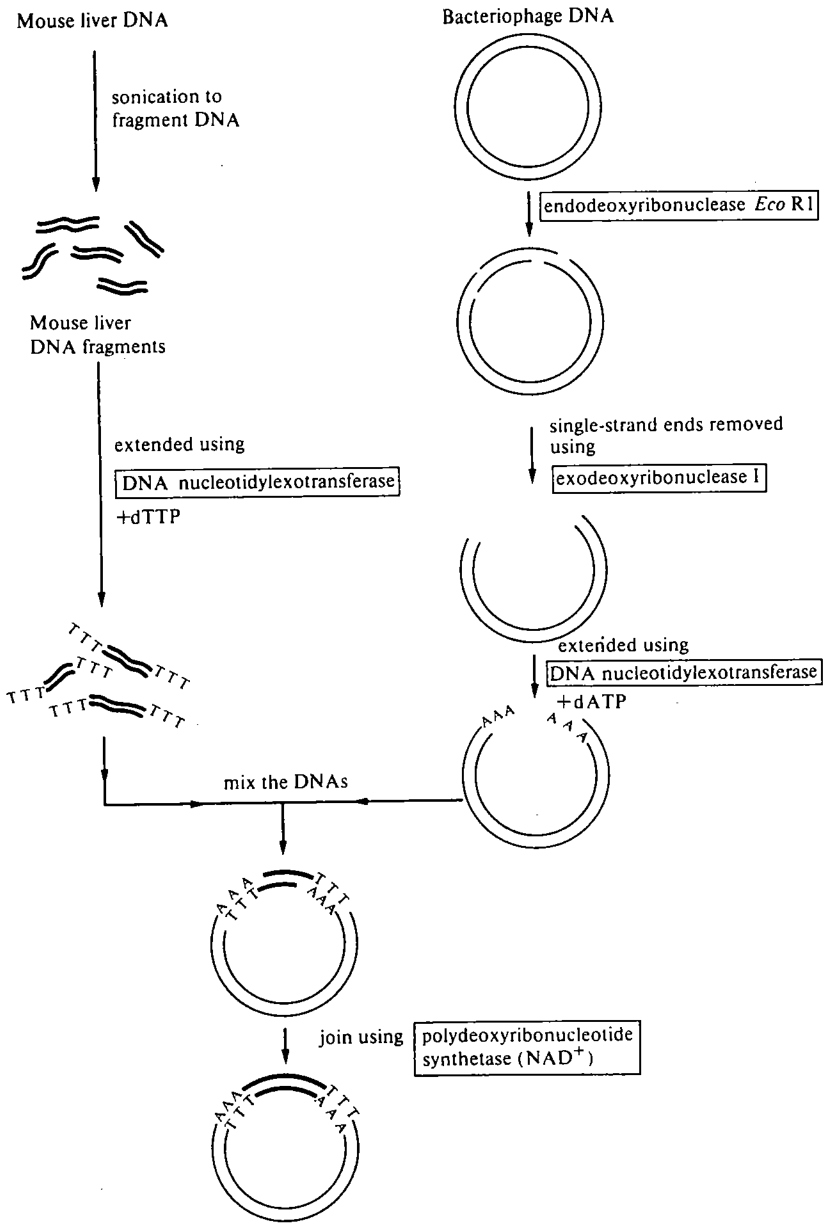


FIG. 11.5. Diagram to illustrate the use of enzymes for the production of recombinant DNA using mouse liver DNA and bacteriophage DNA.

Finally, further developments may be expected by combining the developments in enzyme immobilization with those in genetic engineering, including site-directed mutagenesis. By the process of genetic engineering it will be possible to produce certain proteins on a much larger scale than is possible at present. Proteins produced in relatively small amounts in higher eukaryotes may be produced, as a result of gene transfers, in much larger quantities in prokaryotes or lower eukaryotes, e.g. fungi, since these organisms can multiply much faster. Genetic engineering entails fragmenting DNA from one organism at specific base sequences using restriction endonucleases. The DNA fragments so obtained have to be separated and then joined to a suitable vector, such as plasmid or virus, in order to introduce the DNA into a second species. (For details, see references 19–21) The gene transfer thus entails the use of restriction endonucleases, polydeoxyribonucleotide synthetases (DNA ligases), and DNA nucleotidylexotransferases (terminal transferases) as shown in Fig. 11.5. Using site-directed mutagenesis (see Chapter 5, Section 5.4.5) it is possible to replace specific amino-acid residues; in some cases this may improve the catalytic efficiency of the enzyme. Work is in progress aimed at improving xylose isomerase for commercial applications.[39] The present and future developments in genetic engineering have thus created a demand for these purified enzymes that will undoubtedly be increased by future industrial applications.

There are still problems to be overcome for the successful transcription and translation of genes from higher eukaryotes in lower organisms and for recovery of expressed protein.[41] However, when these technical problems have been overcome, enzymes that are now available in only small quantities will become available in larger quantities. This availability on larger-scale a wider variety of enzymes will further increase the scope for development.

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Appendix
Enzymes referred to in Chapters 1–11

The enzymes referred to in the 11 chapters of this book are listed below in alphabetical order generally under their recommended names as in *Enzyme nomenclature* (Recommendations (1984) of the Nomenclature Committee of the International Union of Biochemistry), except in a few cases where trivial names are more frequently used. A few of the enzymes listed have not yet been classified, usually because insufficient information is at present available for this to be done. In these cases we have indicated what is known about the probable nature of the reaction concerned. The specificities of a number of proteases are given in an abbreviated form, e.g. arg- means the amide bond cleaved is that on the carboxyl side of arginine, whereas -leu means that cleavage occurs on the amino side of leucine.

Enzyme / reaction	EC number
Acetylcholinesterase	
Acetylcholine + H_2O ⇌ choline + acetate	3.1.1.7
Acetyl-CoA acetyltransferase	
Acetyl-CoA + Acetyl-CoA ⇌ CoA + acetoacetyl-CoA	2.3.1.9
Acetyl-CoA carboxylase	
ATP + Acetyl-CoA + CO_2 + H_2O ⇌ ADP + orthophosphate + malonyl-CoA	6.4.1.2
Acetylglutamate kinase	
ATP + *N*-acetyl-L-glutamate ⇌ ADP + *N*-acetyl-L-glutamate 5-phosphate	2.7.2.8
β-N-acetyl-D-hexosaminidase	
R-2-Acetamido-2-deoxy-β-D-hexoside + H_2O ⇌ ROH + 2-acetamido-2-deoxy-hexose	3.2.1.52
Acid phosphatase	
An orthophosphoric monoester + H_2O ⇌ an alcohol + orthophosphate	3.1.3.2
Aconitase	
Citrate ⇌ *cis*-aconitate + H_2O	
The preferred name is aconitate hydratase	4.2.1.3
[Acyl-carrier-protein] acetyltransferase	
Acetyl-CoA + [acyl-carrier protein] ⇌ CoA + acetyl-[acyl-carrier protein]	2.3.1.38
[Acyl-carrier-protein] malonyltransferase	
Malonyl-CoA + [acyl-carrier protein] ⇌ CoA + malonyl-[acyl-carrier protein]	2.3.1.39

[Acyl-carrier-protein] palmityltransferase
A component of certain eukaryote fatty-acid synthase systems
Palmityl-CoA + [acyl-carrier protein] $\rightleftharpoons$ CoA + palmityl-[acyl-carrier protein]

Acyl-CoA desaturase
Stearyl-CoA + AH_2 + O_2 $\rightleftharpoons$ oleyl-CoA + A + $2H_2O$ — 1.14.99.5

Acyl-CoA synthetase
ATP + an acid + CoA $\rightleftharpoons$ AMP + pyrophosphate + An acyl-CoA — 6.2.1.3

Adenosinetriphosphatase
ATP + H_2O $\rightleftharpoons$ ADP + orthophosphate — 3.6.1.3

Adenylate cyclase
ATP $\rightleftharpoons$ 3′:5′ cyclic AMP + pyrophosphate — 4.6.1.1

Adenylate kinase
ATP + AMP $\rightleftharpoons$ ADP + ADP — 2.7.4.3

Alanine aminotransferase
L-Alanine + 2-oxoglutarate $\rightleftharpoons$ pyruvate + L-glutamate — 2.6.1.2

Alcohol dehydrogenase
Alcohol + NAD^+ $\rightleftharpoons$ aldehyde or ketone + NADH — 1.1.1.1

Aldose 1-epimerase
α-D-Glucose $\rightleftharpoons$ β-D-glucose — 5.1.3.3

Alkaline phosphatase
An orthophosphate monoester + H_2O $\rightleftharpoons$ an alcohol + orthophosphate — 3.1.3.1

Amidophosphoribosyltransferase
5-Phospho-β-D-ribosylamine + pyrophosphate + L-glutamate $\rightleftharpoons$ L-glutamine + 5-phospho-α-D-ribose 1-diphosphate + H_2O — 2.4.2.14

Amine oxidase (flavin-containing)
$RCH_2NH_2 + H_2O + O_2 \rightleftharpoons RCHO + NH_3 + H_2O_2$ — 1.4.3.4

Amino-acid acetyltransferase
Acetyl-CoA + L-glutamate $\rightleftharpoons$ CoA + *N*-acetyl-L-glutamate — 2.3.1.1

Aminoacylase
An *N*-acyl-aminoacid + H_2O $\rightleftharpoons$ a fatty acid anion + an aminoacid — 3.5.1.14

Aminoacyl-tRNA synthetases (ligases)
One enzyme for each of the amino acids

ATP + L-amino acid + $tRNA^{AA}$ $\rightleftharpoons$ AMP + pyrophosphate + L-aminoacyl-$tRNA^{AA}$ — 6.1.1.(1–7, 9–12, 14–22)

5-Aminolaevulinate synthase
Succinyl-CoA + glycine $\rightleftharpoons$ aminolaevulinate + CoA + CO_2 — 2.3.1.37

Aminopeptidase (cytosol)
Aminoacyl-peptide + H_2O $\rightleftharpoons$ amino acid + peptide — 3.4.11.1

AMP deaminase
AMP + H_2O $\rightleftharpoons$ IMP + NH_3 — 3.5.4.6

α-Amylase
Endohydrolysis of 1,4-α-D-glucosidic linkages in polysaccharides containing three or more 1,4-α-linked D-glucose units — 3.2.1.1

β-Amylase
Hydrolysis of 1,4-α-D-glucosidic linkages in polysaccharides so as to remove successive maltose units from the non-reducing ends of the chains — 3.2.1.2

Amylo-1,6-glucosidase
Endohydrolysis of 1,6-α-D-glucosidic linkages at points of branching in chains of 1,4-linked α-D-glucose residues. Together with EC 2.4.1.25, these two activities constitute the glycogen *debranching* system — 3.2.1.33

Anthranilate phosphoribosyltransferase
N-(5′-Phospho-D-ribosyl)-anthranilate + pyrophosphate $\rightleftharpoons$ anthranilate + 5-phospho-α-D-ribose 1-diphosphate

In many organisms this enzyme exists as a complex with anthranilate synthase (EC 4.1.3.27) — 2.4.2.18

Anthranilate synthase
Chorismate + L-glutamine $\rightleftharpoons$ anthranilate + pyruvate + L-glutamate

In many organisms this enzyme exists as a complex with anthranilate phosphoribosyltransferase (EC 2.4.2.18) — 4.1.3.27

Arginase
L-Arginine + H_2O $\rightleftharpoons$ L-ornithine + urea — 3.5.3.1

Arginine deiminase
L-Arginine + H_2O $\rightleftharpoons$ L-citrulline + NH_3 — 3.5.3.6

Argininosuccinate lyase
L-Argininosuccinate $\rightleftharpoons$ fumarate + L-arginine — 4.3.2.1

Argininosuccinate synthase
ATP + L-citrulline + L-aspartate $\rightleftharpoons$ AMP + pyrophosphate + L-argininosuccinate — 6.3.4.5

Arginyl-tRNA synthetase (ligase)
ATP + L-arginine + $tRNA^{Arg}$ $\rightleftharpoons$ AMP + pyrophosphate + L-arginyl-$tRNA^{Arg}$ — 6.1.1.19

Aromatic-L-amino acid decarboxylase
L-Tryptophan $\rightleftharpoons$ tryptamine + CO_2 — 4.1.1.28

Arylsulphatase
A phenol sulphate + H_2O $\rightleftharpoons$ a phenol + sulphate — 3.1.6.1

Asparaginase
L-Asparagine + H_2O $\rightleftharpoons$ L-aspartate + NH_3 — 3.5.1.1

Aspartate aminotransferase
L-Aspartate + 2-oxoglutarate $\rightleftharpoons$ oxaloacetate + L-glutamate — 2.6.1.1

Aspartate ammonia-lyase
L-Aspartate $\rightleftharpoons$ fumarate + NH_3 — 4.3.1.1

Aspartate carbamoyltransferase
Carbamoylphosphate + L-aspartate $\rightleftharpoons$ Orthophosphate + *N*-carbamoyl-L-aspartate — 2.1.3.2

Aspartate kinase
ATP + L-aspartate $\rightleftharpoons$ ADP + 4-phospho-L-aspartate 2.7.2.4

ATP citrate (pro-3S)-*lyase*
ATP + citrate + CoA $\rightleftharpoons$ ADP + orthophosphate + acetyl-CoA + oxaloacetate.

Often referred to as citrate cleavage enzyme 4.1.3.8

Branched-chain 2-oxoacid dehydrogenase
3-Methyl-2-oxobutanoate + lipoamide $\rightleftharpoons$ *S*-(2-methylpropanoyl) dihydrolipoamide + CO_2

Part of multienzyme complex 1.2.4.4

Branching enzyme
See 1,4-α-glucan branching enzyme

Bromelain
A thiol protease, cleaving preferentially at lys-, ala-, tyr-, and gly- 3.4.22.4

Carbamoyl-phosphate synthase (ammonia)
2ATP + NH_3 + CO_2 + H_2O $\rightleftharpoons$ 2ADP + orthophosphate + carbamoyl phosphate

This enzyme is present in the livers of ureotelic vertebrates. 6.3.4.16

Carbamoyl-phosphate synthase (glutamine-hydrolysing)
2ATP + glutamine + CO_2 + H_2O $\rightleftharpoons$ 2ADP + orthophosphate + glutamate + carbamoyl phosphate

This enzyme is present in bacteria, fungi, and mammals. Both carbamoyl phosphate synthases from fungi are EC 6.3.5.5, as is the mammalian enzyme that participates in pyrimidine biosynthesis. 6.3.5.5

Carbonate dehydratase
H_2CO_3(or $H^+ + HCO_3^-$) $\rightleftharpoons$ CO_2 + H_2O

Trivial name often used is carbonic anhydrase 4.2.1.1

Carboxylesterase
A carboxylic ester + H_2O $\rightleftharpoons$ an alcohol + a carboxylic acid anion 3.1.1.1

Carboxypeptidase A
Peptidyl-L-amino acid + H_2O $\rightleftharpoons$ peptide + L-amino acid 3.4.17.1

Carboxypeptidase B
Peptidyl-L-lysine (-L-arginine) + H_2O $\rightleftharpoons$ peptide + L-lysine (or L-arginine) 3.4.17.2

Carnitine palmitoyltransferase
Palmitoyl-CoA + L-carnitine $\rightleftharpoons$ CoA + L-palmitoylcarnitine 2.3.1.21

Catalase
H_2O_2 + H_2O_2 $\rightleftharpoons$ O_2 + $2H_2O$ 1.11.1.6

Cathepsin B
Hydrolyses proteins with a specificity resembling that of papain 3.4.22.1

Cathepsin D
Protease with specificity similar to, but narrower than that of pepsin A 3.4.23.5

Cathepsin G
Protease with specificity similar to that of chymotrypsin 3.4.21.20

Cathepsin L
Hydrolysis of proteins: no action on acylamino acid esters 3.4.22.15

Cellulase
Endohydrolysis of 1,4-β-D-glycosidic linkages in cellulose, lichenin and cereal β-D-glucans 3.2.1.4

Chitinase
Random hydrolysis of 1,4-β-acetamido-2-deoxy-D-glucoside linkages in chitin and chitodextrin 3.2.1.14

Chitin synthase
UDP-2-Acetamido-2-deoxy-D-glucose + [1,4-(2-acetamido-2-deoxy-β-D-glucosyl)]$_n$ $\rightleftharpoons$ UDP + [1,4-(2-acetamido-2-deoxy-β-D-glucosyl)]$_{n+1}$ 2.4.1.16

Cholesterol acyltransferase
Acyl-CoA + cholesterol $\rightleftharpoons$ CoA + cholesterol ester 2.3.1.26

Cholesterol esterase
A cholesterol ester + H_2O $\rightleftharpoons$ cholesterol + a fatty-acid anion 3.1.1.13

Cholesterol oxidase
Cholesterol + O_2 $\rightleftharpoons$ 4-cholesten-3-one + H_2O_2 1.1.3.6

Cholinephosphotransferase
CDPcholine + 1,2-diacylglycerol $\rightleftharpoons$ CMP + a phosphatidyl-choline 2.7.8.2

Cholinesterase
An acylcholine + H_2O $\rightleftharpoons$ choline + a carboxylic-acid anion 3.1.1.8

Chorismate synthase
3-Phospho-5-enolpyruvoylshikimate $\rightleftharpoons$ chorismate + orthophosphate 4.6.1.4

Chymosin
Protease, cleaves a single bond in casein κ

Trivial name often used is rennin 3.4.23.4

Chymotrypsin
Protease, preferential cleavage: tyr-, trp-, phe-, leu- 3.4.21.1

Citrate (si) *synthase*
Citrate + CoA $\rightleftharpoons$ acetyl-CoA + H_2O + oxaloacetate 4.1.3.7

Clostripain
Protease, with preferential cleavage of peptides: arg–, especially arg–pro bond 3.4.22.8

Collagenase (vertebrate)
Cleaves preferentially one bond in native collagen leaving a *N*-terminal (75%) and C-terminal (25%) fragment 3.4.24.7

Creatine kinase
ATP + Creatine $\rightleftharpoons$ ADP + phosphocreatine 2.7.3.2

Creatininase
Creatinine + H_2O $\rightleftharpoons$ creatine 3.5.2.10

Crotonyl-[acyl-carrier-protein] hydratase
D-3-Hydroxybutyryl-[acyl-carrier protein] $\rightleftharpoons$ crotonyl-[acyl-carrier protein] + H_2O 4.2.1.58

3′:5′-Cyclic-nucleotide phosphodiesterase
Nucleoside 3′:5′-cyclic phosphate + $H_2O \rightleftharpoons$ nucleoside 5′-phosphate — 3.1.4.17

Cystathionine-γ-lyase
L-Cystathionine + $H_2O \rightleftharpoons$ L-cysteine + NH_3 + 2-oxobutyrate — 4.4.1.1

Cytidylate kinase
ATP + (d)CMP $\rightleftharpoons$ ADP + (d)CDP — 2.7.4.14

Cytochrome b_5 *reductase*
NADH + 2 ferricytochrome $b_5 \rightleftharpoons$ NAD^+ + 2-ferrocytochrome b_5 — 1.6.2.2

Cytochrome c *oxidase*
4 Ferrocytochrome $c + O_2 \rightleftharpoons$ 4 ferricytochrome $c + 2H_2O$ — 1.9.3.1

Debranching enzyme
Consists of two activities: amylo-1,6-glucosidase and 4-α-D-glucanotransferase

3-Dehydroquinate dehydratase
3-Dehydroquinate $\rightleftharpoons$ 3-dehydroshikimate + H_2O — 4.2.1.10

3-Dehydroquinate synthase
7-Phospho-3-deoxy-D-*arabino*-heptulosonate $\rightleftharpoons$ 3-dehydroquinate + orthophosphate — 4.6.1.3

Deoxyribonuclease I
Endonucleolytic cleavage of DNA to 5′-phospho-dinucleotide and 5′-phospho-oligonucleotide end-products — 3.1.21.1

Deoxyribonuclease II
Endonucleolytic cleavage of DNA to 3′-phospho-mononucleotides and 3′-phospho-oligonucleotides — 3.1.22.1

Dihydrolipoamide acetyltransferase
Acetyl-CoA + dihydrolipoamide $\rightleftharpoons$ CoA + S^6-acetylhydrolipoamide

A component of the pyruvate dehydrogenase multienzyme system — 2.3.1.12

Dihydrolipoamide dehydrogenase
Dihydrolipoamide + $NAD^+ \rightleftharpoons$ lipoamide + NADH

A component of the pyruvate, oxoglutarate and branched chain 2-oxoacid dehydrogenase multienzyme complexes — 1.8.1.4

Dihydrolipoamide succinyltransferase
Succinyl-CoA + dihydrolipoamide $\rightleftharpoons$ CoA + S^6-succinyldihydrolipoamide

A component of the oxoglutarate dehydrogenase multienzyme system — 2.3.1.61

Dihydro-orotase
L-5,6-Dihydro-orotate + $H_2O \rightleftharpoons$ *N*-carbamoyl-L-aspartate — 3.5.2.3

Dipeptidyl peptidase
A number of enzymes of differing specificities catalysing the reaction
Dipeptidyl-polypeptide + $H_2O \rightleftharpoons$ dipeptide + polypeptide — 3.4.14.1–5

DNA nucleotidylexotransferase
n Deoxynucleoside triphosphate + $(\text{deoxynucleotide})_m \rightleftharpoons$ *n* pyrophosphate + $(\text{deoxynucleotide})_{m+n}$ — 2.7.7.31

DNA nucleotidyltransferase
n Deoxynucleoside triphosphate $\rightleftharpoons$ *n* pyrophosphate + DNA_n

Often referred to as DNA polymerase — 2.7.7.7

Elastase (pancreatic)
Protease preferentially cleaving bonds involving the carbonyl groups of amino acids bearing uncharged non-aromatic side chains. Hydrolysis of elastin — 3.4.21.36

Endodeoxyribonuclease

A group of endodeoxyribonucleases often referred to as restriction endonucleases. The Type II enzymes (EC 3.1.21.4) catalyse sequence-specific cleavage. There are over 400 enzymes of differing specificity known.

Endo-1,3-β-D-glucanase
Hydrolysis of 1,3-β-D-glucosidic linkages in 1,3-β-D-glucans — 3.2.1.39

Endoproteases A and B (yeast)
These enzymes are classified as EC 3.4.23.6 (*Saccharomyces* aspartic protease) and EC 3.4.21.48, respectively

Endoprotease Glu-C
A serine protease from *Staphylococcus aureus* with preferential cleavage glu-, asp- — 3.4.21.19

Enolase
2-Phospho-D-glycerate $\rightleftharpoons$ phosphoenolpyruvate + H_2O — 4.2.1.11

3-enolpyruvoylshikimate-5-phosphate synthase
Phosphoenolpyruvate + shikimate 5-phosphate $\rightleftharpoons$ orthophosphate + 3-enolpyruvoylshikimate 5-phosphate — 2.5.1.19

Enoyl-[acyl-carrier-protein] reductase (NADPH)
Acyl-[acyl-carrier protein] + $NADP^+ \rightleftharpoons$ 2,3-dehydroacyl-[acyl-carrier protein] + NADPH — 1.3.1.10

Enteropeptidase
Selective cleavage of lys[6]-ile[7] bond in trypsinogen; often referred to as enterokinase — 3.4.21.9

Exodeoxyribonuclease I
Exonucleolytic cleavage of DNA in the 3′ to 5′ direction to yield 5′-phosphomononucleotides — 3.1.11.1

Exo-1,4-α-D-glucosidase
Hydrolysis of terminal 1,4-linked α-D-glucose residues successively from the non-reducing ends of the chains with release of β-D-glucose — 3.2.1.3

Fatty-acid synthase system
A multienzyme protein; for reactions catalysed, see individual catalytic activities by reference to Fig. 7.16. This is now known as Fatty-acyl–CoA synthase; the enzymes from animal sources and from yeast are given EC numbers 2.3.1.85 and 2.3.1.86, respectively

Ficin
A thiol protease preferentially cleaving lys-, ala-, tyr-, gly-, asn-, leu-, val- — 3.4.22.3

β-D-Fructofuranosidase
Hydrolysis of terminal non-reducing β- D-fructofuranoside residues in β-D-fructofuranosides — 3.2.1.26

Fructokinase
ATP + D-fructose $\rightleftharpoons$ ADP + D-fructose 6-phosphate — 2.7.1.4

Fructose-bisphosphatase
D-Fructose 1,6-bisphosphate + H_2O $\rightleftharpoons$ D-fructose 6-phosphate + orthophosphate — 3.1.3.11

Fructose-2,6-bisphosphatase
D-Fructose 2,6-bisphosphate + H_2O $\rightleftharpoons$ D-fructose 6-phosphate + orthophosphate — 3.1.3.46

Fructose-bisphosphate aldolase
D-Fructose 1,6-bisphosphate $\rightleftharpoons$ dihydroxyacetone phosphate + D-glyceraldehyde 3-phosphate — 4.1.2.13

Fumarate hydratase
L-Malate $\rightleftharpoons$ fumarate + H_2O
Often referred to as fumarase — 4.2.1.2

Galactokinase
ATP + D-galactose $\rightleftharpoons$ ADP + α-D-galactose 1-phosphate — 2.7.1.6

Galactose 1-phosphate uridylyltransferase
UTP + α-D-galactose1-phosphate $\rightleftharpoons$ pyrophosphate + UDPgalactose — 2.7.7.10

α-D-Galactosidase
Hydrolysis of terminal, non-reducing α- D-galactose residues in α-D-galactosides, including galactose oligosaccharides, galactomannans and galactolipids — 3.2.1.22

β-D-Galactosidase
Hydrolysis of terminal non-reducing β-D-galactose residues in β-D-galactosides — 3.2.1.23

GDPmannose α-D-mannosyltransferase
GDPmannose + heteroglycan $\rightleftharpoons$ GDP + 1,2(1,3)-α-D-mannosylheteroglycan — 2.4.1.48

1,4-α-Glucan branching enzyme
Transfers a segment of 1,4-α-D-glucan chain to a primary hydroxyl group in a similar glucan chain — 2.4.1.18

4-α-D-Glucanotransferase
Transfers a segment of 1,4-α-D-glucan to a new 4-position in an acceptor which may be glucose or a 1,4-α-D-glucan
This activity forms part of the glycogen debranching system — 2.4.1.25

3-Glucanase (now known as Glucan endo-1,3-glucosidase)
See endo-1,3-β-D-glucanase — 3.2.1.39

Glucokinase
ATP + D-glucose $\rightleftharpoons$ ADP + D-glucose 6-phosphate — 2.7.1.2

Glucose dehydrogenase
β-D-Glucose + NAD(P)$^+$ $\rightleftharpoons$ D-glucono-δ-lactone + NAD(P)H — 1.1.1.47

Glucose oxidase
β-D-Glucose + O_2 $\rightleftharpoons$ D-glucono-δ-lactone + H_2O_2 — 1.1.3.4

Glucose-6-phosphatase
D-Glucose 6-phosphate + H_2O $\rightleftharpoons$ D-glucose + orthophosphate — 3.1.3.9

Glucose-6-phosphate dehydrogenase
D-Glucose 6-phosphate + NADP$^+$ $\rightleftharpoons$ D-glucono-δ-lactone 6-phosphate + NADPH — 1.1.1.49

Glucose-6-phosphate isomerase
D-Glucose-6-phosphate $\rightleftharpoons$ D-fructose 6-phosphate — 5.3.1.9

Glucose-1-phosphate uridylyltransferase
UTP + α-D-glucose 1-phosphate $\rightleftharpoons$ pyrophosphate + UDPglucose — 2.7.7.9

α-D-Glucosidase
Hydrolysis of terminal, non-reducing 1,4-linked α-D-glucose residues with release of α-D-glucose — 3.2.1.20
Often referred to as maltase

β-D-glucuronidase
A β-D-glucuronide + H_2O $\rightleftharpoons$ an alcohol + D-glucuronate — 3.2.1.31

Glutamate dehydrogenase
L-Glutamate + H_2O + NAD$^+$ $\rightleftharpoons$ 2-oxoglutarate + NH_3 + NADH — 1.4.1.2

Glutamate dehydrogenase (NAD(P)$^+$)
L-Glutamate + H_2O + NAD(P)$^+$ $\rightleftharpoons$ 2-oxoglutarate + NH_3 + NAD(P)H — 1.4.1.3

Glutaminase
L-Glutamine + H_2O $\rightleftharpoons$ L-glutamate + NH_3 — 3.5.1.2

Glutamine phosphoribosyl pyrophosphate amidotransferase
5-Phospho-β-D-ribosylamine + pyrophosphate + L-glutamate $\rightleftharpoons$ L-glutamine + 5-phospho-α-D-ribose 1-diphosphate + H_2O — 2.4.2.14

Glutamine synthetase
ATP + L-Glutamate + NH_3 $\rightleftharpoons$ ADP + Orthophosphate + L-Glutamine — 6.3.1.2

γ-Glutamyltransferase
(5-L-Glutamyl)-peptide + An amino acid $\rightleftharpoons$ Peptide + 5-L-Glutamylamino acid — 2.3.2.2

Glutathione-insulin transhydrogenase [recommended name: Protein-disulphide reductase (glutathione)]
2 Glutathione + protein-disulphide $\rightleftharpoons$ oxidized glutathione + protein-dithiol — 1.8.4.2

Glutathione reductase (NAD(P)H)
NAD(P)H + oxidized glutathione $\rightleftharpoons$ NAD(P)$^+$ + 2 glutathione — 1.6.4.2

Glyceraldehyde-3-phosphate dehydrogenase
D-Glyceraldehyde 3-phosphate + orthophosphate + NAD$^+$ $\rightleftharpoons$ 3-phospho-D-glyceroyl phosphate + NADH — 1.2.1.12

Glycerol kinase
ATP + glycerol $\rightleftharpoons$ ADP + *sn*-glycerol 3-phosphate — 2.7.1.30

Glycerol-3-phosphate dehydrogenase
sn-Glycerol 3-phosphate + acceptor $\rightleftharpoons$ dehydroxyacetone phosphate + reduced acceptor — 1.1.99.5

Glycerol-3-phosphate dehydrogenase (NAD$^+$)
sn-Glycerol 3-phosphate + NAD$^+$ $\rightleftharpoons$ dihydroxyacetone phosphate + NADH — 1.1.1.8

Glycerol-3-phosphate acyltransferase
Acyl-CoA + *sn*-glycerol 3-phosphate ⇌ CoA + 1-acylglycerol 3-phosphate 2.3.1.15

Glycogen synthase
UDPglucose + (1,4-α-D-glycosyl)$_n$ ⇌ UDP + (1,4-α-D-glucosyl)$_{n+1}$ 2.4.1.11

Hexokinase
ATP + D-hexose ⇌ ADP + D-hexose 6-phosphate 2.7.1.1

Histidine ammonia-lyase
L-Histidine ⇌ urocanate + NH_3 4.3.1.3

Homogentisate 1,2-dioxygenase
Homogentisate + O_2 ⇌ 4-maleylacetoacetate 1.13.11.5

Homoserine dehydrogenase
L-Homoserine + NAD(P)$^+$ ⇌ L-aspartate β-semialdehyde + NAD(P)H 1.1.1.3

Hyaluronoglucosaminidase
Random hydrolysis of 1,4-linkages between 2-acetamido-2-deoxy-β-D-glucose and D-glucuronate residues in hyaluronate 3.2.1.35

Hyaluronoglucuronidase
Random hydrolysis of 1,3-linkages between β-glucuronate and 2-acetamido-2-deoxy-D-glucose residues in hyaluronate 3.2.1.36

Hydrogenase
2 Reduced ferredoxin + 2H$^+$ ⇌ 2 oxidized ferredoxin + H_2 1.18.99.1

3-Hydroxyacyl-CoA dehydrogenase
(*S*)-3-hydroxyacyl-CoA + NAD$^+$ ⇌ 3-oxoacyl-CoA + NADH 1.1.1.35

3-Hydroxybutyrate dehydrogenase
D-3-Hydroxybutyrate + NAD$^+$ ⇌ acetoacetate + NADH 1.1.1.30

Hydroxymethylglutaryl-CoA reductase
Mevalonate + CoA + 2NAD$^+$ ⇌ 3-hydroxy-3-methylglutaryl-CoA + 2NADH 1.1.1.88

Indole-3-glycerol-phosphate synthase
1-(2′-Carboxyphenylamino)-1-deoxyribulose 5-phosphate ⇌ 1-*C*-(3′-indolyl)-glycerol 3-phosphate + CO_2 + H_2O 4.1.1.48

Iodide peroxidase
Iodide + H_2O_2 ⇌ iodine + 2H_2O

Used to iodinate tyrosine residues 1.11.1.8

Isocitrate dehydrogenase (NAD$^+$)
threo-D$_S$-Isocitrate + NAD$^+$ ⇌ 2-oxoglutarate + CO_2 + NADH 1.1.1.41

Isocitrate lyase
threo-D$_S$-Isocitrate ⇌ succinate + glyoxylate 4.1.3.1

Isoleucyl-tRNA synthetase (ligase)
ATP + L-isoleucine + tRNA^{Ile} ⇌ AMP + pyrophosphate + L-isoleucyl-tRNA^{Ile} 6.1.1.5

Isomaltase
Hydrolysis of 1,6-α-D-glucosidic linkages in isomaltose and dextrins produced from starch and glycogen by α-amylase.

Also known as oligo-1,6-glucosidase.

The enzyme from intestinal mucosa also catalyses the hydrolysis of sucrose and maltose (see Chapter 8, Section 8.4.3) 3.2.1.10

C_{55} isoprenoid-alcohol kinase
ATP + C_{55}-isoprenoid-alcohol ⇌ ADP + C_{55}-isoprenoid-alcohol phosphate 2.7.1.66

Kynureninase
L-Kynurenine + H_2O ⇌ anthranilate + L-alanine 3.7.1.3

Lactate dehydrogenase
L-Lactate + NAD$^+$ ⇌ pyruvate + NADH 1.1.1.27

Lactose synthase
UDPgalactose + D-glucose ⇌ UDP + lactose 2.4.1.22

Lactoylglutathione lyase
(*R*)-*S*-Lactoylglutathione ⇌ glutathione + methylglyoxal 4.4.1.5

Leucine dehydrogenase
L-Leucine + H_2O + NAD$^+$ ⇌ 4-methyl-2-oxopentanoate + NH_3 + NADH 1.4.1.9

Lysozyme
Hydrolysis of 1,4-β-linkages between *N*-acetylmuramic acid and 2-acetamido-2-deoxy-D-glucose residues in a mucopolysaccharide or muropeptide 3.2.1.17

Malate dehydrogenase
L-Malate + NAD$^+$ ⇌ oxaloacetate + NADH 1.1.1.37

Malate dehydrogenase (oxaloacetate-decarboxylating) (NADP$^+$)
L-Malate + NADP$^+$ ⇌ pyruvate + CO_2 + NADPH

Often referred to as the 'malic' enzyme 1.1.1.40

Malonyl-CoA dependent elongase system
A microsomal membrane enzyme system that catalyses the elongation of acyl-CoAs (C_{16}–C_{22}) using malonyl-CoA and NADPH to acyl-CoAs (C_{18}–C_{24}).

Methionine adenosyltransferase
ATP + L-methionine + H_2O ⇌ orthophosphate + pyrophosphate + *S*-adenosyl-L-methionine 2.5.1.6

NADH dehydrogenase
NADH + acceptor ⇌ NAD$^+$ + reduced acceptor 1.6.99.3

NAD$^+$ nucleosidase
NAD$^+$ + H_2O ⇌ nicotinamide + ADPribose 3.2.2.5

NADP$^+$-cytochrome reductase
NADP$^+$ + 2 ferrocytochrome ⇌ NADPH + 2 ferricytochrome 1.6.2.4

Neuraminidase
Hydrolysis of 2,3-, 2,6-, and 2,8-glycosidic linkages joining terminal non-reducing *N*- or *O*-acylneuraminyl residues to galactose, *N*-acetylhexosamine, or *N*- or *O*-acylated neuraminyl residues in oligosaccharides, glycoproteins, glycolipids, or colominic acid 3.2.1.18

Nicotinamide-nucleotide adenylyltransferase
ATP + nicotinamide ribonucleotide ⇌ pyrophosphate + NAD$^+$ 2.7.7.1

Nitrogenase (flavodoxin)
6 Reduced flavodoxin + $6H^+$ + N_2 + nATP $\rightleftharpoons$ 6 oxidized flavodoxin + $2NH_3$ + nADP + n orthophosphate — 1.19.6.1

Nucleosidediphosphatase
A nucleoside diphosphate + H_2O $\rightleftharpoons$ a nucleotide + orthophosphate — 3.6.1.6

Nucleosidediphosphate kinase
ATP + nucleoside diphosphate $\rightleftharpoons$ ADP + nucleoside triphosphate — 2.7.4.6

Nucleosidemonophosphate kinase
ATP + nucleoside monophosphate $\rightleftharpoons$ ADP + nucleoside diphosphate — 2.7.4.4

Nucleoside triphosphatase
NTP + H_2O $\rightleftharpoons$ NDP + orthophosphate — 3.6.1.15

5′-Nucleotidase
A 5′-ribonucleotide + H_2O $\rightleftharpoons$ A ribonucleoside + orthophosphate — 3.1.3.5

Nucleotide pyrophosphatase
A dinucleotide + H_2O $\rightleftharpoons$ 2 mononucleotides
Substrates include NAD^+, $NADP^+$, FAD, CoA, ATP, and ADP — 3.6.1.9

Ornithine carbamoyltransferase
Carbamoylphosphate + L-ornithine $\rightleftharpoons$ orthophosphate + L-citrulline — 2.1.3.3

Ornithine decarboxylase
L-Ornithine $\rightleftharpoons$ putrescine + CO_2 — 4.1.1.17

Ornithine-oxo-acid aminotransferase
L-Ornithine + a 2-oxoacid $\rightleftharpoons$ L-glutamate γ-semialdehyde + an L-amino acid — 2.6.1.13

Orotate reductase (NADH)
L-5,6-Dihydroorotate + NAD^+ $\rightleftharpoons$ orotate + NADH — 1.3.1.14

Orotidine-5′-phosphate decarboxylase
Orotidine 5′-phosphate $\rightleftharpoons$ UMP + CO_2 — 4.1.1.23

3-Oxoacyl-[acyl-carrier-protein] reductase
D-3-Hydroxyacyl-[acyl-carrier protein] + $NADP^+$ $\rightleftharpoons$ 3-oxoacyl-[acyl-carrier protein] + NADPH — 1.1.1.100

3-Oxoacyl-[acyl-carrier-protein] synthase
Acyl-[acyl-carrier protein] + malonyl-[acyl-carrier protein] $\rightleftharpoons$ 3-oxoacyl-[acyl-carrier protein] + CO_2 + acyl-carrier protein — 2.3.1.41

Oxoglutarate dehydrogenase (also known as oxoglutarate decarboxylase)
2-Oxoglutarate + lipoamide $\rightleftharpoons$ S^6-succinyldihydrolipoamide + CO_2
A component of the oxoglutarate dehydrogenase multienzyme system — 1.2.4.2

Papain
Protease preferentially cleaving arg-, lys-, phe-X- (the peptide bond next but one to the carboxyl group of phenylalanine); limited hydrolysis of native immunoglobulins — 3.4.22.2

Penicillin amidase
Penicillin + H_2O $\rightleftharpoons$ 6-aminopenicillinate + a carboxylic acid anion — 3.5.1.11

Penicillinase (recommended name β-lactamase)
Penicillin + H_2O $\rightleftharpoons$ penicilloate
Other β-lactam substrates are also acted on by the enzyme. — 3.5.2.6

Penicillopepsin
An aspartic protease from *Penicillum janthinellum* which resembles pig pepsin structurally. — 3.4.23.6

Pepsin A
Protease causing preferential cleavage at phe-, leu- usually referred to as pepsin. There are also pepsins B and C, which show more restricted specificity — 3.4.23.1

Peroxidase
Donor + H_2O_2 $\rightleftharpoons$ oxidized donor + $2H_2O$ — 1.11.1.7

Phenylalanine ammonia-lyase
L-Phenylalanine $\rightleftharpoons$ *trans*-cinnamate + NH_3 — 4.3.1.5

Phenylalanine 4-monooxygenase
L-Phenylalanine + tetrahydropteridine + O_2 $\rightleftharpoons$ L-tyrosine + dihydropteridine + H_2O — 1.14.16.1

Phosphodiesterase I
Hydrolytically removes 5′-nucleotides successively from the 3′-hydroxy termini of 3′-hydroxy-terminated oligonucleotides — 3.1.4.1

Phosphoenolpyruvate carboxykinase (GTP)
GTP + oxaloacetate $\rightleftharpoons$ GDP + phosphoenolpyruvate + CO_2 — 4.1.1.32

6-Phosphofructokinase
ATP + D-fructose 6-phosphate $\rightleftharpoons$ ADP + D-fructose 1,6-bisphosphate — 2.7.1.11

6-Phosphofructo-2-kinase
ATP + D-fructose 6-phosphate $\rightleftharpoons$ ADP + D-fructose 2,6-bisphosphate — 2.7.1.105

Phosphoglucomutase
α-D-Glucose 1,6-bisphosphate + α-D-glucose 1-phosphate $\rightleftharpoons$ α-D-glucose 6-phosphate + α-D-glucose 1,6-bisphosphate
This enzyme was previously classified as EC 2.7.5.1 — 5.4.2.2

Phosphogluconate dehydrogenase (decarboxylating)
6-Phospho-D-gluconate + $NADP^+$ $\rightleftharpoons$ D-ribulose 5-phosphate + CO_2 + NADPH — 1.1.1.44

Phosphoglycerate kinase
ATP + 3-phospho-D-glycerate $\rightleftharpoons$ ADP + 3-phospho-D-glyceroyl phosphate — 2.7.2.3

Phosphoglycerate mutase
2,3-Bisphospho-D-glycerate + 2-phospho-D-glycerate $\rightleftharpoons$ 3-phospho-D-glycerate + 2,3-bisphospho-D-glycerate — 5.4.2.1
This enzyme was previously classified as EC 2.7.5.3

Phospho-2-keto-3-deoxy-gluconate aldolase
6-Phospho-2-keto-3-deoxy-D-gluconate $\rightleftharpoons$ pyruvate + D-glyceraldehyde 3-phosphate — 4.1.2.14

Phospho-2-keto-3-deoxy-heptonate aldolase
7-Phospho-2-keto-3-deoxy-D-*arabino*-heptonate + orthophosphate ⇌ phosphoenolpyruvate + D-erythrose 4-phosphate + H_2O — 4.1.2.15

Phospholipase A_1
A phosphatidylcholine + H_2O ⇌ 2-acylglycerophosphocholine + a fatty-acid anion — 3.1.1.32

Phospholipase A_2
A phosphatidylcholine + H_2O ⇌ 1-acylglycerophosphocholine + an unsaturated fatty-acid anion — 3.1.1.4

Phospholipase C
A phosphatidylcholine + H_2O ⇌ 1,2-diacylglycerol + choline phosphate — 3.1.4.3

Phosphorylase
(1,4-α-D-Glycosyl)$_n$ + orthophosphate ⇌ (1,4-α-D-glucosyl)$_{n-1}$ + α-D-glucose 1-phosphate — 2.4.1.1

Phosphorylase kinase
4ATP + 2 phosphorylase *b* ⇌ 4ADP + phosphorylase *a* — 2.7.1.38

Phosphorylase phosphatase
Phosphorylase *a* + 4H_2O ⇌ 2 phosphorylase *b* + 4 orthophosphate — 3.1.3.17

Plasmin
Protease cleaving preferentially lys- >arg-; higher selectivity than trypsin — 3.4.21.7

Polydeoxyribonucleotide synthetase (NAD^+)
NAD^+ + (deoxyribonucleotide)$_n$ + (deoxyribonucleotide)$_m$ ⇌ AMP + NMN + (deoxyribonucleotide)$_{n+m}$

Often referred to as DNA ligase — 6.5.1.2

Polynucleotide 5′-hydroxyl-kinase
ATP + 5′-dephospho-DNA ⇌ ADP + 5′-phospho-DNA — 2.7.1.78

Polygalacturonase
Random hydrolysis of 1,4-α-D-galactosiduronic linkages in pectate and other galacturonans — 3.2.1.15

Proline racemase
L-Proline ⇌ D-proline — 5.1.1.4

Protein disulphide-isomerase
Catalyses the rearrangement of —S—S— bonds in proteins — 5.3.4.1

Protein kinase
ATP + a protein ⇌ ADP + a phosphoprotein

The enzyme from rat tissues is stimulated by cyclic AMP and will activate phosphorylase kinase. Other enzymes will phosphorylate other proteins. Some enzymes are activated by cyclic AMP, some by cyclic GMP and some by neither (see Chapter 6, Section 6.4.2). — 2.7.1.37

Protein phosphatase
A phosphoprotein + $n$$H_2O$ ⇌ a protein + *n* orthophosphate

See Chapter 6, Section 6.4.2. for an account of some of the different protein phosphatases — 3.1.3.17

Protein-tyrosine kinase
ATP + protein-tyrosine ⇌ ADP + protein-tyrosine phosphate — 2.7.1.112

Pyruvate carboxylase
ATP + pyruvate + CO_2 + H_2O ⇌ ADP + orthophosphate + oxaloacetate — 6.4.1.1

Pyruvate dehydrogenase (lipoamide) (also known as pyruvate decarboxylase)
Pyruvate + lipoamide ⇌ S^6-acetyldihydrolipoamide + CO_2

A component of the pyruvate dehydrogenase multienzyme system — 1.2.4.1

[Pyruvate dehydrogenase (lipoamide)] kinase
ATP + [pyruvate dehydrogenase (lipoamide)] ⇌ ADP + [pyruvate dehydrogenase (lipoamide)] phosphate — 2.7.1.99

[Pyruvate dehydrogenase (lipoamide)]-phosphatase
[Pyruvate dehydrogenase (lipoamide)] phosphate + H_2O ⇌ [pyruvate dehydrogenase (lipoamide)] + orthophosphate — 3.1.3.43

Pyruvate kinase
ATP-pyruvate ⇌ ADP + phosphoenolpyruvate — 2.7.1.40

Restriction endonuclease
See Endodeoxyribonuclease

Reverse transcriptase
n Deoxynucleoside triphosphate ⇌ *n* pyrophosphate + DNA_n

Also known as DNA nucleotidyltransferase (RNA-directed)) — 2.7.7.49

Ribonuclease II
Exonucleotic cleavage of RNA in the 3′ to 5′ direction to yield 5′ phosphomononucleotides — 3.1.13.1

Ribonuclease (pancreatic)
Endonucleolytic cleavage of RNA to 3′-phosphomonucleotides and 3′ phosphooligonucleotides ending in Cp or Up with 2′,3′-cyclic phosphate intermediates — 3.1.27.5

Ribulosebisphosphate carboxylase
D-Ribulose 1,5-bisphosphate + CO_2 ⇌ 2 3-phospho-D-glycerate — 4.1.1.39

RNA-methyltransferases
S-Adenosyl-L-methionine + tRNA ⇌ *S*-adenosyl-L-homocysteine + tRNA containing methyl groups

A group of enzymes which methylate tRNA on different positions on the purine or pyrimidine bases — 2.1.1.29–2.1.1.36

RNA nucleotidyltransferase
n Nucleoside triphosphate ⇌ *n* pyrophosphate + RNA_n

Usually referred to as RNA polymerase — 2.7.7.6

L-Serine dehydratase
L-Serine + H_2O ⇌ pyruvate + NH_3 + H_2O — 4.2.1.13

Shikimate dehydrogenase
Shikimate + $NADP^+$ ⇌ 3-dehydroshikimate + NADPH — 1.1.1.25

Shikimate kinase
ATP + shikimate ⇌ ADP + shikimate 5-phosphate — 2.7.1.71

Steroid 11β-monooxygenase
A steroid + reduced adrenal ferredoxin + O_2 ⇌ an 11β-hydroxysteroid + oxidized adrenal ferredoxin + H_2O 1.14.15.4

Streptokinase
Although this protein has been purified, it is not clear whether it possesses a catalytic site. Streptokinase interacts with plasminogen and the complex formed shows plasmin activity (see under plasmin). The nature of activation process has to be clarified.

Subtilisin
Hydrolysis of proteins and peptide amides: isolated from *Bacillus subtilis* 3.4.21.14

Succinate-CoA ligase (GDP-forming)
GTP + succinate + CoA ⇌ GDP + orthophosphate + succinyl-CoA
Sometimes misleadingly referred to as succinate thiokinase 6.2.1.4

Succinate dehydrogenase
Succinate + acceptor ⇌ fumarate + reduced acceptor 1.3.99.1

Sucrose α-D-glucohydrolase
Hydrolysis of sucrose and maltose by an α-D-glucosidase-type action often referred to as sucrase or invertase 3.2.1.48

Terminal deoxyribonucleotidyltransferase
n Deoxynucleoside triphosphate + $(\text{deoxynucleotide})_m$ ⇌ n pyrophosphate + $(\text{deoxynucleotide})_{m+n}$
Also known as terminal transferase 2.7.7.31

Tetrahydrofolate dehydrogenase
5,6,7,8-Tetrahydrofolate + $NADP^+$ ⇌ 7,8-dihydrofolate + NADPH 1.5.1.3

Thermolysin
A metalloprotease cleaving preferentially -leu > -phe
The recommended name for this enzyme is *Bacillus thermoproteolyticus* neutral protease but it is usually referred to as thermolysin 3.4.24.4

Thiosulphate sulphurtransferase
Thiosulphate + cyanide ⇌ sulphite + thiocyanate 2.8.1.1

Threonine dehydratase
L-Threonine + H_2O ⇌ 2-oxobutyrate + NH_3 + H_2O 4.2.1.16

Thrombin
Protease cleaving preferentially arg-: activates fibrinogen to fibrin 3.4.21.5

Thymidine kinase
ATP + thymidine ⇌ ADP + thymidine 5′-phosphate 2.7.1.21

Thymidylate synthase
5,10-Methylenetetrahydrofolate + dUMP ⇌ dihydrofolate + dTMP 2.1.1.45

Transketolase
Sedoheptulose 7-phosphate + D-glyceraldehyde 3-phosphate ⇌ D-ribose 5-phosphate + D-xylulose 5-phosphate 2.2.1.1

Triacylglycerol lipase
Triacylglycerol + H_2O ⇌ diacylglycerol + a fatty-acid anion 3.1.1.3

Triosephosphate isomerase
D-Glyceraldehyde 3-phosphate ⇌ dihydroxyacetone phosphate 5.3.1.1

Trypsin
A protease cleaving preferentially arg-, lys- 3.4.21.4

Tryptophan 2,3-dioxygenase
L-Tryptophan + O_2 ⇌ L-formylkynurenine 1.13.11.11

Tryptophan synthase
L-Serine + indoleglycerol 3-phosphate ⇌ L-tryptophan + glyceraldehyde 3-phosphate 4.2.1.20

Tyrosine aminotransferase
L-Tyrosine + 2-oxoglutarate ⇌ 4-hydroxyphenylpyruvate + L-glutamate 2.6.1.5

Tyrosine 3-hydroxylase
L-Tyrosine + tetrahydropteridine + O_2 ⇌ 3,4-dihydroxy-L-phenylalanine + dihydropteridine + H_2O
Also known as tyrosine 3-monooxygenase 1.14.16.2

Tyrosyl-tRNA synthetase (ligase)
ATP + L-tyrosine + $tRNA^{tyr}$ ⇌ AMP + pyrophosphate + L-tyrosyl-$tRNA^{tyr}$ 6.1.1.1

UDPglucose 4-epimerase
UDPglucose ⇌ UDPgalactose 5.1.3.2

UDPglucose-hexose-1-phosphate uridylyltransferase
UDPglucose + α-D-galactose 1-phosphate ⇌ α-D-glucose 1-phosphate + UDPgalactose 2.7.7.12

UDPglucuronosyltransferase
UDPglucuronate + acceptor ⇌ UDP + acceptor β-D-glucuronide 2.4.1.17

Urate oxidase
Urate + O_2 + $2H_2O$ ⇌ allantoin + CO_2 + H_2O_2
The identity of the products of the reaction is uncertain 1.7.3.3

Urease
Urea + H_2O ⇌ CO_2 + $2NH_3$ 3.5.1.5

Urokinase (recommended name: Plasminogen activator)
Protease cleaving preferentially arg-val in plasminogen 3.4.21.31

Uropepsinogen
The enzyme, pepsin, which is secreted in the stomach in mammals, is subsequently absorbed by the intestine into the blood, whence it is excreted by the kidney as an inactive protease, uropepsinogen. Uropepsinogen is activated to uropepsin at pH 1–2

Xanthine oxidase
Xanthine + H_2O + O_2 ⇌ urate + superoxide 1.1.3.22

Xylose isomerase
D-Xylose ⇌ D-xylulose
Sometimes referred to as glucose isomerase, since it will isomerise D-glucose to D-fructose, but D-xylose is the preferred substrate 5.3.1.5

L-Xylulose reductase
Xylitol + $NADP^+$ ⇌ L-xylulose + NADPH 1.1.1.10

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cmp.mcs.anl.gov/cgi-bin/adv-sea...
www.fst.rdg.ac.uk/courses/fs918/lect12.htm